



Consultancy Services for Design and Supervision for STP, Sewer & TE Networks Systems in Ibri

Concept Design Report
October 2026



PO Box 579, Muscat. PC 113
Sultanate of Oman
Tel: (968) 244 84 292
Fax: (968) 244 83 039
e-mail: Renardet-oman@renardet.net

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Abbreviations and Definitions

Term	Meaning
AAF	Average annual flow: the design average of the treatment plant
APSR	Authority for Public Services Regulation
BOD, BOD5	Biochemical oxygen demand; BOD5 is measured over five days
BS EN	British Standard adopting a European Standard
BWh	Hot desert climate, in the Köppen-Geiger classification
CAPEX / OPEX	Capital expenditure / operating expenditure
CESMM3	Civil Engineering Standard Method of Measurement, third edition
COD	Chemical oxygen demand
CRT	Cost Reflective Tariff: the electricity tariff of the large consumers
EIA	Environmental impact assessment
EPSG:32640	The code of the coordinate system used: UTM zone 40 North on the WGS 84 datum
GHS-POP	Global Human Settlement population grid of the European Commission Joint Research Centre
GIS	Geographic information system
H1 to H6	The six flood hazard classes of the Australian classification (Section 1.2.7)
ISO	International Organization for Standardization
kW, MWh	Kilowatt; megawatt hour
l/s, m ³ /d	Litres a second; cubic metres a day
LPCD	Litres per capita per day
MAFWR	Ministry of Agriculture, Fisheries and Water Resources
MD	Ministerial Decision
MOD	Ministry of Defence: a tariff of the electricity dataset
MoHUP	Ministry of Housing and Urban Planning
NAMA	Nama Water Services, in the name of its design guidelines
NASA POWER	The climate data of the United States National Aeronautics and Space Administration, from the MERRA-2 reanalysis, used for the climate and the wind (Section 1.2.5)
NCSI	National Centre for Statistics and Information
NOC	No objection certificate
NPV	Net present value
NWS	Nama Water Services
O1 to O24	The outfalls of the twenty-four subnetworks, the lowest point of each; a treatment plant takes the name of the outfall at which it sits. A pumping station is named by its subnetwork and the manhole at which it sits, as O1-M445
OMR	Omani rial
OR	Occupancy rate: persons per property
PAEW	Public Authority for Electricity and Water
PAM-GUD-201, 202, 203	The Nama Water Services design guidelines: general; water and TSE; wastewater
PE	Population equivalent
PHF	Peak hourly flow
Property	A household connection: one domestic electricity meter

Term	Meaning
Qadf	Average daily flow
Qpdf	Peak flow: the guideline's peak daily flow, equal to its peak hourly flow
S1 to S7	The sewer network options, by where the flow is treated (Section 6.2.6)
SewerGEMS, WaterGEMS	The sewer and the water network modelling software of the Terms of Reference
SRT	Solids retention time
STP	Sewage treatment plant
TE	Treated effluent, as the Terms of Reference use the term (see the note below)
TKN	Total Kjeldahl nitrogen
TSE	Treated sewage effluent, the guidelines' term for treated effluent
TSS	Total suspended solids
UTM 40N	Universal Transverse Mercator zone 40 North, WGS 84 datum

Note on the Term TE

The Terms of Reference and this report use TE to mean treated effluent, the treated product of the sewage treatment plant. The NAMA design guidelines define TE as trade effluent and use TSE for treated sewage effluent. To avoid ambiguity this report writes treated effluent in full, and TSE where a guideline value is quoted.¹

¹ PAM-GUD-201, Table 1, page 16; PAM-GUD-203, page 13. Confirmation of the preferred convention is requested (Section 1.5.4).

Executive Summary

This report presents the concept design for the wastewater and treated effluent systems serving Ibri, together with the sewage treatment plant that will receive the collected flow. It records the data collected and the checks applied to it, sets out the design basis and the flows to saturation, and presents seven options for the sewer network, three of which are recommended for the cost estimate and the appraisal. The design basis was issued on its own as the Design Basis Report and presented to Nama Water Services.

Concept design process

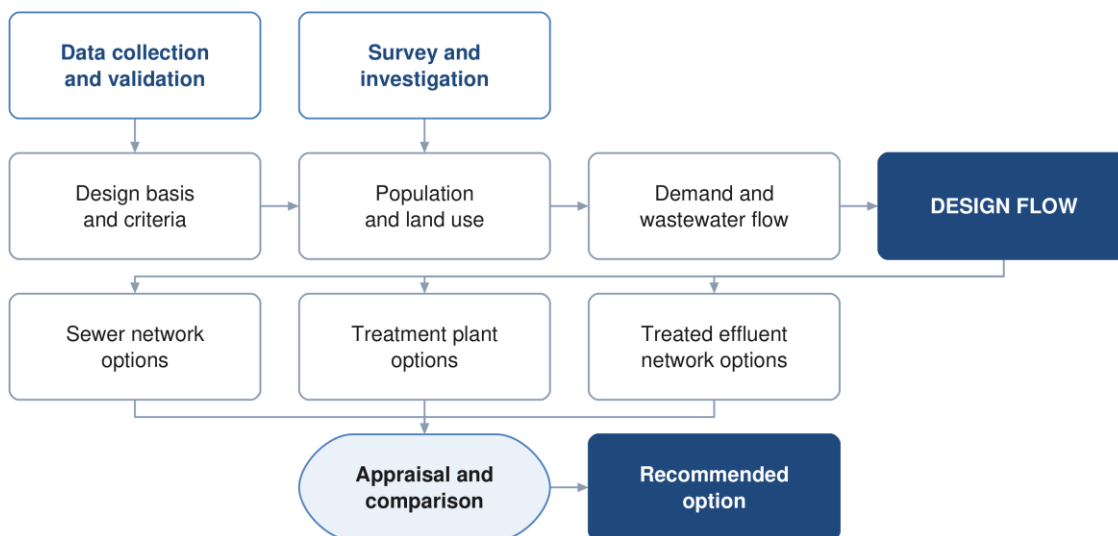


Figure 1. The concept design process, from the data collected to the recommended option.

Project Area

The study area covers 531.4 square kilometres within the Wilayat of Ibri, Governorate of Adh Dhahirah, and contains twenty-five named settlements. The project boundary received covers 439.8 square kilometres; it has been extended so that every cadastral plot of the twenty-five settlements, the planned ones included, lies inside it. All spatial work is carried out in UTM zone 40 North on the WGS 84 datum.¹

The ground falls from the north-east to the south-west, from 489 metres at Bat to 329 metres at the existing STP, and the wadis follow it. Al Makhtibyah and Tanam lie below the existing STP and cannot drain to it by gravity. The climate is hot desert, and the wind blows along one axis, north-west to south-east, across the line from the existing STP to Ibri town. Section 1.2 describes the ground, the climate, the roads and the flood hazard of the study area.

Data Collection

Data has been obtained from Nama Water Services covering the existing wastewater assets, the potable water network, electricity accounts and the cadastral plot layer. The datasets have been loaded into the project geographic information system, checked against the project boundary and assessed for completeness. Section 2.2 records the outcome in full. Four points are carried through the report.

Existing assets — the wastewater dataset holds two networks. The constructed network, dated 2006, comprises 111.6 kilometres of gravity sewer and 10.0 kilometres of force main. A further 199.3 kilometres of gravity sewer, 23.2 kilometres of pumping main and the whole of the 45.7 kilometre treated effluent main are recorded as proposed.

¹ Areas measured in the project geographic information system. Confirmation of the updated boundary is requested (Section 1.5.4).

No treated effluent asset has been built. As the diameters and levels of the existing networks are not recorded, the concept design lays a network that carries all the sewage to the treatment plants; the existing system is assessed as soon as the as-built survey is available (Chapter 5).

Potable water — the PAEW dataset provides 647.8 kilometres of water mains within the study area, and is adopted as the source for utility interfaces.

Electricity accounts — 33,971 meters have been placed on the plots to establish the number of properties on each and its use.

Survey — a topographic and utility survey covering the whole study area is in progress and will confirm levels, diameters and asset condition.

Design Basis

Wastewater generation is derived from water demand in accordance with the NAMA design guidelines. For the Governorate of Adh Dhahirah the domestic consumption rate is 164 litres per person per day, with 22 per cent added for non-domestic and 14 per cent for governmental consumption. Return rates of 85 per cent for domestic and tanker supply and 54 per cent for the rest give the wastewater flow. Sewage brought to the treatment plant by tanker from outside the study area is not in these flows; its volumes and sources are requested from Nama Water Services, so that the plant is designed to receive it (Sections 1.5.5 and 6.6.1).¹

The occupancy rate is set for each settlement as its 2024 population divided by the domestic electricity meters counted in it, with a floor of 4 persons per property, a cap of 6.12, the highest rate among the larger settlements, and 4 for a settlement of fewer than a thousand people. Ibri returns 6.07. Section 4.1 sets out the derivation.²

Population and Wastewater Flows

Every plot in the study area carries a population and an average sewage flow, for 2024 and for every year to the year its settlement is full. The use of each plot is determined from the meters on it and from a satellite image of September 2026; each settlement grows at the rate of the official series and fills its own land before its growth moves to its neighbours. The study area holds 119,893 people in 2024 and generates 20,852 cubic metres of sewage a day. Ibri's land is full in 2056; the last settlement fills in 2070, with 349,029 people and 60,099 cubic metres a day.³

Table 1. The study area in the design years: from the plots to the treatment plants

Year	People	Average sewage flow from the plots, m ³ /d	Average flow at the plants, with infiltration, m ³ /d	Plant design average, with the 10 % margin, m ³ /d
2024, base	119,893	20,852		
2030, opening year	139,535	24,216	24,756	27,232
2040	178,781	30,938	31,415	34,557
2050	218,926	37,815	38,226	42,049
2055, opening year plus 25	244,914	42,266	42,635	46,899
2060	275,608	47,523	47,882	52,671
2070, saturation	349,029	60,099	60,456	66,502

¹ PAM-GUD-201, Table 11, page 60 and Table 19, page 71. The guideline states that the consumption values apply in the absence of updated figures and should be validated by NAMA before design.

² The guideline derives occupancy from population and housing units published by NCSI. Housing units are not published at settlement level, and counted domestic meters have been used in their place. The departure is recorded in Section 3.1.

³ Average flows before infiltration and before the STP margin. The five-year series for every settlement is in Sections 4.1.9 and 4.2.7.

The plants receive the flow the network carries plus its infiltration, 1,079 cubic metres a day over the 1,498 km of sewer, whichever option is chosen. The plants are designed with the guideline's ten per cent margin. At the guideline's minimum loads of 60 grams of BOD and 80 grams of suspended solids per person per day they receive 20,942 kilograms of BOD a day at saturation.¹

Two industrial estates inside the town, at Al Tayyeb and Tanam, were found under the commercial tariff and are treated as special consumption. The army camp, a planned resort and two sources of tankered sewage outside the boundary are recorded in Section 4.3.²

Decisions Requested

Only what the guidelines do not settle is put to Nama Water Services for approval: 6 decisions, listed below and explained in the sections named. 9 further values are the guidelines' own, methods of this design or stated assumptions; they are adopted and reported. No decision had been taken at the date of this report. The largest of them is the overflow: without it 7 settlements never fill, and 75,414 people of the series have nowhere to go by 2070. Section 1.5 gives the decisions, the adopted values, the other matters awaiting confirmation and the 7 data requests.

Table 2. The decisions requested from Nama Water Services

	Decision	Section
1	Settlement boundaries	4.1.2
2	Occupancy rate	4.1.4
3	The overflow	4.1.8
4	The design horizon	4.1.9
5	Growth rate beyond 2050	4.1.7
6	Gradients and self-cleansing at the concept stage	3.3.2 and 6.2.8

Options Development and Evaluation

The guidelines ask for not fewer than three options for each of the sewer network, the treated effluent network and the treatment plant, and describe three characters for them: one advancing sustainability, one representing international best practice, and one based on practice already established in Oman. For the sewer network seven options have been modelled for where the flow is treated, and three of them are recommended. The characters are not tied to an option: each recommended option keeps the same network model and is designed and costed once for each character, changing only the factors the character governs, the treatment process, the energy supply, solar generation included, the reuse and the materials. The three options therefore give nine costed cases. Every case meets the same functional requirement and the same effluent standard, so that the difference between them lies in how the result is achieved. Section 6.1 sets out what distinguishes the characters in design terms.

The cases are compared over a twenty-five year period against seven criteria: total lifetime cost; sustainability, comprising carbon, circular economy and nature-based solutions; social development and in-country value; adaptability and resilience; operability; constructability; and environmental impact. Costs are discounted at five per cent. Nama Water Services sets the weight given to each criterion.

¹ Sections 3.4.1, 4.2.8 and 6.2.7, where the peak flows are also given. The loads are replaced by the laboratory results of the existing STP once received.

² The workforce of the two estates is an assumption, to be replaced by the estates' records.

Where two cases fall within ten per cent of one another on total lifetime cost they are treated as equivalent in cost, and the more sustainable of the two is adopted. Sensitivity is tested by varying the weighting between criteria, the discount rate, and the input design criteria.¹

Sewer Network Options

The sewer network is modelled in SewerGEMS as twenty-four subnetworks, 1,498 km of sewer and 19,083 manholes. Each drains by gravity to its lowest point, its outfall, and the pipes are designed for the flow of 2070. Seven options have been modelled for where the flow is treated, from one plant to six; where an outfall is not at a plant, a pumping station lifts its flow into another subnetwork or to a plant. In every option and every year the pipes stay within the guideline's limits on velocity and depth of flow. The pipes run in the same streets in every option, and the depth hardly changes: 40 manholes are deeper than 12 m, the deepest 15.1 m. The options differ in the trunk sewers, the pumping and the number and size of the plants.

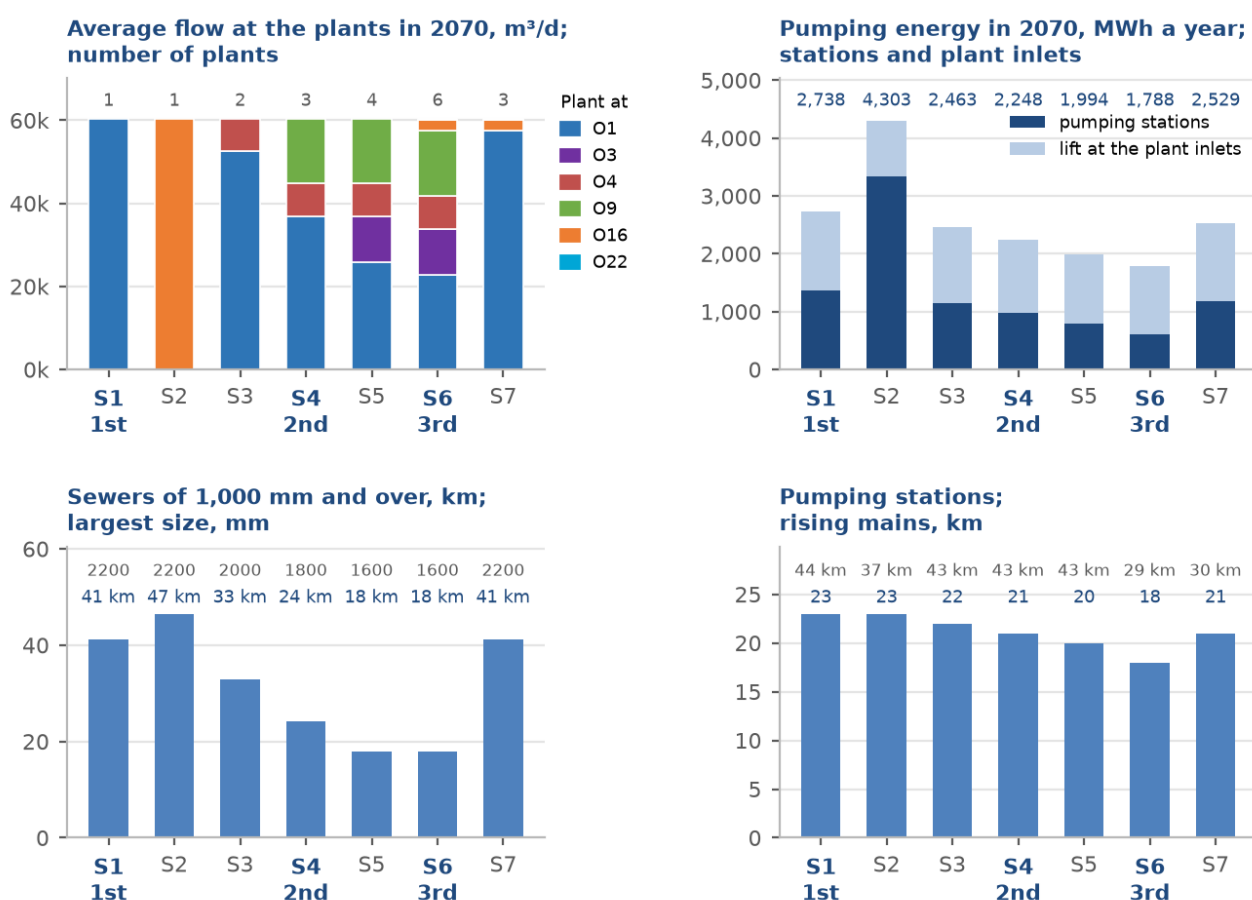


Figure 2. The seven options side by side in 2070: the plants and their flow, the pumping energy of the stations and of the plant inlets, the trunk sewers, and the pumping stations with their rising mains. The recommended options carry their priority.

¹ PAM-GUD-201, Sections 12.6 to 12.9, pages 104 to 106.

Table 3. The sewer network options at a glance, 2070

Option	Plants	Largest plant, m ³ /d	Sewer at the plants, m deep	Pumping stations	Sewers of 1,000 mm and over, km	Pumping energy with the plant inlets, MWh a year
S1	1	60,456	10.3	23	41	2,738
S2	1	60,456	5.9	23	47	4,303
S3	2	52,521	10.1 / 6.9	22	33	2,463
S4	3	36,881	10.0 / 6.9 / 8.8	21	24	2,248
S5	4	25,915	9.8 / 5.8 / 6.9 / 8.8	20	18	1,994
S6	6	22,867	9.8 / 5.8 / 6.9 / 8.8 / 4.9 / 1.5	18	18	1,788
S7	3	57,408	10.3 / 4.9 / 1.5	21	41	2,529

Recommendations

Three options are recommended for the cost estimate and the multi-criteria appraisal, in this order of priority.

S1, first. One plant at O1, receiving 24,756 cubic metres a day on average in 2030, 42,635 in 2055 and 60,456 in 2070, so that it is built in phases on one site, with one plant to staff, one sludge line and one source of treated effluent. Its costs are the largest trunk sewers, 41 km of sewer of 1,000 mm and over, up to 2,200 mm, the deepest arrival at a plant, 10.3 m, and the most pumping of the three, 2,738 MWh a year in 2070 with the lift at the plant.

S4, second. Three plants: O1 with 36,881 cubic metres a day in 2070, O4 with 7,935 and O9 with 15,640. The north-eastern subnetworks O4 to O8 and the eastern subnetworks O9 to O14 are treated at their own outfalls instead of being pumped on towards O1. Against S1 it needs 18 per cent less pumping energy, 2,248 MWh a year in 2070, two fewer pumping stations and 17 km less sewer of 1,000 mm and over; the largest sewer is 1,800 mm. Its cost is two further plant sites, each with its buffer, access, power supply and outlet for the treated effluent.

S6, third. Six plants, at O1, O3, O4, O9, O16 and O22. The least pumping of the seven options, 1,788 MWh a year in 2070, 35 per cent below S1, with the fewest pumping stations, 18, and the shortest rising mains, 28.7 km; the largest sewer is 1,600 mm. The plant at O22, 446 cubic metres a day in 2070, is small enough for the nature-based treatment the guideline allows up to about 500 cubic metres a day. Its cost is six sites and six plants to staff, two of them very small in the early years: O22 receives 77 and O16 receives 657 cubic metres a day in 2030.

The three are selected because together they span the choice the options present: all the flow treated at one plant in S1, at three in S4 and at six in S6, so that the appraisal prices the whole range from central to local treatment. S2 is not selected: it treats all the flow at O16 and has to pump the whole of O1's flow there, 4,303 MWh a year in 2070, 57 per cent more than S1. S3, S5 and S7 lie between the recommended options and offer nothing the three do not: S3 and S5 are steps between S1, S4 and S6, and S7 keeps the trunk sewers of S1 while adding two small plants.

Each of the three is designed and costed under each of the three characters of the guidelines, on the same network model: the characters change the treatment process, the energy supply, solar generation included, the reuse and the materials (Section 6.1.2), and the appraisal compares nine cases. The priority rests on the technical results; the cost estimate and the multi-criteria appraisal confirm it or change it. Under the rule of Section 6.1.4, of two cases whose whole-life costs fall within ten per cent of one another, the more sustainable is preferred. The treated effluent network is designed on the three options, with one source of treated effluent, three or six.¹

Three matters need attention whichever option is chosen. Long rising mains carrying small flows hold the sewage for hours and call for hydrogen sulphide control. The station at O23 lifts a few litres a second against a very high head and is

¹ PAM-GUD-201, Section 12.1, page 95 (the three characters), and Sections 12.6 to 12.9, pages 104 to 106 (the appraisal).

to be reconsidered at the preliminary design. And no year brings every pipe to the self-cleansing velocity, even at 4 per cent: the criterion for the head pipes is the decision of Section 6.2.8. Section 6.2 presents the options in full.

Deliverables

The Terms of Reference set out forty numbered deliverables for the concept stage. Section 1.3.1 lists them, with the section of this report that carries each.

1. Project Background and Scope

This chapter records the project and its scope, the study area and its boundary, the programme and the deliverables, the consultation held, and what is asked of Nama Water Services.

1.1. Introduction

Nama Water Services is developing wastewater collection, treated effluent distribution and sewage treatment for the Wilayat of Ibri. Renardet S.A. & Partners has been appointed to provide consultancy services for the design and supervision of these works.

This report is the concept design deliverable. It establishes the design basis, records and assesses the data collected, sets out the framework within which options for the sewer network, the treated effluent network and the treatment plant are developed, and presents the options of the sewer network.

1.1.1. Project Objectives

The Terms of Reference set three objectives for the work: to verify the Regional Master Plan, to establish the ultimate expected sewage flow in the Ibri catchment, and to develop the concept, preliminary and detailed designs together with the tender documentation.

1.1.2. Report Structure

The report follows the structure approved for the concept stage. Chapter 1 records the project and its process. Chapter 2 presents the data and the assessment of it. Chapter 3 sets out the design basis. Chapter 4 establishes demand and flow. Chapter 5 assesses the existing system. Chapter 6 presents the options. Chapter 7 carries the technical and financial appraisal, and Chapter 8 the delivery arrangements. Appendix A holds the working behind Chapter 4, and Appendix B the quantities, pumping stations and depth of the sewer network options.

1.2. Study Area

This section describes the study area: where it lies and what it contains, its boundary, its ground and drainage, its climate, its roads and the flood hazard across it. The map on the next page shows the study area, its settlements as received and its place in Oman.

Project location and study area boundary

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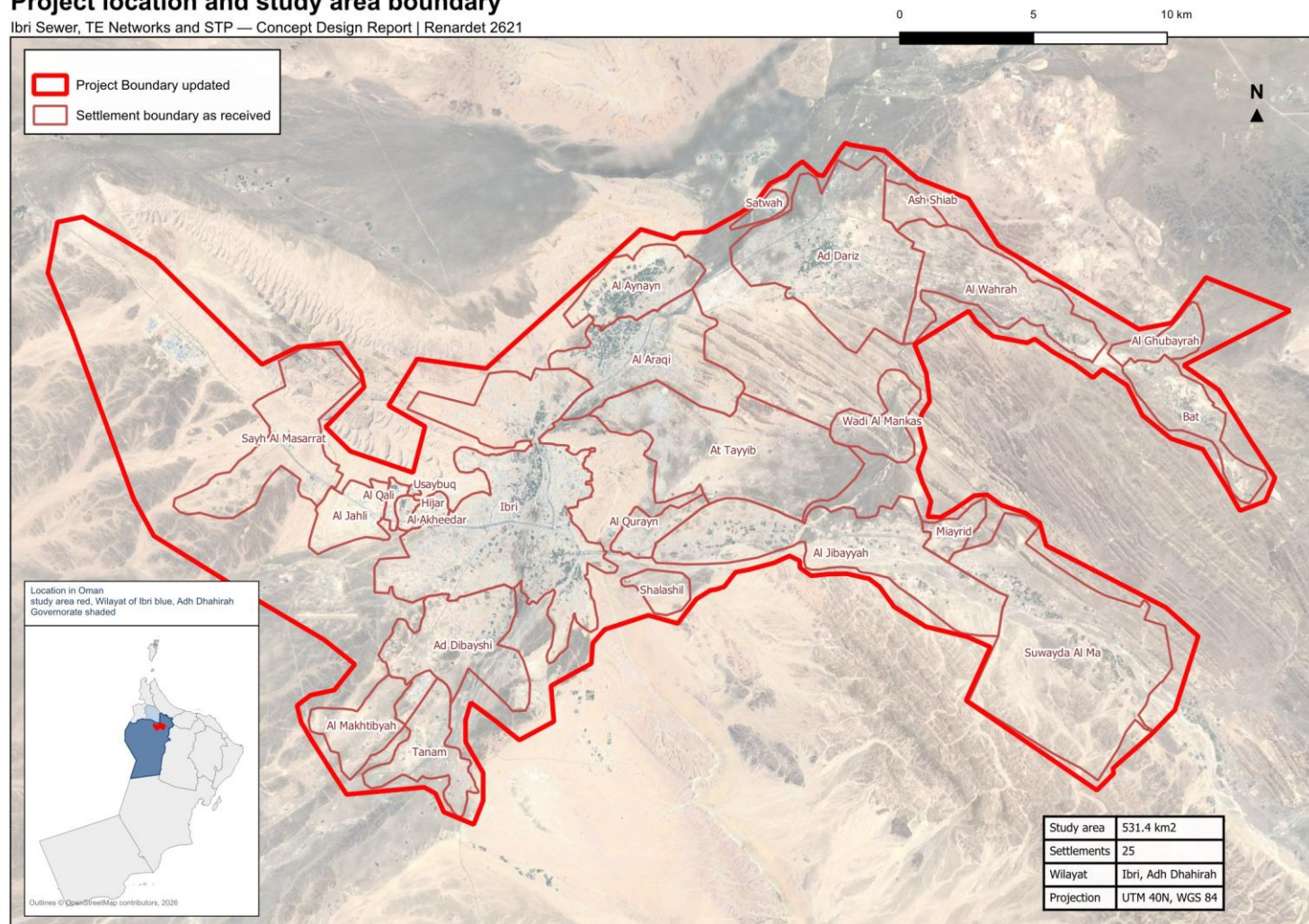


Figure 3. Project location and study area boundary, with the boundaries of the twenty-five settlements as received with the Inception Report. The inset places the study area in the Wilayat of Ibri and in Oman.

1.2.1. Location and Settlements

The study area lies in the Wilayat of Ibri, in Adh Dhahirah Governorate in the north-west of Oman. It covers 531.4 square kilometres and contains twenty-five named settlements. Ibri is the principal settlement; the remainder are distributed along the wadi system and the main road corridors.¹

1.2.2. Project Boundary

The project boundary received covers 439.8 square kilometres. The cadastral plots of the twenty-five settlements, the planned ones included, reach beyond it in several places. The boundary has therefore been updated to take in every plot of those settlements, and covers 531.4 square kilometres. All the work presented here is on the updated boundary, and confirmation of it is requested (Section 1.5.4).²

1.2.3. Coordinate Reference System

All spatial data is held and all measurements are made in Universal Transverse Mercator zone 40 North on the WGS 84 datum. Datasets supplied in geographic coordinates have been reprojected. Levels are referenced to the terrain model described in Section 2.3.

1.2.4. Topography and Drainage

The ground falls from the north-east to the south-west. The plots of the study area lie between 311 and 514 metres. The built plots of Bat, the highest settlement, stand at 489 metres, 31 kilometres from the existing STP, where the ground is at 329 metres: an average fall of about 5 metres in every kilometre.³

The wadis follow the same fall. They gather from the north and the east, join to the south of Ibri town and leave the study area to the south-west through Tanam. The map that follows shows the ground and the wadis of stream order three and above.

¹ Area of the updated project boundary (Section 1.2.2), measured in the project geographic information system in UTM zone 40 North.

² The project boundary received in July 2026. A second boundary, of 521.1 square kilometres, was issued with the Inception Report package; the updated boundary contains both.

³ Terrain model at 0.5 metre resolution (Section 2.3.2), read at every plot. The range quoted is from the 1st to the 99th percentile of the plots; the level of a settlement is the median of its built plots.

Topography and drainage of the study area

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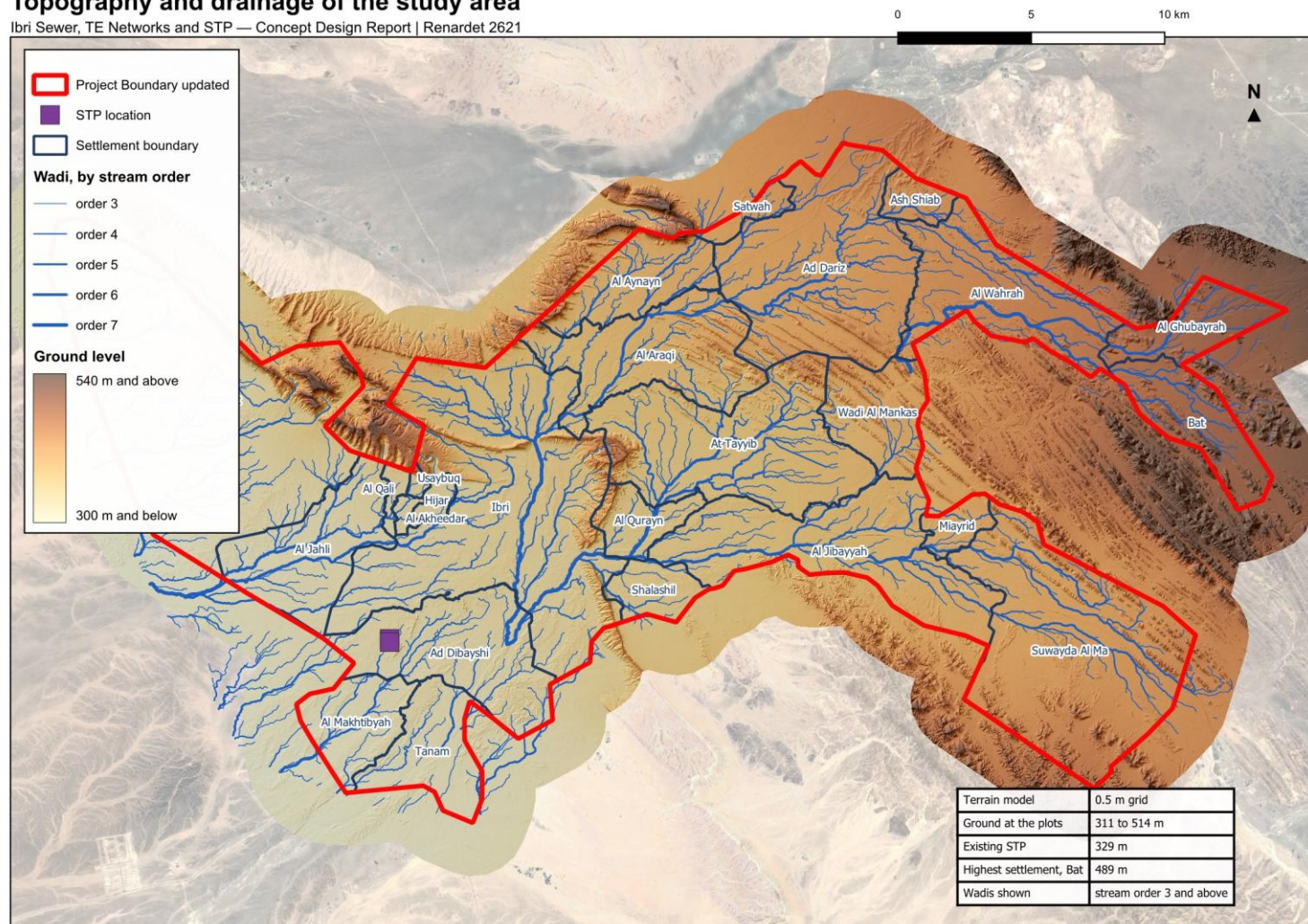


Figure 4. Topography and drainage of the study area: ground level from the 0.5 m terrain model, the wadis of stream order three and above, and the existing STP at the low end of the study area.

The figure below sets the ground of each settlement against the ground at the existing STP, the settlements in order of their distance from it. Twenty-three of the twenty-five settlements stand above the STP, so their sewage can in principle reach it by gravity. Two, Al Makhtibyah and Tanam, lie downstream of it, about 8 metres lower, and cannot drain to the existing STP by gravity. Whether each of the others does so without pumping depends on the ground between, which the network design establishes (Section 6.2).

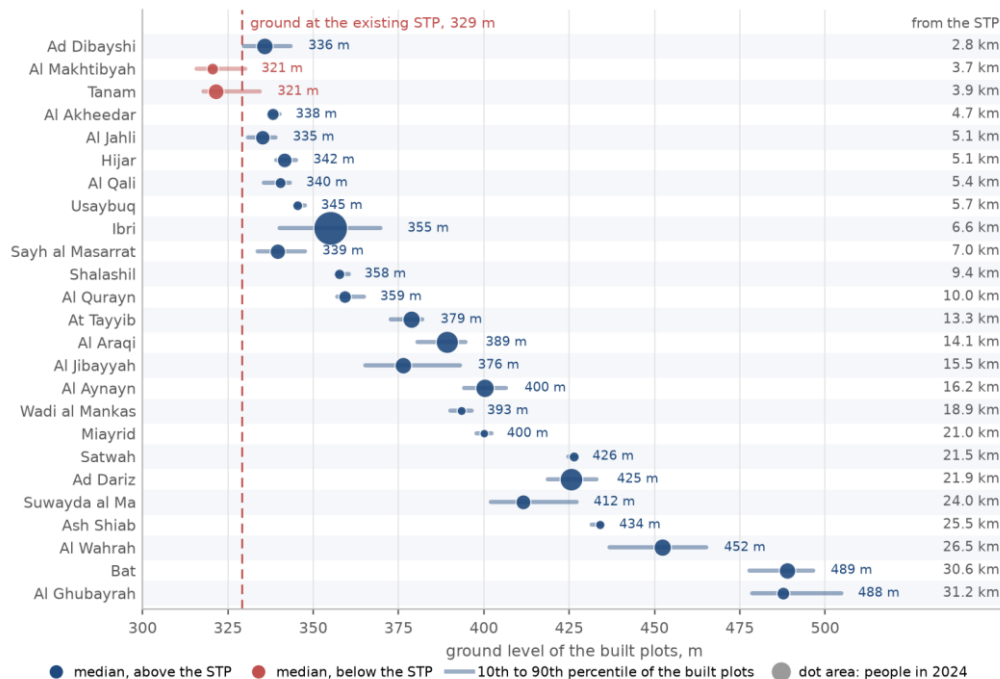


Figure 5. Ground level of the built plots of each settlement, in order of distance from the existing STP. The bar spans the 10th to the 90th percentile of the built plots.

1.2.5. Climate

Ibri has a hot desert climate, class BWh of the Köppen-Geiger classification.¹

No station record has been supplied. The figures below are from the NASA POWER record, a global reanalysis on a grid of about 50 kilometres, read at the existing STP: temperature and rainfall for 2001 to 2020, and hourly wind at 10 metres for 2015 to 2024. The record describes the regime. It does not replace the station record, which the odour dispersion modelling of Section 7.3 requires.²

Temperature and rainfall

The mean daily maximum is 40 °C or more from May to September and reaches 45 °C in June; the mean daily minimum falls to 13 °C in January. The annual mean is 29.7 °C, and the daily maximum reaches 40 °C on about 159 days of the year. The heat bears on the design: sewage turns septic sooner in long sewers and force mains (Section 6.3), and oxygen dissolves less readily in the aeration tanks.

Rain is scarce and irregular: 80 millimetres a year on average, between 17 millimetres in 2001 and 176 in 2017, most of it from a few storms. The sewer system is separate and takes no stormwater, so the rain reaches the design through the wadi floods of Section 1.2.7 and not through the design flow.

¹ Beck, H. E. et al. (2018), Present and future Köppen-Geiger climate classification maps at 1-km resolution, Scientific Data 5, 180214. A desert climate has annual rainfall below half of the aridity threshold, here 735 millimetres, and is hot where the mean annual temperature is 18 °C or more; the record below meets both.

² NASA Prediction Of Worldwide Energy Resources, MERRA-2 reanalysis, power.larc.nasa.gov, read in September 2026. The rainfall record after 2020 is not used: at this grid point it carries an implausible total for August 2024. Storm rainfall on a mountain front is poorly resolved on a grid of this size.

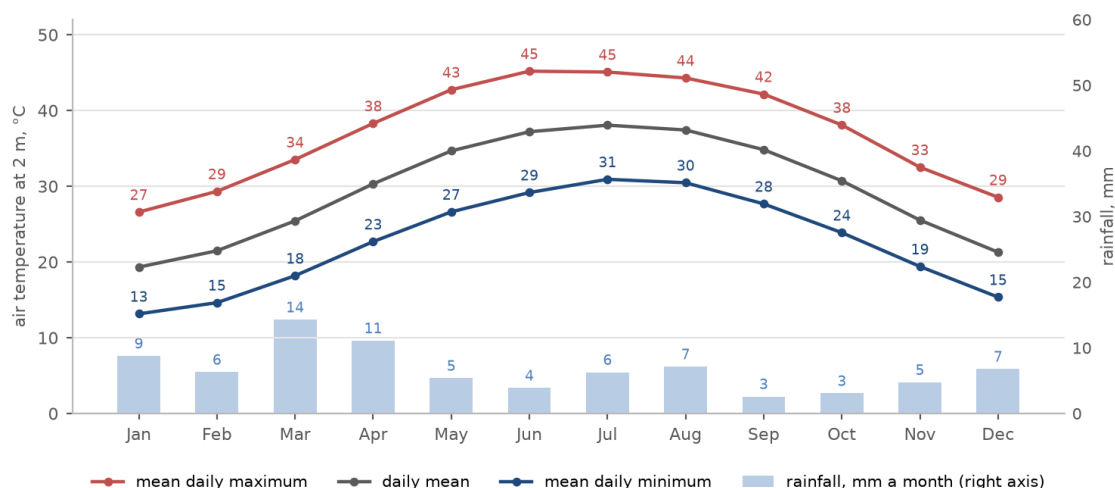


Figure 6. Monthly temperature and rainfall at the existing STP, 2001 to 2020, from the NASA POWER record.

Wind

The wind blows along one axis, north-west to south-east. Winds from the north-west quarter, west-north-west round to north, take 33 per cent of the hours, and winds from the south-east quarter, east-south-east round to south, a further 29 per cent. The north-westerly prevails in winter and the south-easterly in summer; by night the wind comes from the north-north-west and the north, or from the south-east. The mean speed at 10 metres is 3.6 metres a second, and 95 per cent of the hours are below 6.8 metres a second. A reanalysis smooths out the calm hours, which matter most for odour, so the 3 per cent of hours below 1 metre a second shown here is a lower bound. The wind roses are on the next page.

The wind decides where the odour of a treatment plant travels. Ibri town lies 6.6 kilometres to the north-east of the existing STP; the wind that would carry odour from the STP towards it, from the south-west, blows for 5 per cent of the hours and 2 per cent of the night hours, because the prevailing axis runs across that line and not along it. Tanam, to the south-south-east, lies downwind of the night wind from the north-north-west for 13 per cent of the night hours. The table below gives the same reading for every settlement within 8 kilometres of the existing STP. It is applied to each candidate site in the siting assessment (Section 6.5.3); the odour buffer itself is established by dispersion modelling (Section 7.3).

Table 4. Settlements within 8 km of the existing STP, and the wind that carries air from the STP towards them

Settlement	Distance from the STP	Direction from the STP	Carried by wind from	Share of all hours	Share of night hours
Ad Dibayshi	2.8 km	ENE	WSW	5 %	2 %
Al Makhtibyah	3.7 km	S	N	7 %	11 %
Tanam	3.9 km	SSE	NNW	9 %	13 %
Al Akheedar	4.7 km	N	S	7 %	5 %
Al Jahli	5.1 km	N	S	7 %	5 %
Hijar	5.1 km	NNE	SSW	6 %	4 %
Al Qali	5.4 km	N	S	7 %	5 %
Usaybuq	5.7 km	N	S	7 %	5 %
Ibri	6.6 km	NE	SW	5 %	2 %
Sayh al Masarrat	7.0 km	NNW	SSE	9 %	9 %

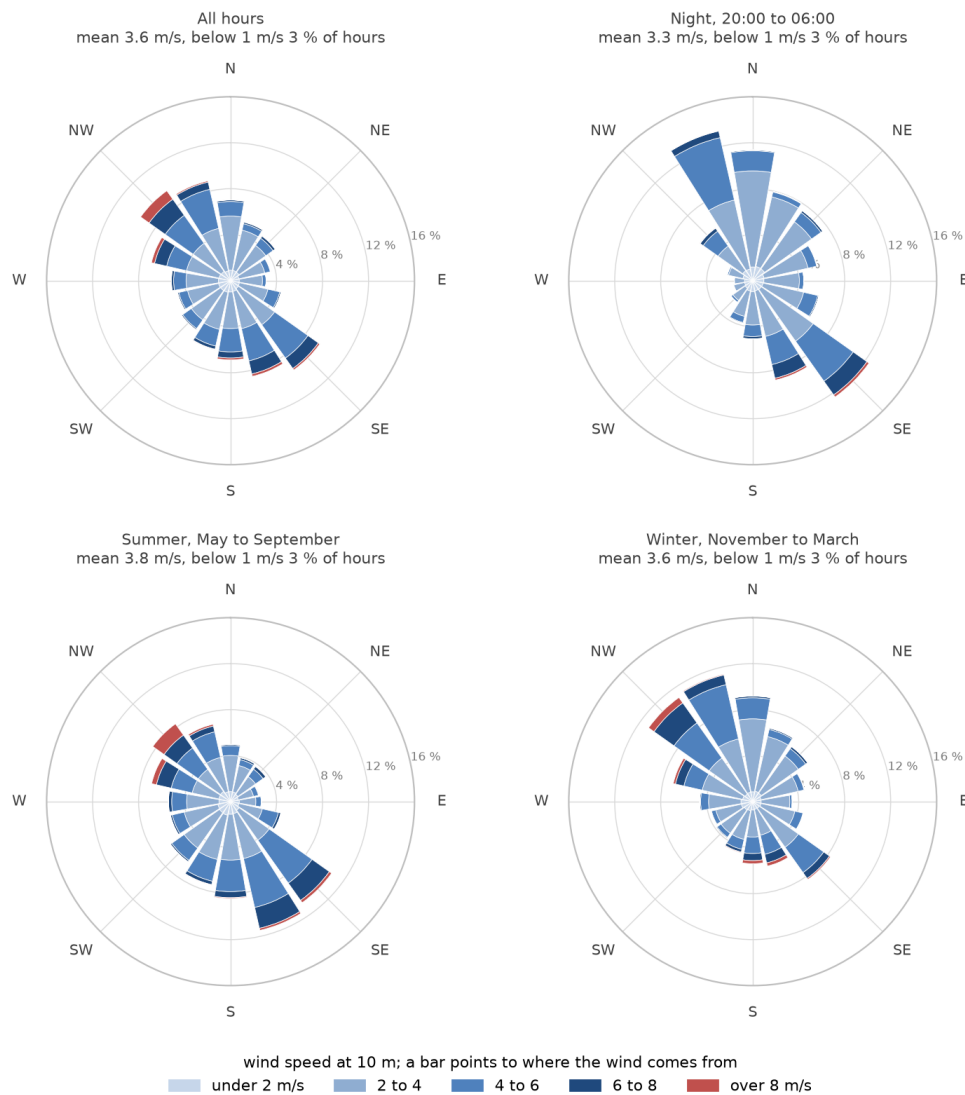


Figure 7. Wind roses at 10 m at the existing STP, 2015 to 2024: the direction the wind blows from and the share of hours in each speed class, for all hours, the night, the summer and the winter.

1.2.6. Road Network

The road centrelines supplied carry a class code, and the map on the next page shows them by that code, as supplied; the chart after it gives the lengths. Inside the study area they total 1,351 kilometres: 112 kilometres of class 01, 50 of class 02, 56 of class 04 and 1,133 of class 05. Classes 01 and 02 are the through routes that cross the study area. Class 05, 84 per cent of the length, is the street grid of the settlements, which the collection network follows.¹

The dual carriageways among them, 142 kilometres of centreline, are not used as sewer corridors and are crossed at right angles only (Section 6.2.2).

¹ The road centrelines supplied, by the road class they carry, clipped to the updated project boundary. A dual carriageway is recorded as two centrelines and is counted on both.

Road network of the study area

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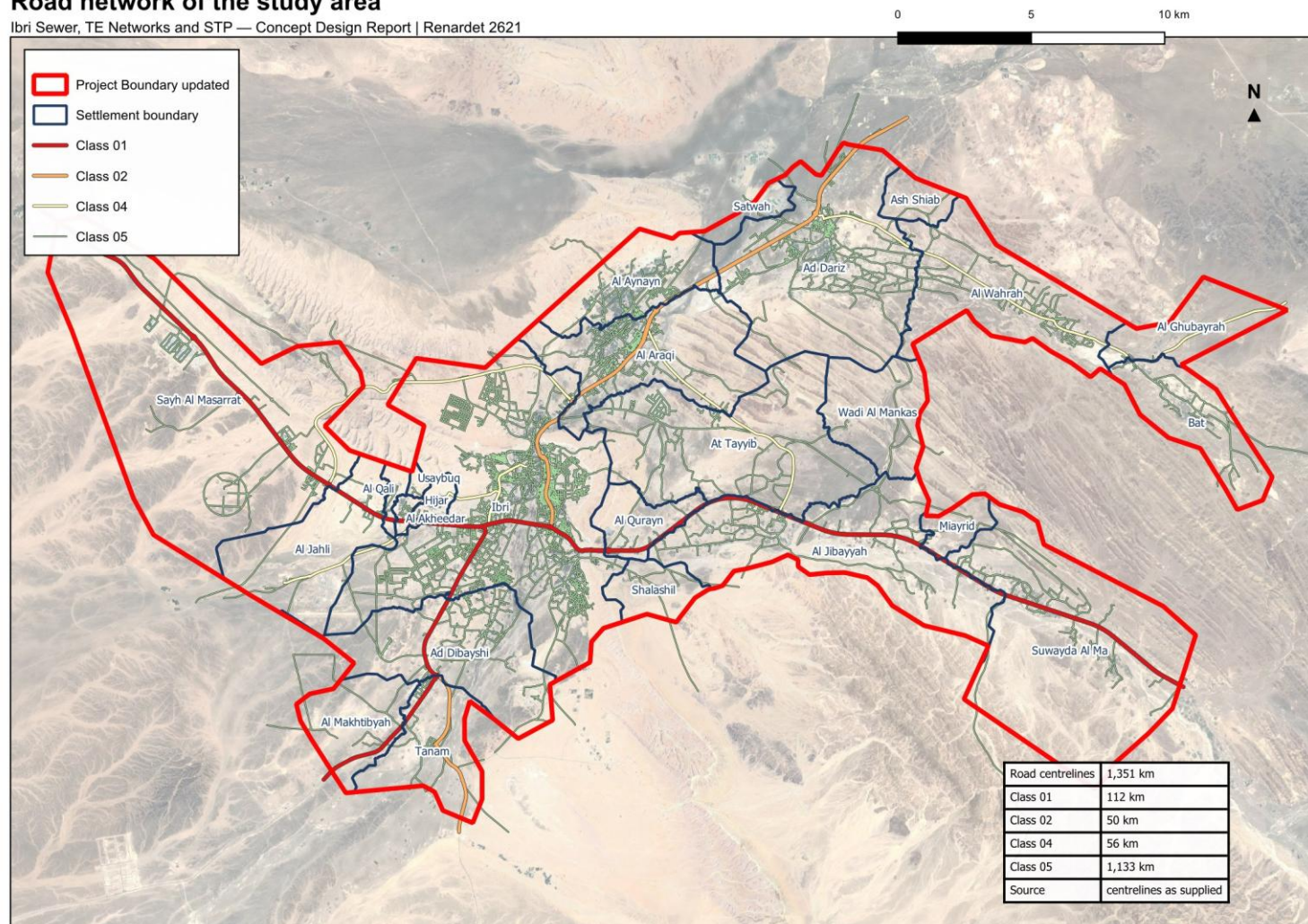


Figure 8. Road network of the study area, by the class code of the centrelines as supplied.

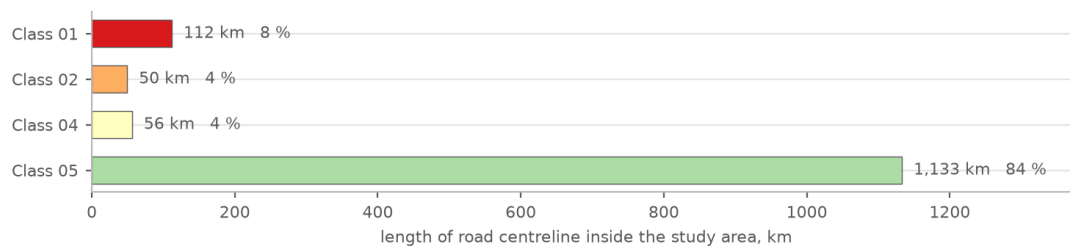


Figure 9. Length of road centreline inside the study area, by the class code supplied.

1.2.7. Flood Hazard

The flood hazard across the study area is taken from the Oman Flood Mapping project of the Ministry of Agriculture, Fisheries and Water Resources, which maps the area on a 3 metre grid. The four maps that follow show the floods of the 10, 25, 50 and 100-year return periods.¹

The hazard is given in the six classes of the Australian classification, which combine the depth of the water and its velocity into what the flood does to people, vehicles and buildings. The table below gives the classes.²

Table 5. Flood hazard classes of the Australian classification

Class	What the flood does	Depth × velocity, m ² /s, up to	Depth, m, up to	Velocity, m/s, up to
H1	Generally safe for people, vehicles and buildings	0.3	0.3	2.0
H2	Unsafe for small vehicles	0.6	0.5	2.0
H3	Unsafe for vehicles, children and the elderly	0.6	1.2	2.0
H4	Unsafe for people and vehicles	1.0	2.0	2.0
H5	Unsafe for people and vehicles; all buildings vulnerable to structural damage, the less robust to failure	4.0	4.0	4.0
H6	Unsafe for people and vehicles; all building types vulnerable to failure	above 4.0	no limit	no limit

In every return period the high classes, H5 and H6, keep to the main wadi channels: the wadi that crosses the south of the study area from Suwayda al Ma through Al Jibayyah to the south of Ibri town, and its continuation to the south-west through Ad Dibayshi and Tanam. Away from the channels the flooding is shallow sheet flow in class H1. The channels widen and the classes rise with the return period.

The maps are used in three places. The treatment plant and the pumping stations are sited clear of the high classes, or protected to the levels of Section 7.2. The wadi crossings of the sewers and force mains are identified from them. And the flood protection assessment of Section 7.2 works from the 25, 50 and 100-year floods.

¹ Oman Flood Mapping project, Ministry of Agriculture, Fisheries and Water Resources; flood hazard grids of the Ibri area.

² Australian Institute for Disaster Resilience (2017), Guideline 7-3, Flood Hazard, supporting Handbook 7, Managing the Floodplain; adopted in Australian Rainfall and Runoff 2019.

Flood hazard, 10-year return period

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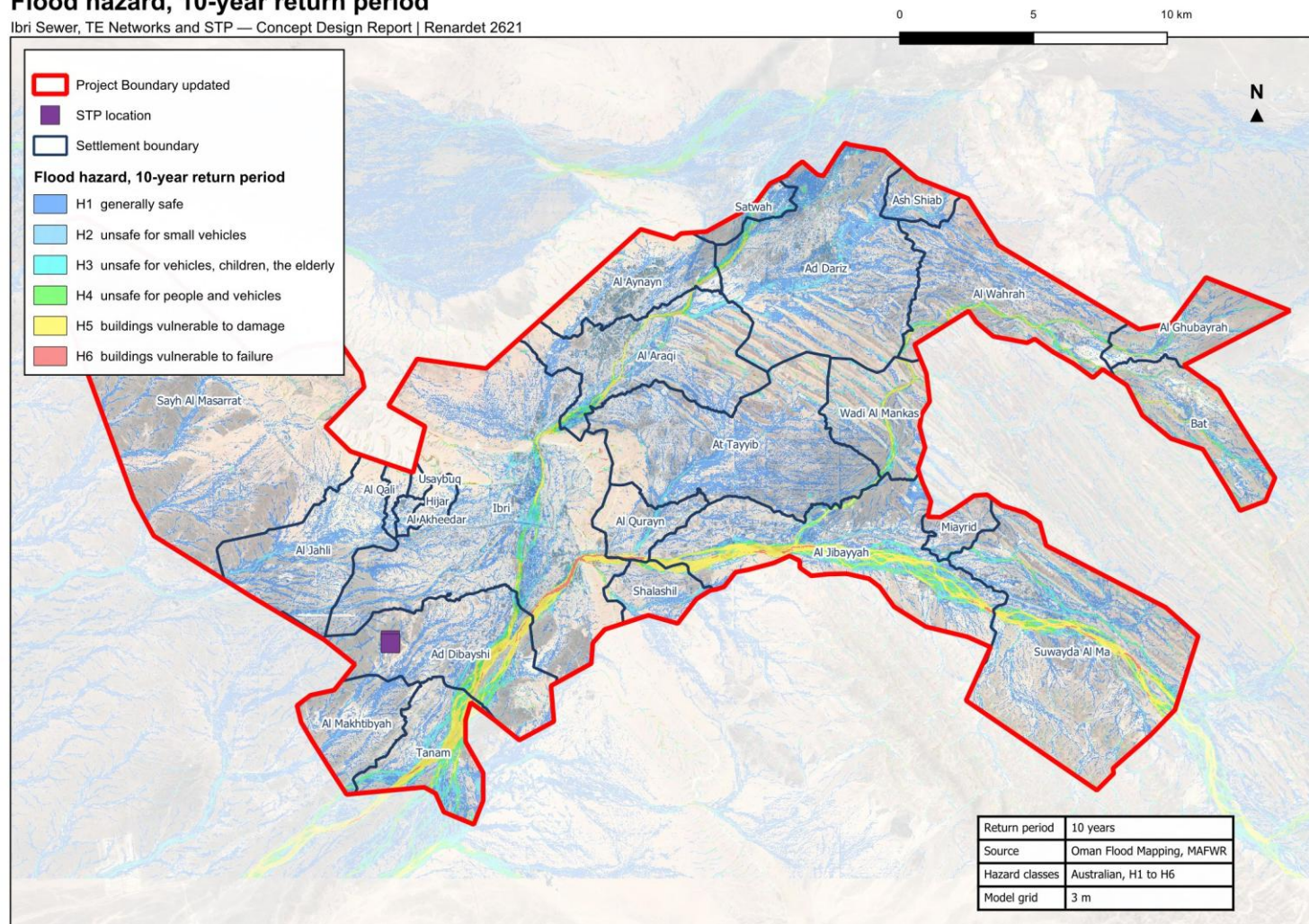


Figure 10. Flood hazard in the study area, 10-year return period. Oman Flood Mapping project, MAFWR; Australian hazard classes.

Flood hazard, 25-year return period

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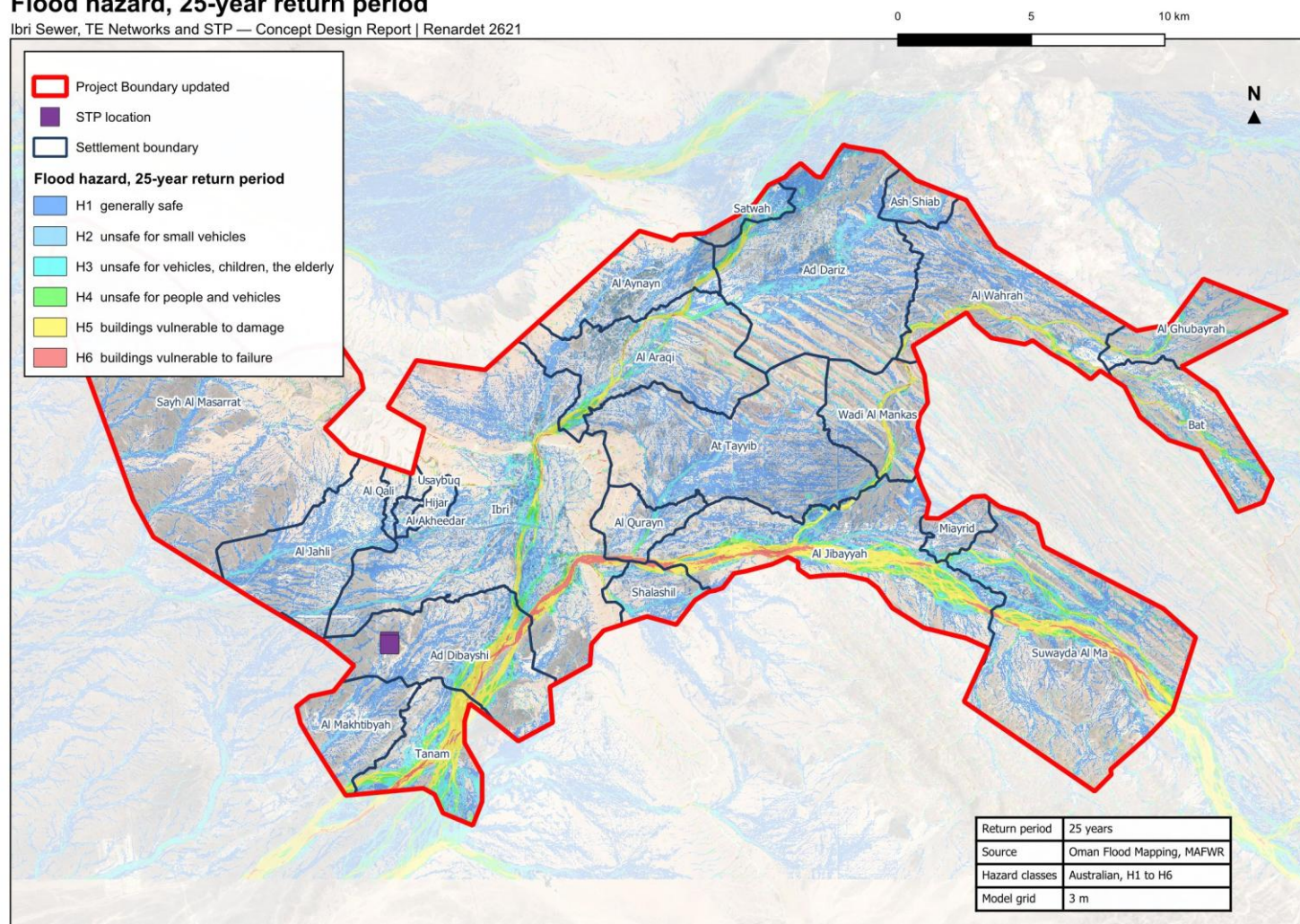


Figure 11. Flood hazard in the study area, 25-year return period. Oman Flood Mapping project, MAFWR; Australian hazard classes.

Flood hazard, 50-year return period

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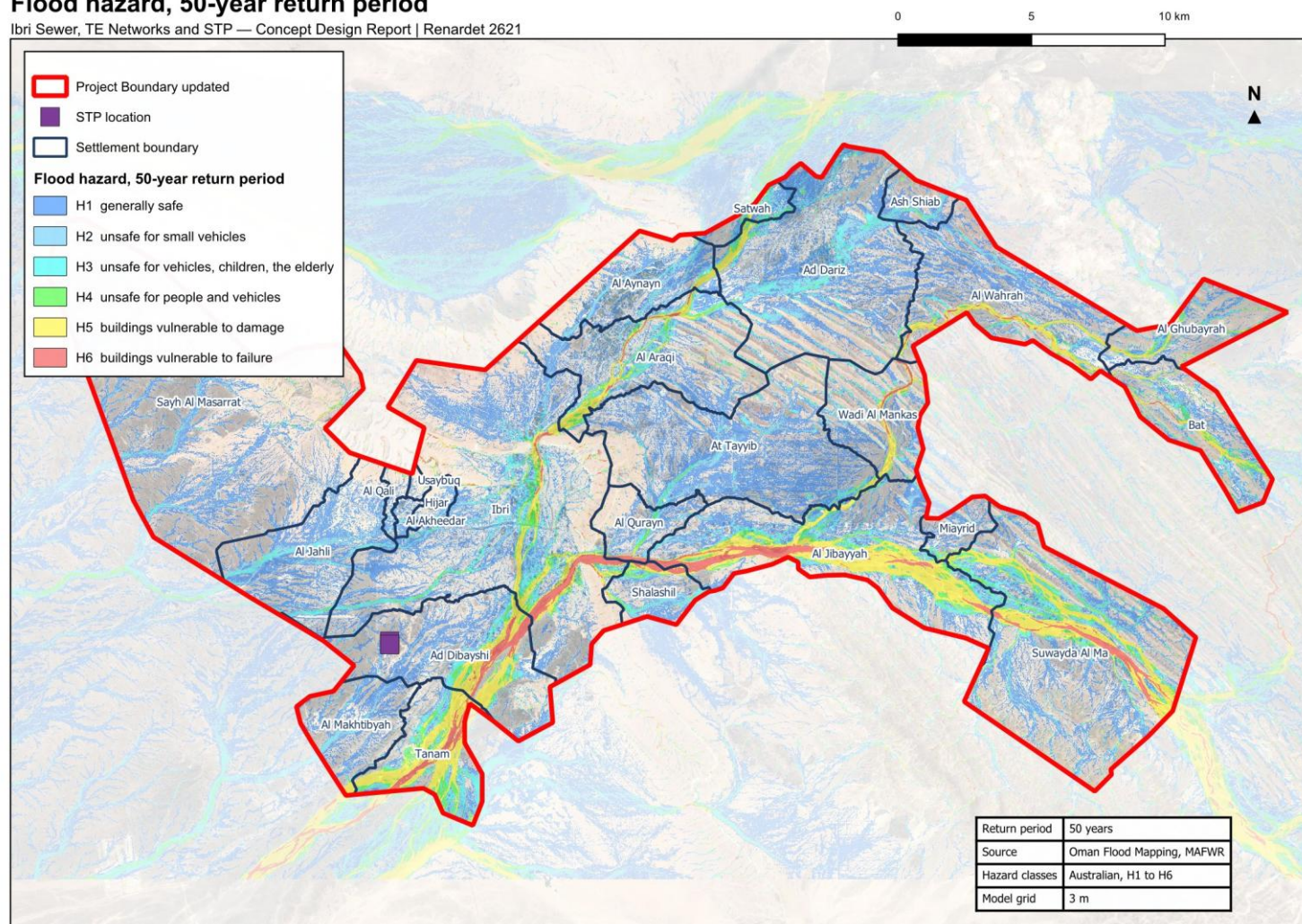


Figure 12. Flood hazard in the study area, 50-year return period. Oman Flood Mapping project, MAFWR; Australian hazard classes.

Flood hazard, 100-year return period

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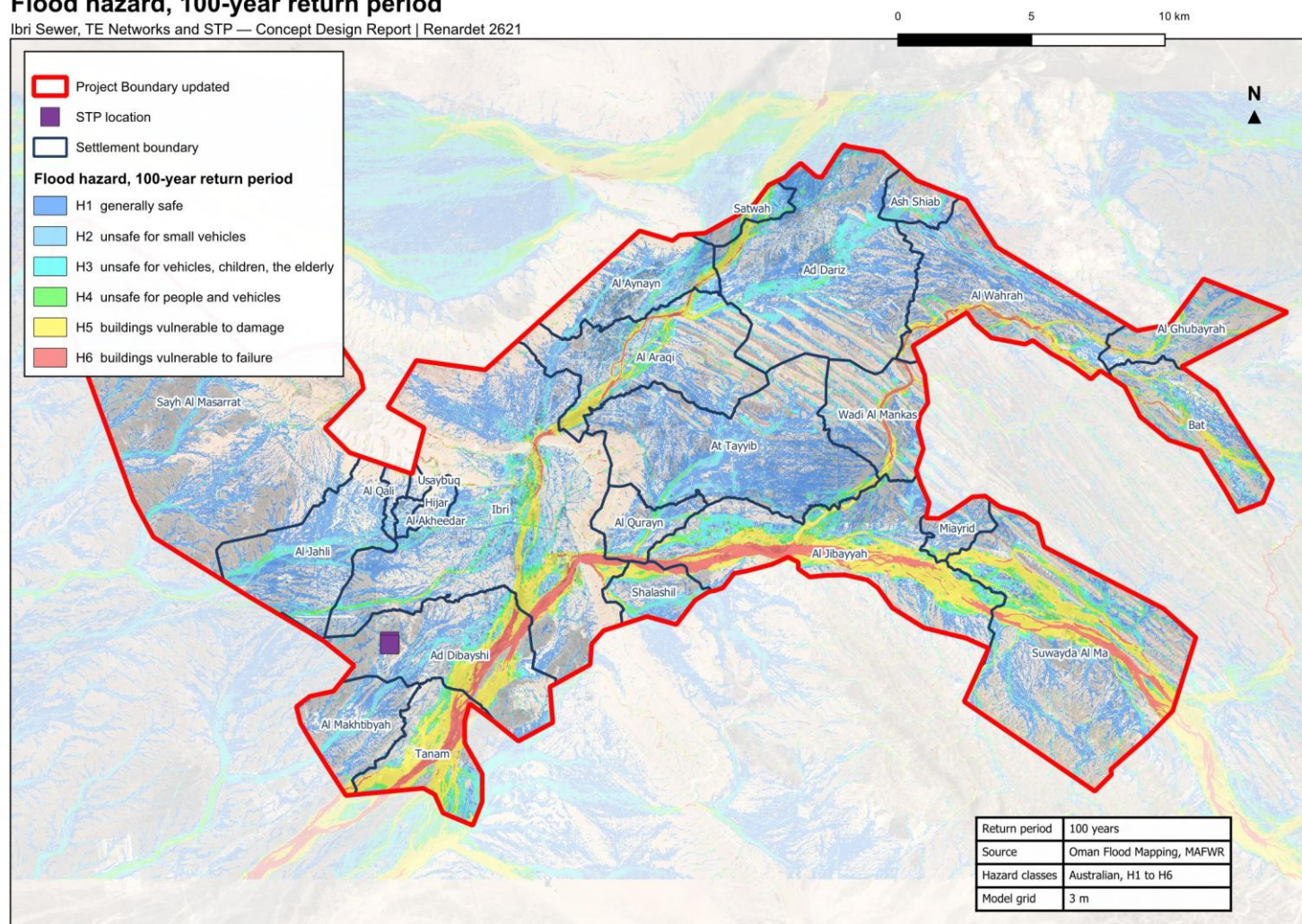


Figure 13. Flood hazard in the study area, 100-year return period. Oman Flood Mapping project, MAFWR; Australian hazard classes.

1.3. Programme and Design Stages

The design comprises four stages: concept design, preliminary design, detailed design and the preparation of tender documents. Each stage is submitted for review and approval before the following stage begins.

The Terms of Reference allow sixty days for the concept design and a further twenty-one days for review. The preliminary design stage begins on approval of the concept design for the treatment plant.¹

1.3.1. Concept Stage Deliverables

The Terms of Reference set out forty numbered deliverables for the concept stage. They are grouped below by subject, with the section of this report that carries each.²

Table 6. Concept stage deliverables

Deliverable	Section of this report
Executive summary and project schedule	Executive Summary; 1.3
Data collection report and assessment of the data	2.1 and 2.2
Design criteria and design basis	3
Population forecasting and flow projection at five-year intervals	4.1 and 4.2
Topographic survey and geotechnical investigation	2.3
As-built records and GIS for the existing systems	2.4
Hydraulic assessment of the existing systems	5.1
Concept hydraulic calculations and capacities	4.2.8, 6.2 and 6.4
Wastewater network options, not fewer than three	6.2
Treated effluent network options, not fewer than three	6.4
Treatment plant options, not fewer than three, with siting and phasing	6.5
Pumping and lifting station concept design	6.3 and Appendix B
Treated effluent and sludge management strategy	6.4, 6.6.2 and 6.7
Excess effluent and emergency overflow provisions	6.6
Environmental impact assessment for the plant location	7.5
Cost estimates and life cycle cost	7.8
Risk analysis and value engineering	7.9 and 7.10
Multi-criteria comparison and recommended option	7.11
Hydraulic models in SewerGEMS and WaterGEMS	5.1.3, 6.2.5 and 6.4
Contracting strategy and implementation plan	8.1 and 8.2
Register of approvals and no objection certificates	7.6.5

1.3.2. Subsequent Stage Deliverables

The preliminary design develops the approved concept to an estimate within ten per cent, with full survey, hazard and operability study, and preliminary bills of quantities. The detailed design completes the engineering, the drawings and the priced bills. Tender documentation follows the approved detailed design.

¹ Appendix to the Form of Bid, page 147, restated as a contract term at page 177 of the tender document.

² Scope of Work, pages 63 and 64 of the tender document.

1.4. Consultation Record

The following meetings have been held with Nama Water Services to the date of this report.

Table 7. Meetings held

Meeting	Date	Outcome
Kick-off meeting	15 July 2026	Project scope, approach, programme and organisation presented
Inception Report submission	August 2026	Design basis, methodology and programme submitted
Design basis meeting	16 September 2026	Progress reported; the Design Basis Report presented

Coordination with the authorities holding assets in the project area is described in Section 7.6, together with the register of approvals and no objection certificates.

1.5. Decisions and Matters for Confirmation

Only what the guidelines do not settle is put to Nama Water Services for a decision. Everything else is adopted and reported: a value the guidelines give, a value of the Inception Report, or an assumption stated as such. Section 1.5.1 lists all 15 items with the status of each. Section 1.5.2 gives the 6 decisions in full; each can be answered with a yes, a no, or one figure, and a no returns the design to the section named. Section 1.5.3 gives the 9 adopted values in full. Section 1.5.4 lists the other matters that await confirmation, and Section 1.5.5 the data requested. No decision had been taken at the date of this report.

1.5.1. Register of Decisions and Adopted Values

Table 8. Status of every item: a decision requested from Nama Water Services, or a value adopted and its basis

	Item	Status	Why a decision is needed, or the basis of the adopted value	Section
1	Settlement boundaries	Decision requested from NWS	The guidelines do not address settlement boundaries; the polygons received leave land unassigned	4.1.2
2	Occupancy rate	Decision requested from NWS	PAM-GUD-201 Section 7.2 (p59) derives occupancy from housing units, which are not published by settlement; the floor, the cap and the small-settlement rule are this design's	4.1.4
3	The overflow	Decision requested from NWS	The guidelines do not say where growth goes once a settlement's land is full	4.1.8
4	The design horizon	Decision requested from NWS	The Terms of Reference name both horizons	4.1.9
5	Growth rate beyond 2050	Decision requested from NWS	PAM-GUD-201 allows ten years of extension beyond a forecast; the series to 2100 goes further	4.1.7
6	Gradients and self-cleansing at the concept stage	Decision requested from NWS	A departure from PAM-GUD-203 Section 4.2.2.1; the tractive tension is not given in the guidelines	3.3.2 and 6.2.8
7	The 126 meters with no plot within 15 metres	Adopted: design choice	No plot lies within 15 m of the meter	4.1.3
8	Land-use layer	Adopted: design method	The cadastre's own land-use field is not usable as it stands; the use is determined from the meters and the satellite image	4.1.5
9	Water demand and return	Adopted: guideline	PAM-GUD-201 Table 11 (p60) and Table 19 (p71)	4.2.1 and 4.2.3
10	The industrial estates	Adopted: stated assumption	Rate from PAM-GUD-201 Table 12 (p61); the workforce is assumed until the estates' records arrive	4.2.2
11	The connection ratio	Adopted: Inception Report	Inception Report R0	4.2.8
12	Infiltration and peak flow	Adopted: guideline	PAM-GUD-201 Sections 7.4.2 and 7.4.3 (p71 to 72)	4.2.5 and 4.2.6
13	Depth	Adopted: guideline recommendation, applied as a limit	PAM-GUD-203 Section 4.6.3 (p33)	3.3.3
14	STP flows and loads	Adopted: guideline	PAM-GUD-201 Section 7.4.5 (p73); PAM-GUD-203 Table 29 (p65) and Section 10.3.1 (p74)	3.4.1 and 4.2.8
15	Treated effluent	Adopted: guideline	PAM-GUD-201 Section 7.4.6 (p73, p76)	4.4

1.5.2. Decisions Requested

Settlements and people

- 1. Settlement boundaries.** Approve the twenty-five settlement boundaries redrawn so that every plot lies in one settlement and there are no gaps between them (Section 4.1.2).

2. **Occupancy rate.** Approve the occupancy rate calculated for each settlement, in persons per property, with the floor of 4.0, the cap of 6.12 and the rule for settlements under a thousand people (Section 4.1.4).

Growth

3. **The overflow.** Approve the overflow: when a settlement's land is full, its further growth moves to its neighbours. Without it 75,000 people of the series have nowhere to go by 2070 (Section 4.1.8).
4. **The design horizon.** Decide the design horizon: 2055, the opening year 2030 plus 25 years, with 244,914 people and 42,266 cubic metres a day; or 2070, the year the land is full, with 349,029 people and 60,099 cubic metres a day (Section 4.1.9).
5. **Growth rate beyond 2050.** Approve the growth rate beyond 2050, where the population series is extended: the rise to 2.40 per cent a year by 2058 and that rate held to 2100. The series to 2050 is the client's own (Section 4.1.7).

The network

6. **Gradients and self-cleansing at the concept stage.** Approve the concept-stage rule: pipes laid at the Table 11 minimum gradient; the tractive-force test at 1 pascal used only to list the pipes needing early washing, with a pipe carrying less than 1.5 l/s checked as if it carried 1.5. The guideline asks for the steeper of the two gradients; this is a departure (Sections 3.3.2 and 6.2.8).

1.5.3. Adopted Values

These are not put to approval. They are the guidelines' own values, a value of the Inception Report, or assumptions stated as such, and the table of Section 1.5.1 says which.

Settlements and people

7. **The 126 meters with no plot within 15 metres.** They are left out of the plot loads: about 140 people (Section 4.1.3).
8. **Land-use layer.** The use of each plot is determined from its meters and the satellite image, and is the basis for placing the loads (Section 4.1.5).

Water and sewage rates

9. **Water demand and return.** 164 litres per person per day, plus 22 per cent for shops and offices and 14 per cent for government, the values the guideline publishes for Adh Dhahirah; 85 per cent of domestic and tanker supply and 54 per cent of the rest return to the sewer. The published ratios are used in place of the unit rates per pupil, bed, employee and floor area until those quantities are held (Sections 4.2.1 and 4.2.3).
10. **The industrial estates.** 4,500 workers at Al Tayyeb and 1,800 at Tanam at 93 litres a day each, an assumption to be replaced by the estates' records when received (Section 4.2.2).

Growth

11. **The connection ratio.** 61 per cent of properties connected in 2030, the Inception Report's ratio, used only for the early-year self-cleansing check (Section 4.2.8).

The network

12. **Infiltration and peak flow.** 720 litres a day per kilometre of sewer, added to each pipe and left out of the early-year check; the Merrimack formula above 100 properties and the Peltier formula at 100 or fewer (Sections 4.2.5 and 4.2.6).
13. **Depth.** 12 metres of cover is the limit at the concept stage, with a pumping station before that point; the preliminary design looks deeper where the quantities allow the cost to be calculated (Section 3.3.3).

The plant and the effluent

- 14. STP flows and loads.** The average annual flow and the peak hourly flow with the 10 per cent margin as the guideline defines them; 60 grams of BOD and 80 grams of suspended solids per person per day, the guideline's minimum, until the laboratory data of the existing STP are received; the maximum day flow from that plant's records (Sections 3.4.1 and 4.2.8).
- 15. Treated effluent.** 95 per cent of the STP inflow produced as treated effluent and 10 per cent of it lost in the distribution network (Section 4.4).

1.5.4. Other Matters for Confirmation

Each of the matters below affects work that would otherwise be carried out twice, and confirmation is requested at the earliest opportunity.

Table 9. Other matters requiring confirmation

Matter	What is to be confirmed	Reference
Project boundary	The updated project boundary of 531.4 square kilometres, which takes in every plot of the twenty-five settlements, and the extent to be surveyed	Section 1.2.2
Project figures	Georeferenced versions of Figures 1, 2 and 3 of the tender, which define the project location, the areas requiring as-built records and the areas subject to each design stage; they are supplied as images without coordinates	Scope of Work
Hydraulic modelling software	The software for the deliverable models: the Scope of Work requires SewerGEMS and WaterGEMS, and the staffing schedule of the tender refers to a different package	Scope of Work, pages 56, 57, 62, 65 and 73; Bidding Form 24, page 123
Environmental impact assessment	The scope required at this stage: the guidelines place the scoping at the preliminary design in one clause and the full assessment at the concept and preliminary stages in another	PAM-GUD-201, Table 2, pages 19 to 22, and Section 6.1.4.3, page 44
Terminology	That TE, as used in the Terms of Reference and in this report, means treated effluent	Abbreviations
Design manual reference	That the treatment plant location report follows Section 10.1 of PAM-GUD-203, Site Selection, which replaced item 2.1 of Section 05 of the former Wastewater Design Manual	PAM-GUD-203 Revision 01, pages 2, 63 and 64
Maximum gradient by tractive force	How the limit of 4 per cent at head pipes, indicated by Nama Water Services on 16 September 2026, is to be applied	Section 3.3.2
Margin and peak at the treatment plant	That the 10 per cent margin is applied to the peak hourly flow as to the average annual flow	Section 3.4.1
Potable water record	That the PAEW dataset is the current record for Ibri	Section 2.2.2

1.5.5. Data Requests

7 items would improve or confirm the numbers in this report. None of them stops the concept design; each replaces an assumption with a record.

Table 10. Data requests

Item	What is asked for	Why
Tanker water and sewage tankers	Filling-station volumes by area and sewage tanker deliveries to the STP, with their source	Water supplied by tanker is not in any flow. It can only raise the loads.
Private wells	Any record of private abstraction inside the study area	The guideline requires it to be assessed. Not held.
Existing STP records	Daily inflow and the laboratory results for the Ibri STP, three years	Sets the maximum day flow and checks the per-person loads.
Industrial estates	Workforce and water use of the Al Tayyeb and Tanam estates	Replaces the assumed 4,500 and 1,800 workers.
Army camp	Occupancy and drainage arrangement of the camp west of the town	296 hectares with no meter in the dataset.
Ibri View	The development plan of the resort at As Sulayf	Two square kilometres announced; no load carried until the plan is issued.
Non-domestic quantities	Pupils, beds, employees and floor areas by plot, where held	Would allow the unit rates of the guideline in place of the ratios.

2. Data Collection and Assessment

This chapter records the data collected, the checks applied to each dataset and what they found, and the surveys in progress.

2.1. Data Collection

Data has been requested from Nama Water Services and from the authorities holding assets in the project area. The table below records what has been requested and the position at the date of this report.



Figure 14. Position of the data register at the date of issue.

Table 11. Data register

Dataset	Received	Position
Constructed sewer network, as-built	Yes	Geometry received, dated 2006. Levels and diameters not recorded; being surveyed
Constructed force main	Yes	Received, dated 2006
Treated effluent network	Proposed only	The dataset contains no constructed treated effluent asset
Lifting station details	Partial	One station located; equipment and duty details being surveyed
Treatment plant records	Partial	Location and nominal capacity received; operating data outstanding
Electricity account data	Yes	Received
Cadastral plot data	Yes	Received, issue of September 2026; the use of each plot determined from the meters (Section 4.1.5)
Potable water network	Yes	Received as part of the PAEW dataset
Topographic survey	In progress	Survey team mobilised
Integrated Master Plan	No	Requested
Plant inflow and tanker records	No	Requested
Electricity, telecom and gas service records	No	To be requested from the respective owners

A topographic and utility survey covering the whole study area is in progress. It will establish cover and invert levels, diameters, materials and condition for the existing sewer, force main and treated effluent networks, together with the lifting stations, the topography and the cadastral boundaries including plot gates. Its results are incorporated in the design as they are received.

2.2. Data Assessment

Each dataset has been loaded into the project geographic information system, reprojected where necessary, and checked against the project boundary. This section records the quantity of each dataset, the proportion falling within the study area, and the limitations identified.

Assessment of the data supplied

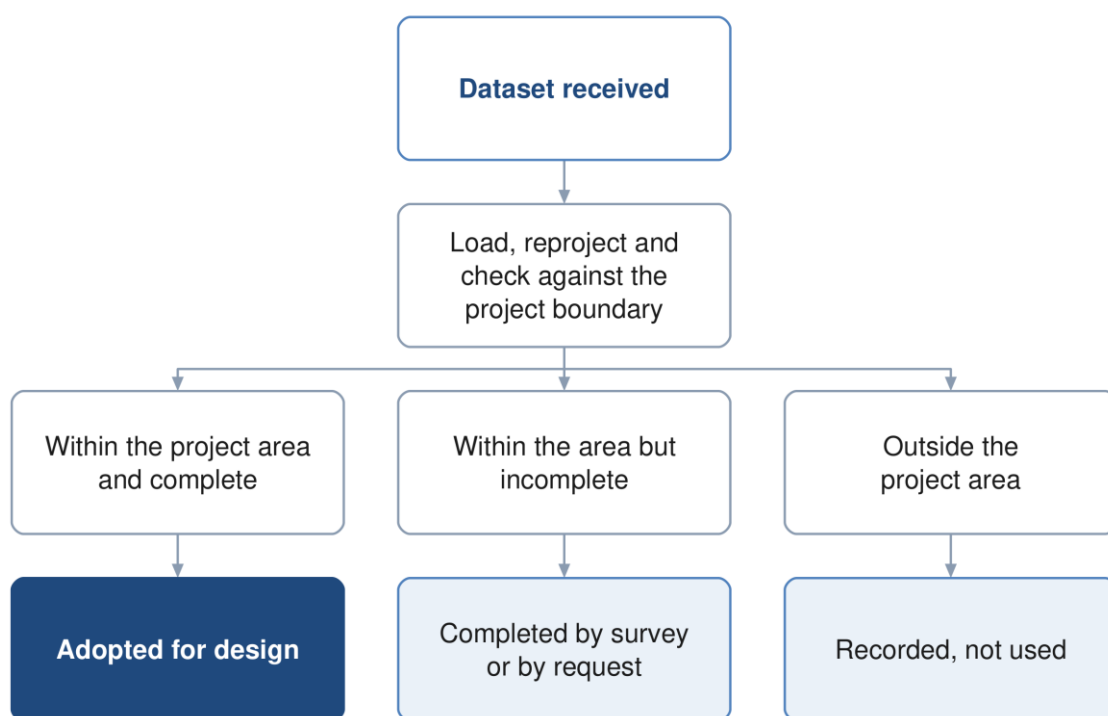


Figure 15. The assessment applied to each dataset supplied.

2.2.1. Wastewater Assets

The wastewater dataset supplied by Nama Water Services holds two networks, not one. They are distinguished by the operational status field, and the distinction is confirmed by four further fields that agree with it on every record.¹

Table 12. The two networks and how they are distinguished

	Constructed network	Proposed network
Operational status	1	0
Installation date	1 January 2006	not recorded
Source	drawings, and closed-circuit television	asset planning
Project code	5A-1 to 5A-5	SUREKHA
Remark	reference data	large urban area gravity, pumping main and treated effluent networks

On that basis the assets within the study area are as follows.

¹ The operational status recorded on every gravity sewer, force main and treated effluent line of the dataset: one value denotes the constructed network and the other the proposed network.

Table 13. Wastewater assets within the study area, by status

Asset	Constructed features	Constructed length	Proposed features	Proposed length
Gravity sewer	3,267	111.6 km	129	199.3 km
Force main	1	10.0 km	8	23.2 km
Treated effluent main	none	none	8	45.7 km
Pumping station	1	1	—	—
Treatment plant	1	1,800 m³/d	1	29,038 m ³ /d, shown as design

The distinction is material to the design. The constructed network extends to 111.6 kilometres of gravity sewer and 10.0 kilometres of force main, serving the central part of the town, and discharges through a single lifting station to the existing treatment plant. It is twenty years old and its condition is not recorded. The remaining 199.3 kilometres of gravity sewer, 23.2 kilometres of pumping main and the whole of the 45.7 kilometre treated effluent main shown in the dataset are proposed, not built. No treated effluent asset has been constructed.¹

Two limitations apply to the constructed network. Neither diameter nor invert level is recorded on any constructed gravity segment, so the hydraulic capacity of the existing system cannot be established from the data supplied. The client's own records carry a remark that the data is not reliable and is to be used for reference only. The survey now in progress will establish diameters, levels and condition, and until it reports, the capacity of the existing network to accept additional flow is treated as unknown.²

¹ Lengths measured within the approved project boundary in EPSG:32640. The proposed alignments are recorded in the dataset as an earlier planning proposal and are not adopted as design input; they are shown for reference and to identify any commitment already made by the client.

² The remarks recorded on the constructed network. The tender records that the existing network layout is based on available information, and that the preparation of complete as-built records and GIS forms part of the consultant's scope.

Wastewater assets: constructed and proposed

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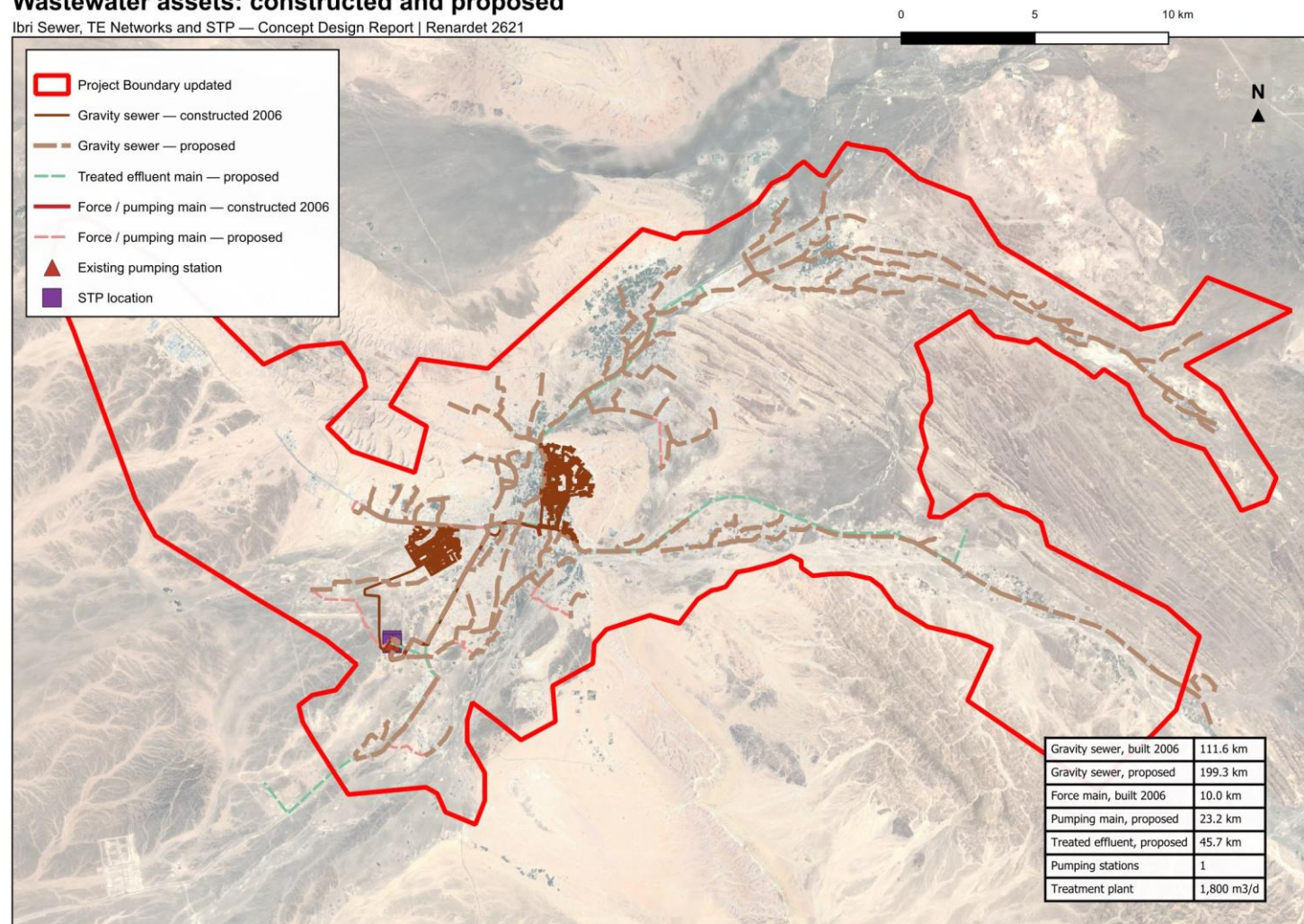


Figure 16. Wastewater assets within the study area. The constructed network, dated 2006, is shown distinctly from the alignments recorded in the dataset as proposed.

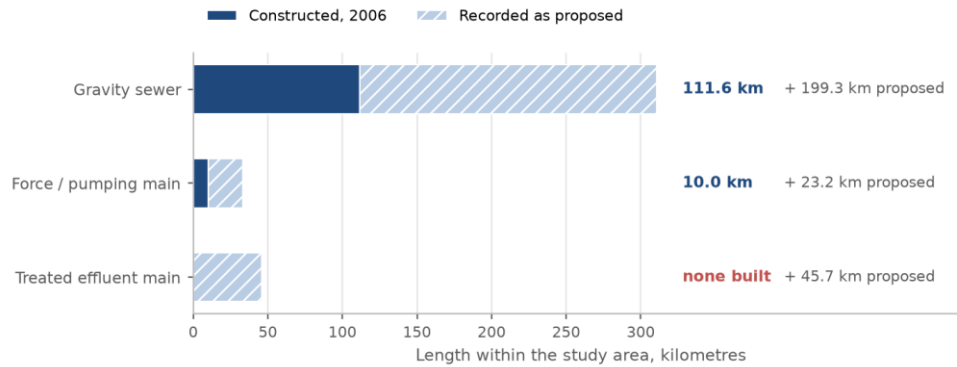


Figure 17. Length of wastewater asset within the study area, separated into constructed and proposed.

2.2.2. Potable Water Network

Two datasets describing potable water assets were supplied. They differ substantially in coverage.

Table 14. Potable water datasets

Dataset	Features	Quantity	Within the study area	Observation
PAEW water mains	6,156	1,315.9 km	647.8 km (49 %)	adopted as the source for utility interfaces
PAEW water laterals	1,954	12.7 km	8.1 km (64 %)	—
PAEW system valves	2,866	2,866 points	1,586	—
PAEW hydrants	965	965 points	568	—
PAEW services	318	318 points	—	—
PAEW facilities	265	265 points	20	—
NAMA water mains extract	90	3.5 km	none	located near 55.80 E, 24.27 N, approximately 130 km north-west of the project area

The PAEW dataset provides 647.8 kilometres of water mains within the study area and is adopted as the source for utility interfaces. The second dataset, supplied as the Ibri water network, contains 3.5 kilometres of mains located approximately 130 kilometres north-west of the project area; none of it falls within the study area. Confirmation is requested that the PAEW dataset is the current record for Ibri.¹

¹ The extent of the second dataset is 55.802 to 55.814 degrees east and 24.269 to 24.292 degrees north, in the vicinity of Al Buraymi.

Potable water network within the study area

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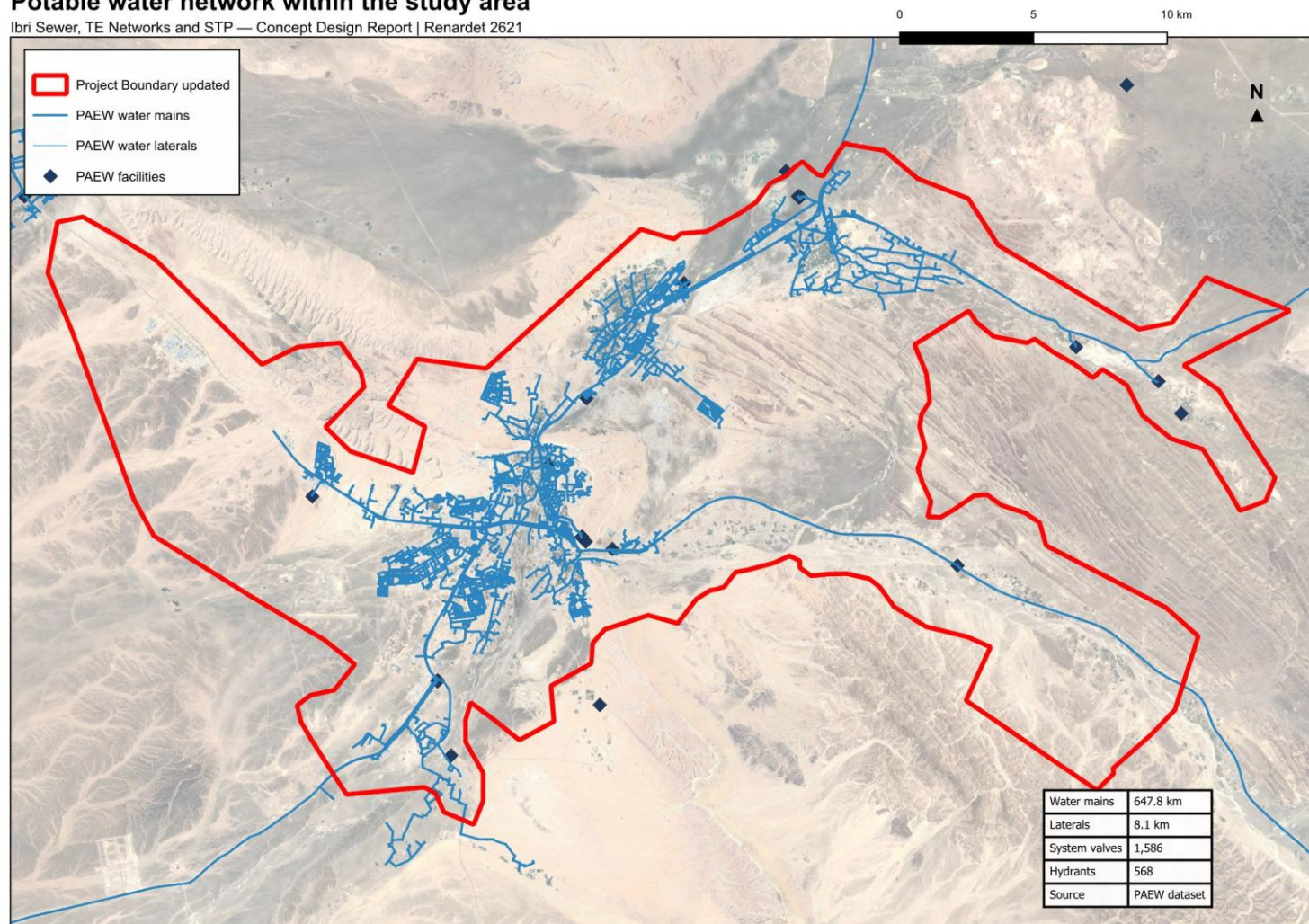


Figure 18. Potable water network within the study area, from the PAEW dataset.

2.2.3. Electricity Accounts

The electricity account dataset contains 33,971 records, each carrying a tariff name and a coordinate. It records neither land use, floor area nor consumption, and the wilayat field is empty on every record. It establishes the number and category of connections at a location, and is used for that purpose in Section 4.1. The data is of 2024, which is the base year of this report.¹

Table 15. Electricity accounts by tariff and the category adopted

Tariff	Accounts	Category adopted
Primary Account	10,973	Domestic: one property
Primary Account with National Subsidy	5,272	Domestic: one property
Additional Account	6,344	Domestic: one property, a further one on the same plot
Commercial, Fisheries, Tourism	9,392	Non-domestic
Government, Defence	967	Governmental
Agricultural	523	Agricultural: an irrigation pump, no sewage
Industrial	1	Special consumption
Cost Reflective Tariff	499	Large consumer; use established in Section 4.3
Total	33,971	

The Cost Reflective Tariff is applied to consumers above a consumption threshold and is therefore a measure of size rather than of use. The 499 accounts carrying it have been placed one by one against public records, and each carries the use found and the evidence for it. Section 4.3.1 gives the result.²

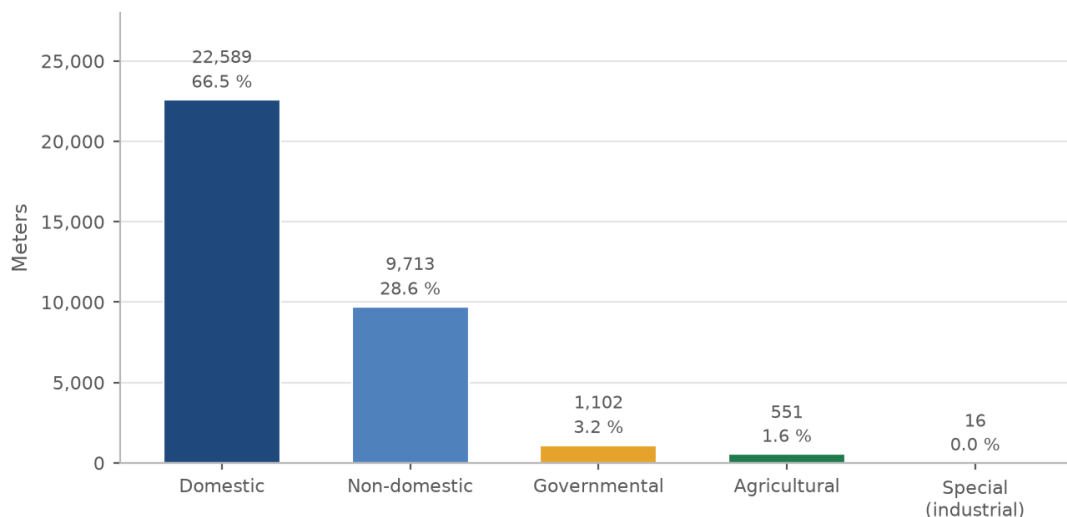


Figure 19. Electricity meters by the category adopted, after the large consumers were placed. Domestic connections, including further properties on the same plot, are two thirds of the total.

¹ Each record carries an identifier, the tariff, coordinates in projected and geographic form, the governorate and the wilayat. The governorate is recorded as Dahira on every record and the wilayat is blank on every record. The positions were adjusted onto the cadastre by the client before issue.

² The tariff comprises three variants in the dataset: fixed rate (9 accounts), seasonal (298) and time of use (192).

Electricity meters by category, placed on the plots

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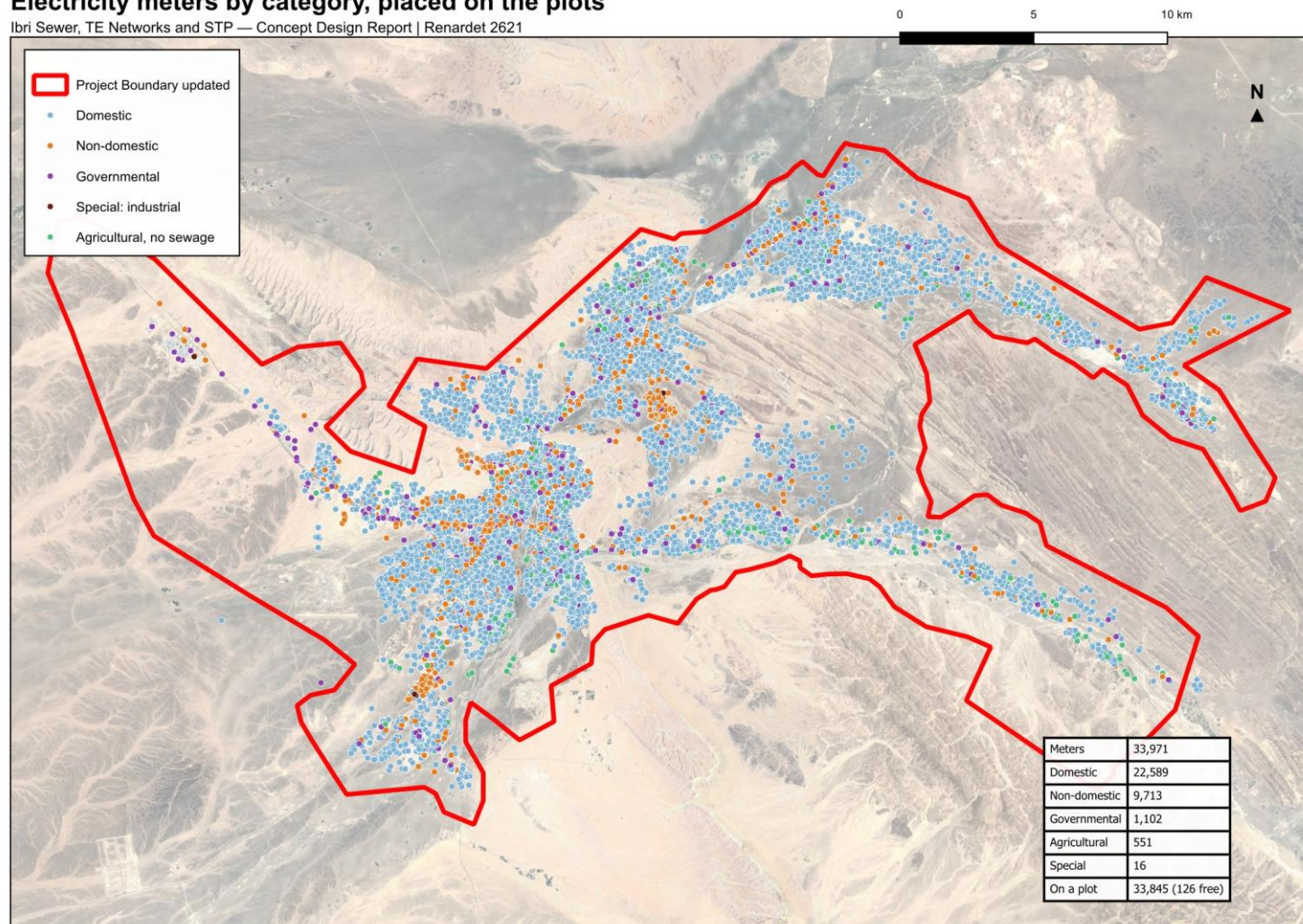


Figure 20. Electricity meters by category, placed on the plots. The pattern of connections defines the developed extent of each settlement.

2.2.4. Cadastral and Settlement Data

The cadastral plot layer supplied by the Ministry of Housing and Urban Planning, in the issue of September 2026, holds 77,265 plots with a built-or-future flag and a land-use class. The flag follows the electricity meters and is reliable. The class is not: it carries no government category, records whole districts under codes that mean unclassified, and disagrees with the meters on 24 per cent of the built plots. The use of each plot is therefore determined from the meters and the satellite image, as Section 4.1.5 sets out. The survey now in progress will establish the cadastral boundaries including plot gates.

The settlement boundaries supplied with the Inception Report are twenty-six polygons for twenty-five settlements, Al Aynayn being drawn as two. They outline the built cores and leave the land between them unassigned. Section 4.1.2 describes how every plot has been assigned to a settlement and the boundaries redrawn to meet.

2.2.5. Other Datasets

Table 16. Further datasets received

Dataset	Extent	Observation
Electricity accounts	33,970 points	Tariff, coordinates and governorate. No land use, floor area or consumption is recorded.
Ibri Bypass scheme	13,404 placemarks	A drawing export carrying 11,715 sublayers of blocks, hatches and dimensions. Suitable as a positional reference; conversion is required before it can be used as structured data.
Al Raybah pipelines	3 files	Pipeline alignments at 110, 180 and 225 mm, within the project area.
Rehabilitation packages	6 archives	Work-order submissions covering Al Sad, Khadil, Yanqul, Dhank and Hay Al Aqabah, comprising drawings, meter schedules and geodatabases.
House connection packages	4 archives	Contractor submissions dated May and June 2026.

Five of the datasets supplied relate to areas outside the project boundary and have not been used. They are recorded here for completeness.

Table 17. Datasets relating to areas outside the project

Dataset	Extent	Location
Mudhaibi zone 6	544 placemarks	Wilayat of Mudhaibi, Ash Sharqiyah
Mudhaibi zone 8	461 placemarks	Wilayat of Mudhaibi, Ash Sharqiyah
Qabil Al Rukha	5 placemarks	Wilayat of Mudhaibi
Qunaib	4 placemarks	Wilayat of Mudhaibi
New location	8 placemarks	Approximately 30 km north of the project area

2.3. Surveys and Investigations

2.3.1. Topographic and Utility Survey

A survey team is mobilised and working across the study area. The survey covers the existing sewer network, lifting stations, force mains and treated effluent network as built; the topography; and the cadastral boundaries including plot gates. Its outputs will establish the levels, diameters and condition that the supplied datasets do not carry, and are incorporated in the design as they are received.

2.3.2. Digital Terrain Model

A bare-earth terrain model at 0.5 metre resolution covering the study area is in use for the work presented here. It will be superseded by the topographic survey for design purposes.¹

2.3.3. Geotechnical Investigation and Trial Pits

Geotechnical investigation and trial pits form part of the scope. Fifty trial pits are to be carried out at critical locations proposed by the consultant and approved by Nama Water Services. The programme of pits will be set out following the desk study of utility records described in Section 7.6.

2.4. As-Built Records and GIS of the Existing Systems

The preparation of as-built records and geographic information for the existing sewer and treated effluent systems forms part of the scope of work. The datasets supplied provide the geometry of those systems; the survey in progress will establish the levels, diameters, materials and condition required to complete the records.

The records will be prepared to the Nama Water Services specification and uploaded to the client's geographic information system for acceptance.

¹ The model excludes buildings and is held in UTM zone 40 North.

3. Basis of Design

This chapter sets out the documents that govern the design and the departures from them, the level of service, and the design criteria for the collection network and the treatment plant.

3.1. Codes, Standards and Departures

The design follows the Nama Water Services design guidelines and standard specifications. Where a value is taken from a guideline it is cited to the guideline and page.

Table 18. Governing documents

Reference	Title
PAM-GUD-201	General Design Guidelines, Revision 01, March 2026
PAM-GUD-202	Water and TSE Design Guidelines, Revision 01, March 2026
PAM-GUD-203	Wastewater Design Guidelines, Revision 01, March 2026
MD 145/1993	Ministerial Decision, treated effluent and sludge reuse
MD 159/2005	Ministerial Decision, discharge to the marine environment
MD 41/2017	Ministerial Decision, ambient air quality

3.1.1. Departures from the Guidelines

The departures from the guidelines that arise from the data available, and the items that await data, are set out below with the reason and the position adopted. Those the guidelines do not settle are put to Nama Water Services as decisions in Section 1.5; the others are adopted and reported there.

Table 19. Departures from the design guidelines

Subject	Guideline position	Position adopted	Reason
Occupancy rate	Population divided by housing units, both from NCSI	Settlement population divided by the domestic electricity meters counted in the settlement, with a floor and a cap (Decision 2)	Housing units are not published at settlement level
Non-domestic and governmental demand	Unit rates per pupil, bed, employee and floor area where detailed land use allocation is available	The published governorate ratios	The quantities the unit rates require are not recorded in any dataset held

Subject	Guideline position	Position adopted	Reason
Allocation of non-domestic demand	Distributed across the served population	Placed on the plots whose meters generate it	A residential street generates no commercial flow; the total is unchanged and only its distribution differs
Industrial estates	Determined case by case with the developer	An assumed workforce at the dry-industry unit rate	No workforce or discharge record has been supplied; the assumption is stated and is to be replaced by the record
Water supplied by tanker	Assessed from the tanker filling station records	Households on tanker supply carry the domestic rate like any other metered household; no separate tanker term	No filling-station record is held; requested
Other water sources	Private wells and other non-network abstraction assessed	Not assessed; every flow is on the network-accounted basis	No record of private abstraction is held; requested
Connection to the sewer	Coverage rising to the whole served area by the end of the period	Every metered property counted as connected; pipes and the STP are sized on the full load, and the early-year self-cleansing case takes the connection ratio of the Inception Report, 61 per cent in 2030	Sizing on the full load is the safe side; the ratio is carried separately where over-estimation would be the risk
Design flow standard	Where detailed land use is available, design flows calculated to a stated international standard such as BS EN 752	The planning ratios and return rates of the guideline, applied plot by plot	The quantities a detailed method requires are not held; the standard will be stated when they are
Minimum gradient	The steeper of the self-cleansing and the minimum tractive force gradients	Table 11 minimum gradients; the tractive force used at the concept stage only to list the pipes needing early washing, at a tractive tension of 1 Pa and with a floor of 1.5 l/s (Decision 6)	The tractive tension is not stated in the guidelines; the method sets gradients at the preliminary design once it is confirmed
Depth of cover	About 10 to 12 m recommended as the point where excavation cost justifies pumping	12 m applied as the limit at the concept stage, with a pumping station before it	The excavation cost cannot be calculated without the detailed quantities
Design horizon	Completion plus twenty-five years, or the saturation of the area	Both carried: 2055 and the saturation year 2070 (Decision 4)	The Terms of Reference name both; the choice is for Nama Water Services
Growth beyond the forecast	Not extrapolated more than ten years beyond the available forecast	The Inception Report series to 2100, used for its growth rates only (Decision 5)	The land fills after 2050; the saturation year rests on the capacity of the cadastre, and the horizon to 2100 was instructed by Nama Water Services

The unit rates will be adopted for non-domestic and governmental demand as soon as the quantities they require become available.¹

3.2. Level of Service and Resilience

The systems are designed to convey and treat the flows arising across the design horizon without surcharge in the collection network and without loss of treatment capacity at the STP.

Redundancy — sufficient capacity is provided that the design peak flow can be achieved with any one unit out of service.

¹ PAM-GUD-201, Table 12, page 61. The rates are expressed per pupil, per bed, per employee and per square metre of floor area.

Design margin — a margin of ten per cent is applied to the STP design flow, over and above redundancy.

Flood resilience — structures are sited and set above the flood levels described in Section 7.2, and the STP is to remain operational during floods.

Failure — emergency provisions are described in Section 6.6.

3.3. Design Criteria for the Collection Network

The criteria below govern the gravity network. All are taken from the wastewater design guideline.

Table 20. Gravity sewer design criteria

Criterion	Value	Reference
Self-cleansing velocity	not less than 0.75 m/s at peak flow	PAM-GUD-203 p26
Preferred velocity	0.90 m/s at peak flow	PAM-GUD-203 p26
Maximum velocity	3.0 m/s at the design depth of flow	PAM-GUD-203 p27
Depth of flow, up to 350 mm	0.65 of the diameter at peak flow	PAM-GUD-203 Table 10
Depth of flow, above 350 mm	0.50 of the diameter at peak flow	PAM-GUD-203 Table 10
Roughness, Colebrook-White	1.5 mm for all sizes and materials	PAM-GUD-203 p24
Minimum cover	1.3 m to the crown of the pipe	PAM-GUD-203 p33
Minimum cover with protection	0.5 m	PAM-GUD-203 p33
Recommended maximum cover	approximately 10 to 12 m	PAM-GUD-203 p33
Minimum diameter, laterals and mains	200 mm outside diameter	PAM-GUD-203 Table 6
Maximum lateral length	45 m	PAM-GUD-203 Table 6
Manhole spacing, 200 to 315 mm	100 m	PAM-GUD-203 Table 12
Manhole spacing, 350 to 900 mm	120 m	PAM-GUD-203 Table 12
Backdrop required	where inverts differ by more than 600 mm	PAM-GUD-203 p30
Inlet angle at a manhole	not less than 90 degrees to the flow	PAM-GUD-203 p19 and p30

3.3.1. Hydraulic Formulation

Full-bore velocity is computed by the Colebrook-White equation.

$$V = -2\sqrt{2gDS} \log_{10} \left[\frac{k_s}{3.7D} + \frac{2.51\nu}{D\sqrt{2gDS}} \right] \quad (1)$$

V : full-bore velocity, m/s

g : acceleration due to gravity, m/s²

D : internal diameter of the pipe, m

S : hydraulic gradient, m/m

k_s : roughness coefficient, 1.5 mm, m

ν : kinematic viscosity of the sewage, m²/s

Gradients are set so that the self-cleansing velocity is achieved at peak flow, and are checked against the minimum gradients tabulated in the guideline. The tabulated gradients correspond to a full-bore velocity of 0.75 m/s; the velocity at the design depth of flow is verified separately for each run.¹

¹ PAM-GUD-203, Table 11, page 29. The tabulated values have been reproduced using the Colebrook-White equation with a roughness of 1.5 mm at 15 degrees Celsius, and correspond to full-bore flow.

3.3.2. Pipe Gradients and Self-Cleansing

At the concept stage every pipe is laid at the guideline's minimum gradient for its size, rounded up to a step of 0.05 per cent, or 0.025 per cent for trunks of 500 mm and above. The steps are our rounding, so that the gradients are round figures on the drawings. The guideline asks for a second test, the minimum tractive force, and for the steeper of the two gradients to govern. At this stage that test is used only to list the pipes that will need washing in the early years, not to steepen them, because the tractive tension it needs is not given in the guideline. One pascal is used until Nama Water Services confirms a value. That is a departure from the guideline, stated here for approval.¹

Table 21. Minimum gradients at the concept stage

Pipe size, mm	200	250	315	400	500	600	700	800	900 and above
Minimum gradient, mm/m	5.00	3.75	2.70	2.05	1.55	1.25	1.00	0.85	0.75

The tractive-force test gives the minimum gradient that moves the deposited particle at a given flow.

$$S_{\min} = K\tau^{1.23}Q^{-0.461} \quad (2)$$

$S_{\min}$: minimum gradient to move the deposited particle, m/m

τ : tractive tension, 1 pascal at the concept stage, Pa

Q : flow in the pipe, m³/s

K : coefficient, 2.33×10^{-4} for the flow in m³/s

The chart on the next page shows what the test asks of a pipe. It is a curve: the smaller the flow, the steeper the pipe must be. A 200 mm pipe at its minimum gradient of 0.5 per cent passes the test at 1.29 l/s and above. Below that, at the very head of a street, no practical gradient passes; those pipes go on the washing list. A design floor is proposed: a pipe carrying less than 1.5 l/s is checked as if it carried 1.5, for which the curve asks 0.47 per cent, so a 200 mm pipe at the Table 11 gradient always passes.²

At the meeting of 16 September 2026 Nama Water Services indicated that the gradient obtained from the tractive-force method is limited to a maximum of 4 per cent at the head pipes. How that limit is applied will be confirmed with Nama Water Services (Section 1.5.4).

Decision 6. Approve the concept-stage rule: pipes laid at the Table 11 minimum gradient; the tractive-force test at 1 pascal used only to list the pipes needing early washing, with a pipe carrying less than 1.5 l/s checked as if it carried 1.5. The guideline asks for the steeper of the two gradients; this is a departure (Sections 3.3.2 and 6.2.8).

¹ PAM-GUD-203, Table 11, page 29; Sections 4.2.2.1 and 4.3, pages 25 to 29.

² The relationship of Mara, Sleigh and Taylor, PAM-GUD-203, page 27. No value of tractive tension in pascals is stated in PAM-GUD-203 or PAM-GUD-201. Section 6.2.8 gives the result for the whole network.

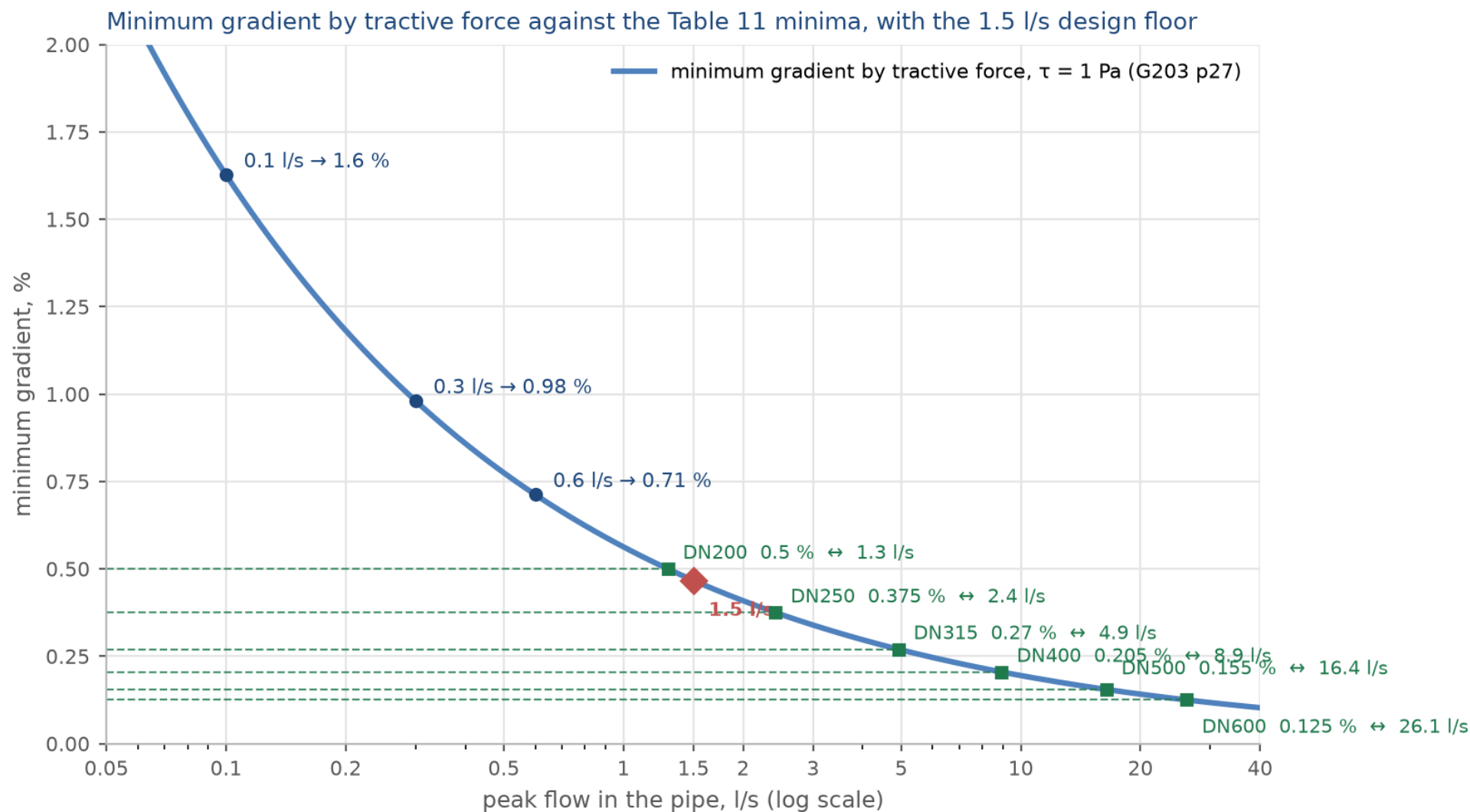


Figure 21. The minimum gradient by tractive force at one pascal against the flow in the pipe, with the Table 11 minimum of every size marked where it meets the curve, and the 1.5 l/s floor.

3.3.3. Depth of Cover and Pumping

A gravity sewer goes deeper the further it runs. The guideline recommends about 10 to 12 metres of cover as the point where the cost of excavation justifies a pumping station. That cost cannot be calculated without the detailed quantities, so 12 metres is applied as the limit at this stage: where a sewer would pass it, a station is placed before that point and the sewer restarts at normal cover. The preliminary design looks deeper where the quantities allow the cost to be calculated.¹

The station's duty flow is the peak flow of the catchment it drains plus the infiltration of its sewers, at saturation. The rising main is sized on that duty: at least 0.75 metres a second, 1.0 where the pumps start and stop, and at most 2.5 metres a second.²

Adopted (13). 12 metres of cover is the limit at the concept stage, with a pumping station before that point; the preliminary design looks deeper where the quantities allow the cost to be calculated (Section 3.3.3).

3.4. Design Criteria for the Treatment Plant

Table 22. Treatment plant design criteria

Criterion	Value	Reference
Design horizon	not less than 15 years	PAM-GUD-203 p65
Planning life cycle	25 years	PAM-GUD-201 p57
Design margin	10 per cent	PAM-GUD-201 p73
Size category	large where 20,000 m ³ /d or above	PAM-GUD-203 p65
Organic load	not less than 60 g BOD per person per day; the measured strength at the existing STP governs once received	PAM-GUD-203 p74
Solids load	not less than 80 g suspended solids per person per day	PAM-GUD-203 p74
COD to BOD ratio, domestic	1.8 to 2.2	PAM-GUD-203 p74
Effluent standard	Class A of MD 145/1993	PAM-GUD-203 p69
Total nitrogen	less than 15 mg/l as N	PAM-GUD-203 p71
Chlorine residual at the STP discharge	0.3 to 1.0 mg/l	PAM-GUD-203 p71
Chlorine residual at the consumer	more than 0.3 and less than 1.0 mg/l	PAM-GUD-203 p130
Treated effluent produced	95 per cent of the inflow	PAM-GUD-201 p73
Sludge produced	0.25 kg per cubic metre of inflow, indicative	PAM-GUD-201 p78

The total nitrogen limit is the governing criterion for process selection. Class A of Ministerial Decision 145/1993 does not itself set a total nitrogen limit, and the individual nitrogen limits it does set would together permit a higher concentration than 15 mg/l. Full nitrification and denitrification are therefore required.³

¹ PAM-GUD-203, Section 4.6.3, page 33.

² PAM-GUD-203, Section 8.1, page 50. Which chamber the main discharges into, and whether neighbouring stations are joined, are layout decisions of the options (Section 6.2.6 and Appendix B).

³ Class A permits ammoniacal nitrogen at 5 mg/l, organic nitrogen at 5 mg/l and nitrate at 50 mg/l as NO₃, equivalent to 11.3 mg/l as N.

3.4.1. Design Flows and Loads

The STP is sized on three flows, and the guideline names what each one sizes: the average annual flow for the biological treatment, with the load; the peak hourly flow for everything the flow passes through; and the maximum day flow, which the guideline defines but gives no factor for. The maximum day flow is read from a year of daily records at the existing STP, requested in Section 1.5.5.¹

$$Q_{AAF} = 1.10(Q_{avg} + Q_{inf}) \quad (3)$$

Q_{AAF} : average annual flow, the design average of the STP, m³/d

Q_{avg} : average flow of the settlements upstream, m³/d

Q_{inf} : infiltration of the whole network upstream, 720 l/d per km, m³/d

1.10: the 10 per cent margin the guideline requires for a new STP, over and above any standby

$$Q_{PHF} = 1.10(Q_{peak} + Q_{inf}) \quad (4)$$

Q_{PHF} : peak hourly flow, for the structures the flow passes through, m³/d

Q_{peak} : the Merrimack peak of the average flow of the settlements upstream (Section 4.2.6), m³/d

The margin is applied to the peak as to the average. The guideline does not state the order, and this reading is for confirmation. Section 4.2.8 gives the flows by year.²

The biological treatment is sized on the load, not only the flow. No laboratory data for Ibri's sewage is held. The guideline sets a minimum for a new STP of 60 grams of BOD and 80 grams of suspended solids per person per day, and gives the strength of raw sewage below. The per-person loads are used until the laboratory results of the existing STP are received, and the sewage brought by tanker, which is far stronger, is provided for separately once its records arrive.³

Table 23. Raw sewage strength the guideline gives, for the network and for tankers, mg/l

Parameter	Network sewage	Sewage by tanker, average to maximum
BOD	350 to 400	350 to 1,050
COD	700 to 900	1,350 to 5,000
Suspended solids	400 to 500	900 to 4,300
Total Kjeldahl nitrogen	60 to 80	115 to 265
Ammonia as nitrogen	40 to 50	70 to 125
Total phosphorus	10 to 15	16 to 35

¹ PAM-GUD-203, Table 29, page 65, and Section 10.2.2.1, page 66.

² PAM-GUD-201, Section 7.4.5, page 73.

³ PAM-GUD-203, Section 10.3.1, page 74, "at least 60 g of BOD5 per capita per day and 80 g of suspended solids"; Table 30, page 67; Table 31, page 68. Peak factors for the aeration design at a large STP: 1.2 on BOD and COD, 1.5 on TKN (page 74).

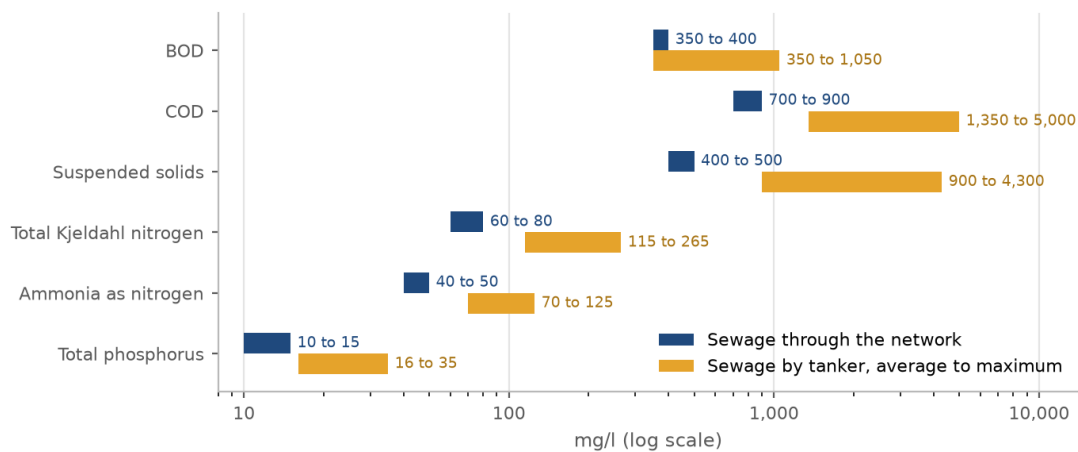


Figure 22. The strength of raw sewage the guideline gives: through the network, and by tanker. The tanker ranges run from the average to the maximum.

Adopted (14). The average annual flow and the peak hourly flow with the 10 per cent margin as the guideline defines them; 60 grams of BOD and 80 grams of suspended solids per person per day, the guideline's minimum, until the laboratory data of the existing STP are received; the maximum day flow from that plant's records (Sections 3.4.1 and 4.2.8).

4. Population, Demand and Wastewater Flows

This chapter establishes the population, the use of the land and its growth to saturation, the wastewater flow that every element of the system is designed for, the non-domestic discharges and the treated effluent available.

4.1. Population and Land Use

4.1.1. Methodology

The population of the study area, its distribution and its growth are established by counting rather than by assumption. Every domestic electricity meter is a property. Every plot takes its use from the meters on it. Every empty plot takes its future from the built plots around it. The official population series supplies the growth rate of each settlement and nothing else. The result is a population and an average sewage flow for every plot, today and in every year to the year the land is full, which is the saturation the Terms of Reference ask for.

The distribution of population across the study area is shown below from the Global Human Settlement population grid. The grid is a third-party product and is used to show where population lies, not how much of it there is.¹

¹ GHS-POP, European Commission Joint Research Centre, 2025 epoch at 100 m resolution, licensed under Creative Commons Attribution 4.0. The grid distributes census counts across built-up surface and is therefore modelled rather than observed.

Where the population is concentrated

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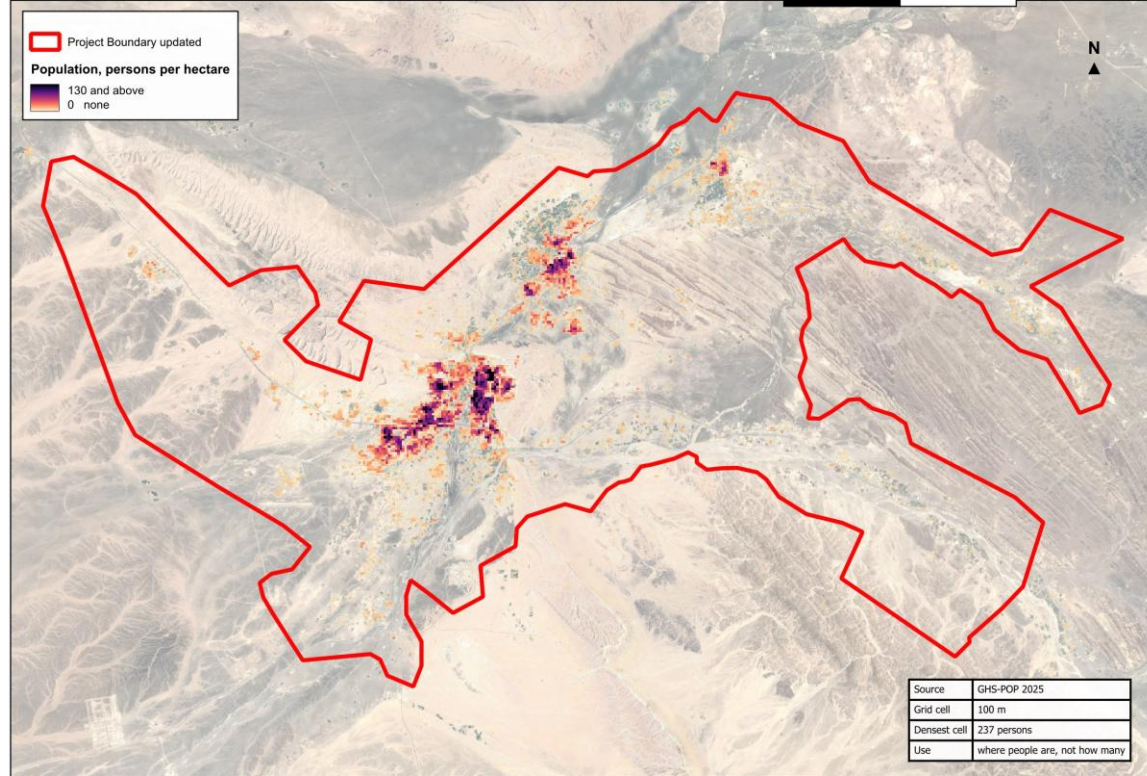


Figure 23. Where the population of the study area lies. Density is concentrated in Ibri town and along the wadi corridors, and falls away sharply beyond them.

4.1.2. Settlements and Settlement Boundaries

Every rule in this chapter is applied settlement by settlement: the persons per property, the share of empty land that becomes housing, the growth rate and the year the land is full are all set for each settlement. So every plot must belong to exactly one settlement.

The Inception Report supplied twenty-six polygons for the twenty-five settlements, Al Aynayn being drawn as two. The polygons are drawn around the built cores. They do not touch, so the land between them belongs to nobody, and four of them, at Tanam, Satwah, Al Makhtibiyah and Bat, miss part of the plots that carry the settlement's name. The map on the next page shows them in red.¹

The boundaries were therefore redrawn. Each of the 77,265 plots was given the settlement whose polygon contains it, or, where none does, the nearest one. The land was then divided so that every point belongs to the settlement of the nearest plot, and the shared edges were smoothed. The result, in blue on the map, covers the whole study area with twenty-five settlements and no gaps. Nothing else changed: the names, the population series and the plots are as received.²

The twenty-five settlements inside the study area carry 63.4 per cent of the wilayat's population in every year of the series. The other settlements of the wilayat lie outside the boundary and are not part of the design.

¹ The boundary issued with the Inception Report R0.

² A Voronoi division of the plot centroids by settlement, merged, clipped to the study area boundary and smoothed as a coverage so that neighbours share one line.

Settlement boundaries: as received, and redrawn as a partition

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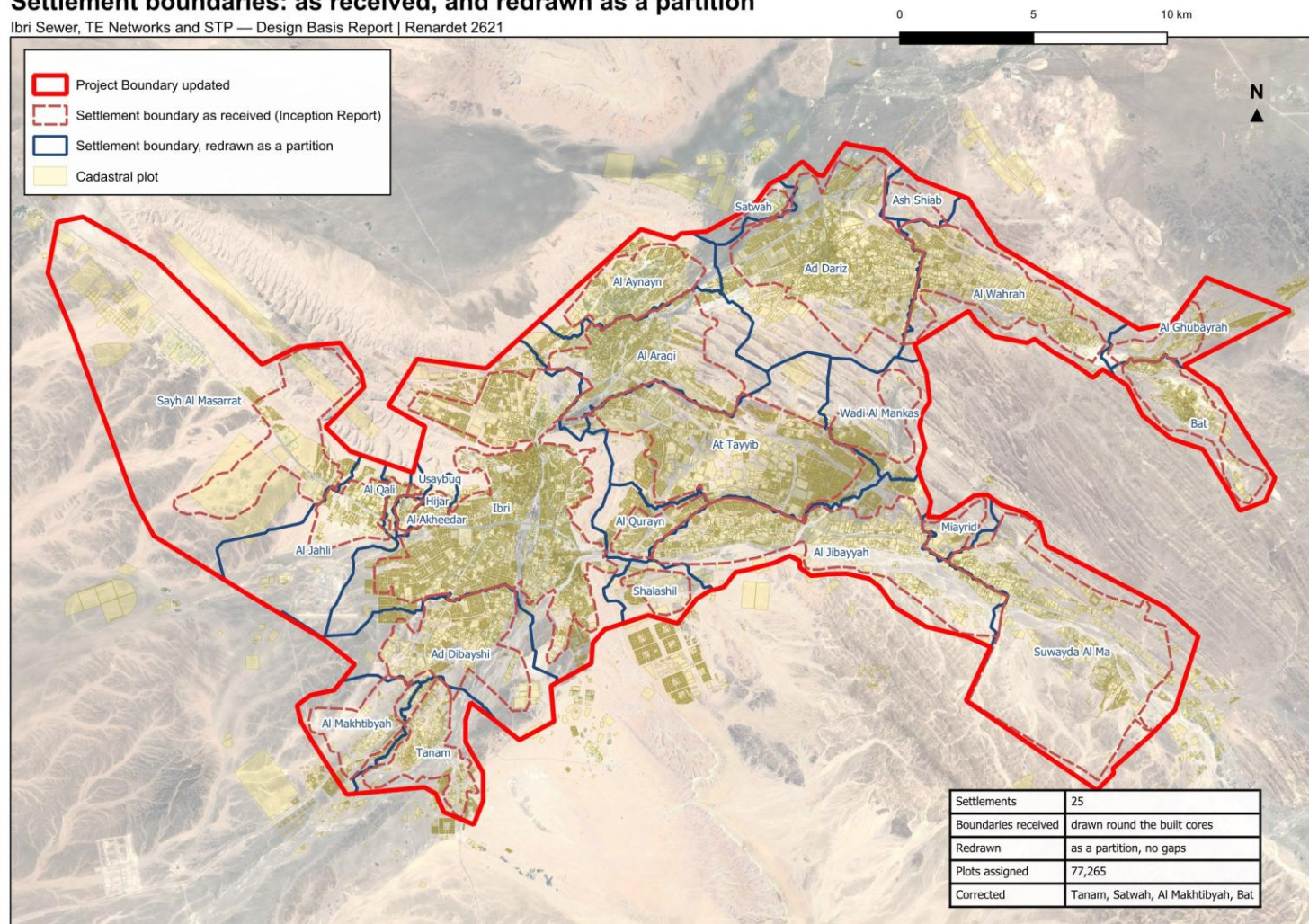


Figure 24. The settlement boundaries as received (red, dashed) and as redrawn for the design (blue). Every plot lies in one settlement and the boundaries meet without gaps.

Decision 1. Approve the twenty-five settlement boundaries redrawn so that every plot lies in one settlement and there are no gaps between them (Section 4.1.2).

4.1.3. Properties from Electricity Meters

Each domestic electricity meter is one property. Of the 33,971 meters, 30,931 fall inside a plot. 2,914 lie on a road or in a gap between plots and were given the nearest plot within fifteen metres. 126 lie further than that from any plot and are shown below. The plots hold 22,559 domestic properties. Shop, office, government, farm and large-consumer meters carry no people.¹

Properties per home plot

The number of properties on a built home plot is measured in each settlement over the plots that are homes and nothing else, excluding the few plots that carry fifteen or more domestic meters, which are blocks of flats and workers' housing and would distort the ratio. Plots that also carry shops, farms or offices are counted for their people but not for the ratio.

Table 24. Properties per home plot, the principal settlements

Settlement	Domestic properties	Home plots measured	Properties per home plot
Ibri	11,052	5,853	1.33
Ad Dariz	2,439	1,567	1.20
Al Araqi	2,466	1,282	1.32
Al Aynayn	889	560	1.14
Al Wahrah	716	420	1.22
At Tayyib	833	623	1.09
Al Jibayyah	524	291	1.26
Bat	418	239	1.17

Ibri carries 1.33 properties per home plot; the principal settlements lie between 1.1 and 1.4. Four small settlements with institutional housing on a handful of plots return two or more; the thirteen settlements under a thousand people take one property per plot, as Section 4.1.4 sets out. The full table is given in Appendix A.3.²

Meters More Than 15 Metres from a Plot

126 meters lie more than fifteen metres from any plot. 30 of them are domestic meters, about 140 people at their settlement's occupancy; the rest are government, shop and farm meters. Some stand on buildings the cadastre has not yet captured. Others stand on nothing: a road reserve, open ground, a pumping station. All of them are left out of the plot loads. The map on the next page shows where they are, and the figure after it shows nine of them close up, with the plots around them.³

¹ The primary, subsidised primary and additional-account tariffs count as properties. The 499 large-consumer accounts were placed by public records; their use decides only where the shop and government water is placed, not how much there is.

² 12,443 home plots were measured across the twenty-five settlements.

³ By tariff: Government 46, Commercial 32, Primary Account Tariff 18, CRT Seasonal 9, Primary Account Tariff (with National Subsidy) 6, Additional Account Tariff 6, Agricultural 5, CRT Time of Use 4.

Table 25. The 126 meters by settlement

Settlement	Meters	Settlement	Meters
Ibri	29	Ad Dibayshi	3
Sayh al Masarrat	14	Al Jahli	3
Al Qurayn	14	Al Makhtibyah	2
Al Araqi	9	Al Wahrah	2
Suwayda al Ma	9	Shalashil	1
Al Jibayyah	9	Ash Shiab	1
Tanam	8	Al Qali	1
Ad Dariz	6	Bat	1
Al Ghubayrah	5	Al Akheedar	1
At Tayyib	4	Wadi al Mankas	1
Al Aynayn	3		

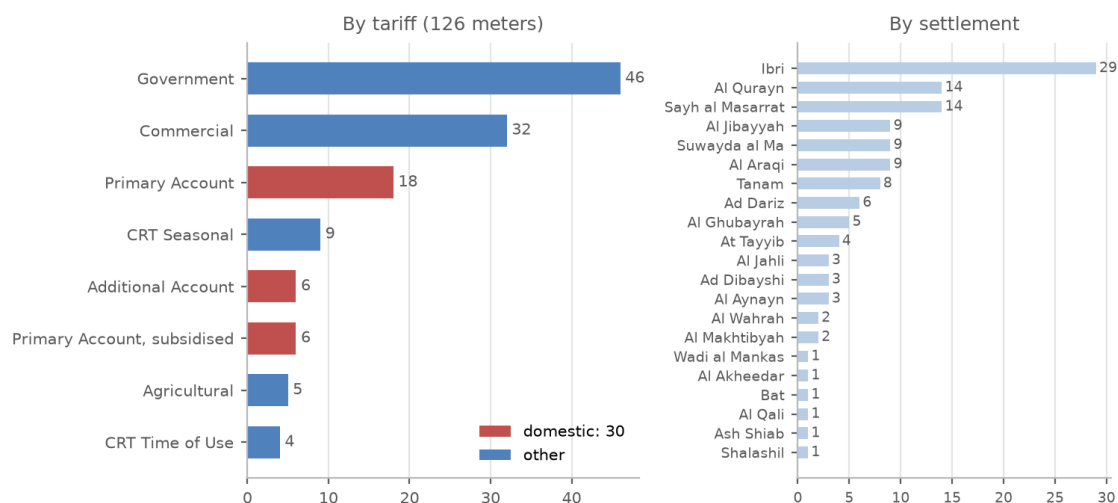


Figure 25. The 126 meters by tariff and by settlement. Thirty are domestic meters.

Adopted (7). They are left out of the plot loads: about 140 people (Section 4.1.3).

The meters more than 15 metres from any plot

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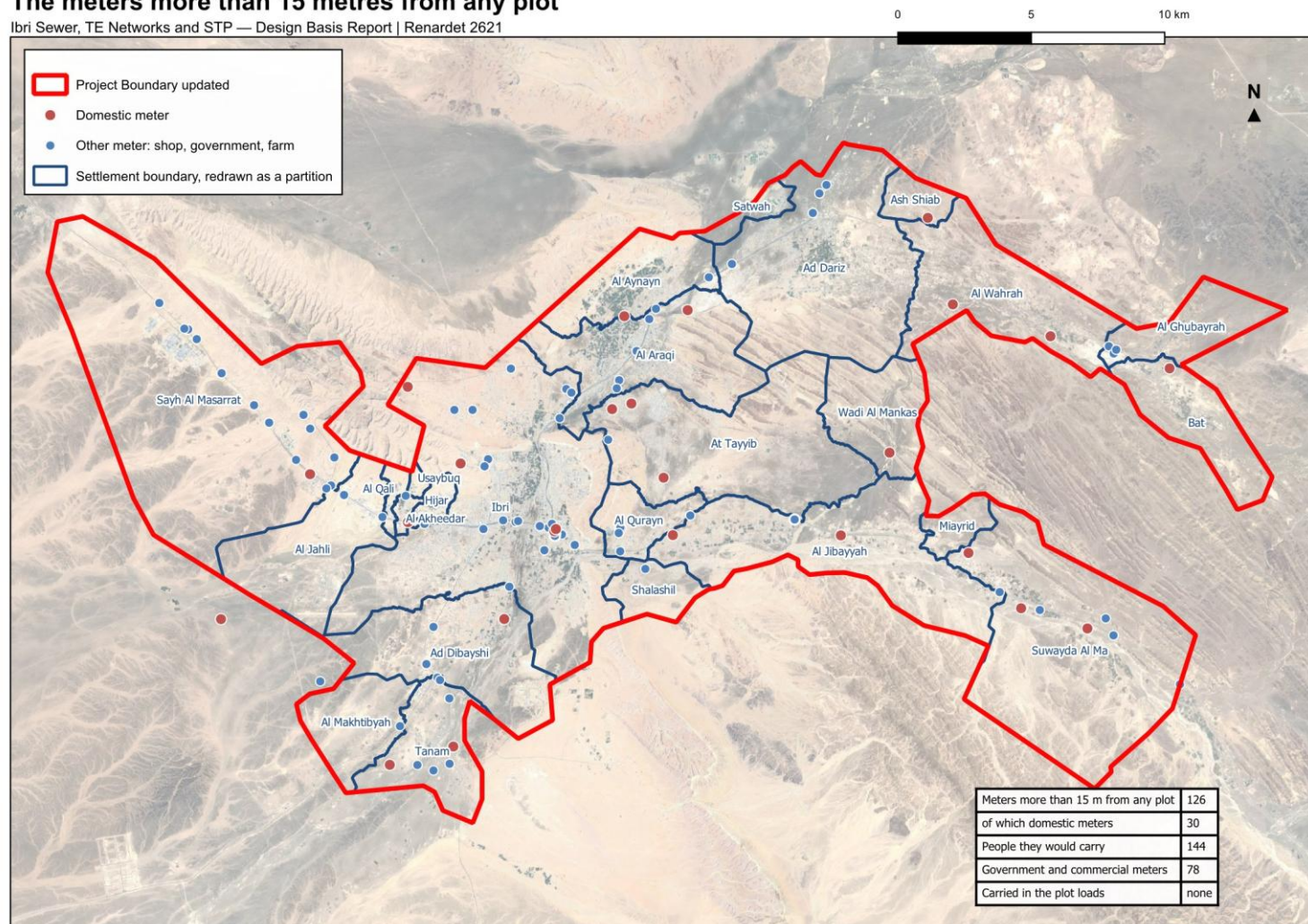


Figure 26. The 126 meters more than fifteen metres from any plot. Red: a domestic meter; blue: any other meter.



Figure 27. Nine of the 126 meters close up: on a road, in open ground, at a pumping station, on a building the cadastre does not carry. The plots are outlined in yellow.

4.1.4. Occupancy Rate

The occupancy rate of each settlement is its population in 2024, the year of the electricity data, divided by the domestic properties counted in it.

$$OR = \frac{\text{Settlement population in 2024}}{\text{Domestic properties}} \quad (5)$$

OR: occupancy rate, persons per property

Three rules bound the result. A floor of 4.0: some settlements would otherwise return between one and four persons per property, because their meters include the housing of the police, the college and similar institutions, whose occupants the census does not count as residents. Those properties discharge to the sewer whatever the census says, so the floor keeps them in. A cap of 6.12, the highest rate among the settlements of two thousand people or more. And a settlement of fewer than a thousand people takes 4.0 outright, one property per home plot and a home share of 0.9 (Section 4.1.6), because it has too few meters to measure on. That rule applies to 13 settlements.¹

Ibri returns 6.07 persons per property. Over the whole study area the census gives 5.16. The guideline derives occupancy from population and housing units published by the statistics centre; housing units are not published at settlement level, and the counted properties are used in their place.²

Table 26. Occupancy rate by settlement, 2024

Settlement	Domestic properties	Population 2024	Rate calculated	Rate adopted	People in the design
Ibri	11,052	67,106	6.07	6.07	67,106
Ad Dariz	2,439	11,850	4.86	4.86	11,850
Al Araqi	2,466	10,696	4.34	4.34	10,696
Al Aynayn	889	4,554	5.12	5.12	4,554
Al Wahrah	716	3,351	4.68	4.68	3,351
At Tayyib	833	3,348	4.02	4.02	3,348
Al Jibayyah	524	2,686	5.13	5.13	2,686
Bat	418	2,557	6.12	6.12	2,557
Ad Dibayshi	681	2,411	3.54	4.00	2,724
Tanam	403	2,116	5.25	5.25	2,116
Suwayda al Ma	301	1,559	5.18	5.18	1,559
Hijar	352	1,033	2.93	4.00	1,408
Al Ghubayrah	161	631	3.92	4.00	644
Al Qurayn	147	549	3.73	4.00	588
Sayh al Masarrat	457	466	1.02	4.00	1,828
Al Jahli	344	415	1.21	4.00	1,376
Satwah	27	263	9.73	4.00	108
Al Akheedar	120	198	1.65	4.00	480
Al Qali	56	191	3.41	4.00	224
Shalashil	38	130	3.42	4.00	152
Al Makhtibyah	91	122	1.34	4.00	364

¹ 4.0 is the rate measured at At Tayyib, 3,300 people, the smallest settlement with a sample worth measuring. The cap is Bat's. The thirteen small settlements: Al Ghubayrah, Al Qurayn, Sayh al Masarrat, Al Jahli, Satwah, Al Akheedar, Al Qali, Shalashil, Al Makhtibyah, Usaybuq, Wadi al Mankas, Ash Shiab, Miayrid.

² PAM-GUD-201, Section 7.2, page 59. 116,452 divided by 22,559 properties gives the area-wide 5.16. The Inception Report used a single rate for all settlements; the rate per settlement replaces it.

Settlement	Domestic properties	Population 2024	Rate calculated	Rate adopted	People in the design
Usaybuq	21	91	4.33	4.00	84
Wadi al Mankas	8	52	6.49	4.00	32
Ash Shiab	11	47	4.27	4.00	44
Miayrid	4	34	8.49	4.00	16
Total	22,559	116,452			119,893

With the floor and the cap the study area holds 119,893 people, against 116,452 in the official series for the same settlements. The difference is the institutional housing.

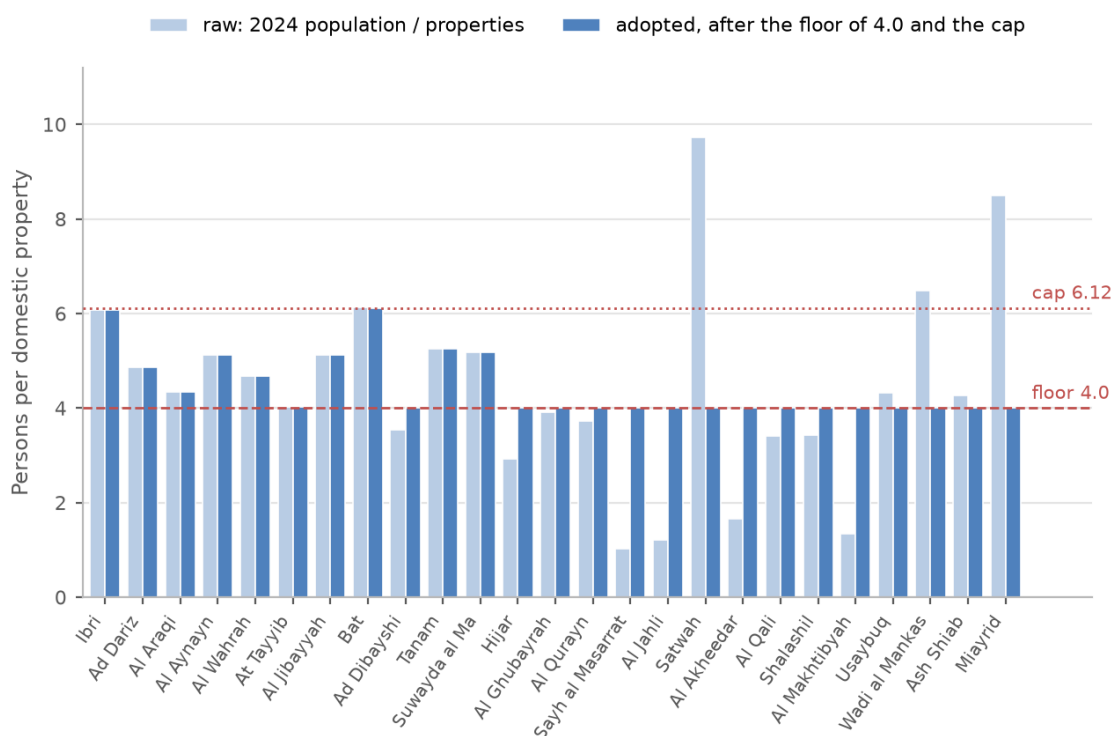


Figure 28. Occupancy rate by settlement: the value calculated from the census and the meters, and the value adopted after the floor and the cap.

Decision 2. Approve the occupancy rate calculated for each settlement, in persons per property, with the floor of 4.0, the cap of 6.12 and the rule for settlements under a thousand people (Section 4.1.4).

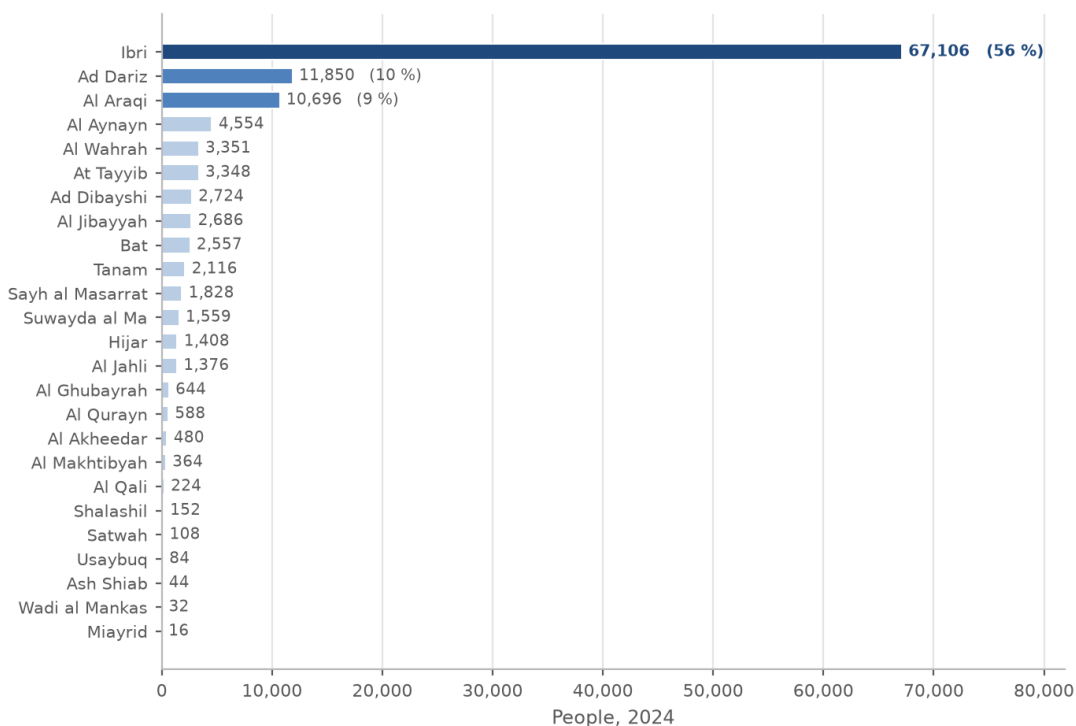


Figure 29. People in each settlement in 2024, from the domestic meters at the settlement's occupancy rate. Ibri holds 56 per cent of the total.

4.1.5. Land Use

The cadastral layer received in September 2026 holds 77,265 plots. Its own land-use field cannot be used as it stands: it has no government class, it records whole districts under codes that mean unclassified, and on the plots that carry meters it disagrees with the meters on 24 per cent of them. The use of every plot has therefore been determined from what is on it, in a fixed order.¹

¹ The cadastral plots of the Ministry of Housing and Urban Planning, 77,265 plots, received in September 2026.

From the electricity meter to the use of each plot

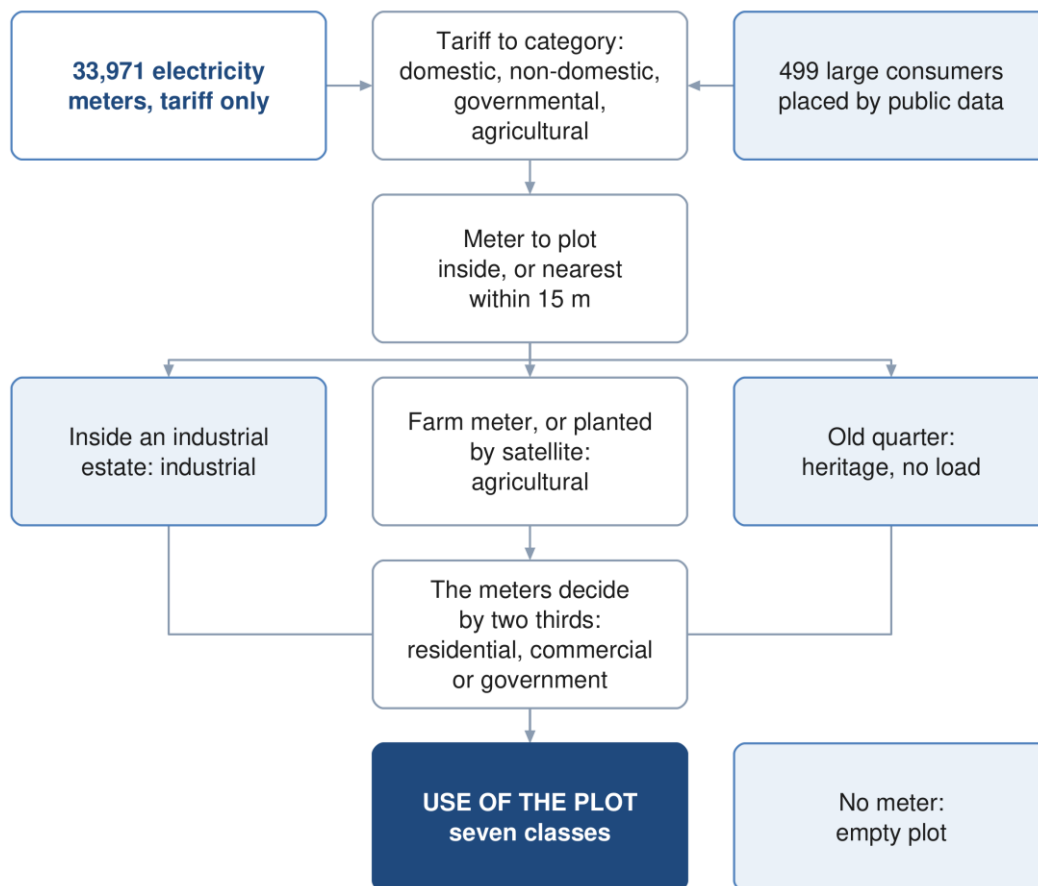


Figure 30. How the use of each plot is determined from the electricity meters, the satellite image and the plot itself.

1. a plot inside one of the two industrial estates is industrial;
2. the 177 plots of the old quarter of Ibri, recorded as tourism in the cadastre, are heritage and carry no load;
3. a plot with a farm meter is agricultural;
4. a plot the satellite image shows as planted is agricultural, unless two thirds of its meters are shops or most are government;
5. a plot whose industrial meters equal or outnumber its domestic meters is industrial;
6. otherwise the meters decide by proportion: more than two thirds domestic makes it residential, two thirds or more government makes it government, two thirds or more shops makes it commercial, and a mix is residential-commercial, or government where the government meters are the more;
7. a plot without a meter is empty.

The satellite image is a cloud-free Sentinel-2 image of 9 September 2026. It is used for one purpose: to tell a planted plot, a date-palm plantation, from a bare one, because a farm meter alone misses the plantations that pump from elsewhere.¹

¹ Sentinel-2 Level 2A, tile 40QDL, red and near-infrared bands at 10 m. 38 per cent of the plots with a farm meter show no planting: pump sites without a crop. The treatment plant's own plot carries farm meters for its pumps and is classed as a government site.

Table 27. Use of the plots

Use	Plots	How it was decided
Residential	12,464	more than two thirds of the meters are domestic
Residential-Commercial	614	a mix, with at least as many shop as government meters
Commercial	1,321	two thirds or more of the meters are shops
Government	410	two thirds or more government; or a mix with more government than shop meters; the treatment plant
Agricultural	1,684	a farm meter (488) or planted on the satellite image (1,196)
Industrial	263	inside an industrial estate, or industrial meters that match the domestic meters
Heritage	177	the old quarter of Ibri; no meter, no load
Empty	60,509	no meter

The use decides where the shop and government water is placed and which empty plots can become homes. It does not change the total: that comes from the people and the rates of Section 4.2.

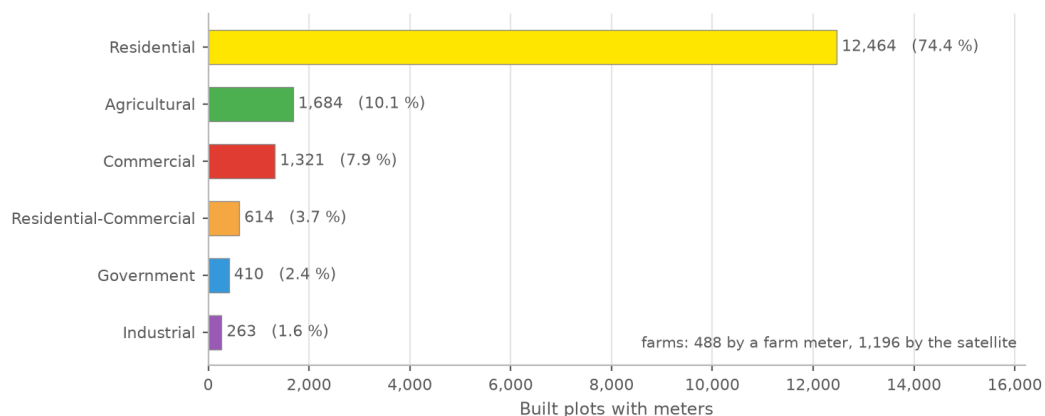


Figure 31. Use of the built, metered plots. Residential plots are 74 per cent of them; agricultural plots, 10 per cent, are found by the meter or by the satellite image.

Adopted (8). The use of each plot is determined from its meters and the satellite image, and is the basis for placing the loads (Section 4.1.5).

Use of each plot, derived from the meters and the satellite

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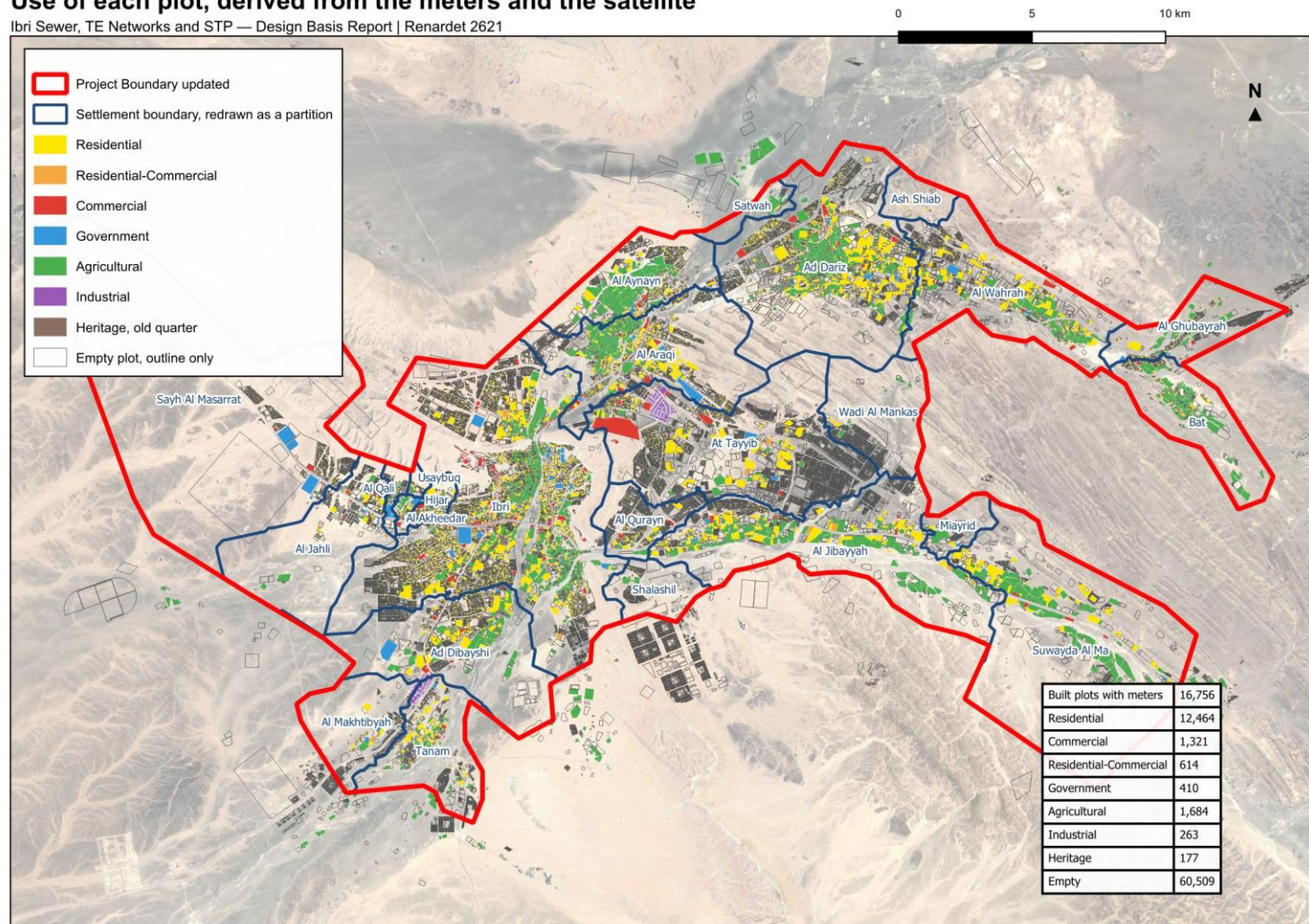


Figure 32. The use of each plot. Coloured plots are built and metered; empty plots are shown by their outline only.

4.1.6. Development Capacity of Vacant Plots

Of the 77,265 plots, 60,509 carry no meter. Not all of them will become homes. A future home plot must look like the built ones: between 200 and 1,000 square metres, compact, not a strip. Very small plots below 200 square metres, odd shapes and everything above 1,000 square metres are left out, as are plantations, industrial plots and the heritage quarter. 46,109 empty plots pass that test.¹

A district also needs mosques, shops and offices, so not every home-shaped plot becomes a home. The share that does is measured in each settlement over its built, metered, home-shaped plots: residential plots divided by all. Ibri returns 0.87; among the settlements of a thousand people or more the measured share runs from 0.67 in Tanam, with its workshops, to 0.94 in Al Aynayn.

The capacity of a settlement is then its home-shaped empty plots, multiplied by its home share, by its properties per home plot and by its occupancy rate.

$$\text{Capacity} = \text{Empty home-shaped plots} \times \text{Home share} \times \text{Properties per plot} \times \text{OR} \quad (6)$$

Capacity: people the empty land of the settlement can hold, persons

Home share: share of the home-shaped plots that become homes

OR: occupancy rate of the settlement, persons per property

The empty land of the study area holds 229,136 people on this basis, against 119,893 living in it in 2024.²

¹ Compactness is the plot's area against that of its smallest enclosing rectangle, not less than 0.6; the rectangle's sides may not differ by more than three to one. Built home plots have a median area of 718 square metres and a median compactness of 0.98. Shapes above 2,000 square metres are whole future districts drawn as one plot and take nothing until they are subdivided.

² The thirteen settlements under a thousand people take a home share of 0.9 and one property per plot, as Section 4.1.4 sets out. Capacity by settlement: Appendix A.3.

4.1.7. Population Growth Rate

From the empty plot to the saturation year

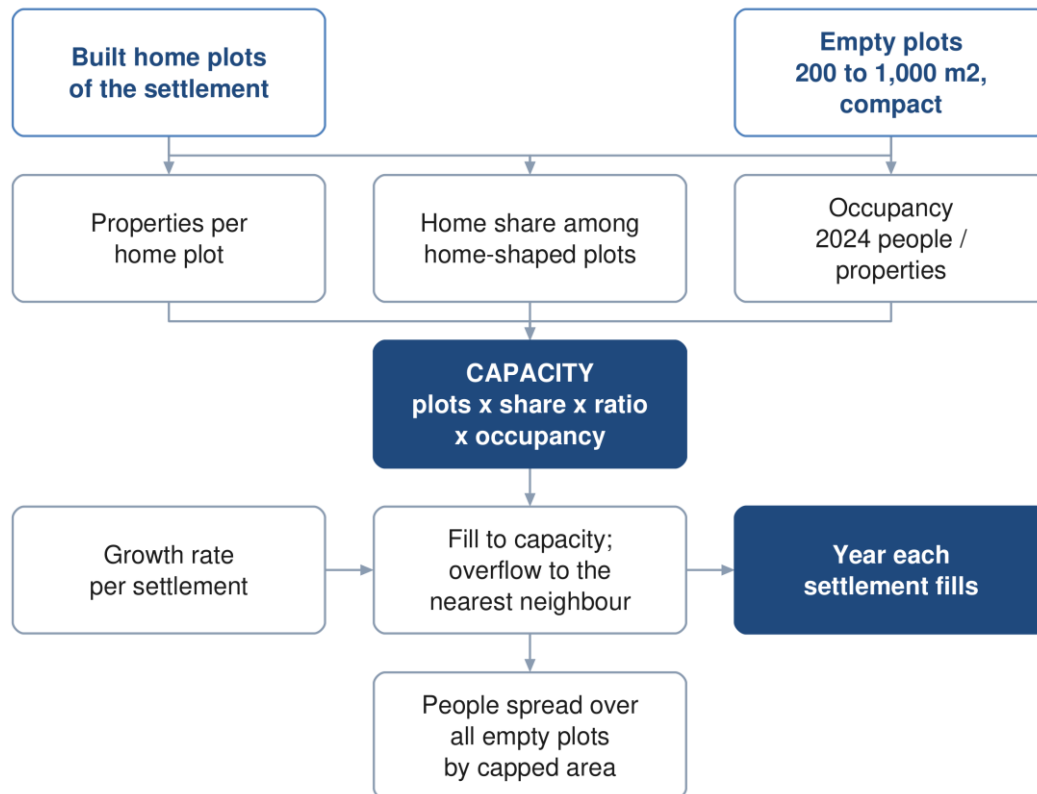


Figure 33. From the empty plot to the year each settlement is full.

The growth rate comes from the population series of the Inception Report, and nothing else is taken from it. To 2040 the series is the official forecast for the wilayat, from the statistics centre and Nama Water Services; from 2041 to 2050 it is the extension of that forecast in the Inception Report. Both are the client's own figures. From 2051 the series rises from 2.2 per cent a year to 2.40 by 2058 and stays there to 2100, the horizon Nama Water Services instructed; that part goes beyond the ten years of extension the guideline allows and is put to approval. Every settlement carries the same annual rate, applied to its counted population. The rate averages 2.6 per cent a year to 2030, 2.5 in the 2030s, 2.0 in the 2040s, 2.3 in the 2050s and 2.4 after 2060.¹

The series' total for 2100 is not a target. The land fills long before then.

¹ Inception Report R0, Section 6, and its demand calculation for the wilayat and for the project settlements. Wilayat population 183,564 in 2024; the twenty-five settlements are 63.4 per cent of it in every year. PAM-GUD-201 allows an extrapolation of ten years beyond the available forecast.

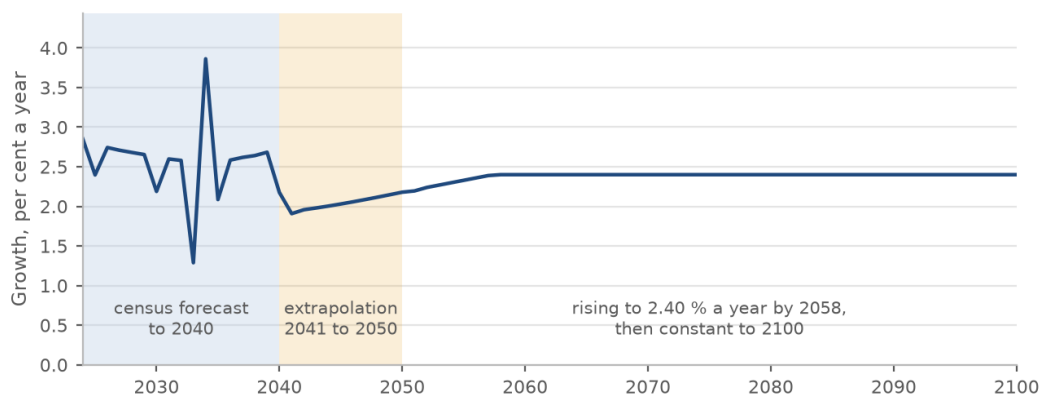


Figure 34. The annual growth rate of the series: the forecast to 2040, the extension to 2050, and the rise to a constant 2.40 per cent.

Decision 5. Approve the growth rate beyond 2050, where the population series is extended: the rise to 2.40 per cent a year by 2058 and that rate held to 2100. The series to 2050 is the client's own (Section 4.1.7).

4.1.8. Settlement Saturation and Population Overflow

Each settlement grows at the rate of the series until its empty plots are built, its growth filling them together and in proportion. Then its further growth has to go somewhere. Two readings are possible, and the choice between them is the largest single decision in this report.

With the overflow: when a settlement is full, its further growth moves to a neighbour with room. Ibri's goes in parallel to Al Araqi, Al Qurayn and Shalashil, in the proportions seventy, twenty and ten per cent, then to Ad Dariz, then to the nearest settlement with room. At Tayyib's goes first to Miayrid. Every other settlement overflows to its nearest neighbour with room.

Without: each settlement grows on its own rate and stops when it is full. The growth it cannot house is lost to the study area.

The difference is large. With the overflow every settlement fills by 2070 and the study area holds 349,029 people. Without it, 7 settlements never fill before 2100, the study area holds 273,616 people in 2070, and 75,414 people of the series have nowhere to go. A sewer in those settlements sized on their own growth would be sized for land that never fills.¹

¹ The settlements that never fill on their own growth: At Tayyib, Al Ghubayrah, Al Qurayn, Al Makhtibyah, Shalashil, Wadi al Mankas, Miayrid. The overflow routes reflect local knowledge of where Ibri's growth is taking place.

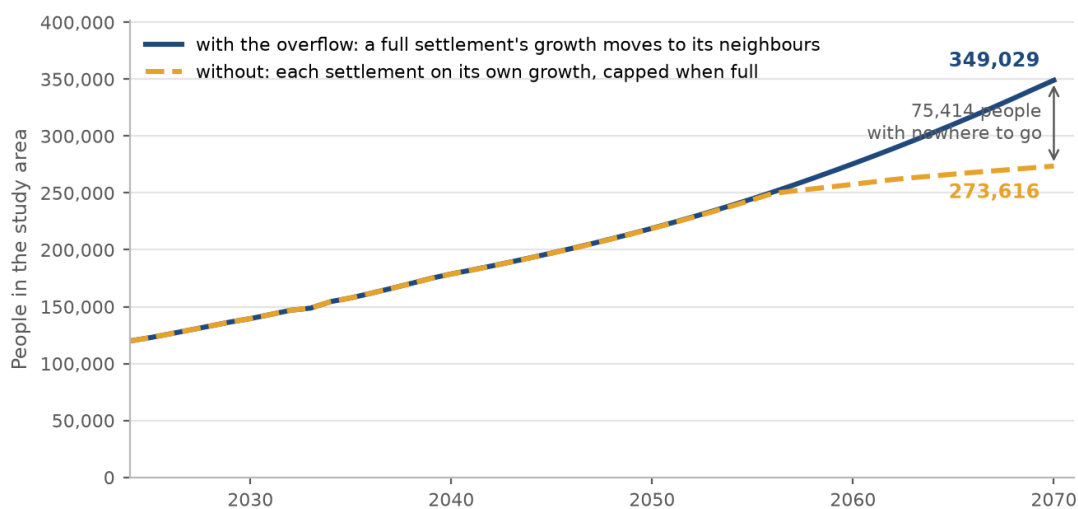


Figure 35. People in the study area by year, with the overflow and without it.

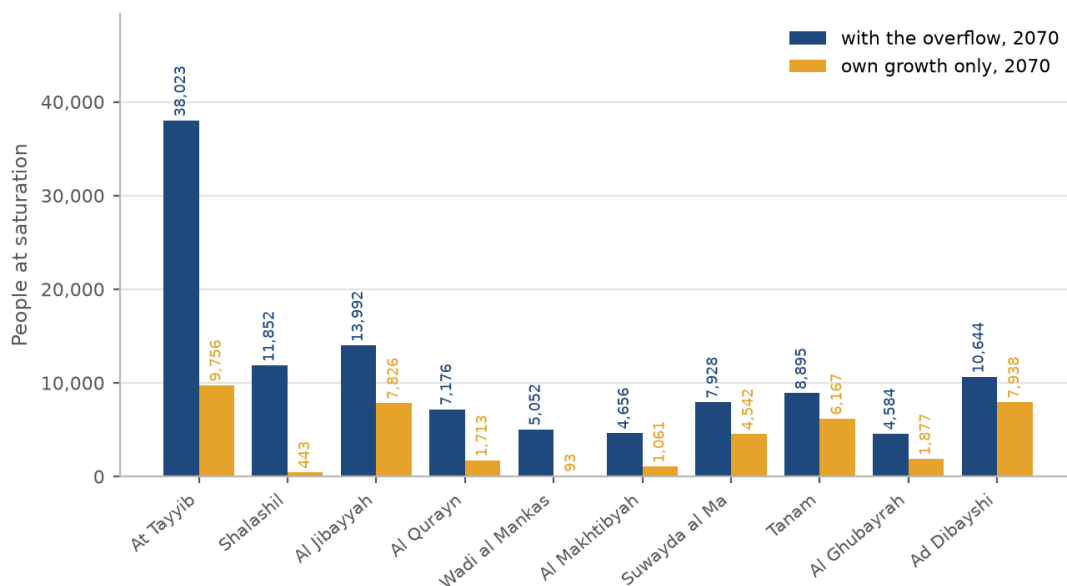


Figure 36. People at saturation in the ten settlements the choice affects most, with the overflow and without.

Table 28. Every settlement with and without the overflow: people in 2070 and the year it is full

Settlement	Capacity, people	With the overflow: people 2070	full in	Own growth only: people 2070	full in
Ibri	73,299	140,405	2056	140,405	2057
Ad Dariz	16,854	28,704	2062	28,704	2063
Al Araqi	11,832	22,528	2057	22,528	2057
Al Aynayn	7,459	12,013	2060	12,013	2066
Al Wahrah	7,109	10,460	2070	9,765	2073
At Tayyib	34,675	38,023	2068	9,756	not before 2100
Ad Dabayshi	7,920	10,644	2063	7,938	2083
Al Jibayyah	11,306	13,992	2070	7,826	2095
Bat	2,226	4,783	2052	4,783	2052
Tanam	6,779	8,895	2069	6,167	2086
Sayh al Masarrat	4,984	6,812	2069	5,327	2081

Settlement	Capacity, people	With the overflow: people 2070	full in	Own growth only: people 2070	full in
Suwayda al Ma	6,369	7,928	2070	4,542	2094
Hijar	1,288	2,696	2053	2,696	2053
Al Jahli	3,684	5,060	2063	4,010	2080
Al Ghubayrah	3,940	4,584	2070	1,877	not before 2100
Al Qurayn	6,588	7,176	2060	1,713	not before 2100
Al Akheedar	160	640	2036	640	2036
Al Makhtibyah	4,292	4,656	2069	1,061	not before 2100
Al Qali	564	788	2054	653	2078
Shalashil	11,700	11,852	2061	443	not before 2100
Satwah	284	392	2060	315	2080
Usaybuq	140	224	2054	224	2067
Ash Shiab	48	92	2056	92	2056
Wadi al Mankas	5,020	5,052	2068	93	not before 2100
Miayrid	616	632	2070	47	not before 2100
Total	229,136	349,029	2070	273,616	

The rule sets the order; the land sets the result. Al Araqi fills in 2057, a year after Ibri, so it takes only 163 of Ibri's people, and most of Ibri's overflow passes on: the largest route is to At Tayyib, 22,078 people, and 14 settlements take some of Ibri's growth in all. The routes carrying more than a thousand people are set out below; the full list is in Appendix A.4.

Table 29. The main overflow routes

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Ibri	At Tayyib	22,078	2056	2060	2068
Ibri	Shalashil	11,471	2056	2056	2061
Ibri	Al Qurayn	5,856	2056	2056	2060
Ad Dariz	Wadi al Mankas	4,965	2062	2062	2068
Al Araqi	At Tayyib	4,873	2057	2060	2068
Ibri	Ad Dibayshi	4,078	2056	2062	2063
Ibri	Al Jibayyah	3,927	2056	2068	2070
Bat	Al Ghubayrah	2,493	2052	2052	2070
Ibri	Al Makhtibyah	2,441	2056	2063	2069
Ibri	Tanam	2,010	2056	2063	2069
Al Araqi	Al Aynayn	1,790	2057	2057	2060
Al Aynayn	At Tayyib	1,639	2060	2060	2068
Al Araqi	Al Jibayyah	1,410	2057	2068	2070
Ibri	Sayh al Masarrat	1,225	2056	2063	2069
Ad Dibayshi	Tanam	1,004	2063	2063	2069

Decision 3. Approve the overflow: when a settlement's land is full, its further growth moves to its neighbours. Without it 75,000 people of the series have nowhere to go by 2070 (Section 4.1.8).

Where the growth goes once a settlement is full

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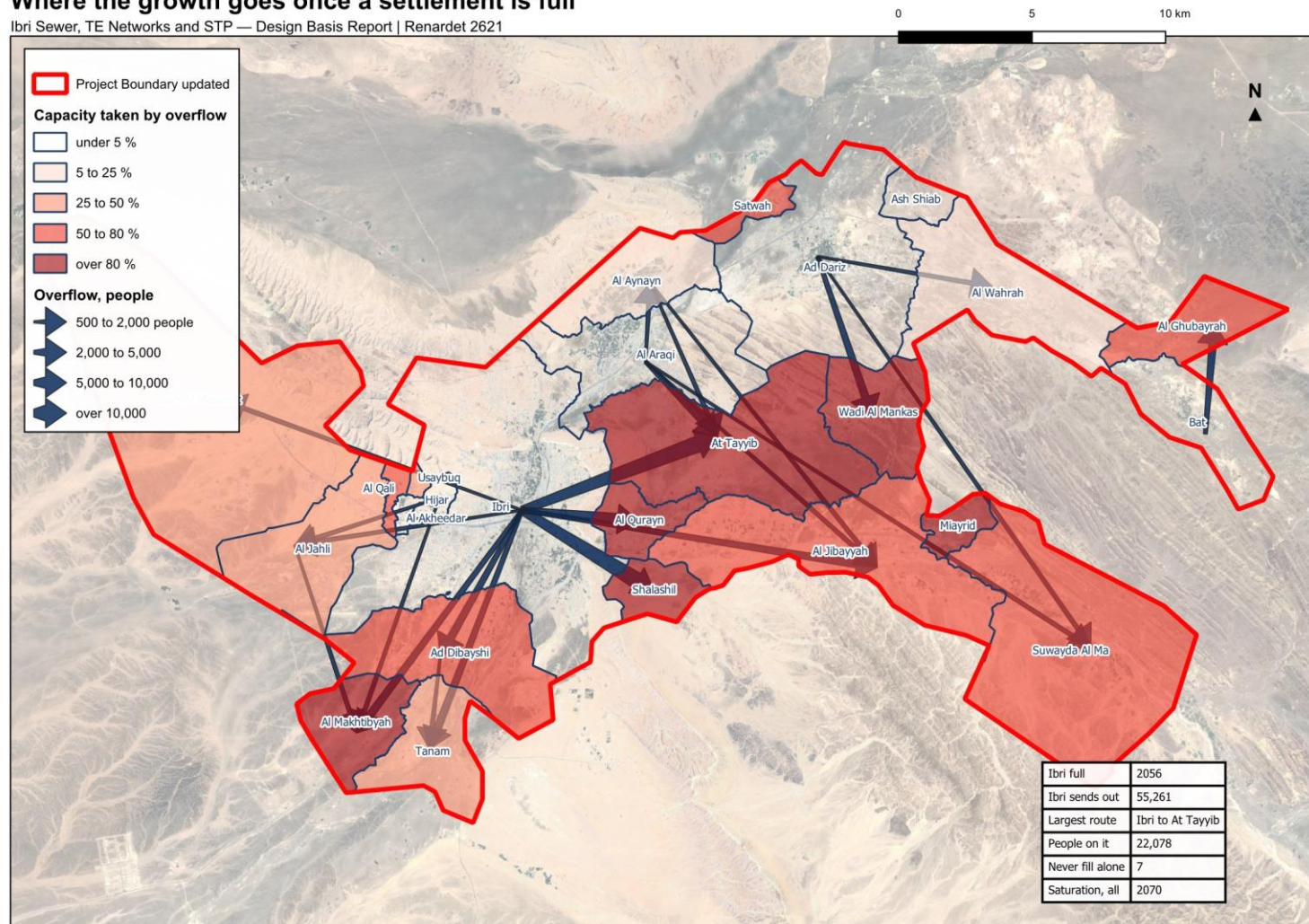


Figure 37. Where the growth goes once a settlement is full. Arrows show the routes carrying five hundred people or more at saturation; the shading shows how much of each settlement's capacity is taken by its neighbours' overflow.

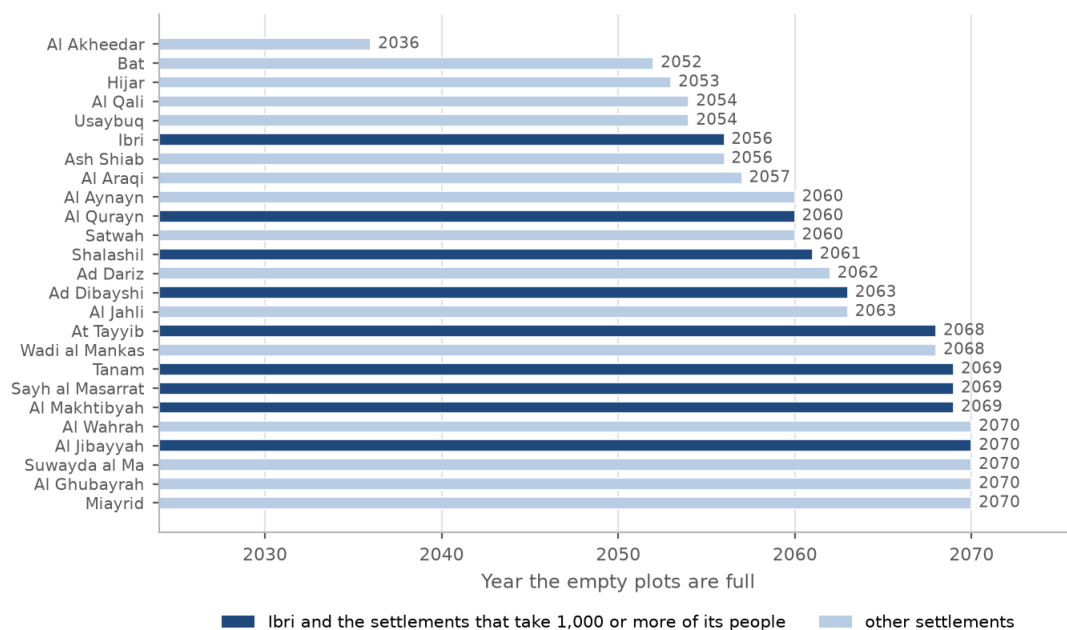


Figure 38. The year each settlement's empty plots are full. Ibri and the settlements that take a thousand or more of its people are marked.

4.1.9. Design Horizon

The Terms of Reference name a horizon of completion plus twenty-five years. With the opening year taken as 2030, that is 2055: 244,914 people and 42,266 cubic metres a day. The Terms of Reference also ask for the saturation of the area. With the overflow, Ibri fills in 2056, Al Araqi in 2057, Ad Dariz in 2062, and the last settlement in 2070: 349,029 people and 60,099 cubic metres a day, 42 per cent more than in 2055.¹

The two horizons ask different things of the design. A network sized on 2055 is cheaper to build and reaches its capacity while the plots around it are still being built; the pipes under those streets would then need to be replaced or relieved. A network sized on the year the land is full is laid once, and its sewers run at low flow for longer, which the self-cleansing check of Section 4.2.8 is there to catch. The STP is staged either way, on the series. The horizon is for Nama Water Services to decide.

Table 30. The study area in the design years

Year	People	Average sewage flow, m ³ /d	What it is
2024	119,893	20,852	today, from the meters
2030	139,535	24,216	opening year
2055	244,914	42,266	opening year plus 25 years
2070	349,029	60,099	saturation: the land is full

¹ Terms of Reference, page 51 of the tender document. Average flows before infiltration and before the STP margin.

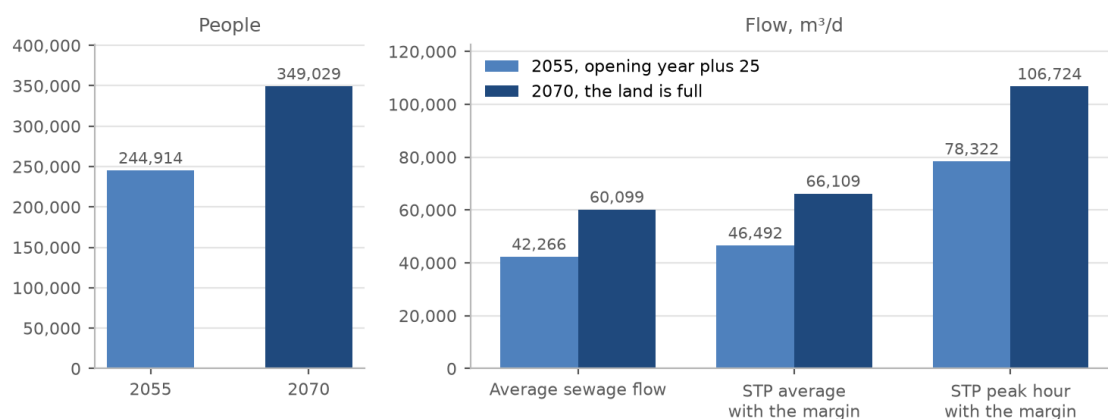


Figure 39. The two horizons side by side: people, average sewage flow, and the STP's average and peak-hour flow with the margin.

Decision 4. Decide the design horizon: 2055, the opening year 2030 plus 25 years, with 244,914 people and 42,266 cubic metres a day; or 2070, the year the land is full, with 349,029 people and 60,099 cubic metres a day (Section 4.1.9).

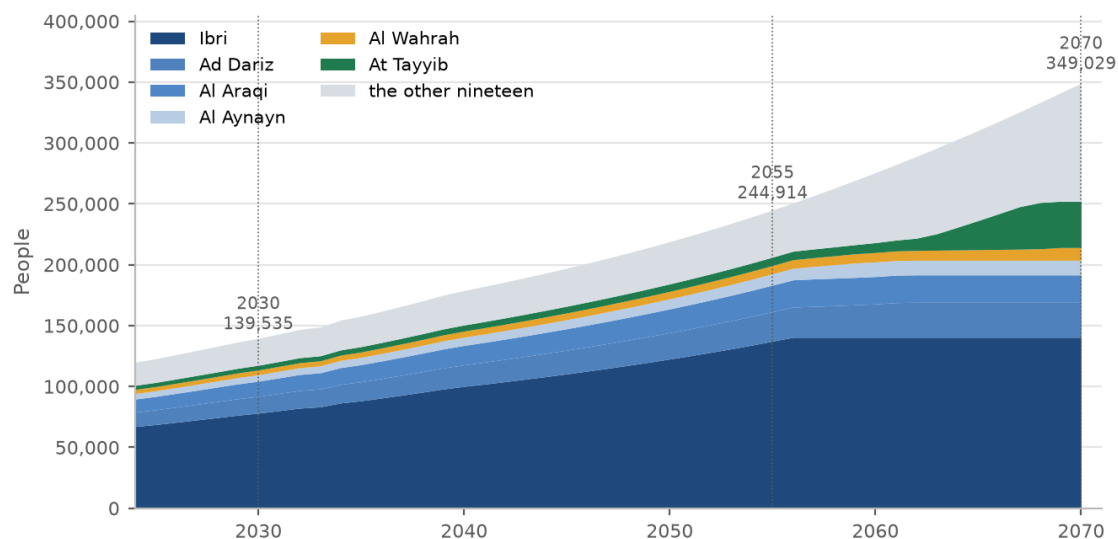


Figure 40. People by year to saturation, the six largest settlements shown separately. The flat top of each band is the year that settlement fills.

The people of every settlement at five-year steps, to the year it is full. A cell is blank once the settlement is full.¹

¹ Totals are rounded from the unrounded values and may differ by one from the sum of the rows. A settlement's population after its saturation year is its capacity plus what it held in 2024; its growth beyond that is housed elsewhere or not at all.

Table 31. People by settlement at five-year intervals

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Full in	At saturation
Ibri	67,106	68,713	78,099	88,272	100,066	110,349	122,535	137,198				2056	140,405
Ad Dariz	11,850	12,134	13,791	15,587	17,670	19,486	21,638	24,206	27,350			2062	28,704
Al Araqi	10,696	10,952	12,448	14,069	15,949	17,588	19,531	21,849				2057	22,528
Al Aynayn	4,554	4,663	5,300	5,990	6,791	7,489	8,316	9,303	12,013			2060	12,013
Al Wahrah	3,351	3,431	3,900	4,408	4,997	5,511	6,119	6,845	7,703	8,684	10,460	2070	10,460
At Tayyib	3,348	3,428	3,897	4,404	4,993	5,506	6,114	6,839	8,130	23,972		2068	38,023
Ad Dibayshi	2,724	2,789	3,170	3,583	4,062	4,479	4,974	5,564	6,262			2063	10,644
Al Jibayyah	2,686	2,750	3,126	3,533	4,005	4,416	4,904	5,486	6,174	6,951	13,992	2070	13,992
Bat	2,557	2,618	2,976	3,363	3,813	4,204	4,669					2052	4,783
Tanam	2,116	2,167	2,463	2,784	3,156	3,480	3,864	4,323	4,865	5,961		2069	8,895
Sayh al Masarrat	1,828	1,872	2,127	2,405	2,726	3,006	3,338	3,734	4,202	4,786		2069	6,812
Suwayda al Ma	1,559	1,596	1,814	2,050	2,324	2,563	2,846	3,184	3,583	4,034	7,928	2070	7,928
Hijar	1,408	1,442	1,639	1,852	2,100	2,315	2,571					2053	2,696
Al Jahli	1,376	1,409	1,601	1,810	2,052	2,263	2,513	2,831	3,746			2063	5,060
Al Ghubayrah	644	659	750	847	960	1,059	1,176	1,756	2,575	3,501	4,584	2070	4,584
Al Qurayn	588	602	684	773	877	967	1,074	1,201	7,176			2060	7,176
Al Akheedar	480	492	559	631								2036	640
Al Makhtibyah	364	373	424	479	543	599	665	744	837	1,522		2069	4,656
Al Qali	224	229	261	295	410	518	646					2054	788
Shalashil	152	156	177	200	227	250	278	311	8,335			2061	11,852
Satwah	108	111	126	142	161	178	197	221	392			2060	392
Usaybuq	84	86	98	110	125	138	153					2054	224
Ash Shiab	44	45	51	58	66	72	80	90				2056	92
Wadi al Mankas	32	33	37	42	48	53	58	65	74	2,859		2068	5,052
Miayrid	16	16	19	21	24	26	29	33	37	41	632	2070	632
Total	119,893	122,766	139,535	157,710	178,781	197,153	218,926	244,914	275,608	310,307	349,029	2070	349,029

4.1.10. Spatial Distribution of Growth

For the network every empty plot matters, not only the ones counted as future homes, because the sewer that serves a street serves every plot on it. The people a settlement houses in a given year are therefore spread over all its empty plots up to 2,000 square metres, in proportion to plot area capped at 1,000 square metres. A very small plot takes its small share and a large plot takes no more than a full-sized home plot. The sum over the plots equals the settlement's housed population by construction, so the network total and the STP total are the same figure.

4.2. Wastewater Generation and Design Flows

Derivation of the design flow

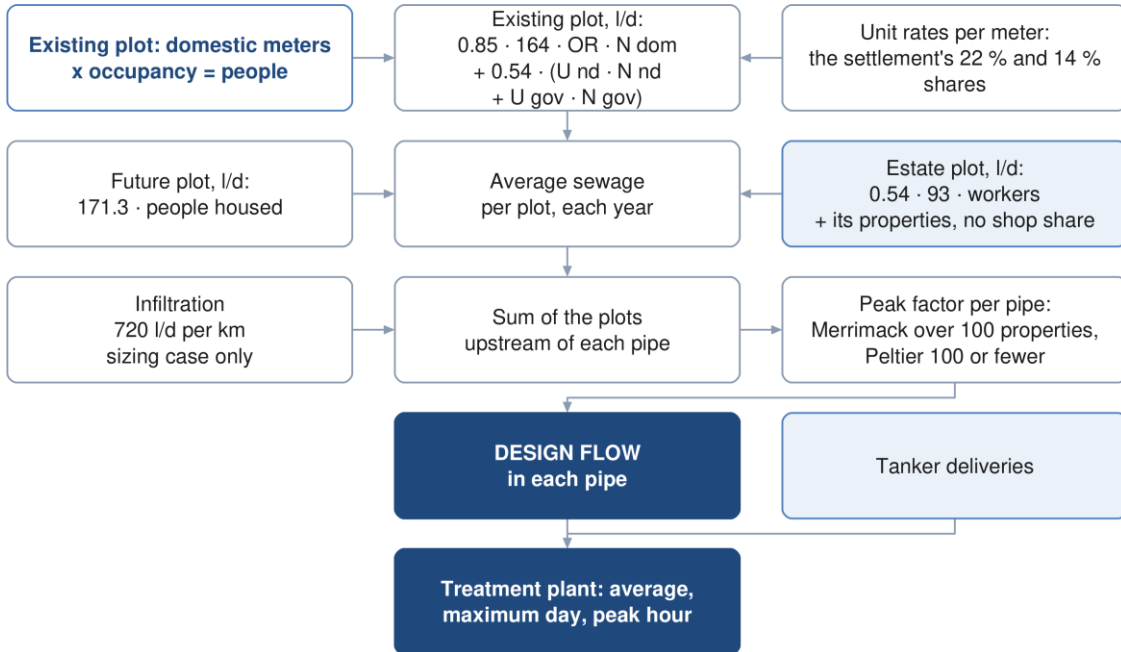


Figure 41. Derivation of the design flow, from the plot to the pipe and the STP.

4.2.1. Water Demand

Wastewater generation is derived from water demand. Four components are carried: domestic, non-domestic, governmental and special consumption. A household supplied by tanker is metered for electricity like any other and carries the domestic rate; the tanker volumes themselves are a data request and no separate term is carried (Section 3.1.1).

$$Q_{\text{dom}} = \text{Population} \times \text{LPCD} \quad (7)$$

Q_{dom} : domestic water demand, l/d

LPCD: unit consumption, 164 for Adh Dhahirah, l per person per day

Non-domestic and governmental demand are established as proportions of domestic demand.

$$Q_{\text{ND}} = 0.22Q_{\text{dom}} \quad (8)$$

Q_{ND} : non-domestic water demand of the settlement: shops and offices, l/d

$$Q_{\text{gov}} = 0.14Q_{\text{dom}} \quad (9)$$

Q_{gov} : governmental water demand of the settlement, l/d

The proportions are the values published for the Governorate of Adh Dhahirah. The guideline calls them distributed. They come from the governorate's water balance of 2021 to 2023, all the non-domestic water divided by all the domestic water, so they stand for the everyday shops, mosques, schools and offices that come with any population, and they are added whether or not a particular village has a shop today. They do not cover named projects such as economic zones, which are added separately. Per person they are 36 and 23 litres a day, and the three are applied as separate streams,

never combined into one rate. Each settlement's non-domestic and governmental water is placed on its shop and government meters in proportion to their number.¹

4.2.2. Special Consumption: Industrial Estates

Two industrial estates lie inside the study area, at Al Tayyeb in the north-east of the town and at Tanam beside the treatment plant. Their meters are recorded on the commercial tariff. The guideline places such estates outside the population ratios, and they are treated as special consumption: a workforce of 4,500 at Al Tayyeb and 1,800 at Tanam, spread over their industrial plots in proportion to area, at the dry-industry rate of 93 litres per employee per day. Their meters take no share of the settlement's non-domestic or governmental water: an estate plot carries its workers, and the few properties on the estates carry the domestic rate like any other. The workforce is an assumption from the number of workshop meters and is to be replaced by the estates' own records. Section 4.3 describes the estates and the other identified projects.²

Adopted (10). 4,500 workers at Al Tayyeb and 1,800 at Tanam at 93 litres a day each, an assumption to be replaced by the estates' records when received (Section 4.2.2).

4.2.3. Return to Sewer

Not all supplied water reaches the sewer. The proportion that does is applied by category.

$$Q_{ww} = 0.85Q_{dom} + 0.54(Q_{ND} + Q_{gov} + Q_{special}) \quad (10)$$

Q_{ww} : wastewater generated, before infiltration, l/d

Q_{dom} : domestic demand; tanker supply is added when its records are received, l/d

$Q_{special}$: water of the special consumption, the guideline's term; in the study area the two industrial estates, l/d

Water supplied by tanker returns to the sewer at the same proportion as piped supply. The domestic rate of 164 litres a day measures network water only, so tanker supply is not inside the domestic stream; it is added when the filling-station records are received (Section 3.1.1).³

Farm meters supply irrigation pumps and are taken to return nothing, and the estates' dry-industry demand is taken to return 54 per cent, as non-domestic supply does. Both are rules of this design, not of the guideline.⁴

Adopted (9). 164 litres per person per day, plus 22 per cent for shops and offices and 14 per cent for government, the values the guideline publishes for Adh Dhahirah; 85 per cent of domestic and tanker supply and 54 per cent of the rest return to the sewer. The published ratios are used in place of the unit rates per pupil, bed, employee and floor area until those quantities are held (Sections 4.2.1 and 4.2.3).

4.2.4. Plot Flows

Applied plot by plot, the rules above give every plot an average daily sewage flow for 2024. An existing plot is built from its meters and the rates of its settlement:

$$Q_{plot} = 0.85 \times 164 \times OR \times N_{dom} + 0.54 \times (U_{nd}N_{nd} + U_{gov}N_{gov} + 93N_w) \quad (11)$$

Q_{plot} : average sewage flow of the plot in 2024, l/d

¹ PAM-GUD-201, Table 11, page 60, and page 59: the ratios "refer to spatially distributed non-domestic consumption that are to be added to the domestic consumption" and "do not apply to ... specific identified non-domestic projects such as economic zones".

² PAM-GUD-201, Section 7.3.1, page 59, and Table 12, page 61, dry industry. Al Tayyeb: 94 hectares, 202 industrial plots, 1,035 meters. Tanam: 61 hectares, 105 plots, 409 meters. About five meters per plot and three to six workers per workshop give the assumed figures.

³ PAM-GUD-201, Table 19, page 71, which gives a single discharge ratio for domestic and tanker supply.

⁴ On 260 of the 489 plots with a farm meter a household is metered separately from the pump.

OR: occupancy rate adopted for the settlement (Section 4.1.4), persons per property

N dom: domestic meters on the plot, one property each

U nd: non-domestic water per shop meter in the settlement, l/d per meter

N nd: non-domestic meters on the plot; none counted on an estate plot

U gov: governmental water per government meter in the settlement, l/d per meter

N gov: government meters on the plot; none counted on an estate plot

N w: workers on an industrial-estate plot

The unit rates are the settlement's non-domestic and governmental shares, divided equally among the meters that carry them:

$$U_{nd} = \frac{0.22 \times 164 \times P_s}{N_{nd,s}} \quad (12)$$

P s: people of the settlement in 2024

N nd,s: shop meters of the settlement outside the industrial estates

$$U_{gov} = \frac{0.14 \times 164 \times P_s}{N_{gov,s}} \quad (13)$$

N gov,s: government meters of the settlement outside the industrial estates

The estates' meters take no share of the pool; an estate plot carries its workers at 93 litres a day, and the properties on it, if any, at the domestic rate. A settlement with no shop or government meter outside the estates keeps that share on its houses, in proportion to their people. A future plot carries the people it houses in the year at 171.3 litres a day each, all three streams together, since where its shops will stand is not known.¹

The shares and the unit rates of every settlement are set out below. The sum of the plots equals the settlement in every column.²

Table 32. Water shares and unit rates by settlement, 2024

Settlement	People	Domestic m ³ /d	Non-domestic share m ³ /d	Governmental share m ³ /d	Shop meters	Government meters	l/d per shop meter	l/d per government meter	Estates m ³ /d	Sewage m ³ /d
Ibri	67,106	11,005	2,421	1,541	4,747	461	510	3,342	–	11,494
Ad Dariz	11,850	1,943	428	272	820	82	521	3,318	–	2,030
Al Araqi	10,696	1,754	386	246	1,018	96	379	2,558	–	1,832
Al Aynayn	4,554	747	164	105	204	26	805	4,022	–	780
Al Wahrah	3,351	550	121	77	132	23	916	3,345	–	574
At Tayyib	3,348	549	121	77	179	19	675	4,046	419	799
Al Jibayyah	2,686	440	97	62	96	20	1,009	3,083	–	460
Bat	2,557	419	92	59	51	26	1,809	2,258	–	438
Ad Dibayshi	2,724	447	98	63	236	17	416	3,679	27	481
Tanam	2,116	347	76	49	124	32	616	1,518	105	419
Suwayda al Ma	1,559	256	56	36	34	13	1,654	2,753	–	267
Hijar	1,408	231	51	32	63	4	806	8,082	–	241
Al Ghubayrah	644	106	23	15	105	4	221	3,697	–	110
Al Qurayn	588	96	21	14	14	7	1,515	1,929	–	101
Sayh al Masarrat	1,828	300	66	42	128	99	515	424	–	313
Al Jahli	1,376	226	50	32	143	87	347	363	–	236
Satwah	108	18	4	2	5	0	779	on the houses	–	19
Al Akheedat	480	79	17	11	69	6	251	1,837	–	82

¹ $164 \times 0.85 + 36 \times 0.54 + 23 \times 0.54 = 171.3$ litres per person per day; 36 and 23 are 22 and 14 per cent of 164.

² The Tanam estate's plots lie across Tanam, Ad Dibayshi and Al Makhtibyah in the settlement partition, so its workforce appears under all three. Where a settlement has no shop or government meter outside the estates, the table shows that share as placed on the houses. Two plots outside the estates carry a special meter each, one on the industrial tariff and one large consumer identified as industrial; they are loaded through their shop meters, their own consumption being unknown. Totals are rounded from the unrounded values and may differ by one from the sum of the rows.

Settlement	People	Domestic m ³ /d	Non-domestic share m ³ /d	Governmental share m ³ /d	Shop meters	Government meters	l/d per shop meter	l/d per government meter	Estates m ³ /d	Sewage m ³ /d
Al Qali	224	37	8	5	5	22	1,616	234	–	38
Shalashil	152	25	5	3	2	1	2,742	3,490	–	26
Al Makhtibiyah	364	60	13	8	45	4	292	2,089	36	82
Usaybuq	84	14	3	2	0	1	on the houses	1,929	–	14
Wadi al Mankas	32	5	1	1	0	0	on the houses	on the houses	–	5
Ash Shiab	44	7	2	1	0	1	on the houses	1,010	–	8
Miayrid	16	3	1	0	4	2	144	184	–	3
Total	119,893	19,663	4,326	2,753	8,224	1,053	526	2,614	586	20,852

In 2024 the study area generates 20,852 cubic metres a day, of which 16,713 is domestic, 2,336 non-domestic, 1,486 governmental and 316 from the two estates.

20,852 m³/d today

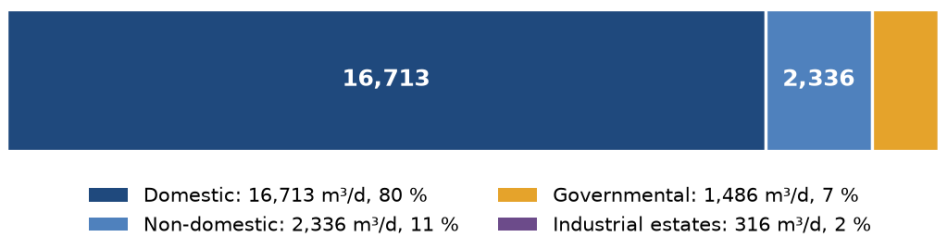


Figure 42. Average sewage flow in 2024 by stream.

4.2.5. Infiltration

For newly constructed networks an allowance of 720 litres per day per kilometre of sewer is included. It is a property of the pipe, not of the plot, and is added along each run. Infiltration from stormwater is not considered.¹

4.2.6. Peak Flow

Peak flow belongs to the pipe, not to the plot: it is applied to the sum of the plots upstream of each pipe. Above 100 properties the Merrimack formula applies; at 100 or fewer, the Peltier formula.

$$Q_{\text{peak}} = 2.65 Q_{\text{avg}}^{0.879} \quad (14)$$

Merrimack: above 100 properties; both flows in Ml/d

Q peak: peak flow: the guideline's peak daily flow, Qpdf, equal to its peak hourly flow, Ml/d

Q avg: average flow of the plots upstream, the guideline's Qadf, Ml/d

$$Q_{\text{peak}} = \left(1.5 + \frac{1}{\sqrt{Q_m}} \right) Q_{\text{avg}} \quad (15)$$

Peltier: at 100 properties or fewer

Q m: the average flow of the plots upstream, l/s

The guideline recommends that the hourly peak factor not exceed 5.0. It is a recommendation, applied as such.²

¹ PAM-GUD-201, Section 7.4.3, page 72.

² PAM-GUD-201, Section 7.4.2, pages 71 and 72. PAM-GUD-203, Table 29, page 65, defines the peak hourly flow and gives it the symbol QPDF.

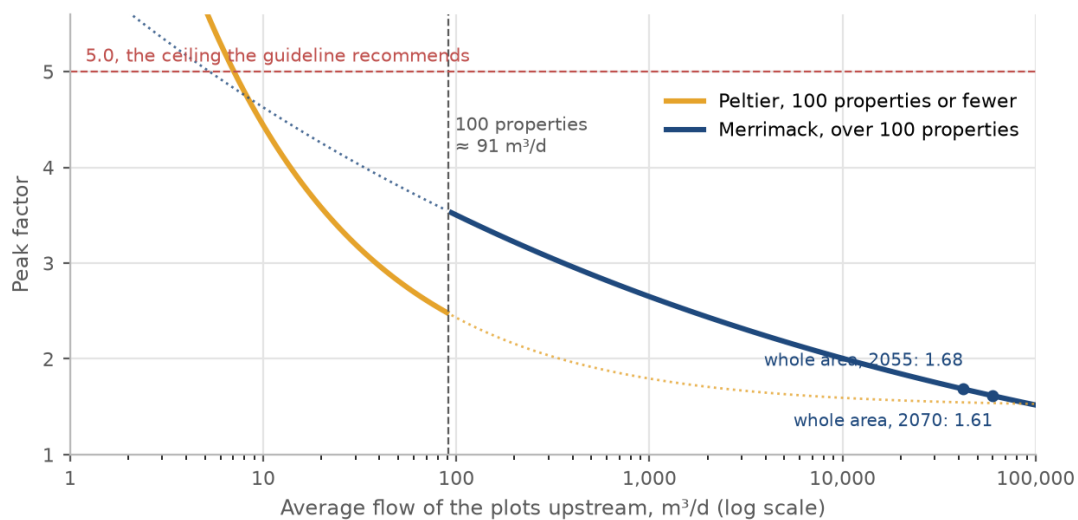


Figure 43. The peak factor against the average flow of the plots upstream. The Peltier formula applies to the head of a street, the Merrimack formula beyond 100 properties; for the whole study area the factor is about 1.6.

Adopted (12). 720 litres a day per kilometre of sewer, added to each pipe and left out of the early-year check; the Merrimack formula above 100 properties and the Peltier formula at 100 or fewer (Sections 4.2.5 and 4.2.6).

4.2.7. Flow Projection over the Design Period

Flows are established annually and reported at five-year intervals to saturation, as the Terms of Reference require. The study area generates 20,852 cubic metres a day in 2024, 24,216 in 2030, 42,266 in 2055 and 60,099 at saturation in 2070.¹

¹ Average daily flows before infiltration and before the STP margin. The ultimate flow of about 49,700 cubic metres a day stated in the Inception Report was built on the connected population and a single occupancy rate; the figures here supersede it.

Table 33. Average sewage flow by settlement at five-year intervals, m³/d

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Full in	At saturation
Ibri	11,494	11,769	13,377	15,119	17,140	18,901	20,988	23,500				2056	24,049
Ad Dariz	2,030	2,078	2,362	2,670	3,027	3,338	3,706	4,146	4,685			2062	4,917
Al Araqi	1,832	1,876	2,132	2,410	2,732	3,013	3,345	3,742				2057	3,859
At Tayyib	800	813	893	980	1,081	1,169	1,273	1,398	1,618	4,332		2068	6,739
Al Aynayn	780	799	908	1,026	1,163	1,283	1,424	1,593	2,058			2060	2,058
Al Wahrah	574	588	668	755	856	944	1,048	1,173	1,320	1,487	1,792	2070	1,792
Ad Dibayshi	481	492	557	628	710	782	866	967	1,087			2063	1,838
Al Jibayyah	460	471	535	605	686	756	840	940	1,057	1,191	2,397	2070	2,397
Bat	438	448	510	576	653	720	800					2052	819
Tanam	419	428	479	534	597	653	719	797	890	1,078		2069	1,580
Sayh al Masarrat	313	321	364	412	467	515	572	640	720	820		2069	1,167
Suwayda al Ma	267	273	311	351	398	439	488	545	614	691	1,358	2070	1,358
Hijar	241	247	281	317	360	397	440					2053	462
Al Jahli	236	241	274	310	351	388	430	485	642			2063	867
Al Ghubayrah	110	113	128	145	165	181	201	301	441	600	785	2070	785
Al Qurayn	101	103	117	133	150	166	184	206	1,229			2060	1,229
Al Akheedar	82	84	96	108								2036	110
Al Makhtibyah	82	83	92	101	112	122	133	147	163	280		2069	817
Al Qali	38	39	45	51	70	89	111					2054	135
Shalashil	26	27	30	34	39	43	48	53	1,428			2061	2,030
Satwah	19	19	22	24	28	30	34	38	67			2060	67
Usaybuq	14	15	17	19	22	24	26					2054	38
Ash Shiab	8	8	9	10	11	12	14	15				2056	16
Wadi al Mankas	6	6	6	7	8	9	10	11	13	490		2068	865
Miayrid	3	3	3	4	4	5	5	6	6	7	108	2070	108
Total	20,852	21,344	24,216	27,329	30,938	34,085	37,815	42,266	47,523	53,466	60,099	2070	60,099

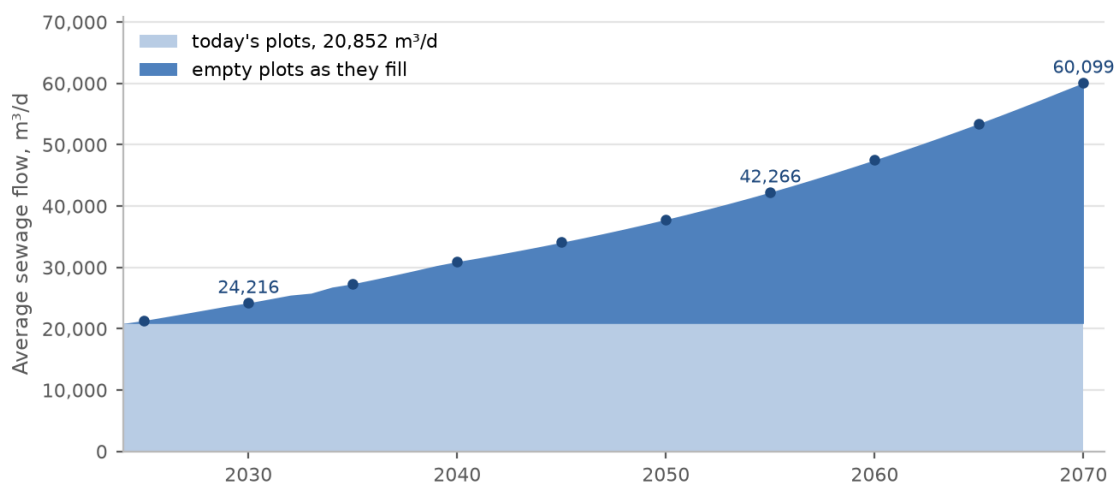


Figure 44. Average sewage flow of the study area by year, from today's plots and from the empty plots as they fill. The points mark the five-year reporting intervals.

The first map below gives the average flow of every plot at saturation. The second gives every settlement's average flow in 2055 and at saturation, and the four steps to an STP flow, so that any grouping of settlements into one or more plants can be calculated by hand. The table carries the same values.

Table 34. Average sewage flow by settlement in 2055 and at saturation, m³/d

Settlement	2055	2070	Settlement	2055	2070
Ibri	23,500	24,049	Hijar	462	462
Ad Dariz	4,146	4,917	Al Ghubayrah	301	785
Al Araqi	3,742	3,859	Al Qurayn	206	1,229
Al Aynayn	1,593	2,058	Al Makhtibyah	147	817
At Tayyib	1,398	6,739	Al Qali	135	135
Al Wahrah	1,173	1,792	Al Akheedar	110	110
Ad Dibayshi	967	1,838	Shalashil	53	2,030
Al Jibayyah	940	2,397	Usaybuq	38	38
Bat	819	819	Satwah	38	67
Tanam	797	1,580	Ash Shiab	15	16
Sayh al Masarrat	640	1,167	Wadi al Mankas	11	865
Suwayda al Ma	545	1,358	Miayrid	6	108
Al Jahli	485	867	Total	42,266	60,099

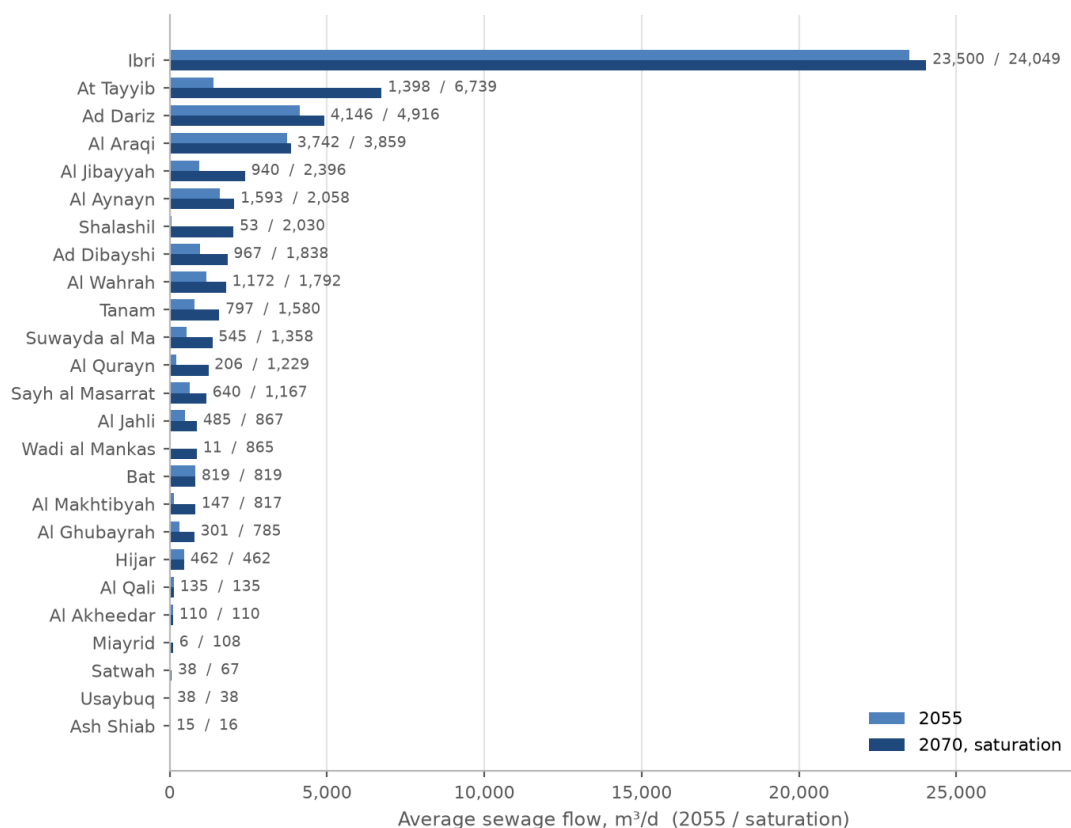


Figure 45. Average sewage flow of every settlement in 2055 and at saturation. Ibri is full before 2055; the growth after it lands on its neighbours.

Average sewage flow per plot at saturation

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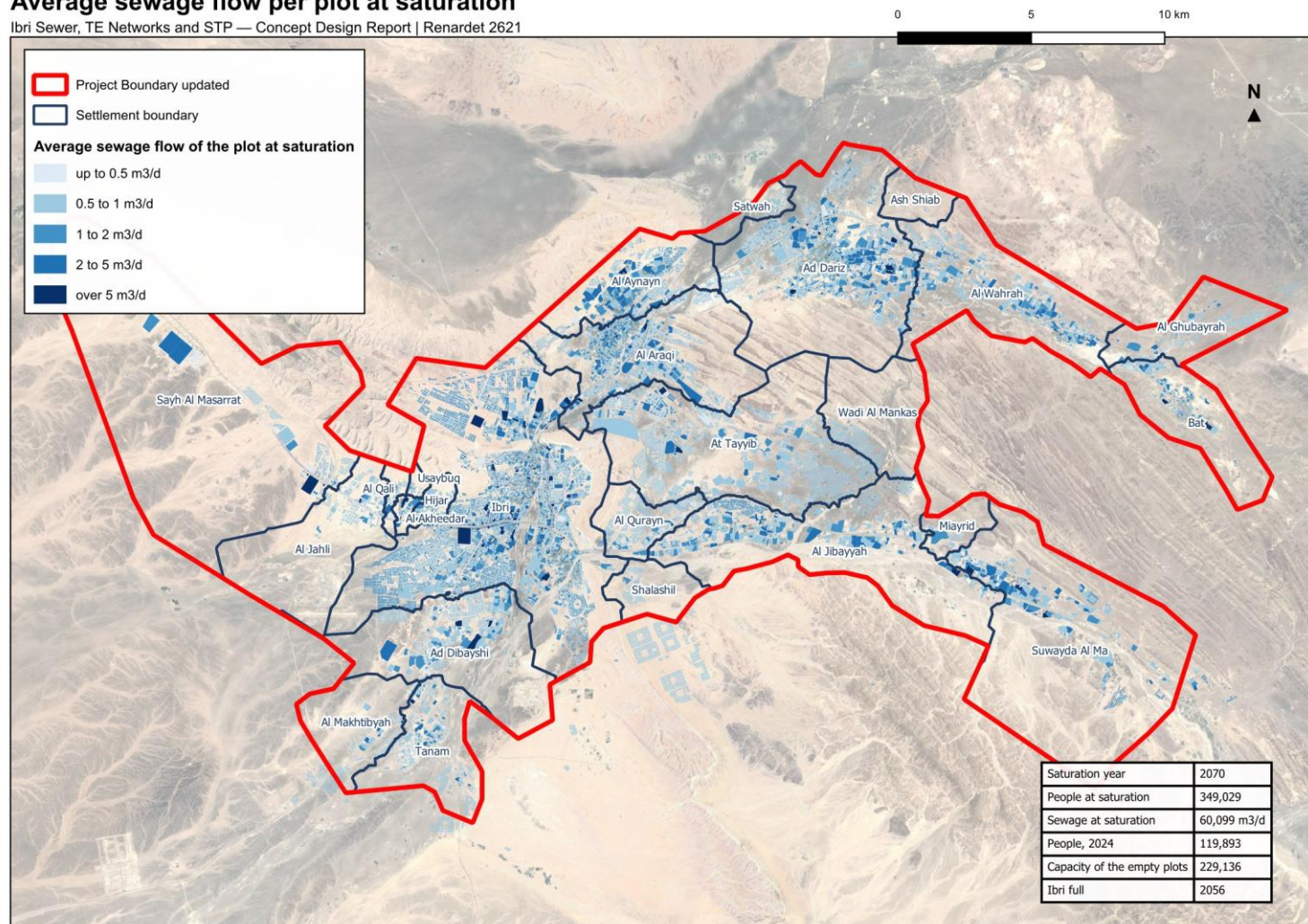


Figure 46. Average sewage flow of every plot at saturation.

Average sewage flow of each settlement in 2055 and at saturation, 2070

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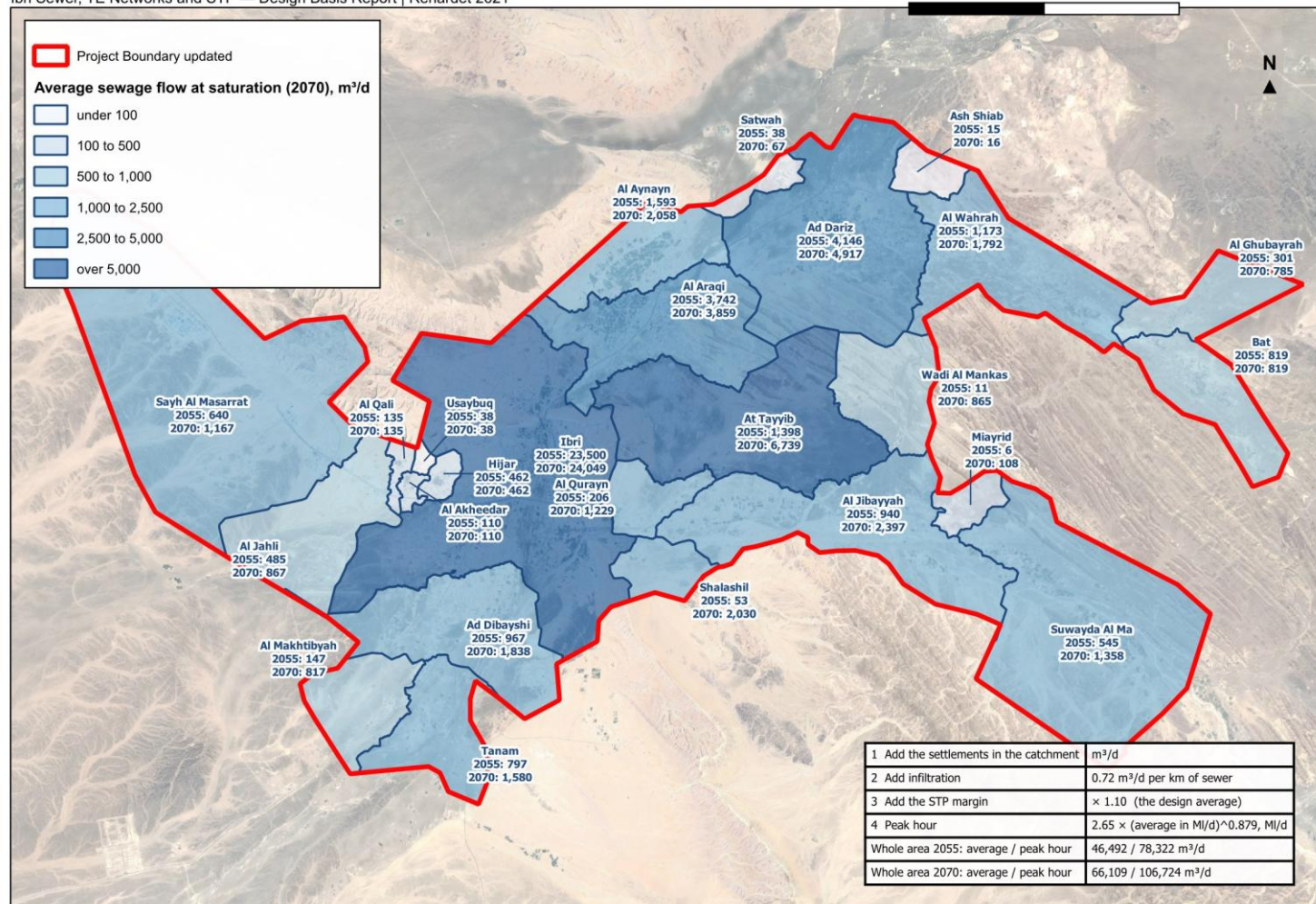


Figure 47. Average sewage flow of each settlement in 2055 and at saturation, and the four steps from the settlements in a catchment to the flow of their STP.

4.2.8. Design Flows by System Element

Every plot carries an average sewage flow for 2024, 2030, 2055 and the saturation year. What each element of the system is designed for follows from those plot flows by fixed rules. The rules are the same wherever the element sits, so any one of them can be checked.

Plots

The plot carries the average flow of its people and meters, and nothing else: no peak, no infiltration, no tankers. Those belong to the pipe and the STP.

Sewers

A pipe is checked twice, on two different flows, because the two checks fail in opposite directions.

Size and capacity: the flow at saturation of every plot upstream, added up, peaked, plus infiltration along the pipe. Under-estimating it surcharges the pipe.

Self-cleansing: the flow in 2030 of the plots upstream, with 61 per cent of the properties connected, peaked, and no infiltration. Over-estimating it declares a pipe self-cleansing while it silts.

The 61 per cent is the connection ratio of the Inception Report, reached by 2028. It is never applied to the saturation flow: that would give a pipe in a new district most of its final flow in a year when it carries almost nothing.¹

Adopted (11). 61 per cent of properties connected in 2030, the Inception Report's ratio, used only for the early-year self-cleansing check (Section 4.2.8).

Pumping Stations and Rising Mains

A station's duty flow is the peak flow of the catchment it drains plus the infiltration of its sewers, at saturation. Section 3.3.3 gives the depth rule that places it and the velocities of its rising main.

Trunk Sewers and Treatment Plant

The trunk sewers and the STP carry the sum of the settlements upstream. If the whole study area drains to one STP, the flows are as in the table, on the definitions of Section 3.4.1. The network of Section 6.2, 1,498 km of sewer, adds 1,079 cubic metres a day of infiltration, 1,187 with the margin; Section 6.2.7 gives the flow arriving at each plant of each option. Tanker deliveries are added when their records arrive.²

¹ Whole area: $24,216 \times 0.61 = 14,772 \text{ m}^3/\text{d}$ in 2030, against 36,660 if the ratio were applied to saturation.

² PAM-GUD-201, Section 7.4.5, page 73, the margin.

Table 35. Flows at one STP for the whole area, before infiltration and tankers, m³/d

Flow	2024	2030	2055	2070
People	119,893	139,535	244,914	349,029
Average flow from the plots	20,852	24,216	42,266	60,099
Peak flow before the margin	38,263	43,639	71,202	97,022
Merrimack peak factor	1.83	1.80	1.68	1.61
Average annual flow with the margin	22,937	26,638	46,492	66,109
Peak hourly flow with the margin	42,089	48,003	78,322	106,724
Infiltration with the margin, to add to both	1,187 m ³ /d for the network of Section 6.2			
Maximum day flow	from the existing STP's records			

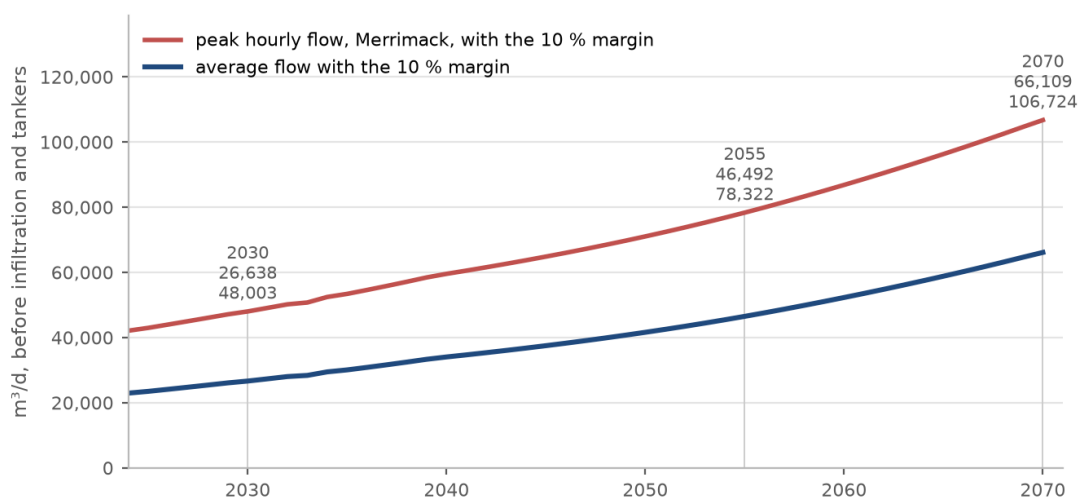


Figure 48. Average and peak-hour flow at one STP for the whole area, by year, with the margin.

At the guideline's minimum loads per person the STP receives the loads below. That is 348 milligrams of BOD per litre at saturation, at the low end of the range the guideline gives for raw sewage, 350 to 400.¹

Table 36. Loads at one STP for the whole area, from the per-person minima

	2024	2030	2055	2070
People	119,893	139,535	244,914	349,029
BOD, kg/d	7,194	8,372	14,695	20,942
Suspended solids, kg/d	9,591	11,163	19,593	27,922
BOD as a concentration at the average flow, mg/l	345	346	348	348
Suspended solids as a concentration, mg/l	460	461	464	465
BOD at the peak factor of 1.2, kg/d	8,632	10,047	17,634	25,130

¹ PAM-GUD-203, Section 10.3.1, page 74, and Table 30, page 67.

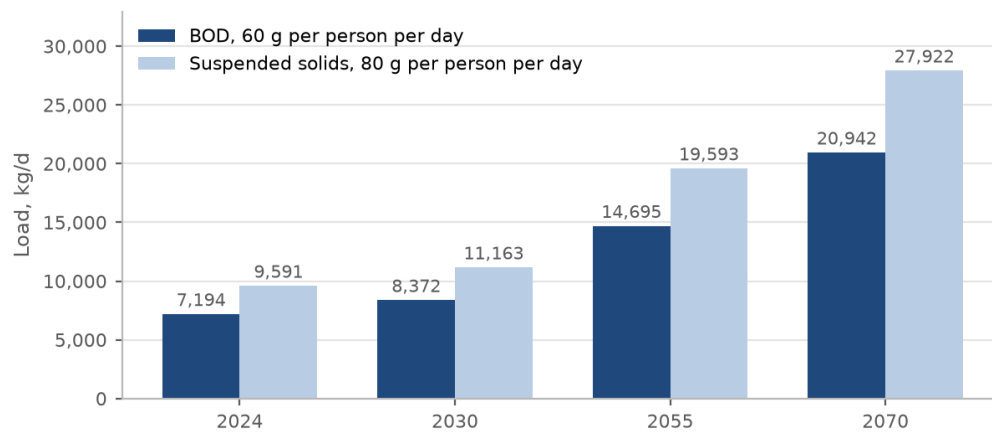


Figure 49. BOD and suspended solids reaching one STP for the whole area, by year, at the guideline's minimum loads per person.

4.3. Non-Domestic Discharges and Identified Projects

Sewage from industry and commerce, called trade effluent in the guidelines, affects the STP's load and how treatable the sewage is. Where its chemical to biochemical oxygen demand is above the domestic range, a separate treatment line is needed if the volume is significant. In Ibri the two industrial estates are workshops on the commercial tariff; no wet industry has been found. The estates are carried at the dry-industry rate (Section 4.2.2), and no separate line is foreseen at this stage.¹

4.3.1. Large-Consumer Accounts

The electricity dataset carries 499 accounts on the Cost Reflective Tariff, which applies above a consumption threshold and says nothing about use: a shopping centre, a hospital, a mosque and a factory all fall in it. Each account has been placed by public records: named features in OpenStreetMap, reverse geocoding, the aerial imagery and the press. 205 are commercial, 135 governmental, educational, medical or religious, 15 industrial, 28 farm pumps and 6 telecommunications or utility; 110 stand more than eighty metres from any recorded feature and are carried as commercial. Appendix A.2 gives the result by use.²

4.3.2. Identified Projects

Beyond the accounts, the public record was searched for the projects the guideline keeps outside the population ratios: economic zones, camps and high-water users. Those found are listed below with the way each is treated.

Table 37. Identified projects and special consumption

Site	What it is	Treatment in the design
Al Tayyeb industrial area	94 ha, 202 industrial plots, 1,035 meters, in the north-east of the town	Special consumption, 4,500 workers at 93 l/d
Tanam industrial area	61 ha, 105 plots, 409 meters, beside the treatment plant	Special consumption, 1,800 workers at 93 l/d
Army camp, Northern Frontier Regiment	296 ha, west of the town, no meter in the dataset	Recorded; no load. Its occupancy and drainage are to be confirmed (Section 1.5.5)
Ibri View resort	A tourism development of 2 square kilometres at As Sulayf, announced, not yet in the cadastre	Recorded; no load until its plan is issued
Ibri Industrial City (Madayn)	10 square kilometres, 8.6 km south-west of the boundary, first phase opened 2024 on a collection tank	Outside the network; a source of tankered sewage to be confirmed from delivery records
Power station and solar plants	7.5 to 11 km north-west of the boundary, with a construction camp	Outside the network; a source of tankered sewage to be confirmed
Central slaughterhouse	Announced; site not yet fixed	High-strength discharge requiring pre-treatment; to be included when sited

Sewage delivered by tanker is materially stronger than sewage arriving through the network. It is to be provided for separately in the STP load once the delivery records are received.³

¹ PAM-GUD-203, page 74: a COD to BOD ratio of 1.8 to 2.2 is typical of domestic sewage.

² The largest groups are the Bawadi shopping centre with twenty accounts, the commercial street of Al Murtafa with nineteen, the police headquarters, the college campus and the hospital.

³ PAM-GUD-203, Table 31, page 68, gives tankered sewage at 350 to 1,050 mg/l BOD against 350 to 400 mg/l for network sewage (Section 3.4.1).

Identified projects and special consumption

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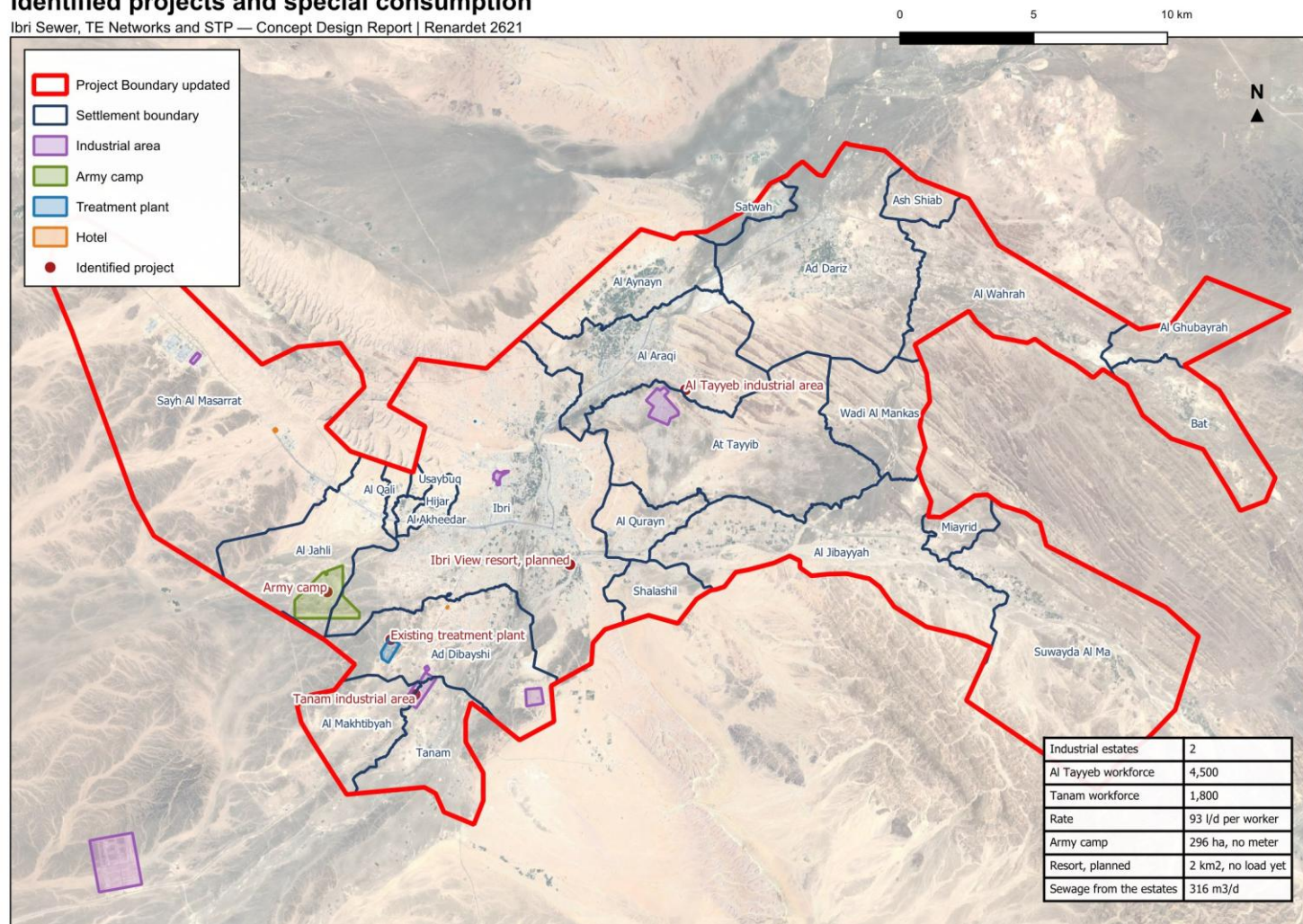


Figure 50. The identified projects and special consumers: the industrial areas, the army camp, the treatment plant and the hotel sites recorded in OpenStreetMap, the two estates among them.

4.4. Treated Effluent Demand and Customers

Treated effluent is produced at 95 per cent of the STP inflow. A further ten per cent is lost within the treated effluent distribution network, so 85.5 per cent of the inflow reaches the customers.¹

$$Q_{\text{delivered}} = 0.90 \times 0.95 \times Q_{\text{inflow}} \quad (16)$$

$Q_{\text{delivered}}$: treated effluent available to customers, m³/d

Q_{inflow} : flow entering the STP, the design average, m³/d

Table 38. Treated effluent available, one STP for the whole area, m³/d

	2024	2030	2055	2070
STP inflow, design average, before infiltration	22,937	26,638	46,492	66,109
Produced, 95 per cent	21,790	25,306	44,168	62,803
Delivered, less 10 per cent in the network	19,611	22,776	39,751	56,523

The infiltration of the network of Section 6.2 adds 1,187 cubic metres a day to the inflow with the margin, and 1,015 to the treated effluent delivered.

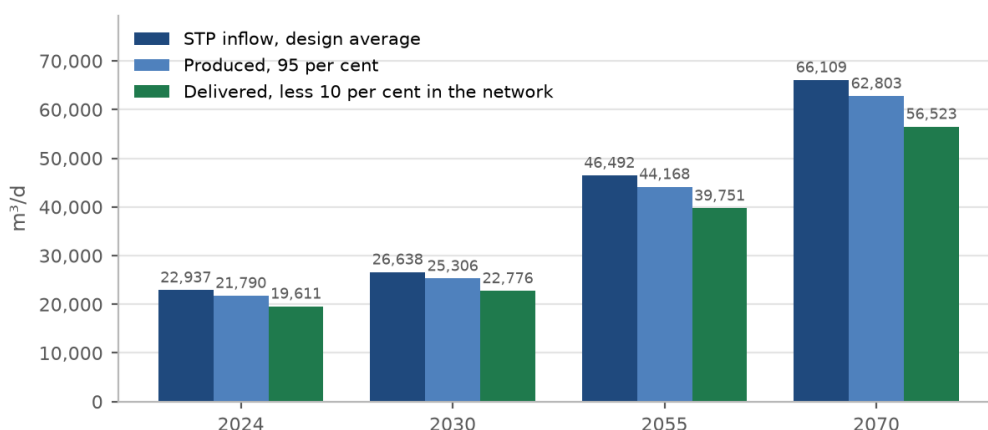


Figure 51. STP inflow, treated effluent produced and treated effluent delivered to customers, by year.

Adopted (15). 95 per cent of the STP inflow produced as treated effluent and 10 per cent of it lost in the distribution network (Section 4.4).

Customers are classified as public or private. Public consumers comprise the landscaping of highways, secondary roads, interchanges and roundabouts, and public parks. Private consumers comprise community parks, golf courses, private gardens, and nurseries and farms.

Table 39. Treated effluent demand for concept planning

Planting	Summer demand	Planting	Summer demand
Shrubs	20 to 40 l per plant per day	Ground cover	10 l/m ² /d
Palm trees	120 to 165 l per plant per day	Seasonal flowers	10 l/m ² /d
Other trees	40 to 80 l per plant per day	Grass	12 l/m ² /d
Hedges	10 l per metre per day	Roads and junctions	10 l/m ² /d

¹ PAM-GUD-201, Section 7.4.6.1, page 73, and Section 7.4.6.3 (b), page 76: "a system loss of 10 percent of all produced TSE shall be assumed".

Demand varies seasonally, at 100 per cent from June to August, 75 per cent in spring and autumn and 50 per cent from December to February. The system is sized for the summer peak. Consumers whose average demand exceeds 500 cubic metres per day are assessed individually.¹

The customers, their present and potential demand and their development plans are registered with the treated effluent network options (Section 6.4).

¹ PAM-GUD-201, Tables 21 to 23, pages 75 and 76. The rate for roads and junctions applies in the absence of specific vegetation information and is subject to municipality approval.

5. Existing Wastewater System

This chapter covers the existing wastewater and treated effluent systems: their hydraulic assessment, their rehabilitation, and the verification of the Regional Master Plan. It follows the survey in progress.

5.1. Hydraulic Assessment of the Existing Networks

5.1.1. Purpose

The existing sewer, force main and treated effluent networks are assessed against the flows arising across the design period, to establish which assets have capacity for the future flow, which require upgrading, and where the new works connect to them.

5.1.2. Methodology

The assessment is carried out by hydraulic model. The existing network is verified before it is modelled: the datasets supplied provide geometry but not levels, and the diameter is recorded on part of the network only. The survey now in progress establishes both, and the model will be built on the surveyed data. Until then the concept design does not rely on the existing networks: the network of Section 6.2 carries all the sewage to the treatment plants, and the existing system is assessed as soon as the as-built survey is available.

Each asset is then classified as having capacity for the design flow, requiring upgrading, or requiring replacement. Where an asset is retained, the point and manner of connection to the new works is designed.

5.1.3. Hydraulic Model

The wastewater network is modelled in SewerGEMS and the treated effluent network in WaterGEMS. Models are submitted in native editable format with the calculations and assumptions, and are updated following construction.¹

The model is run for the design year at peak flow, for the ultimate condition, and for the opening years at low flow. The last of these establishes the period during which the network requires assisted cleansing, described in Section 6.2.4.

5.2. Rehabilitation and Upgrading

Rehabilitation of the existing systems forms part of the concept scope. The extent of it follows from the assessment in Section 5.1 and from the condition established by survey and closed-circuit television inspection.

¹ The software to be used is the subject of Section 1.5.4.

Rehabilitation work received to date comprises contractor submissions for completed work orders at Al Sad, Khadil, Yanqul, Dhank and Hay Al Aqabah. These have been reviewed and recorded; they describe work already carried out rather than work required.

5.3. Verification of the Regional Master Plan

The Terms of Reference require the Regional Master Plan to be verified. The verification compares the flows established in Chapter 4 against those on which the master plan was based, and identifies and explains any difference.

The asset data supplied includes a treatment plant record shown as a design case with a capacity of 29,038 cubic metres per day, annotated as arising from the Regional Master Plan concept design and not yet approved. That figure is taken as the master plan position for comparison.¹

The comparison depends on the design horizon, which is the subject of Section 1.5.2, and on the flow series described in Section 4.2.7.

The data required by Nama Water Services Asset Management Planning for the updating of the master plan will be provided in the format the client specifies.

¹ The treatment plant record in the asset data supplied, with the status Design and the source Asset Planning.

6. Options Development

This chapter sets out how the options are developed and compared, presents the seven options of the sewer network with their pumping stations, and sets the framework for the options of the treated effluent network, the treatment plant, the excess effluent and the sludge.

6.1. Options Methodology

Development and selection of options

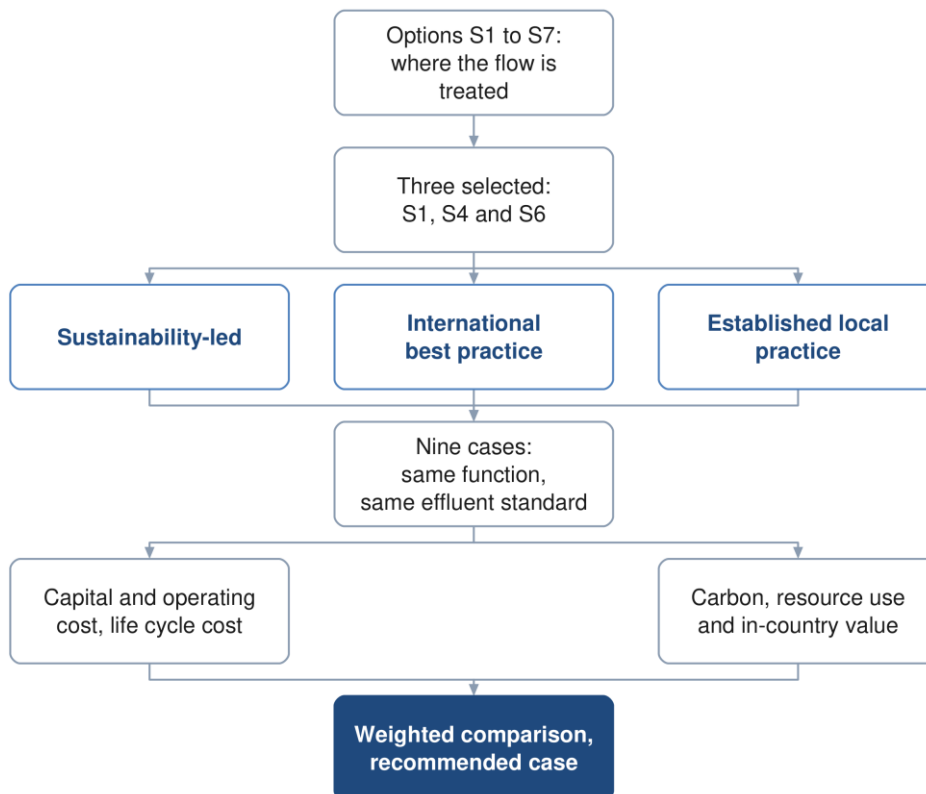


Figure 52. Development of the options: the three selected, each designed under the three characters, and the basis on which the nine cases are compared.

6.1.1. Number and Character of the Options

Not fewer than three options are developed for each of the sewer network, the treated effluent network and the treatment plant. The guidelines indicate the character the options should take: one advancing environmental sustainability including nature-based solutions, one representing international best practice, and one based on established practice and technology available in the Sultanate.¹

For the sewer network, seven options have been modelled for where the flow is treated (Section 6.2), and further options can be added on the same model. Three of them, S1, S4 and S6, are selected for the appraisal, for the reasons given in Section 7.11. The characters are not tied to an option. Each selected option keeps the same network model and is designed and costed once for each character, changing only the factors the character governs, so that the three options give nine costed cases.

Each option is developed to equivalent functional requirements with comparable reliability and redundancy, so that the comparison between them is not influenced by differences in scope.

6.1.2. Proposed Approach by Character

The guidelines describe the three characters but do not prescribe what distinguishes them in design terms. The following approach is proposed and is offered for comment. The network of an option, its pipes, pumping stations and plant sites, is the same in all three characters; what changes is the treatment process, the energy supply, solar generation included, the reuse and the materials.

Table 40. Proposed approach for each character

	Sustainability-led	International best practice	Established local practice
Treatment process	Lower-energy process, with nature-based polishing where the scale permits	Highest-performing process, smallest footprint	Process already established in Oman, simple to operate and maintain
Energy	Solar generation on site, with a self-sufficiency target	High-efficiency plant and advanced process control	Grid supply with standby generation
Reuse	Reuse maximised, discharge to wadi minimised	High reuse at high effluent quality	Reuse to the demand identified, surplus discharged
Materials	Low-carbon and locally sourced where performance permits	Specified for performance and durability	Standard specification
Consequence	Higher capital cost, lowest carbon and operating cost	Highest capital cost, least land, best effluent quality	Lowest capital cost, most land, highest operating cost

Two constraints apply across all three characters. Nature-based treatment is limited by the guidelines to small plants, so it is available for outlying settlements or for polishing rather than for the main plant. And every option is developed to the same functional requirement and the same effluent standard, so that the difference between them lies in how the result is achieved and not in what is achieved.²

6.1.3. Basis of Comparison

The options are compared on capital and operating cost, life cycle cost, carbon footprint over the project lifetime, resource efficiency, in-country value and the degree to which they employ nature-based solutions. The evaluation period is twenty-five years. The weighting applied to each parameter is set by Nama Water Services. The figure on the following page shows how each case is costed and compared.

¹ PAM-GUD-201, Section 12.1, page 95.

² PAM-GUD-203, Section 10.5, page 101, limits constructed wetlands to plants of approximately 500 cubic metres per day.

How each case is costed and compared

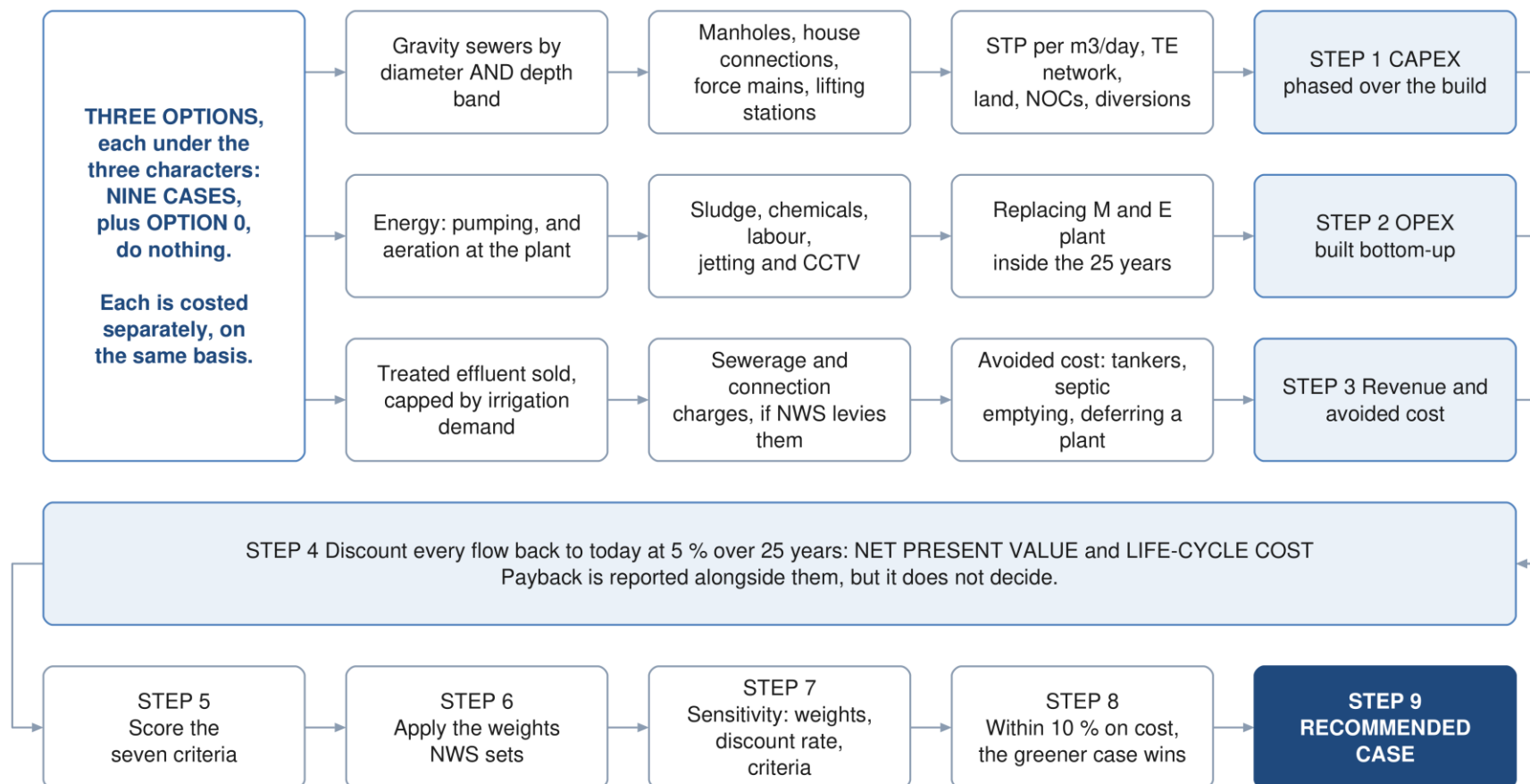


Figure 53. How each case is costed and compared. The three cost streams are established separately for every case, discounted to present value together, and only then scored against the seven criteria.

6.1.4. Selection Criteria

A weighted multi-criteria analysis compares the options against total lifetime cost, sustainability, social development and in-country value, adaptability and resilience, operability, constructability and environmental impact. Sensitivity is tested by varying the weighting between categories, the discount rate and the input design criteria. Where two cases fall within ten per cent of one another on total lifetime cost, the more sustainable is adopted.¹

6.2. Sewer Network Options

Network design approach

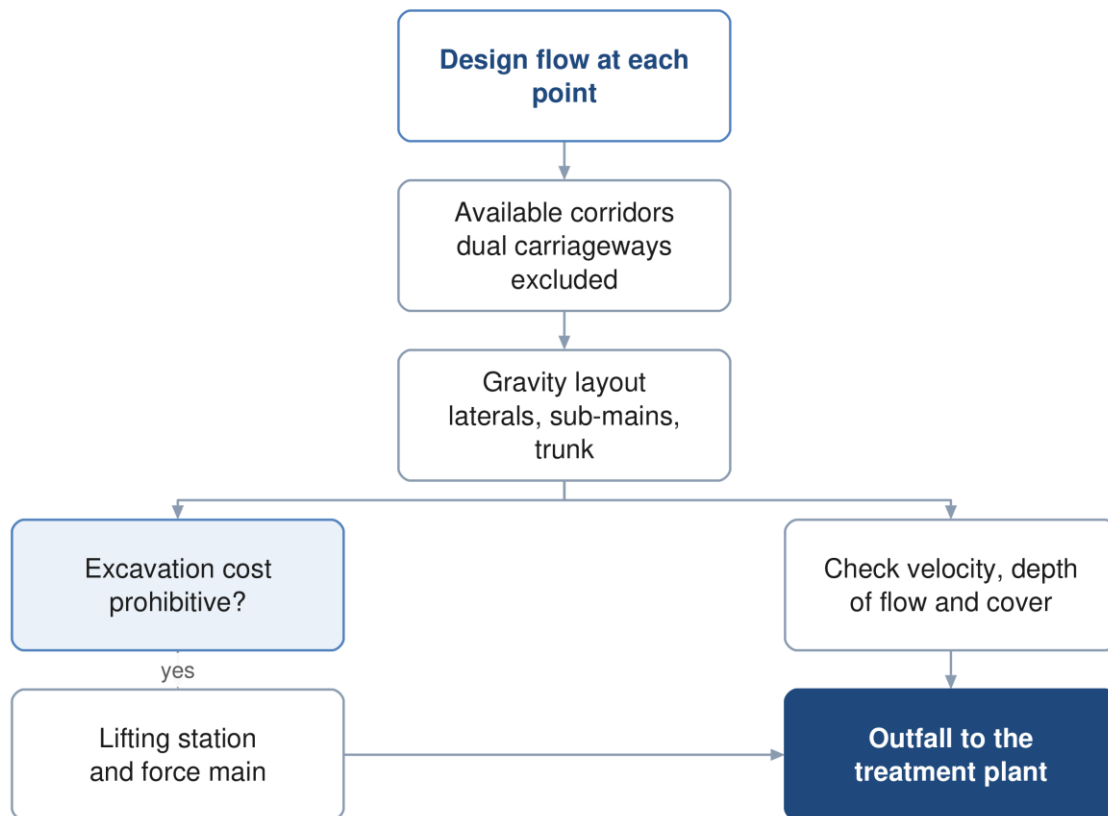


Figure 54. The network design approach, and the point at which a lifting station becomes necessary.

6.2.1. Design Principles

The network is laid out to convey the flow by gravity wherever that is feasible and cost effective, and to keep pumping to the minimum that the topography requires. The layout follows the road corridors, and each part of the network drains to its lowest point. The existing treatment plant site lies low relative to the developed areas; the options differ in whether the flow is treated there or at plants nearer the outlying settlements (Section 6.2.6).

6.2.2. Corridor Constraints

Dual carriageways are excluded as sewer corridors, as they cannot be taken out of service for construction or maintenance. Crossings of a dual carriageway are made perpendicular and, where available, through an existing underpass. Wadi crossings are designed to the cover required by the guideline.

¹ PAM-GUD-201, Sections 12.6 to 12.9, pages 104 to 106.

6.2.3. Network Hierarchy

The network is arranged in three tiers, following the arrangement of the existing network in Ibri: laterals collecting from properties, sub-mains collecting from laterals, and a trunk main conveying the collected flow to the plant. Arranging the network this way limits the number of connections made directly to the trunk main and keeps each tier to a manageable size.

6.2.4. Early-Year Operation

In the years following commissioning the connected population is a fraction of the design population, and the flow in the network is correspondingly lower. Velocities in that period may fall below the self-cleansing value, and the network requires assisted cleansing until the flow is sufficient to scour the pipes.

The period during which this applies is established for each pipe from the flow series and the connection ratio, and is presented as a schedule of the pipes affected and the years concerned, so that the operating requirement is known before the network is handed over. Section 6.2.8 sets out the year from which each pipe of the network reaches the self-cleansing velocity.

6.2.5. Hydraulic Model

The network is modelled in SewerGEMS as twenty-four subnetworks. Each collects the flow of its own catchment by gravity and ends at its lowest point, its outfall. Where an outfall is not at a treatment plant, a pumping station there lifts the flow through a rising main into a manhole of another subnetwork, or directly to a plant. The subnetworks take the names of their outfalls, O1 to O24, as shown in the figure on the following page.

The sewer network: twenty-four subnetworks and their outfalls

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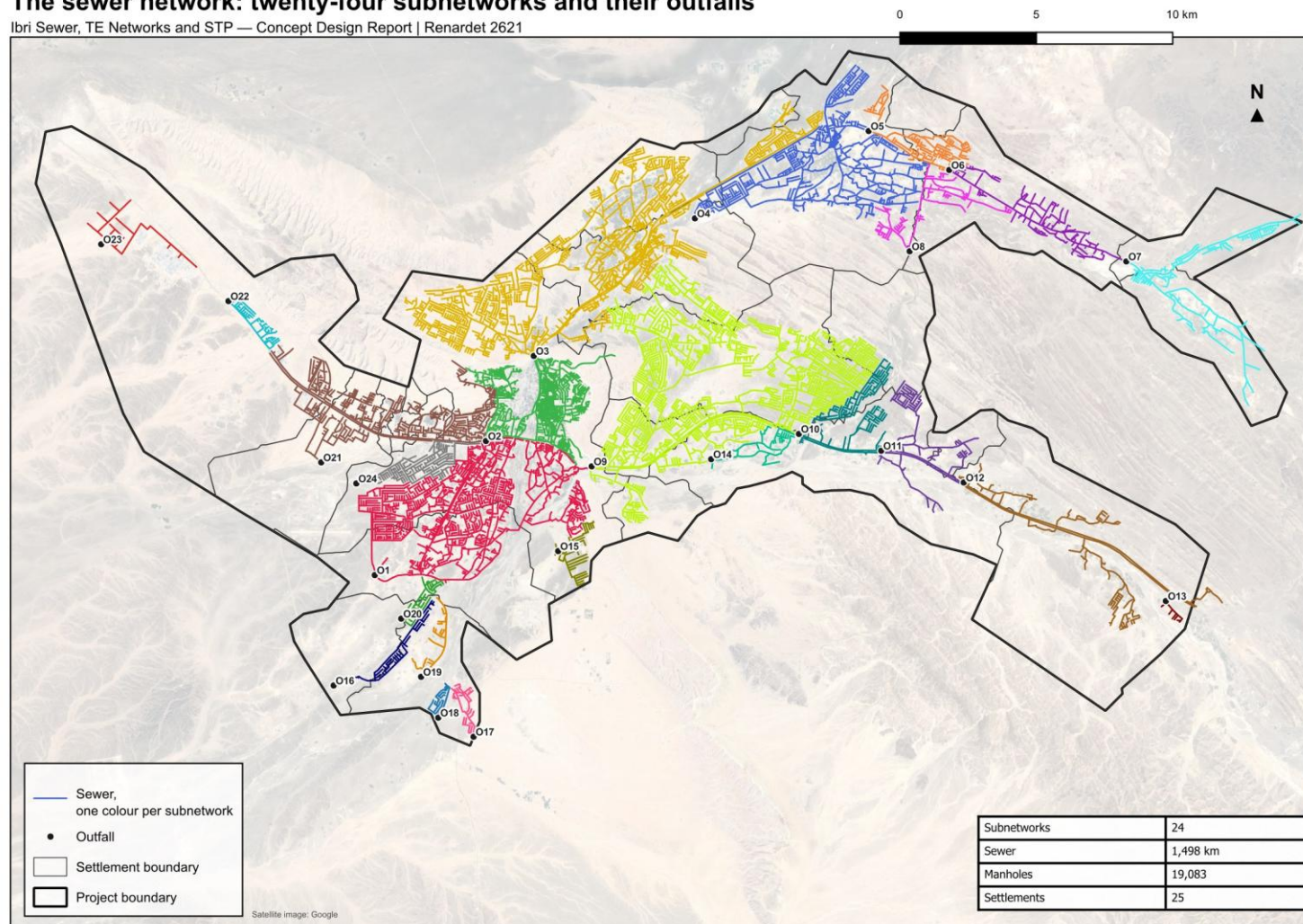


Figure 55. The twenty-four subnetworks of the sewer network, each with its outfall, and the settlement boundaries.

The flow of every plot, established in Chapter 4, is placed at the manhole nearest to it for each of the six model years: 2030, 2040, 2050, 2055, 2060 and 2070. The network carries 23,677 cubic metres a day of it in 2030 and 59,377 in 2070. A further 539 cubic metres a day in 2030, rising to 722 in 2070, lies at manholes that are not part of these options: branches in planted areas, where the ground levels are not reliable enough to lay a sewer until the survey is complete, and three small outlying catchments, where a sewer would be deep for little flow and an on-site solution such as an absorption well is proposed.

Infiltration of 720 litres a day for every kilometre of sewer is added along the pipes: 1,079 cubic metres a day over the 1,498 km of the network.¹

The model sets the peak flow in each pipe from the average flow upstream of it, with the peak factor of Section 4.2.6. The flow pumped from one subnetwork into another enters the receiving manhole as the peak outflow of the subnetwork that sends it, for the same year, and is not peaked a second time.

Pipe friction follows the Colebrook-White equation with a roughness of 1.5 mm for every size.²

The pipes are designed for the flow of 2070, the year the land is full, and the design is repeated until no pipe changes size. The years 2030 to 2060 are then analysed on the designed pipes, which gives the flow arriving at each plant as the area develops. In every option and every year, no pipe runs faster than 3.0 m/s, and no pipe runs deeper than 65 per cent of its diameter up to 350 mm, or 50 per cent above.³

The design horizon is still to be decided (Decision 4). The pipes are designed for 2070 so that the network is laid once and is not relieved while the plots around it are still being built (Section 4.1.9). The flow at each plant is given for every model year, 2055 among them, so that the plants can be phased on either horizon.

6.2.6. Description of the Options

The options differ in where the flow is treated. The pipes run in the same streets in all of them; what changes is where each subnetwork's flow is sent, the sizes of the pipes that carry it, the pumping stations and their rising mains, and the number and size of the plants.

Table 41. The options and the subnetworks each plant serves

Option	Plants	Plants and the subnetworks they serve
S1	1	O1: 24 subnetworks
S2	1	O16: 24 subnetworks
S3	2	O1: 19 subnetworks; O4: O4, O5, O6, O7, O8
S4	3	O1: 13 subnetworks; O4: O4, O5, O6, O7, O8; O9: O9, O10, O11, O12, O13, O14
S5	4	O1: 12 subnetworks; O3: its own subnetwork only; O4: O4, O5, O6, O7, O8; O9: O9, O10, O11, O12, O13, O14
S6	6	O1: O1, O2, O15, O21, O24; O3: its own subnetwork only; O4: O4, O5, O6, O7, O8; O9: O9, O10, O11, O12, O13, O14; O16: O16, O17, O18, O19, O20; O22: O22, O23
S7	3	O1: 17 subnetworks; O16: O16, O17, O18, O19, O20; O22: O22, O23

Option S1. All the flow is treated at one plant at O1: 24,756 cubic metres a day on average in 2030 and 60,456 in 2070. The sewer arrives at the plant 10.3 m below ground. It needs 23 pumping stations with 422 kW of pumps in duty; the largest lifts 636 l/s from O2. The longest rising main runs 9.5 km from O23, against 301 m of head.

¹ PAM-GUD-201, Section 7.4.3, page 72.

² PAM-GUD-203, pages 24 and 28.

³ PAM-GUD-203, page 27 and Table 10.

Option S2. All the flow is treated at one plant at O16: 24,756 cubic metres a day on average in 2030 and 60,456 in 2070. The sewer arrives at the plant 5.9 m below ground. It needs 23 pumping stations with 884 kW of pumps in duty; the largest lifts 1,451 l/s from O1. The longest rising main runs 9.5 km from O23, against 301 m of head.

Option S3. The flow is divided between two plants. The largest, at O1, receives 52,521 cubic metres a day on average in 2070, its sewer arriving 10.1 m below ground; O4 receives 7,935 at 6.9 m. It needs 22 pumping stations with 361 kW of pumps in duty; the largest lifts 414 l/s from O2. The longest rising main runs 9.5 km from O23, against 301 m of head.

Option S4. The flow is divided between three plants. The largest, at O1, receives 36,881 cubic metres a day on average in 2070, its sewer arriving 10.0 m below ground; O4 receives 7,935 at 6.9 m, O9 receives 15,640 at 8.8 m. It needs 21 pumping stations with 319 kW of pumps in duty; the largest lifts 414 l/s from O2. The longest rising main runs 9.5 km from O23, against 301 m of head.

Option S5. The flow is divided between four plants. The largest, at O1, receives 25,915 cubic metres a day on average in 2070, its sewer arriving 9.8 m below ground; O3 receives 10,966 at 5.8 m, O4 receives 7,935 at 6.9 m, O9 receives 15,640 at 8.8 m. It needs 20 pumping stations with 277 kW of pumps in duty; the largest lifts 150 l/s from O2. The longest rising main runs 9.5 km from O23, against 301 m of head.

Option S6. The flow is divided between six plants. The largest, at O1, receives 22,867 cubic metres a day on average in 2070, its sewer arriving 9.8 m below ground; O3 receives 10,966 at 5.8 m, O4 receives 7,935 at 6.9 m, O9 receives 15,640 at 8.8 m, O16 receives 2,602 at 4.9 m, O22 receives 446 at 1.5 m. It needs 18 pumping stations with 198 kW of pumps in duty; the largest lifts 150 l/s from O2. The longest rising main runs 7.5 km from O17; the highest head, 225 m, is at O23.

Option S7. The flow is divided between three plants. The largest, at O1, receives 57,408 cubic metres a day on average in 2070, its sewer arriving 10.3 m below ground; O16 receives 2,602 at 4.9 m, O22 receives 446 at 1.5 m. It needs 21 pumping stations with 343 kW of pumps in duty; the largest lifts 636 l/s from O2. The longest rising main runs 7.5 km from O17; the highest head, 225 m, is at O23.

Three figures follow for each option: a diagram of where the flow of every subnetwork goes, with the peak flow of each transfer in 2070; a map of the sewers by the plant they drain to, with the depth of the sewer arriving at each plant; and a map of the pumping, on which each pumping station carries the depth of its outfall, the average flow it receives in 2070, its pump head and its duty.

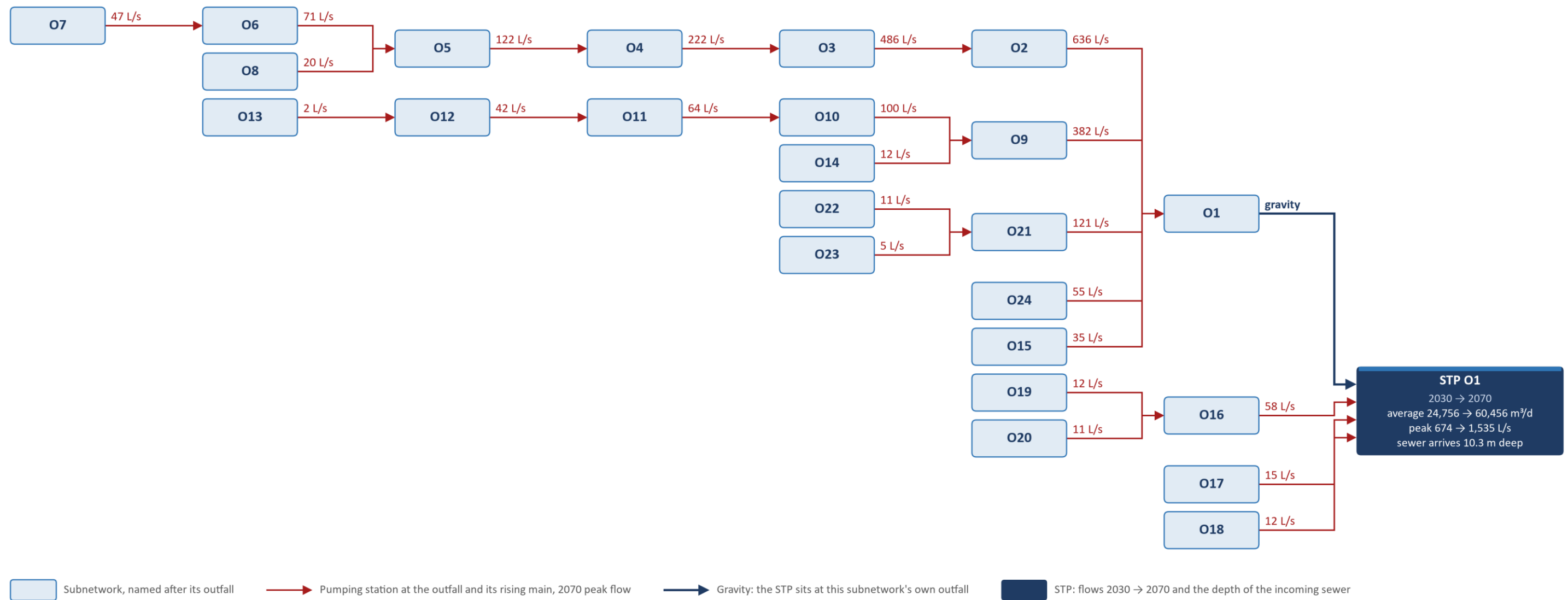


Figure 56. Option S1, one STP for the whole area, at O1: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S1: one STP for the whole area, at O1

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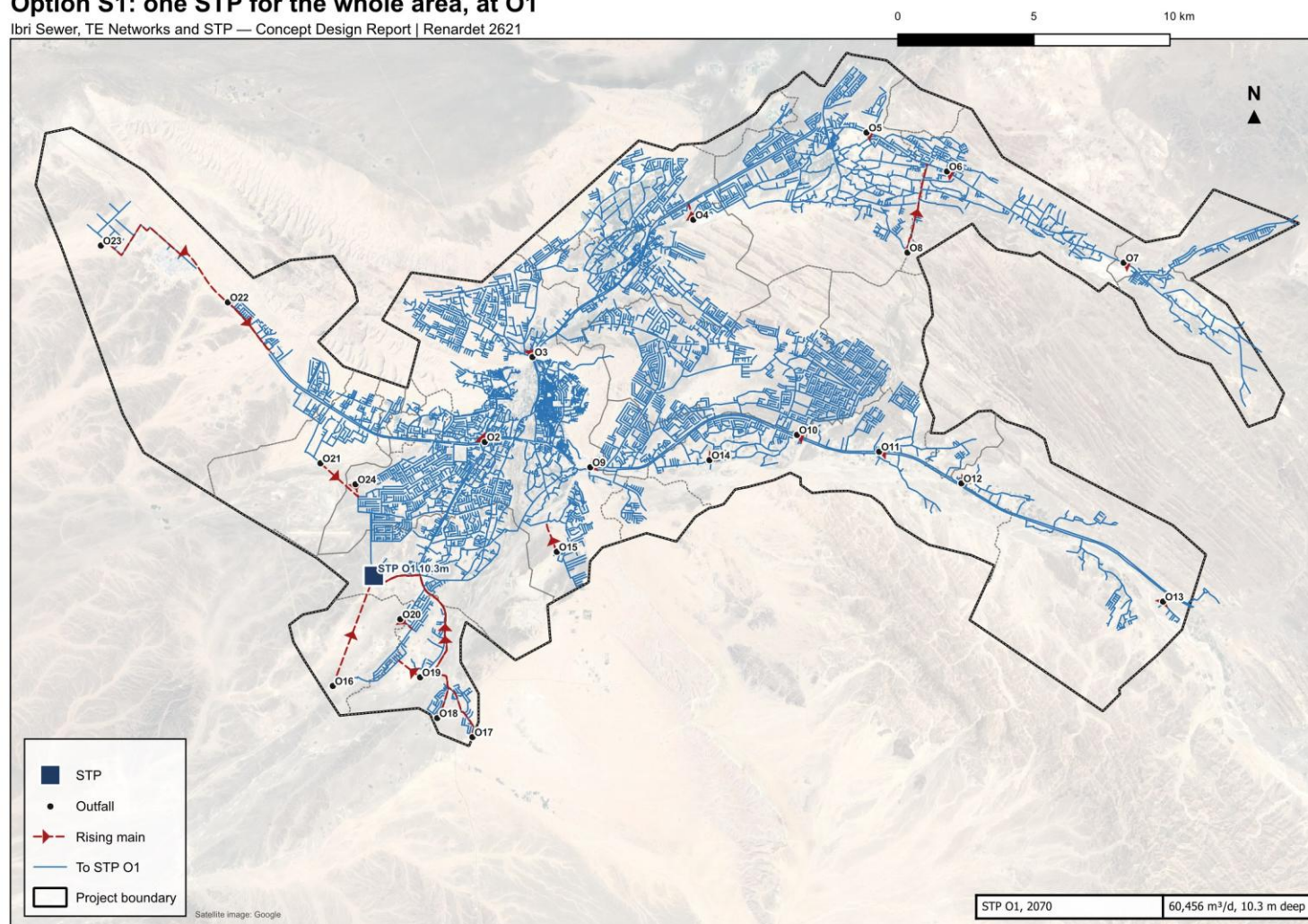


Figure 57. Option S1: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S1: sewers, pumping stations and rising mains

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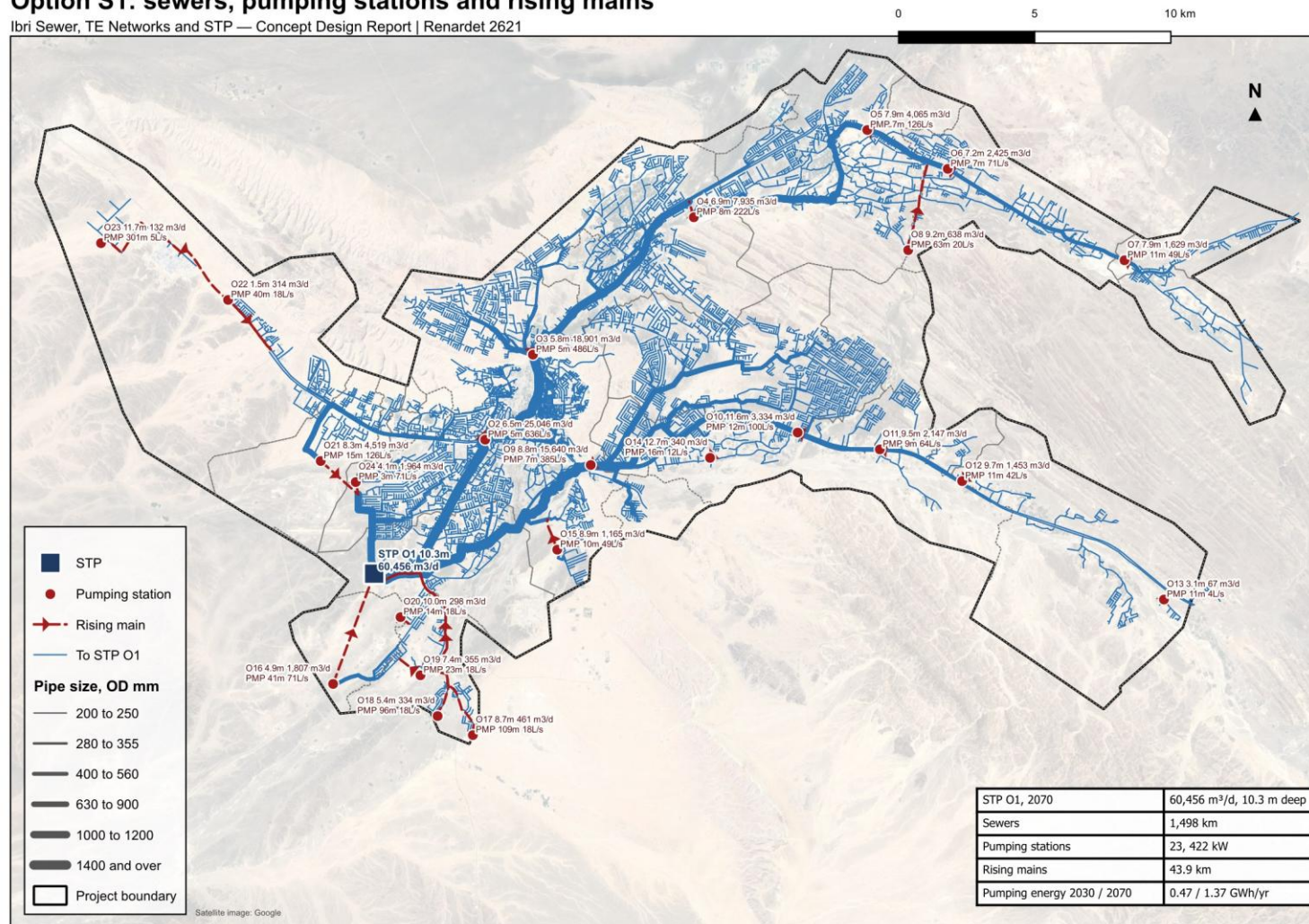


Figure 58. Option S1: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

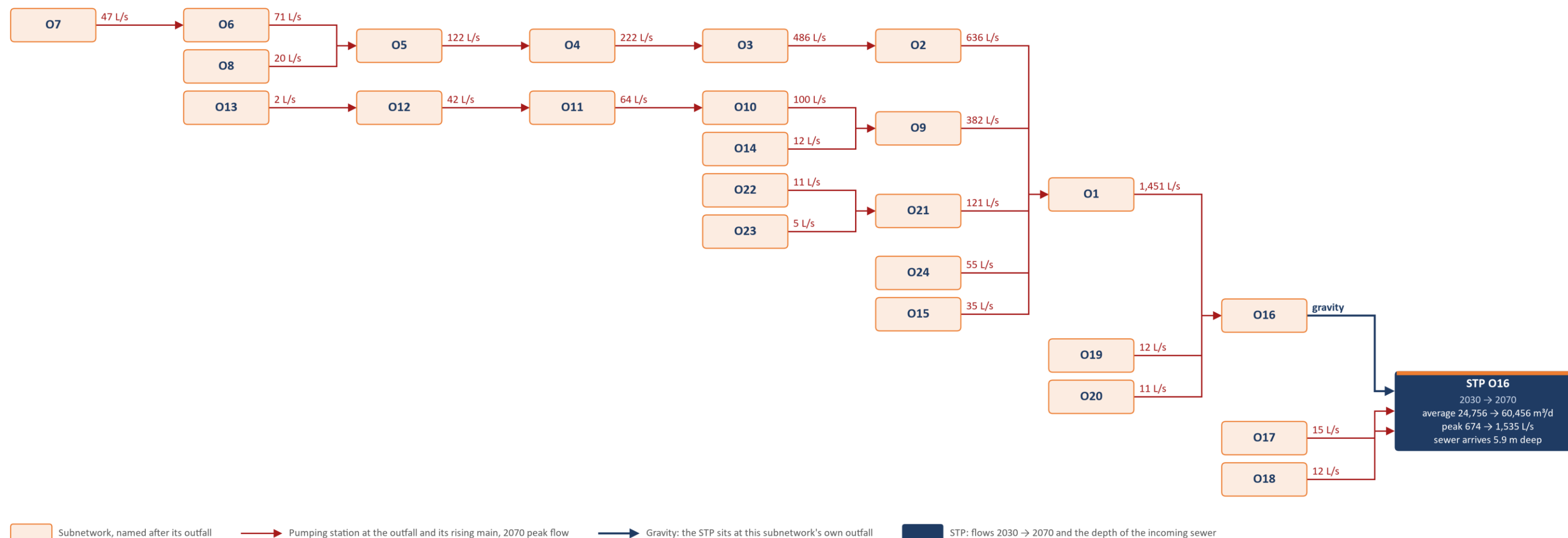


Figure 59. Option S2, one STP for the whole area, at O16: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S2: one STP for the whole area, at O16

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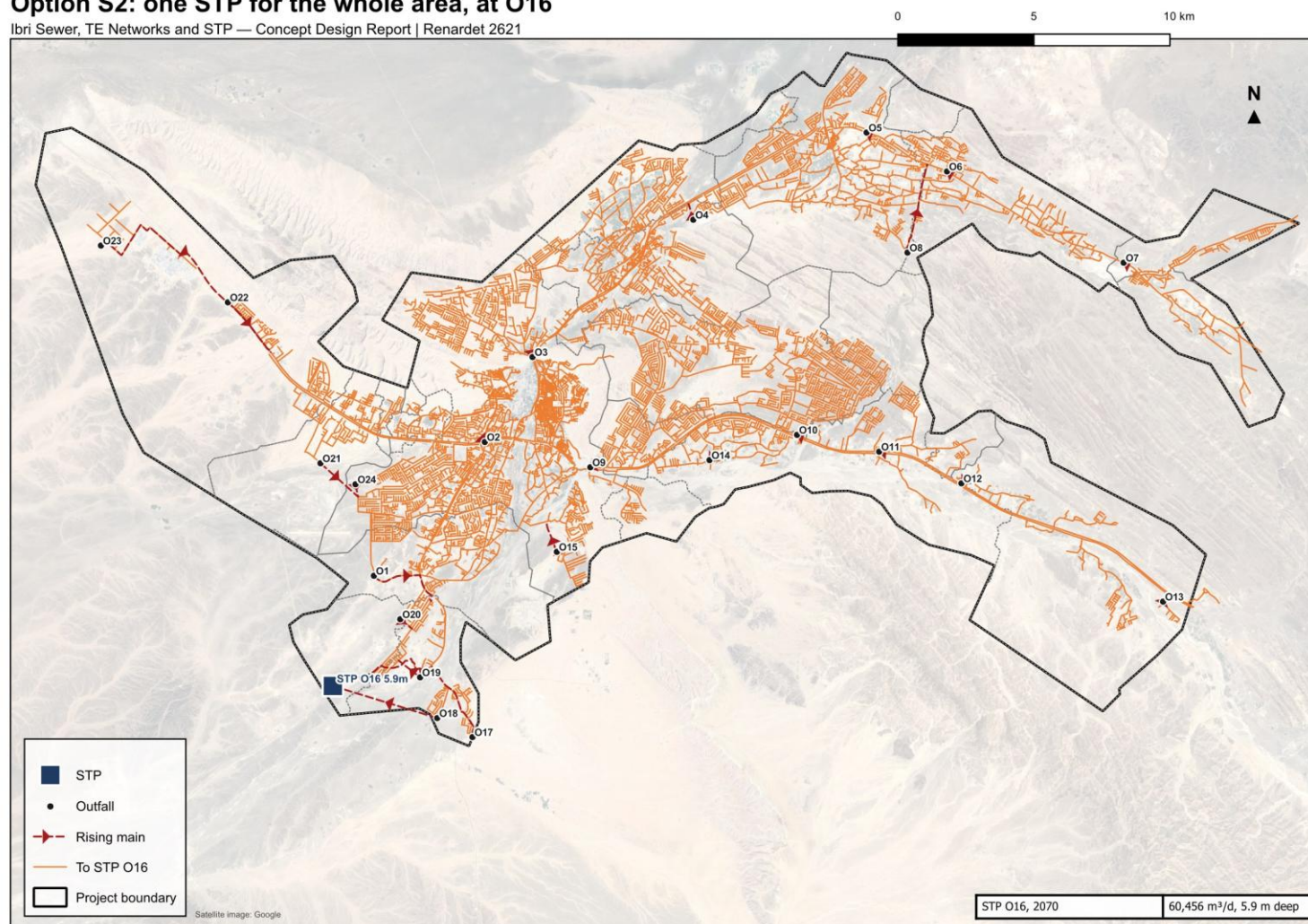


Figure 60. Option S2: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S2: sewers, pumping stations and rising mains

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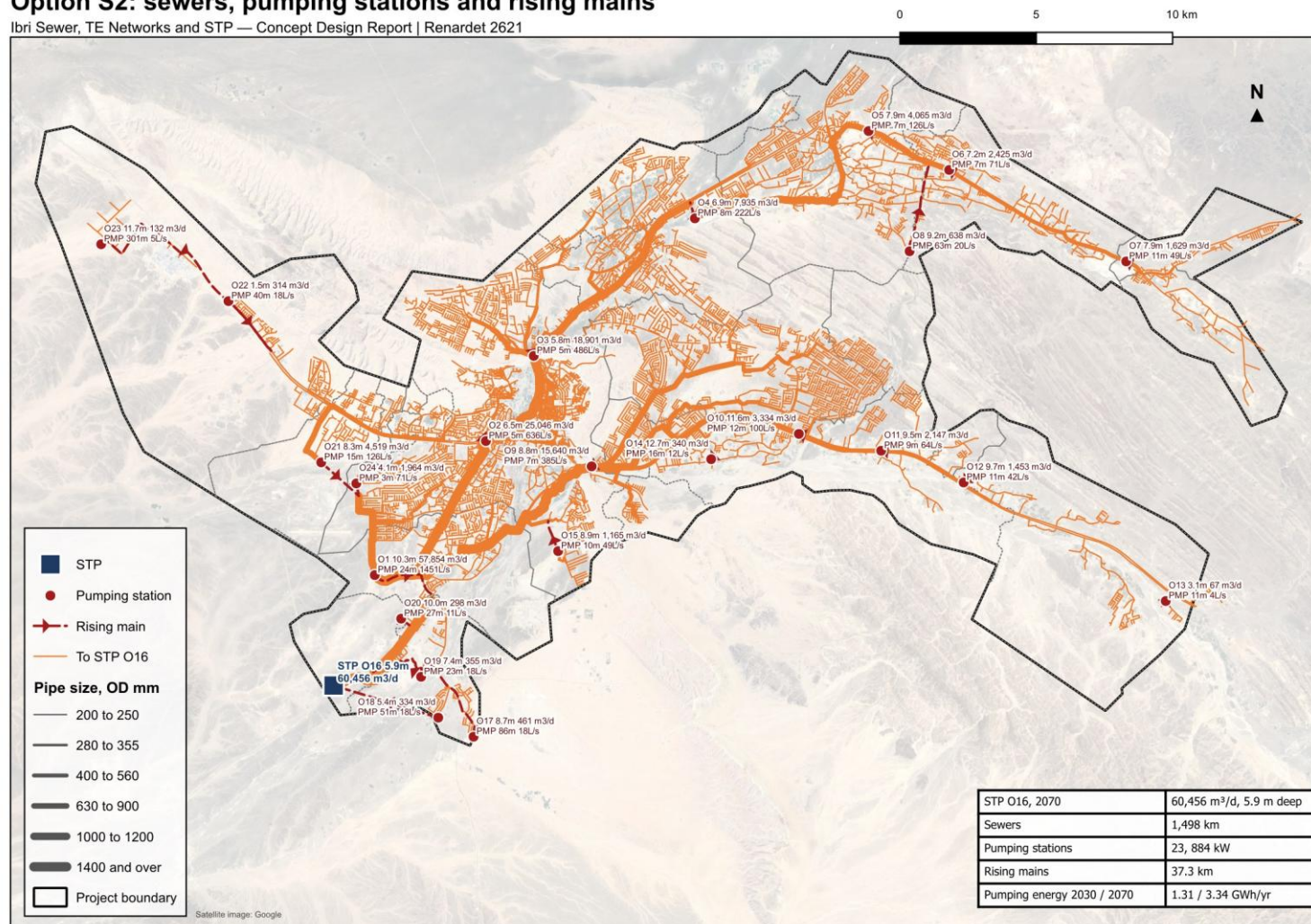


Figure 61. Option S2: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

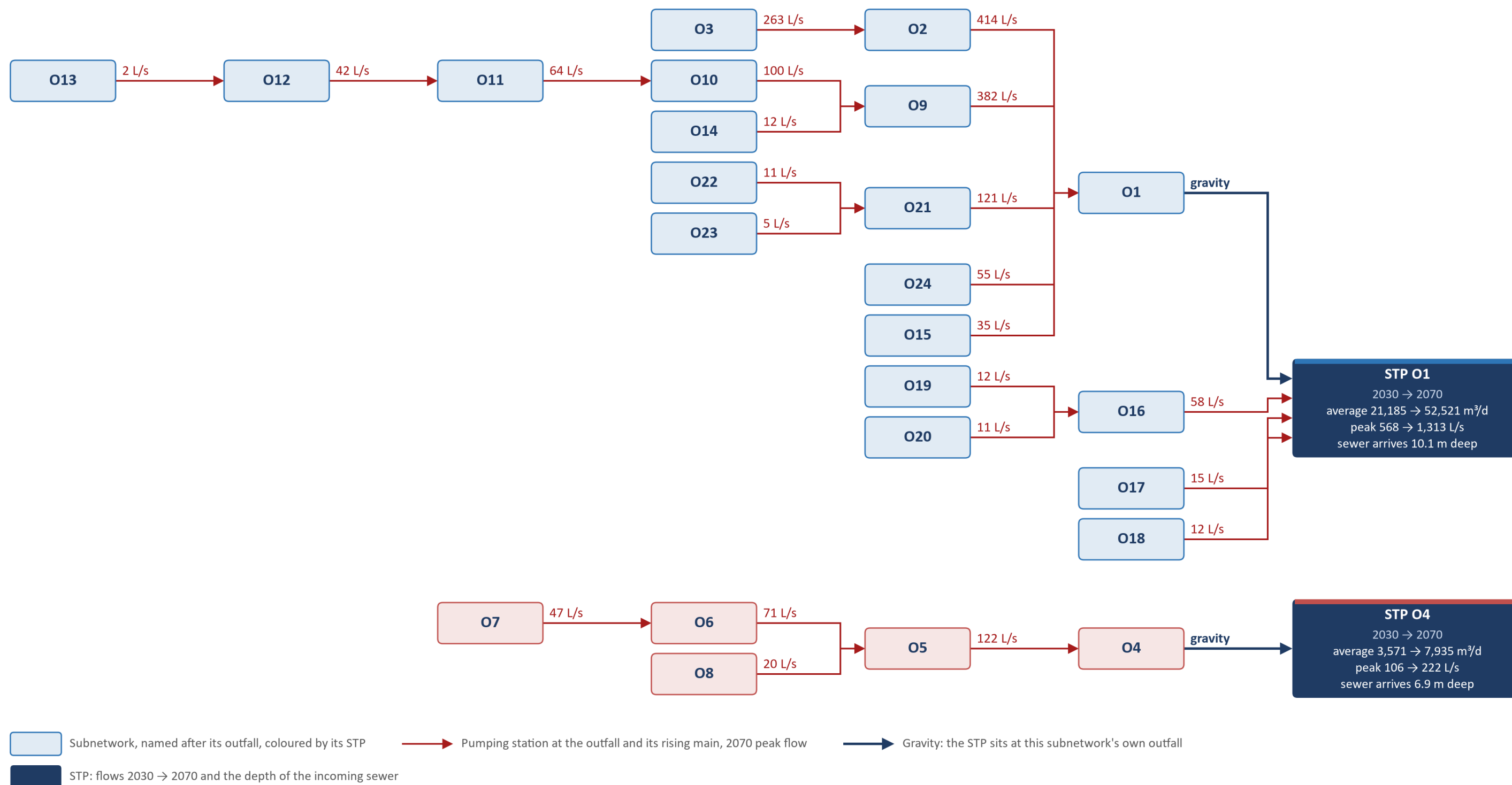


Figure 62. Option S3, two STPs, at O1 and O4: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S3: two STPs, at O1 and O4

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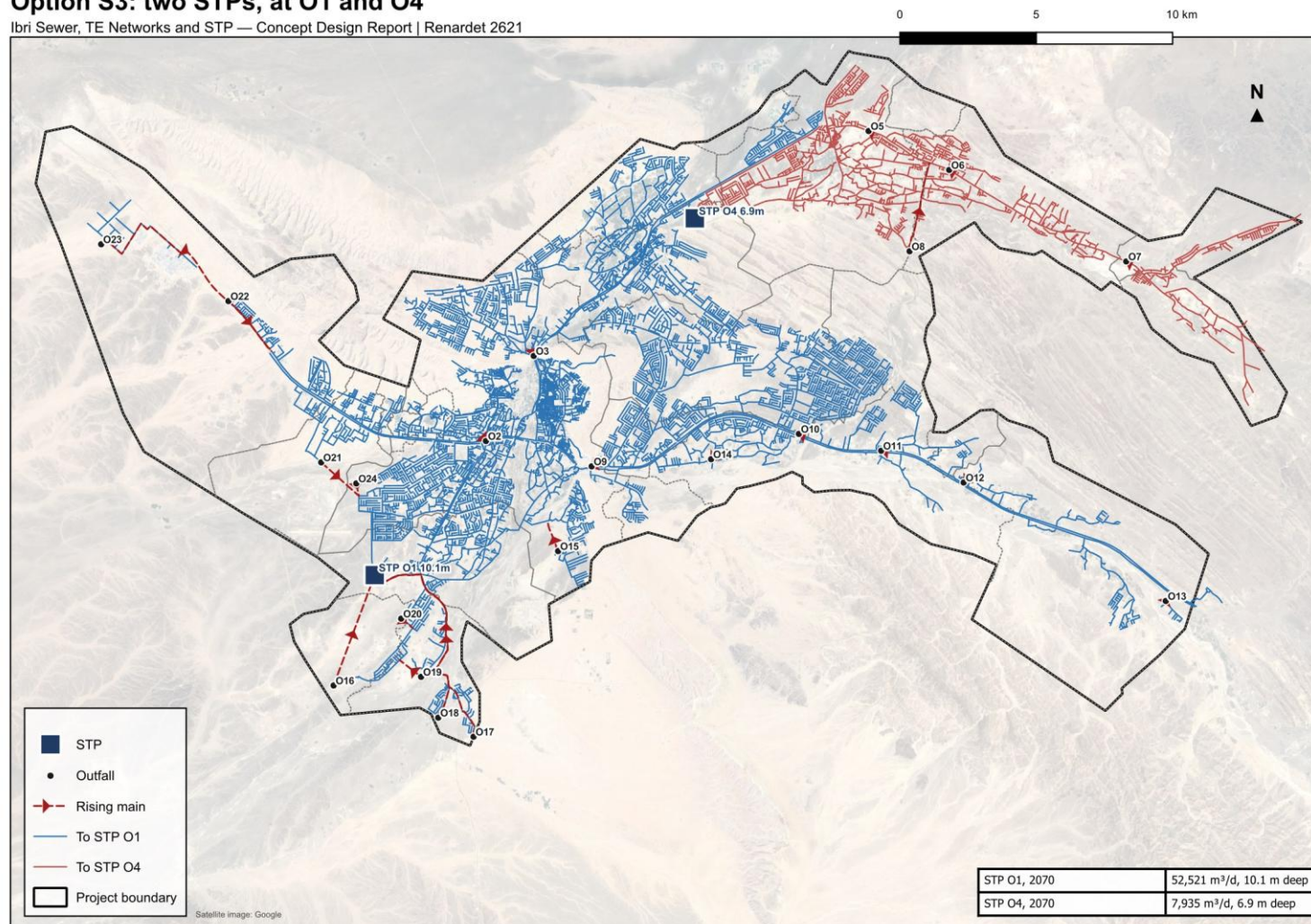


Figure 63. Option S3: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S3: sewers, pumping stations and rising mains

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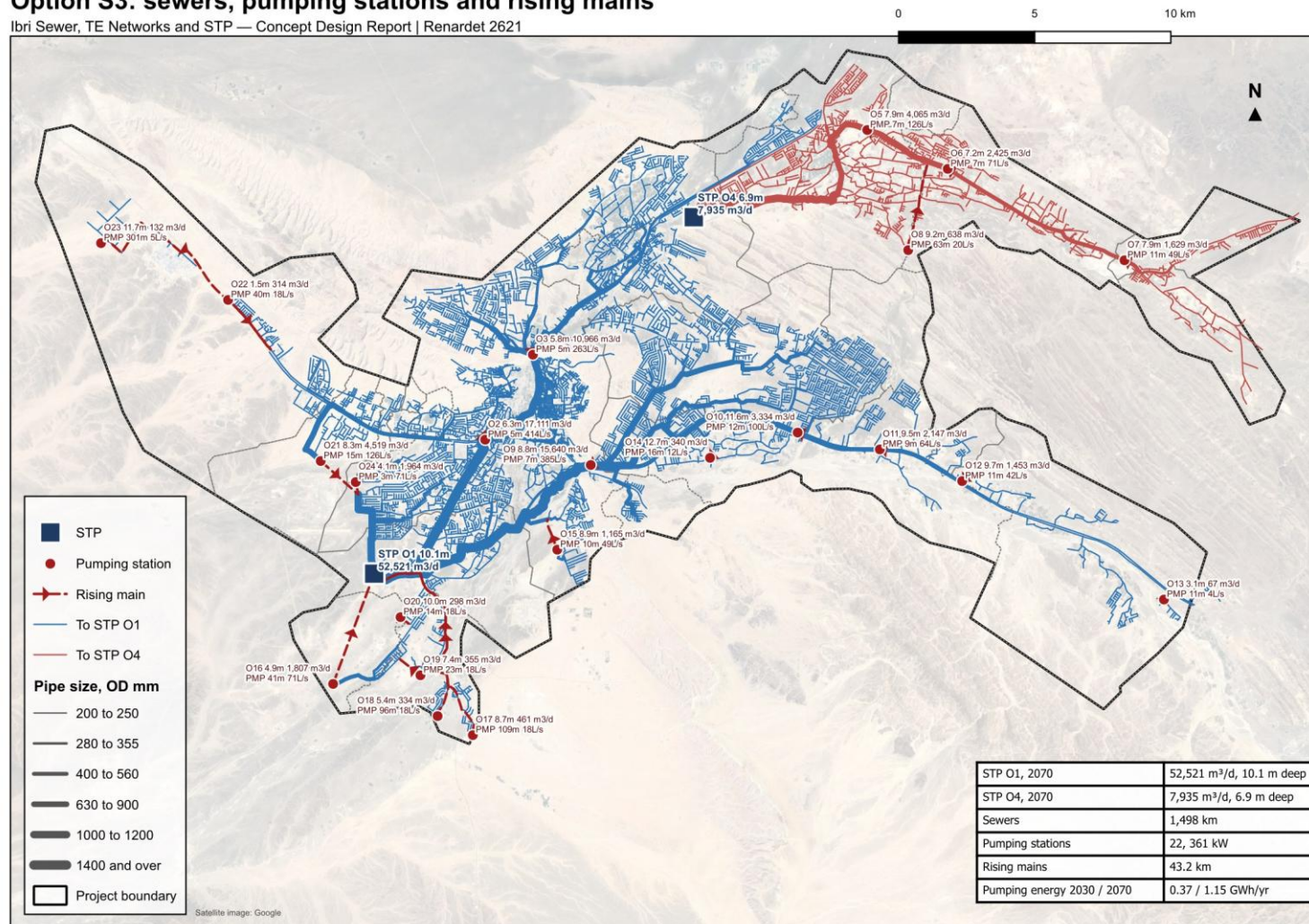


Figure 64. Option S3: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

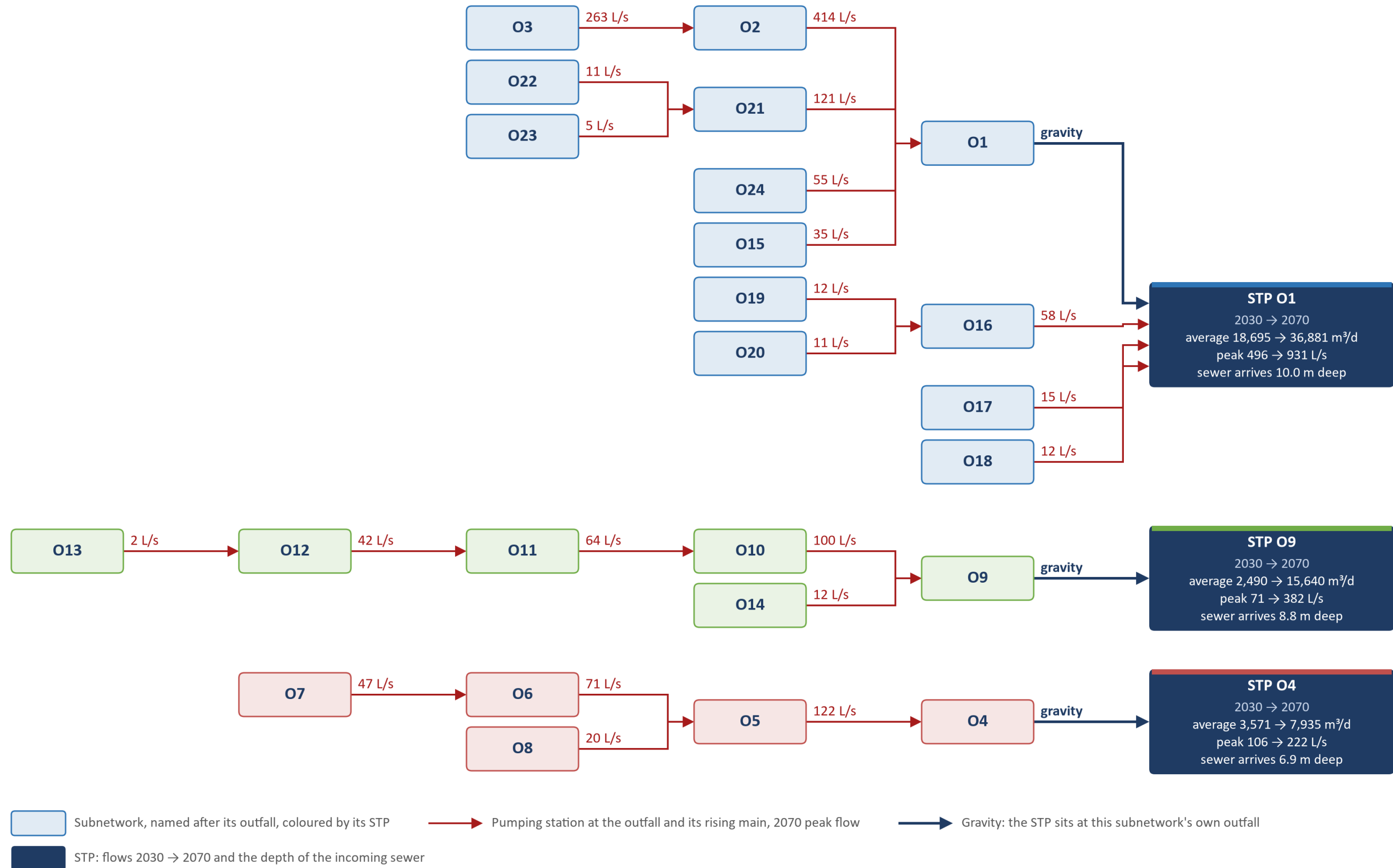


Figure 65. Option S4, three STPs, at O1, O9 and O4: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S4: three STPs, at O1, O9 and O4

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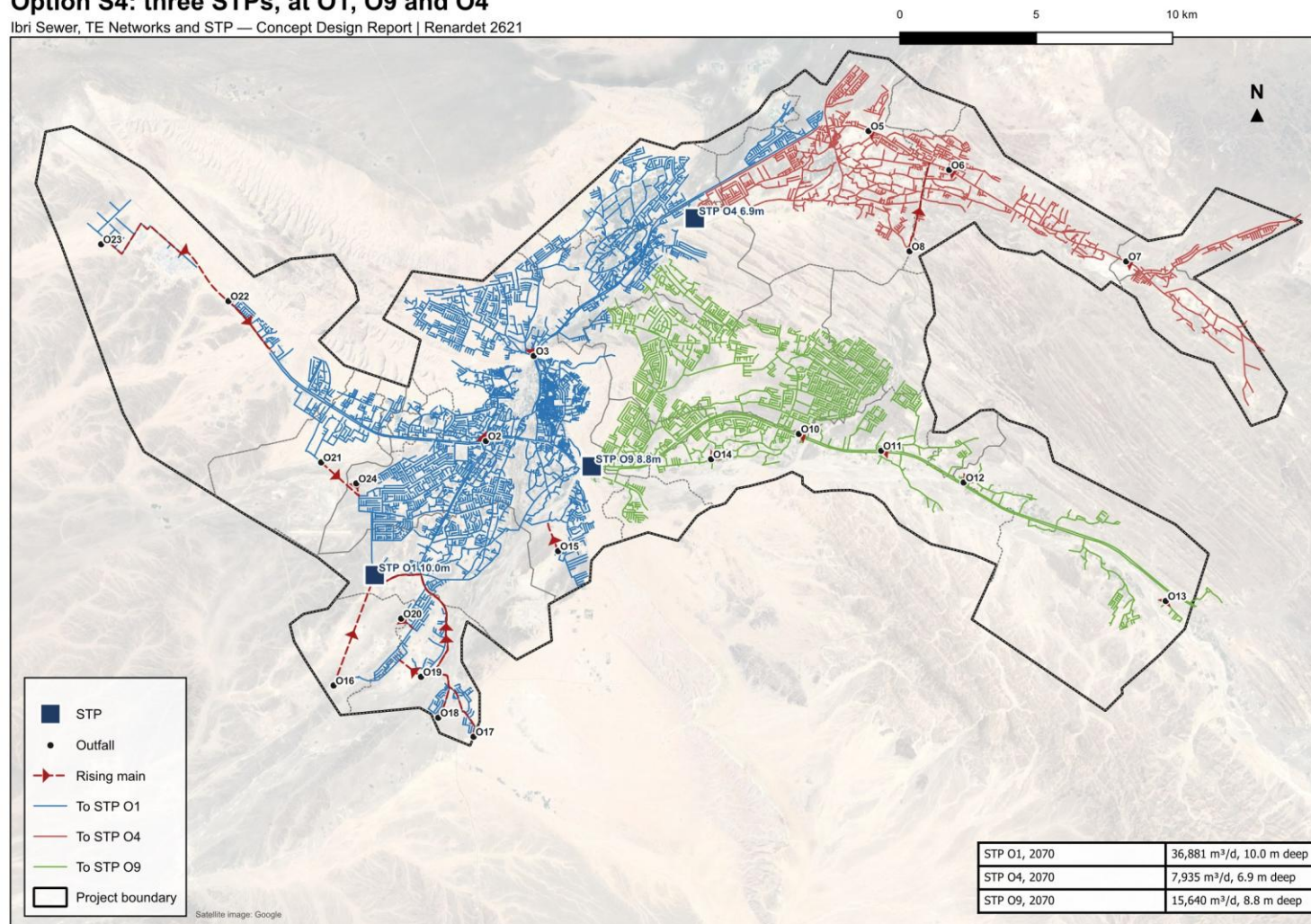


Figure 66. Option S4: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S4: sewers, pumping stations and rising mains

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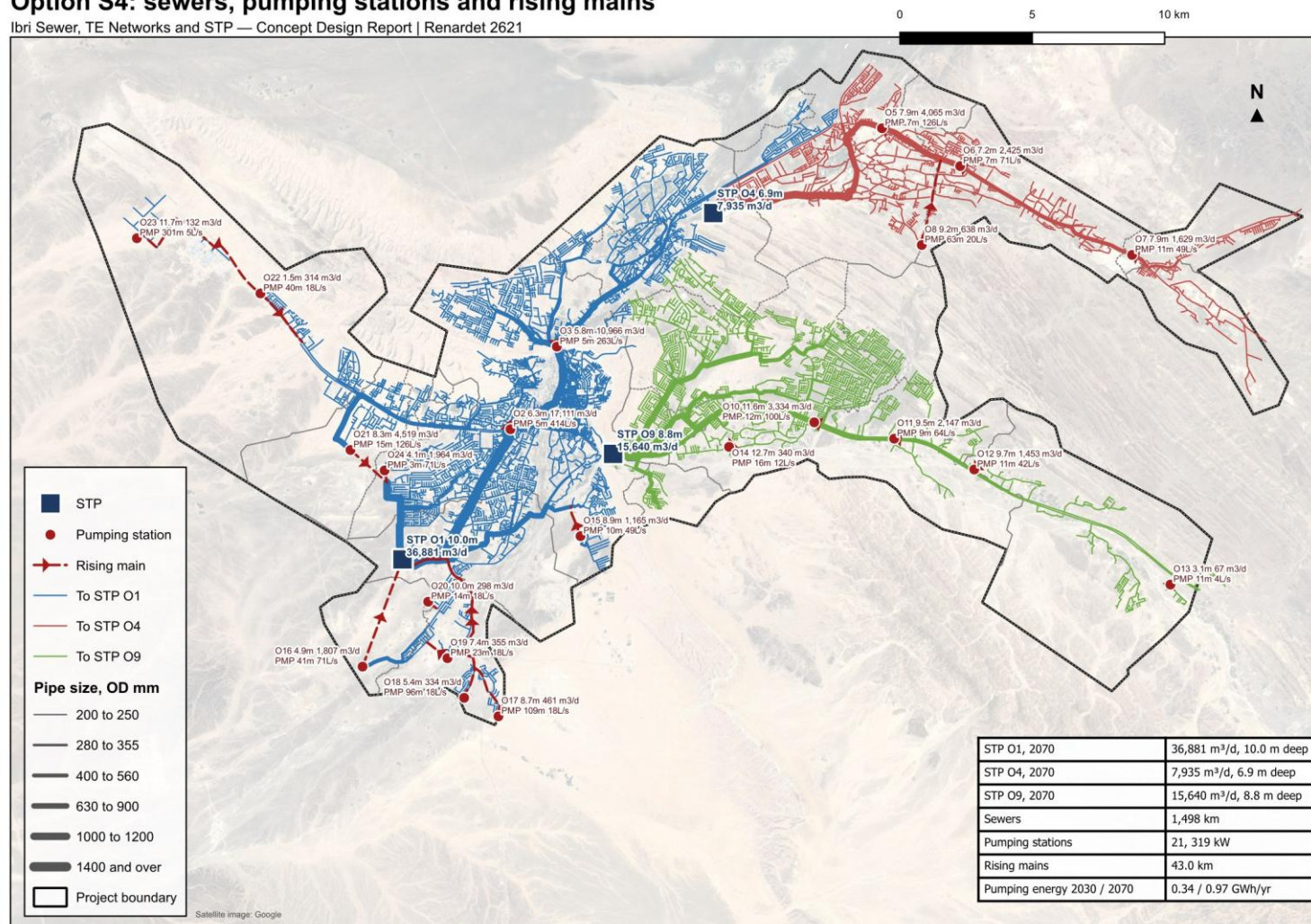


Figure 67. Option S4: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

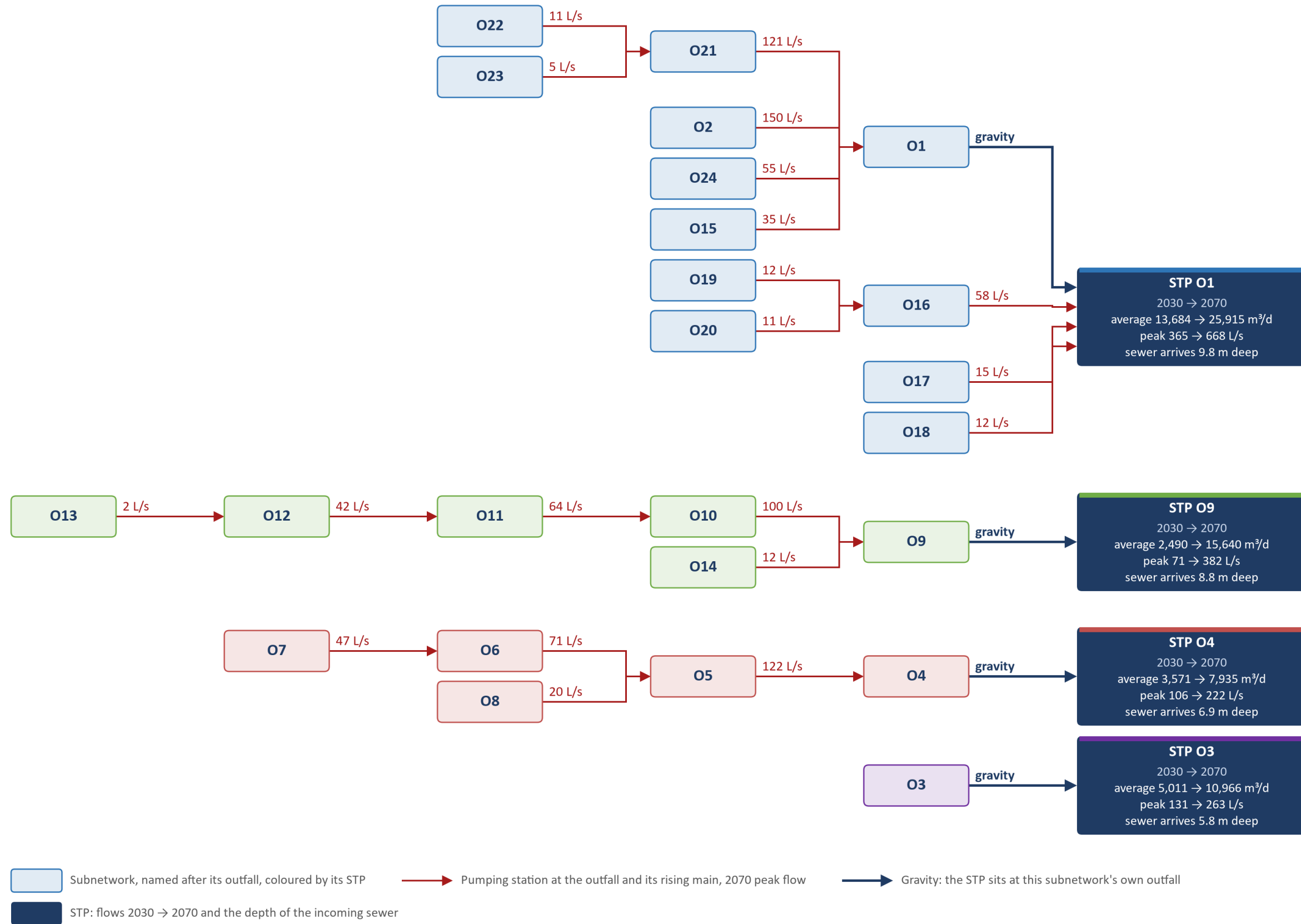


Figure 68. Option S5, four STPs, at O1, O3, O9 and O4: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S5: four STPs, at O1, O3, O9 and O4

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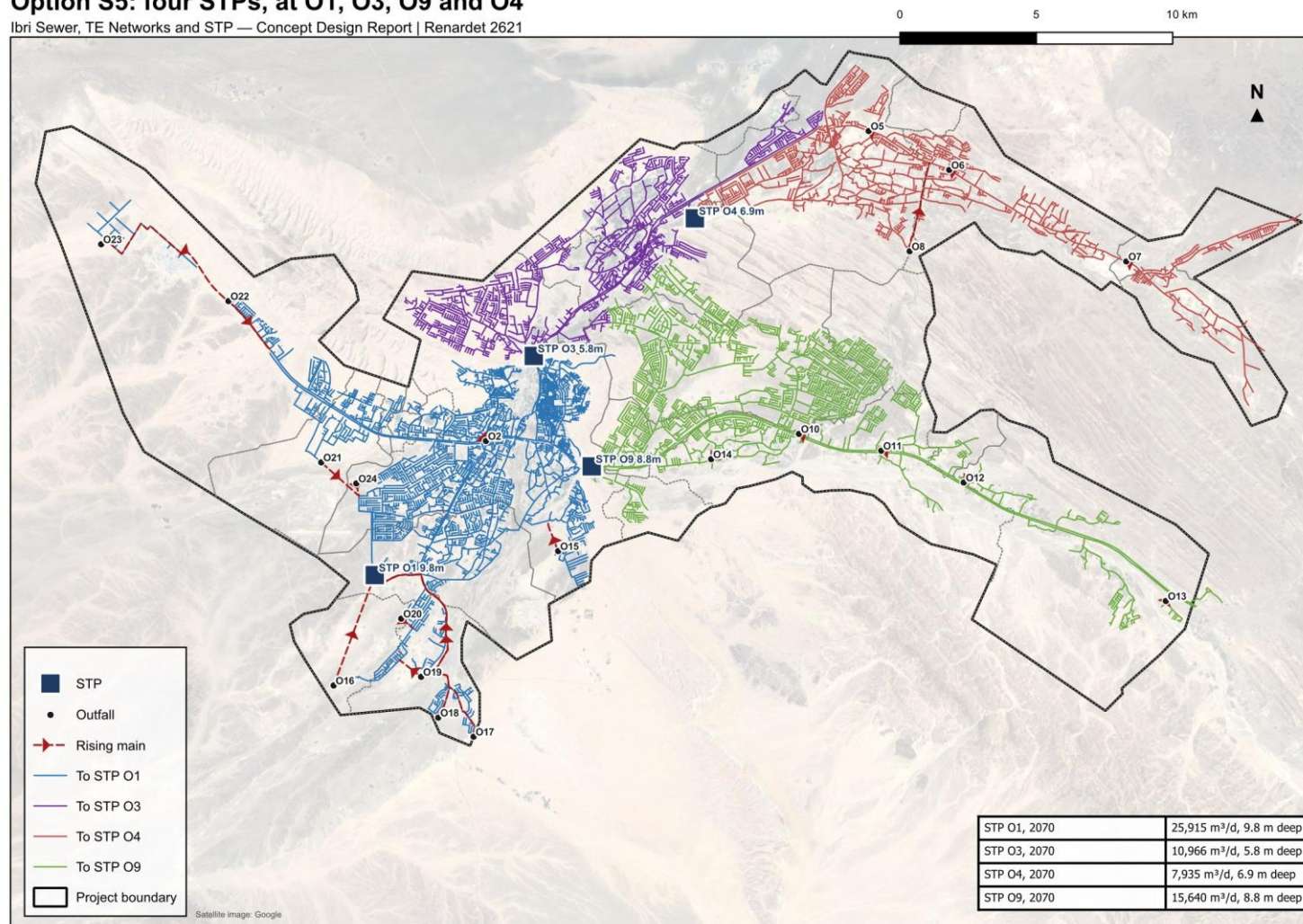


Figure 69. Option S5: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S5: sewers, pumping stations and rising mains

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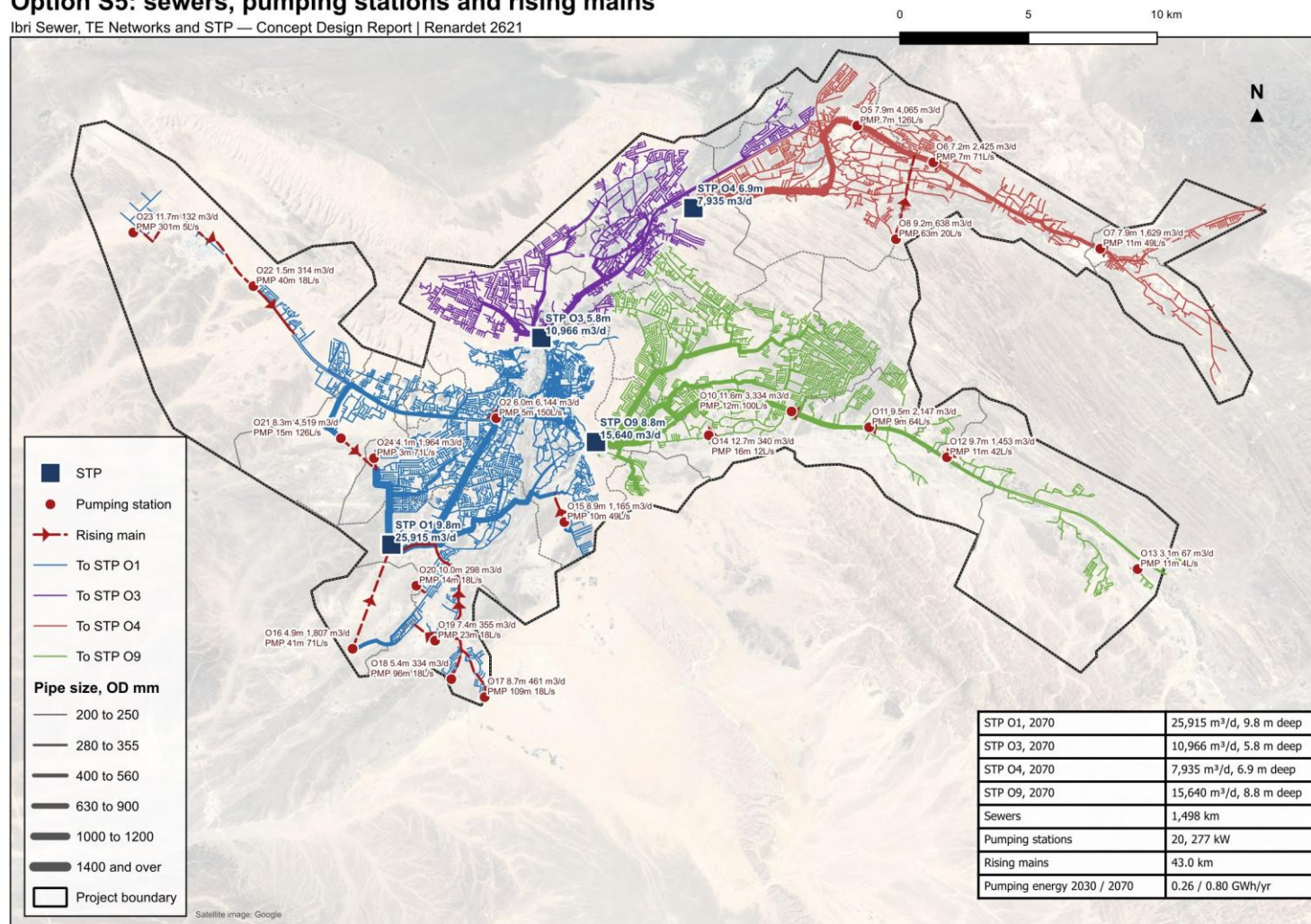


Figure 70. Option S5: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

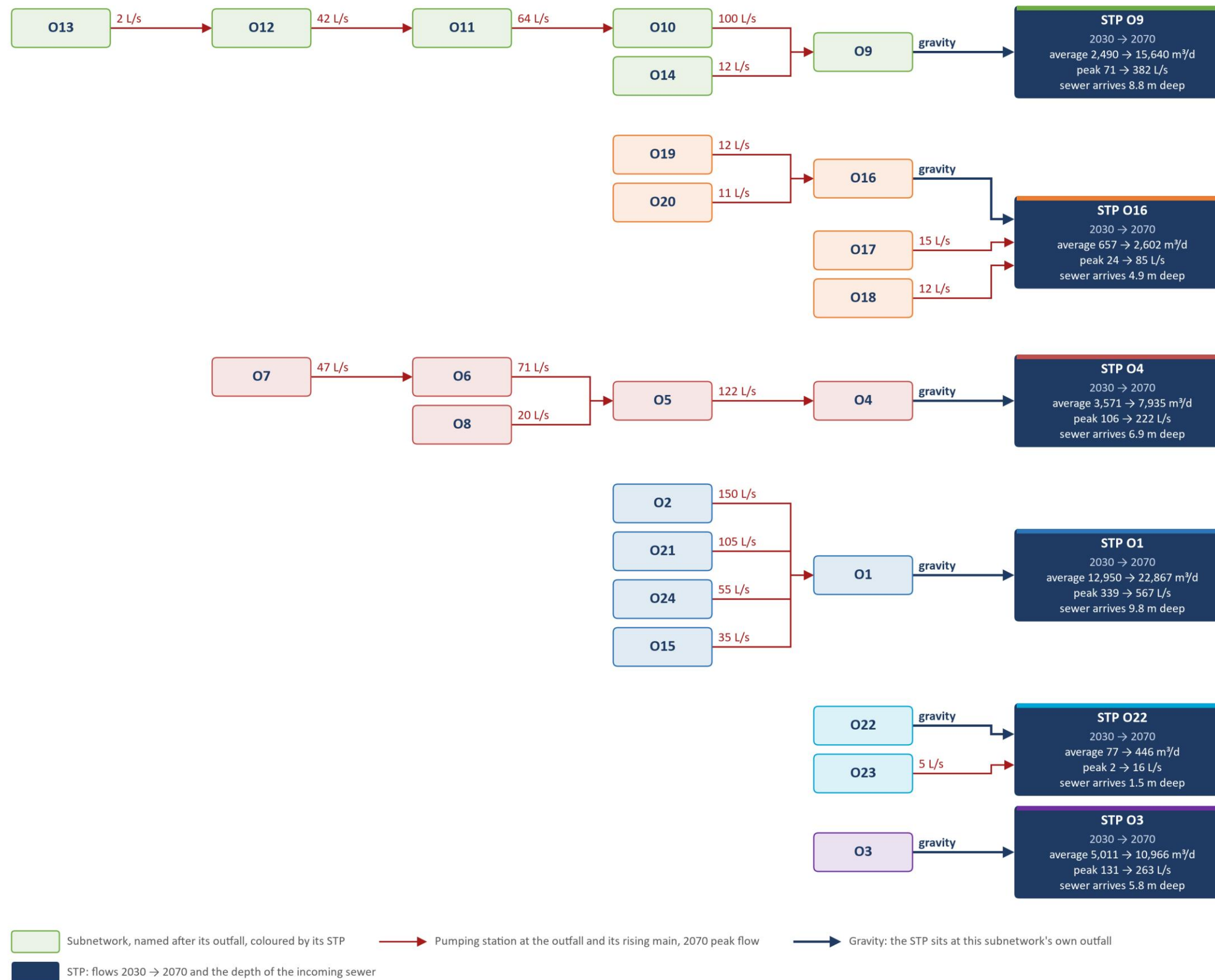


Figure 71. Option S6, six STPs, at O1, O3, O9, O4, O16 and O22: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S6: six STPs, at O1, O3, O9, O4, O16 and O22

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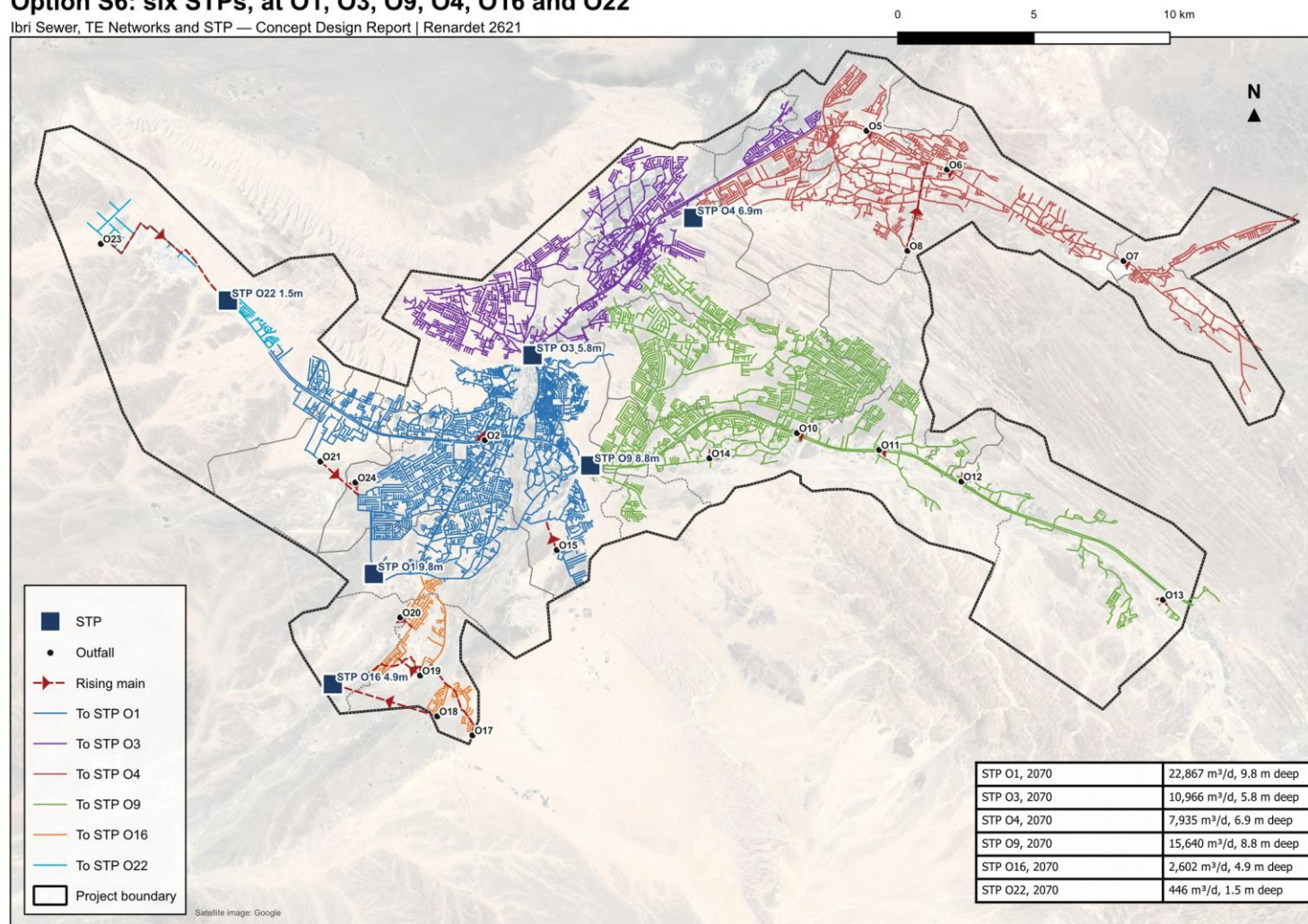


Figure 72. Option S6: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S6: sewers, pumping stations and rising mains

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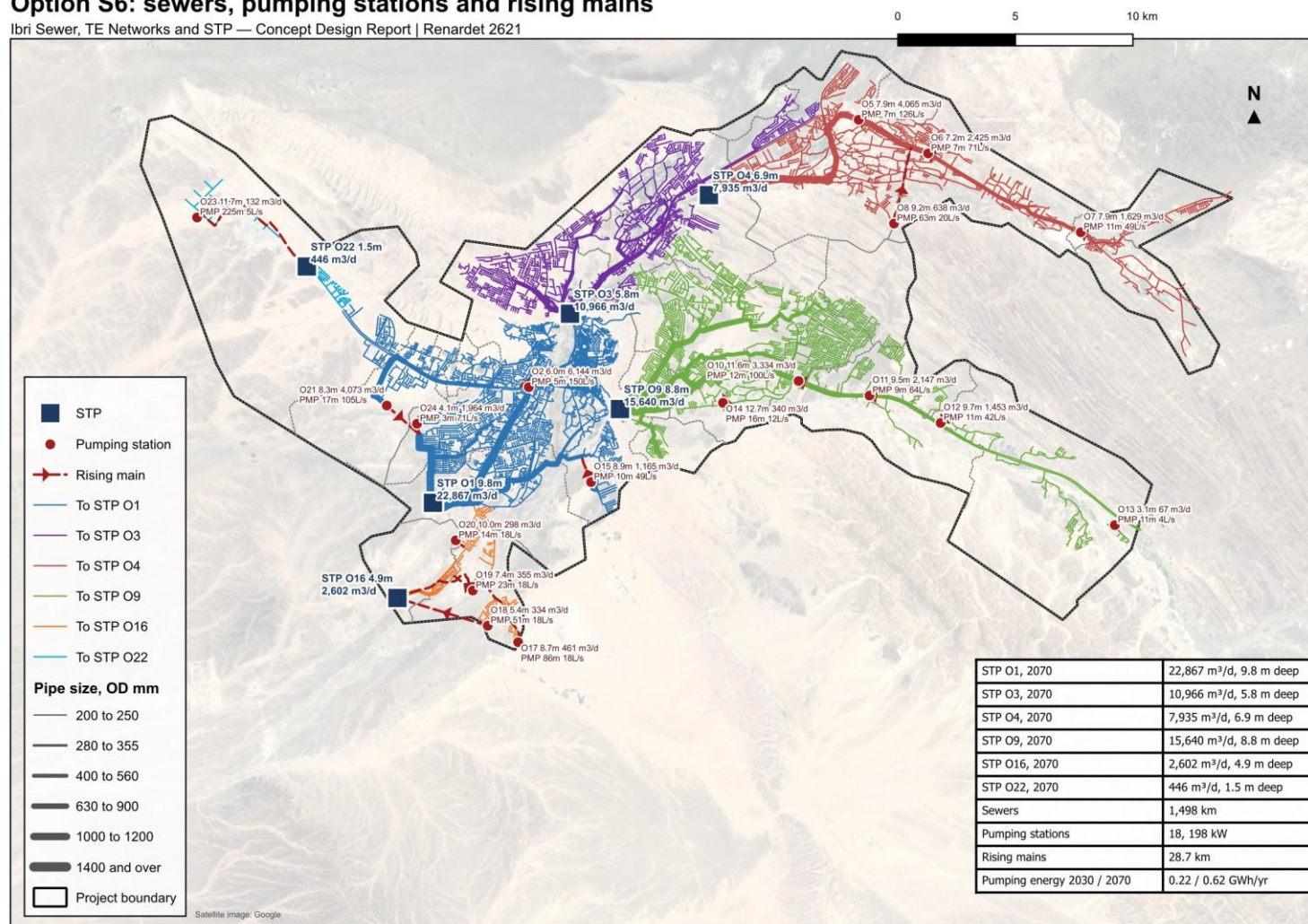


Figure 73. Option S6: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

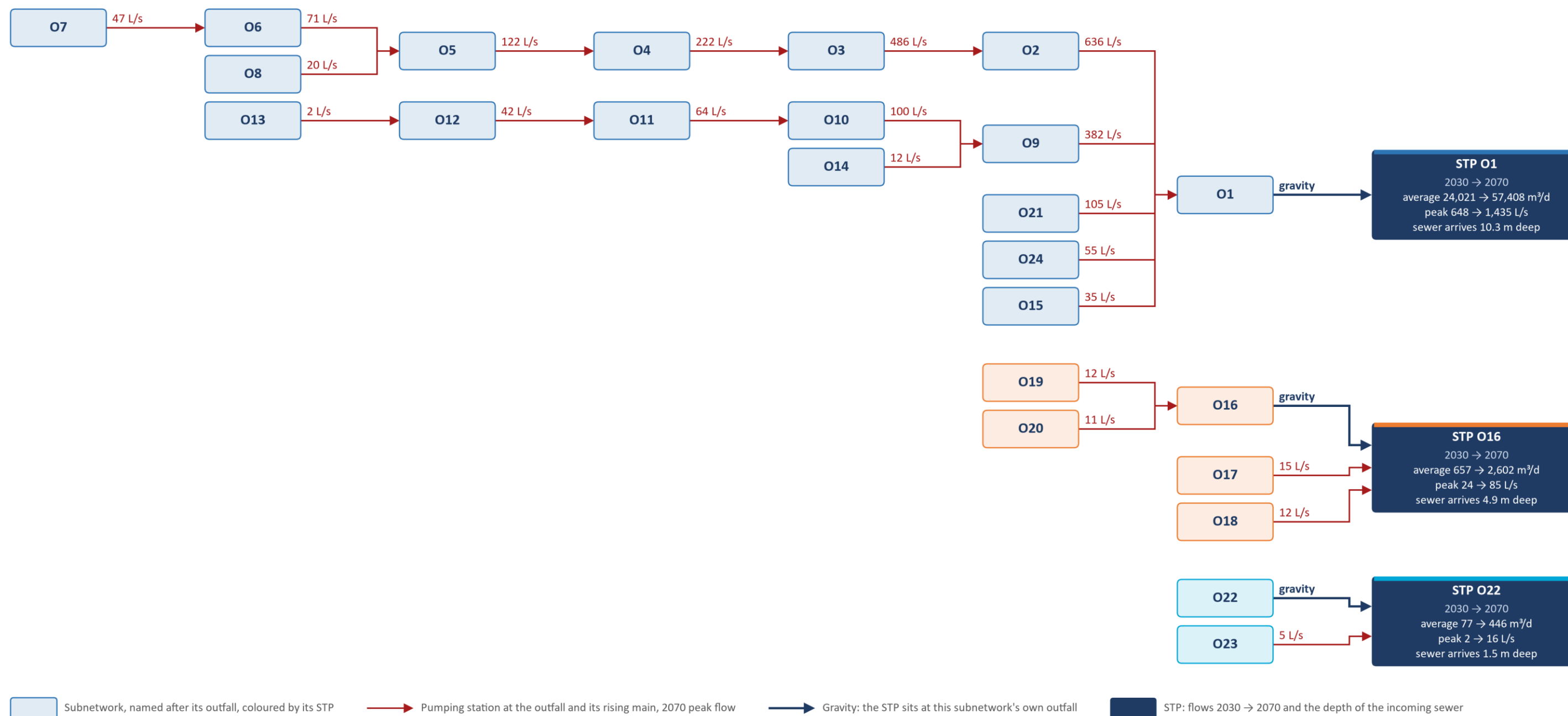


Figure 74. Option S7, three STPs, at O1, O16 and O22: where the flow of every subnetwork goes, with the peak flow of each transfer in 2070.

Option S7: three STPs, at O1, O16 and O22

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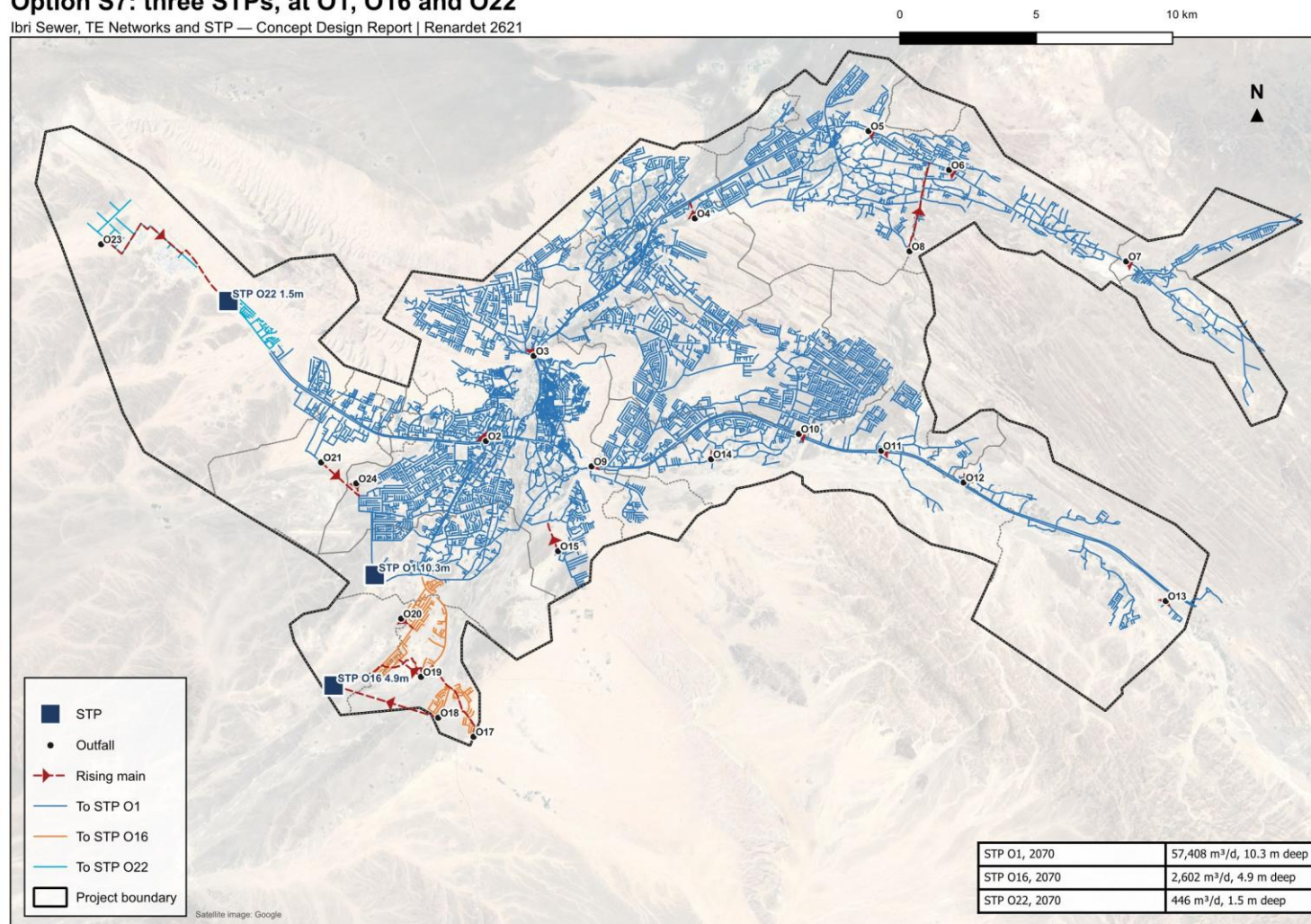


Figure 75. Option S7: the sewers by the plant they drain to, the outfalls, and the depth of the sewer arriving at each plant.

Option S7: sewers, pumping stations and rising mains

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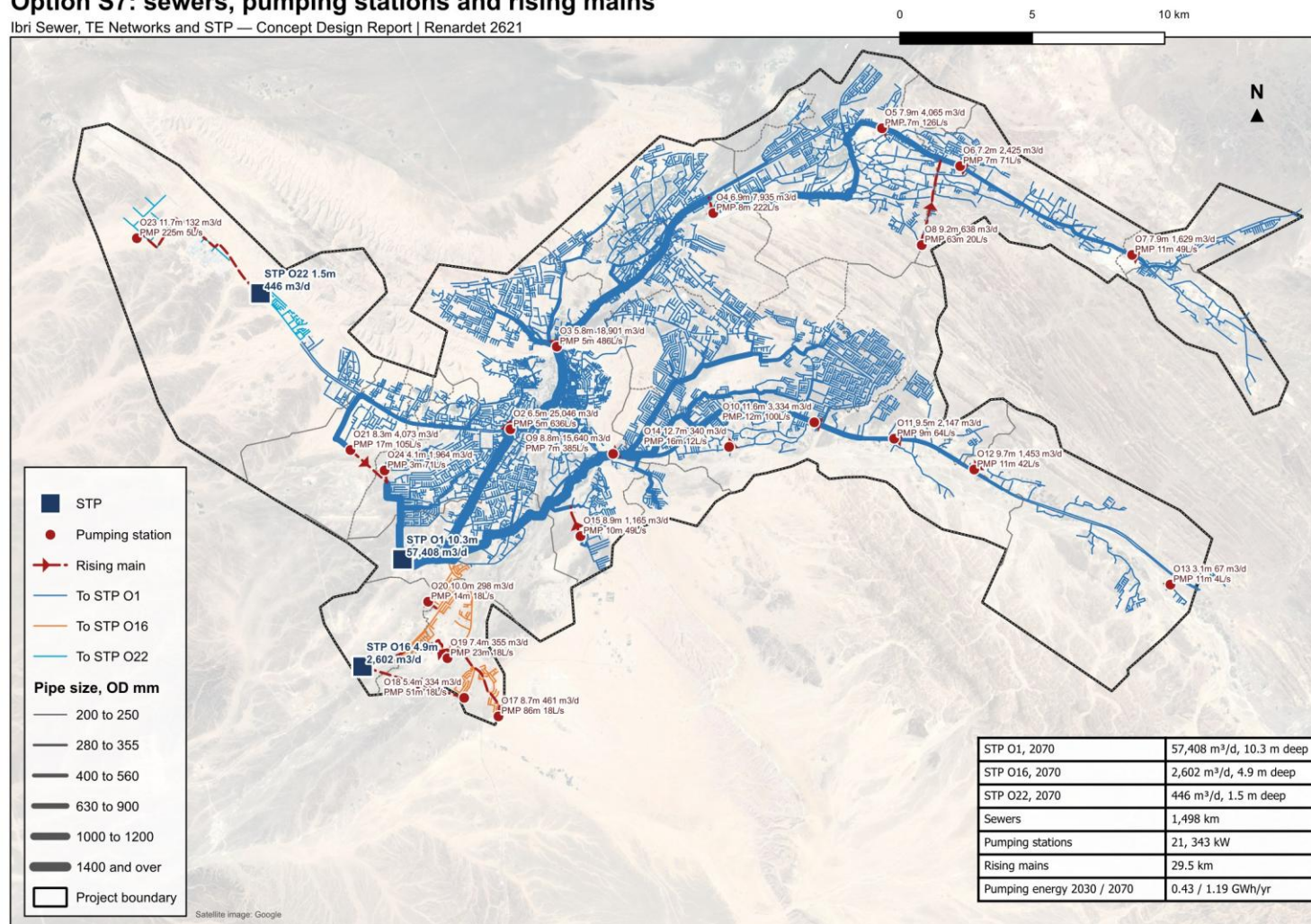


Figure 76. Option S7: the sewers by plant and size, the pumping stations and the rising mains. A station label gives the outfall, its depth and the average flow in 2070, then the pump head and the duty.

6.2.7. Comparison of the Options

Table 42. The options compared: network, depth and pumping

Option	Plants	Sewer at the plants, m deep	Manholes deeper than 12 m	Deepest manhole, m	Pumping stations	Pumps in duty, kW	Energy 2030, MWh a year	Energy 2070, MWh a year	Rising mains, km
S1	1	10.3	40	15.1	23	422	465	1,367	43.9
S2	1	5.9	40	15.1	23	884	1,311	3,337	37.3
S3	2	10.1 / 6.9	40	15.1	22	361	367	1,148	43.2
S4	3	10.0 / 6.9 / 8.8	40	15.1	21	319	339	974	43.0
S5	4	9.8 / 5.8 / 6.9 / 8.8	40	15.1	20	277	259	797	43.0
S6	6	9.8 / 5.8 / 6.9 / 8.8 / 4.9 / 1.5	40	15.0	18	198	220	616	28.7
S7	3	10.3 / 4.9 / 1.5	40	15.0	21	343	426	1,186	29.5

Every option has the same 1,498 km of sewer and 19,083 manholes; they differ in the sizes of the trunk sewers, in the pumping and in the plants. The depth of the sewer at a plant is the depth below ground of the gravity sewer arriving at the plant site, one figure for each plant in the order of the plants table; the plant's inlet works lift the flow from that depth. Manhole depths are measured from the cover to the invert.

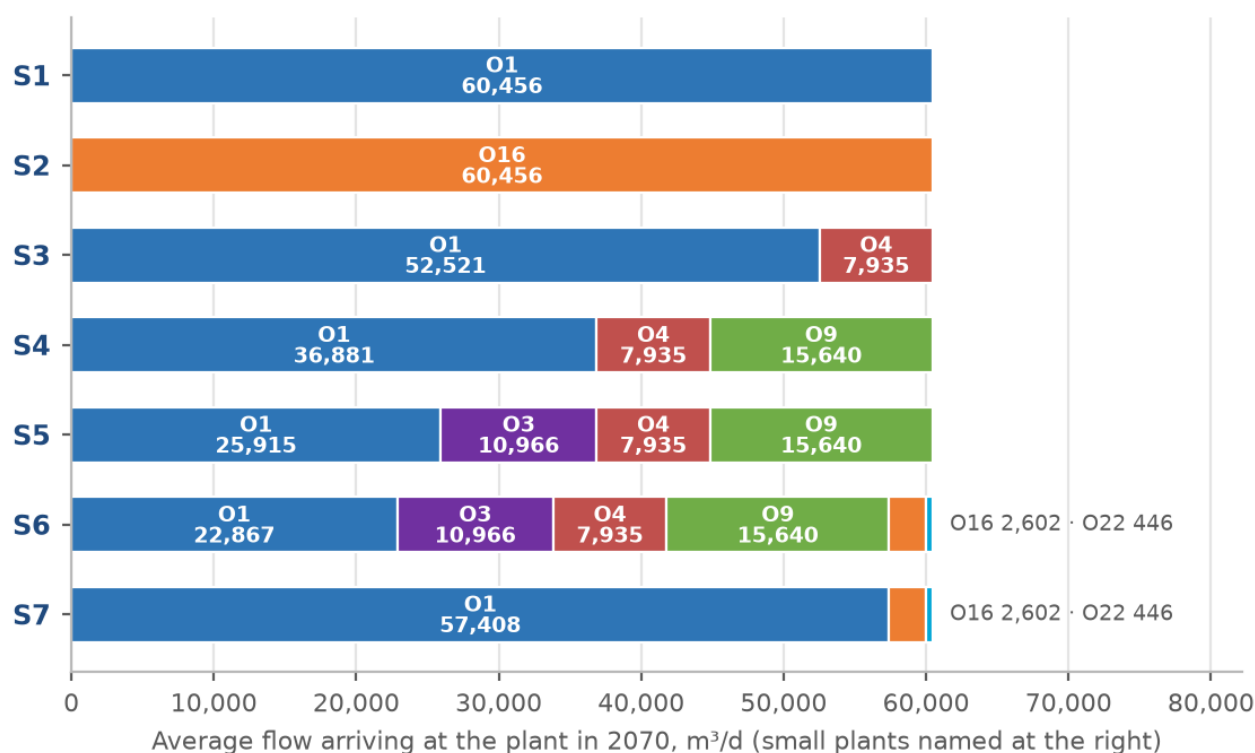


Figure 77. The average flow arriving at each plant in 2070, by option. Colours as on the maps and diagrams.

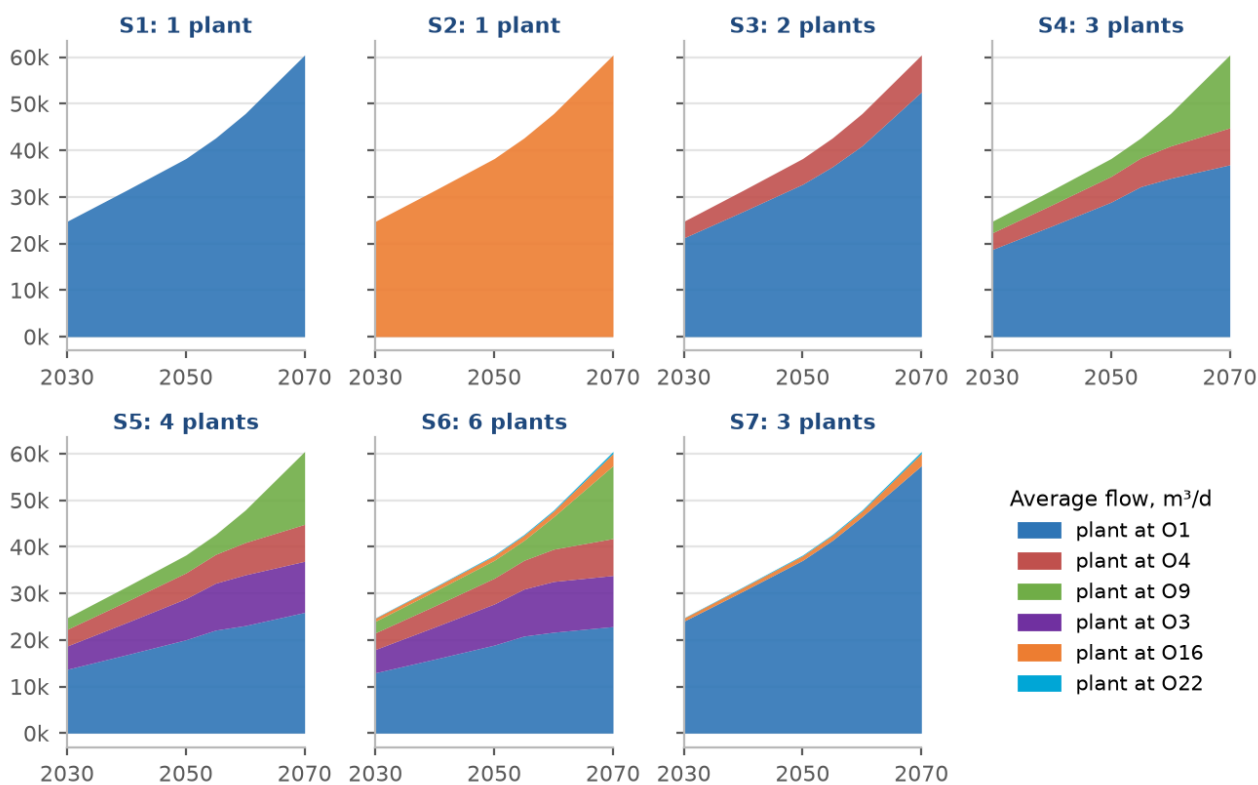


Figure 78. The average flow arriving at each plant, 2030 to 2070, by option.

Table 43. Average flow arriving at each plant, cubic metres a day, with the 2070 peak flow and the depth of the incoming sewer

Option	Plant	2030	2040	2050	2055	2060	2070	Peak 2070, l/s	Sewer at the plant, m deep
S1	O1	24,756	31,415	38,226	42,635	47,882	60,456	1,535	10.3
S2	O16	24,756	31,415	38,226	42,635	47,882	60,456	1,535	5.9
S3	O1	21,185	26,875	32,695	36,463	40,952	52,521	1,313	10.1
S3	O4	3,571	4,540	5,531	6,172	6,930	7,935	222	6.9
S4	O1	18,695	23,720	28,860	32,188	33,980	36,881	931	10.0
S4	O4	3,571	4,540	5,531	6,172	6,930	7,935	222	6.9
S4	O9	2,490	3,155	3,835	4,275	6,972	15,640	382	8.8
S5	O1	13,684	16,821	20,030	22,100	23,082	25,915	668	9.8
S5	O3	5,011	6,899	8,831	10,087	10,898	10,966	263	5.8
S5	O4	3,571	4,540	5,531	6,172	6,930	7,935	222	6.9
S5	O9	2,490	3,155	3,835	4,275	6,972	15,640	382	8.8
S6	O1	12,950	15,886	18,890	20,828	21,653	22,867	567	9.8
S6	O3	5,011	6,899	8,831	10,087	10,898	10,966	263	5.8
S6	O4	3,571	4,540	5,531	6,172	6,930	7,935	222	6.9
S6	O9	2,490	3,155	3,835	4,275	6,972	15,640	382	8.8
S6	O16	657	810	967	1,069	1,188	2,602	85	4.9
S6	O22	77	124	173	204	241	446	16	1.5
S7	O1	24,021	30,480	37,086	41,363	46,453	57,408	1,435	10.3
S7	O16	657	810	967	1,069	1,188	2,602	85	4.9
S7	O22	77	124	173	204	241	446	16	1.5

The flows are the average daily flow of the plots served plus the infiltration of the sewers that drain to the plant. In every option the plants together receive 60,456 cubic metres a day in 2070: the 59,377 the network carries and 1,079 of infiltration. The ten per cent margin for a new plant is not included; it is applied when the plant is sized (Section 6.5), and brings the plants together to 66,502. The peak is the sum of the peak flows of the outfalls that reach the plant, the flow its inlet works receive if every catchment peaks at the same time. It is higher than the peak hourly flow of Section 3.4.1, which applies one peak factor to the plant's whole average flow, because the peak factor falls as the flow grows.

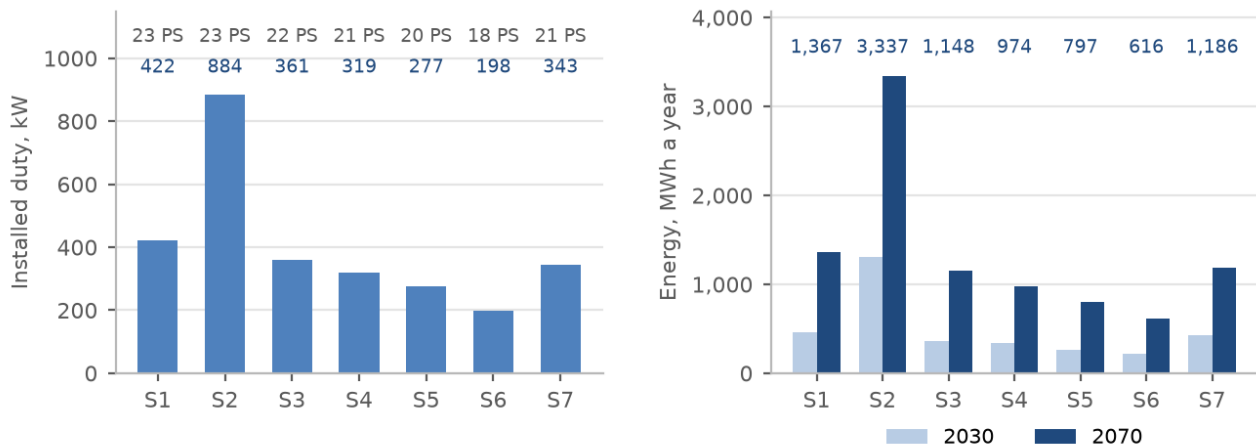


Figure 79. Pumping stations by option: the pumps in duty, and the energy they use in 2030 and 2070. The count of stations is above each bar.

The depth hardly changes between the options: in every option 40 manholes are deeper than 12 m, 3.2 km of sewer lies deeper than 12 m and the deepest manhole is at 15.0 to 15.1 m. The deep sewers are set by the ground within the subnetworks, which the options do not change; where the flow is treated changes the trunk sizes and the pumping, not the depth.

Pumping figures. The rising mains follow the roads and streets, and run across open ground where that route is more than 1.6 times the straight line, at 1.1 times its length. Each station pumps the 2070 peak flow of everything that reaches it, raised where needed to keep 1.0 m/s in a main of at least 75 mm; the main is the size that needs the least power within 2.5 m/s. Friction in the mains is taken with a Hazen-Williams coefficient of 120 and 10 per cent for fittings; the wet well is 1.5 m below the incoming sewer, the discharge 0.3 m above the invert of the receiving manhole, or 3 m above the ground at a plant; the pumps run at 65 per cent wire-to-water efficiency. These are concept values, to be replaced by the routes, the levels and the pumps of the preliminary design.

The stations are not the only lift. Each plant's inlet works lift the arriving flow from the depth of its sewer to the inlet channels. On the same concept values, the wet well 1.5 m below the arriving sewer and the inlet works 3 m above the ground, the energy of the stations and of the plant inlets together is as follows.

Table 44. Pumping energy of the stations and of the plant inlets, MWh a year

Option	Stations 2030	Plant inlets 2030	Together 2030	Stations 2070	Plant inlets 2070	Together 2070	Together 2070 against S1
S1	465	561	1,027	1,367	1,371	2,738	
S2	1,311	395	1,706	3,337	966	4,303	+57 %
S3	367	537	904	1,148	1,315	2,463	-10 %
S4	339	527	867	974	1,274	2,248	-18 %
S5	259	492	751	797	1,197	1,994	-27 %
S6	220	486	705	616	1,172	1,788	-35 %
S7	426	555	981	1,186	1,343	2,529	-8 %

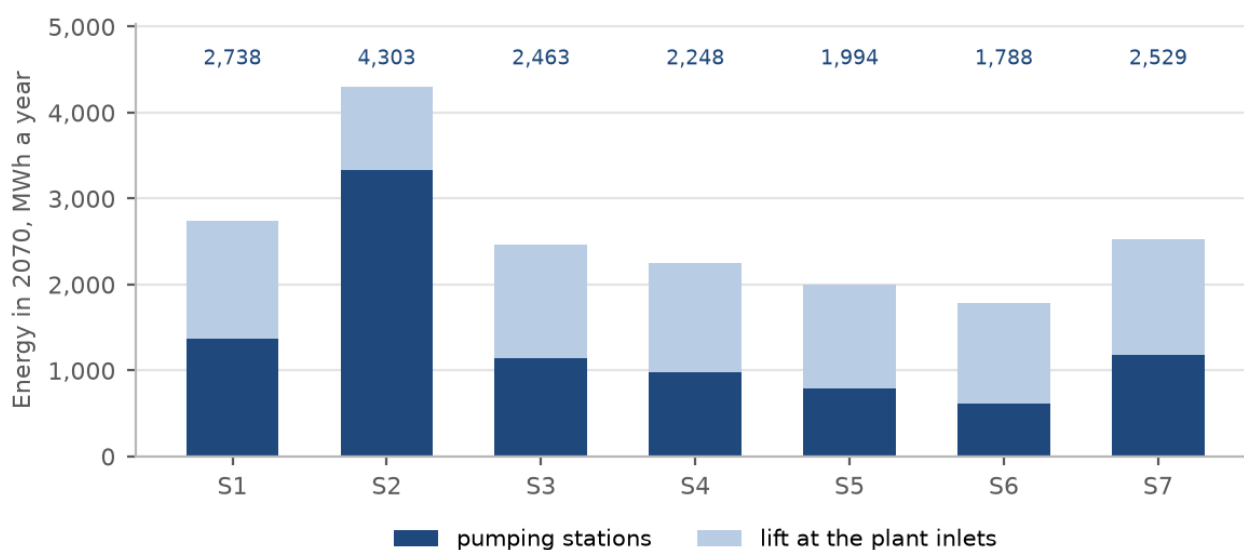


Figure 80. Pumping energy in 2070 by option: the network's pumping stations and the lift at the plant inlets.

The lift at the plants narrows the differences between the options but does not change their order. In option S1 the whole flow arrives 10.3 m below ground, and lifting it at the plant uses 1,371 MWh a year in 2070, against 1,367 for the 23 stations of the network.

Two features of the pumping need attention whichever option is chosen. First, a long rising main carrying a small flow holds the sewage for hours. In 2030, between 10 and 14 stations, depending on the option, hold it longer than the 30 minutes the guideline aims for, the longest 73 hours in the main from O18. Sulphide forms in that time, and these mains need the hydrogen sulphide evaluation the guideline asks of the designer, with dosing or a shorter route.¹

Second, the station at O23 lifts a few litres a second through a long main against 220 to 300 m of head in every option, depending on where it discharges. A connection of that kind is to be reconsidered at the preliminary design: a shorter route to a nearer subnetwork, or local treatment.

6.2.8. Self-Cleansing Assessment

This section sets out the year from which each pipe reaches the self-cleansing velocity of 0.75 m/s at its peak flow, with the head pipes allowed as steep as 4 per cent, the steepest gradient indicated by Nama Water Services. For every pipe

¹ PAM-GUD-203, page 50 (retention) and Section 11.1, page 162 (the evaluation).

and every year, the gradient at which that year's peak flow reaches 0.75 m/s in the pipe's own size is calculated, and the pipe counts as self-cleansing from the first year in which that gradient is 4 per cent or less.¹

The flows are those of the model, with every property connected and the infiltration included, which is the most favourable reading of the early years. With 61 per cent of the properties connected in 2030 (Section 4.2.8) and without infiltration, fewer pipes reach the velocity than the table shows.

Table 45. Pipes that reach 0.75 m/s at their peak flow at a gradient of 4 per cent or less

Year	Pipes	Share of the pipes	Length, km	Share of the length	Head pipes
2030	5,562	29 %	438	29 %	21 of 3,612
2040	6,304	33 %	500	33 %	23 of 3,612
2050	6,993	37 %	555	37 %	26 of 3,612
2055	7,347	38 %	584	39 %	29 of 3,612
2060	7,746	41 %	616	41 %	34 of 3,612
2070	8,822	46 %	708	47 %	38 of 3,612

In no year does every pipe reach it. The share grows from 29 per cent of the pipes in 2030 to 46 per cent in 2070 and then stops, because at saturation the number of houses on each street is fixed by the plots. The other 10,261 pipes, 790 km of 200 mm sewer at the heads of the network, never carry enough: a 200 mm sewer needs a peak flow of 0.93 l/s, the flow of about 32 houses, to reach 0.75 m/s at 4 per cent, and about 4.5 l/s at 1 per cent. Whether a pipe can clean itself by velocity depends on how many houses drain into it, not on the year.

The guideline provides for this case. At the head of the system, where the self-cleansing velocity may not be attainable, the minimum gradient is set by the minimum tractive force. At a tractive stress of 1 pascal the head pipes need a median gradient of 2.5 per cent in 2030 and 1.3 per cent in 2070, and more than 4 per cent only below 0.0135 l/s, pipes carrying little more than their own infiltration. The guideline also asks for more frequent inspection and cleansing in the early years of development.²

Decision required. The year from which the network is to be self-cleansing (2030, the opening year, or a later year); the tractive stress for the head pipes (1 pascal, or another value); and whether 4 per cent is the steepest gradient for every pipe or for the head pipes only. A steeper network is deeper and needs more pumping, and the cost of each choice will be given once it is made. This extends Decision 6 (Section 3.3.2).

6.3. Pumping and Lifting Stations

A lifting station is provided where the cost of excavation to maintain gravity flow becomes prohibitive. Each station lifts the flow to a level from which gravity conveyance resumes, discharging through a force main to a receiving manhole or to the treatment plant. The pumping stations of the seven options, with their duty, rising main, head and power, are listed in Appendix B.

¹ PAM-GUD-203, page 26 (0.75 m/s at peak flow); Colebrook-White with a roughness of 1.5 mm. Calculated on option S1; the head pipes, which decide the answer, carry the same flow in every option.

² PAM-GUD-203, page 27 (tractive force at the head of the system) and Section 4.2.6 (early development).

Table 46. Force main design criteria

Criterion	Value
Minimum velocity, raw sewage	0.75 m/s at minimum flow
Minimum velocity, intermittent flow	1.0 m/s
Minimum velocity, vertical mains	1.2 m/s
Maximum velocity	2.5 m/s
Minimum internal diameter	75 mm
Gradient, rising	1 in 500
Gradient, falling	1 in 300
Retention time	30 minutes or less, ideally

Retention in a force main allows sulphide to form, which is both a corrosion and an odour risk in this climate. Alignments are therefore kept short, discharges are submerged to limit turbulence, and the need for dosing is assessed for each station.¹

Surge analysis is carried out for each force main in approved software, and the protection required is established from it.

6.4. Treated Effluent Network Options

The treated effluent network conveys the product of the treatment plant to the customers identified in Section 4.4. The network is sized for the summer peak demand, with storage of not less than twenty-four hours.

Where demand is below the volume produced, provision is made for the disposal of the excess, described in Section 6.6.

The treated effluent network options are developed on the three sewer network options recommended in Section 7.11, which give one source of treated effluent at O1, three, or six.

6.5. Treatment Plant Options

6.5.1. Capacity and Phasing

Each plant is sized on the flow arriving at it in the option chosen, given for every model year in Section 6.2.7, with the ten per cent design margin applied, and is built in phases so that capacity follows demand. The capacity of the first phase, and the years at which subsequent phases are required, follow from that series and are presented with the plant options.

6.5.2. Process Selection

Process selection is governed by the effluent standard. The total nitrogen limit described in Section 3.4 requires full nitrification and denitrification, which narrows the technologies that can be considered. Land area, energy consumption, operating complexity and whole life cost distinguish the remaining options.

¹ PAM-GUD-203, Sections 8.2.1 and 11.5.3, pages 50 and 181.

Table 47. Indicative land requirement by process

Process	Area, m ² per m ³ /d
Membrane bioreactor	0.45 to 0.9
Sequencing batch reactor	0.9 to 1.8
Moving bed biofilm reactor	0.9 to 1.8
Integrated fixed film activated sludge	1.2 to 2.5
Conventional activated sludge and extended aeration	1.8 to 3.6

The areas are indicative and are used for planning. The land required includes the process units, the sludge facilities, the buffer zone and provision for the energy-efficiency measures described in Section 7.7.¹

6.5.3. Site Selection

The site is assessed against the criteria in the wastewater guideline, covering access, physical characteristics, environmental and climatic impact, social impact and cost. Two criteria carry stated requirements: the site is assessed against the twenty-five and one hundred year flood levels, and the plant is to remain operational during floods.

The buffer distance to residential areas for a plant of this size is established from odour dispersion modelling rather than from a fixed figure, and lies between 300 and 1,000 metres measured to the five odour unit contour. The modelling is described in Section 7.3 and precedes the confirmation of the site.²

6.6. Excess Effluent, Emergency Provisions and Tanker Reception

6.6.1. Tanker Reception

A proportion of the wastewater arising in the area is collected by tanker and delivered to the treatment plant. Sewage delivered this way is stronger than sewage arriving through the network, and its arrival is concentrated in the working day. A dedicated reception facility is provided, with screening, grease removal, sampling before acceptance and flow equalisation. The volumes delivered from outside the study area, and their sources, are requested from Nama Water Services (Section 1.5.5), so that the reception facility and the plant are designed for them.

6.6.2. Excess Treated Effluent

Where the effluent produced exceeds the demand, provision is made for its disposal. Discharge to a wadi requires the effluent to meet Class A of Ministerial Decision 145/1993, and where the wadi discharges to the sea the limits for ammonia, nitrogen and phosphorus of Ministerial Decision 159/2005 apply in addition. Any such discharge is subject to the approval of the Environment Authority and the Authority for Public Services Regulation.³

6.6.3. Emergency Provisions

Provision is made for the diversion of raw sewage in the event that the plant is out of operation, and for emergency storage. Every pumping station is provided with an emergency overflow to prevent flooding of the station or of connected properties; overflows are subject to the approval of the Environment Authority.

6.7. Sludge Management Strategy

The Oman Sludge Management Plan, as reproduced in the wastewater design guideline, identifies a sludge treatment centre performing composting at Ibri as the solution for the Governorate of Adh Dhahirah. The plant is therefore to be

¹ PAM-GUD-203, Table 28, page 64.

² PAM-GUD-201, Table 8, pages 43 and 44.

³ PAM-GUD-203, Section 10.2.4.3, pages 72 and 73. The substituted limits are materially tighter than Class A for phosphorus and ammoniacal nitrogen.

designed to receive and process sludge arising beyond its own catchment, and the land area, buffer distance and vehicle access are established accordingly.¹

Sludge is to be reused unless no form of reuse is possible, and its quality is to comply with the heavy metal limits of Ministerial Decision 145/1993. Where disposal to landfill is required, the receiving authority sets acceptance criteria including a minimum solids content of eighty per cent. That content is above what mechanical dewatering alone achieves, and drying or composting is therefore required.

Dewatering facilities are provided on site. The strategy sets out the quantities of sludge arising and the disposal route for each.

¹ PAM-GUD-203, Table 67, page 136.

7. Technical and Financial Appraisal

This chapter carries the technical and financial appraisal: ground conditions, flood, odour, climate, environment, utility interfaces, sustainability, cost, risk, value engineering and the comparison of the options.

7.1. Ground Conditions

Geophysical investigation is required for all infrastructure and geotechnical investigation for all above-ground assets. At this stage the purpose is to establish a site-wide ground model and a range of parameters sufficient to select foundation types and platform levels, and to identify the risks that the detailed investigation must resolve.

Boreholes at the treatment plant and lifting station sites are taken to a minimum depth of fifteen metres or to a competent stratum, and are closely spaced at wet wells, valve chambers and electrical rooms. Groundwater monitoring is required, as the depth to groundwater governs both the excavation method and the infiltration allowance.

7.2. Flood Risk and Protection

The study area is crossed by a wadi system, and flood behaviour governs both the alignment of the works and the siting of the treatment plant. The site is assessed against the twenty-five and one hundred year flood levels, and the plant is to remain operational during floods. The flood hazard across the study area for the 10, 25, 50 and 100-year floods is mapped in Section 1.2.7.

Pumping station floor levels, transformers and standby generators are set not less than 300 millimetres above the one in fifty year flood level. Wadi crossings are designed with a minimum cover of 1.5 metres to the crown of the pipe.

Sewers and Manholes

The guideline requires sewers and their manholes to be kept out of wadis and of ground subject to washout in heavy storms. The hazard grids give that ground a measure: the high classes, H4 to H6, where the flood is unsafe for people and vehicles and damages buildings, are taken as the ground to be avoided.¹

The sewers and the manholes lie in the same streets in every option, so their exposure does not depend on the option chosen. Each of the 1,498 km of sewer has been sampled every metre along its length, and each of the 19,083 manholes at its centre, on the grids of the 10, 25, 50 and 100-year floods. The tables below give the length of sewer and the number of manholes in each class.

¹ PAM-GUD-203, Section 4.4.1, page 30, and page 33. The guideline sets no hazard class for washout; taking H4 to H6 is this design's measure, to be confirmed by a scour check at the preliminary design.

Table 48. Length of sewer in each flood hazard class, km

Hazard class	10-year	25-year	50-year	100-year
Not flooded	1,014.6	878.8	766.3	657.6
H1	376.7	435.4	460.6	462.7
H2	52.9	81.8	105.1	124.1
H3	30.1	55.1	80.5	103.0
H4	10.1	16.6	33.2	56.5
H5	13.4	27.2	46.1	81.0
H6	0.5	3.4	6.5	13.3
H4 to H6	23.9	47.2	85.8	150.8
H4 to H6, per cent of the length	1.6	3.1	5.7	10.1

Table 49. Manholes in each flood hazard class

Hazard class	10-year	25-year	50-year	100-year
Not flooded	13,054	11,339	9,982	8,626
H1	4,702	5,446	5,705	5,735
H2	652	1,034	1,328	1,577
H3	370	684	1,031	1,306
H4	135	204	405	700
H5	166	337	551	976
H6	4	39	81	163
H4 to H6	305	580	1,037	1,839
H4 to H6, per cent of the manholes	1.6	3.0	5.4	9.6

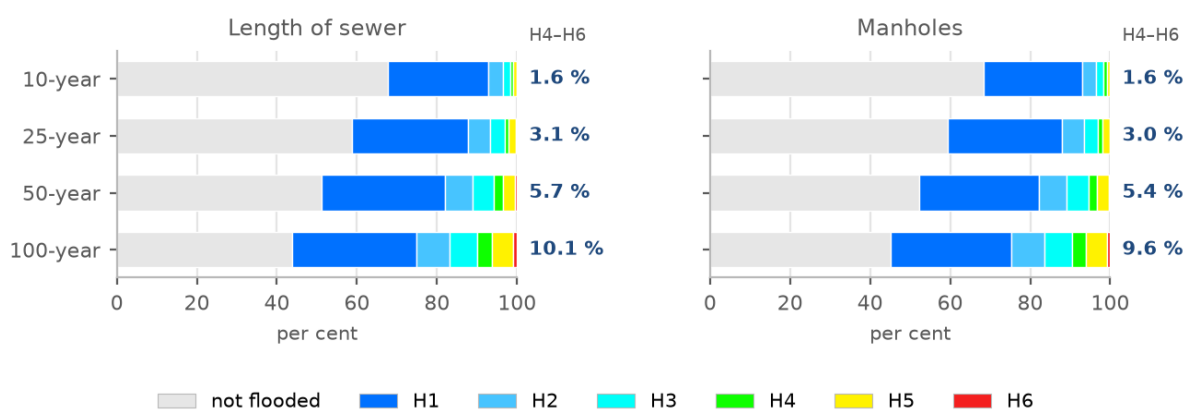


Figure 81. Share of the sewer length and of the manholes in each flood hazard class. The figure at the right of each bar is the share in the high classes, H4 to H6.

In the 100-year flood 150.8 km of sewer, 10.1 per cent of the length, and 1,839 manholes, 9.6 per cent, lie in H4 to H6. In the 25-year flood the shares are 3.1 per cent of the length and 3.0 per cent of the manholes; H6, the highest class, holds 13.3 km of sewer in the 100-year flood.

At the preliminary design each sewer and manhole in these classes is reviewed on the levels of the survey. The sewer is moved out of the channel where the streets allow. Where it has to cross, the crossing is designed as a wadi crossing, with 1.5 metres of cover to the crown, protection against scour and no manhole in the wadi bed or its embankments. Where

no other route exists, the length is recorded as a justified exception, and the manholes that remain in the high classes are sealed against the entry of flood water.¹

The treatment plants, the pumping stations and the rising mains are assessed on the same grids once their sites are fixed.

7.3. Odour Assessment

Odour governs the buffer distance between the treatment plant and residential development, and for a plant of this size the buffer is established from dispersion modelling rather than from a fixed distance. The modelling establishes the distance to the five odour unit contour, using site meteorological records and the treatment processes proposed.²

The prevailing winds are described in Section 1.2.5 from a regional record, which serves the comparison of sites; the modelling itself uses the record of the nearest meteorological station.

Odour control is provided at the inlet works, the sludge facilities and the pumping stations, with the treatment train selected from the assessment. Compliance is verified by continuous monitoring at the site boundary.

7.4. Climate Resilience

The works are assessed for resilience to the climate conditions anticipated over their service life. The assessment considers a fifty-year horizon at two and at four degrees Celsius of warming, and addresses the effect of changes in rainfall frequency and intensity on site selection and on the design of the infrastructure.³

7.5. Environmental and Social Assessment

An environmental impact assessment is required for the treatment plant and its associated works. The Environment Authority is informed of the project, its objectives and its anticipated impact, and is consulted on the location of the facilities.

Each option is assessed against resource use, ecosystem disruption, pollution risk, climate resilience, socio-economic impact and land use, and the assessment forms part of the comparison described in Section 7.11.

The scope required at this stage is the subject of Section 1.5.4.

7.6. Utility Interfaces and Approvals

7.6.1. Methodology

The proposed alignments are superimposed on the service records of each authority holding assets in the area. Where a conflict cannot be avoided by re-routing within the corridor, the cost of relocating the proposed works is compared with the cost of relocating the existing service, and the agreement of the owning authority is obtained for whichever is adopted.

7.6.2. Utility Records

The potable water network is available from the dataset described in Section 2.2.2 and provides 647.8 kilometres of mains within the study area. Records for electricity distribution, telecommunications and, where present, gas and fuel pipelines are being requested from the respective owners.

¹ PAM-GUD-201, Section 9.3, pages 85 and 86; PAM-GUD-203, Section 8.2.4, page 52.

² PAM-GUD-201, Table 8, page 43; PAM-GUD-203, Table 90, page 170, which sets five odour units per cubic metre at the site boundary where the surrounding area is sensitive.

³ PAM-GUD-201, Section 4.3, page 33.

The electricity data held records the position of consumer connections. It does not record the routes of the distribution cables, which are required for clash assessment and are requested separately.¹

7.6.3. Clearances

Table 50. Separation from other services

Situation	Requirement
Force main to water main, horizontal	3.0 m
Force main crossing a water main	the force main passes beneath, with 450 mm vertical clearance
Shallow sewer beneath a major road	3.0 m horizontal clearance
Another service in the same trench	placed on a separate bench on undisturbed ground

Beyond these, the clearance applied is that specified by the authority owning the service.

7.6.4. Trial Pits

Fifty trial pits are provided for. Their locations are selected at road intersections, along the routes of major existing services and along the expected routes of the trunk sewers and force mains. The programme is agreed with Nama Water Services and the municipal excavation approvals obtained before work begins.

7.6.5. Approvals Register

Approvals and no objection certificates are required from the authorities listed below. A register is maintained recording the authority, the consent required, the date of application and the current position, and is reported to Nama Water Services.

Table 51. Authorities from which approvals are required

Authority
Ministry of Housing and Urban Planning
Environment Authority
Ministry of Agriculture, Fisheries and Water Resources
Directorate General of Roads
Regional electricity distribution company
Oman Telecommunications Company
Royal Oman Police
Ministry of Heritage and Tourism
Municipality of Ibri

7.7. Sustainability

The carbon footprint of each option is evaluated for both construction and operation in accordance with recognised greenhouse gas accounting standards, and is expressed in tonnes of carbon dioxide equivalent per year and per cubic metre of effluent produced.²

Because the greater part of operational emissions arises from electricity consumption, the measures that reduce carbon are largely the same as those that reduce operating cost: conveying the flow by gravity wherever possible, minimising

¹ The account dataset described in Section 2.2.3 comprises point locations of metered connections.

² PAM-GUD-201, Section 12.5.1, pages 98 to 100, which cites ISO 14064 and the Greenhouse Gas Protocol and gives a benchmark of 1.17×10^{-3} tonnes of carbon dioxide equivalent per cubic metre of treated effluent.

lift, selecting efficient equipment, and generating renewable energy on site. Provision for photovoltaic generation within the plant perimeter is included in the land requirement.

Resource efficiency, in-country value and the use of nature-based solutions are assessed for each option and carried into the comparison in Section 7.11.

7.8. Cost Estimate

7.8.1. Basis of the Estimate

Item	Basis
Accuracy	plus or minus twenty per cent at concept stage
Measurement	CESMM3
Rates	recent tender and contract rates, and rates from Nama Water Services pre-investment appraisals, escalated to the base date of the estimate
Price basis	current prices at the date of the estimate
Presentation	by system element

7.8.2. Scope of the Estimate

The estimate covers the works, the associated costs and the provisions, as set out below.

Collection network — excavation by depth and by ground condition, trench support, bedding and backfill, pipework, manholes, property connections, road reinstatement, traffic management, crossings, and testing.

Lifting stations — land, civil structures, pumps and station pipework, electrical supply and standby generation, control and telemetry, odour control, and the force main with its chambers and valves.

Treatment plant — land, site preparation and flood protection, inlet works, tanker reception, biological treatment, clarification, tertiary treatment, disinfection, the sludge line, chemical systems, odour control, electrical and control installations, buildings, and commissioning.

Treated effluent network — pipework, storage, boosting, filling stations and customer connections.

Associated costs — design and supervision, survey and investigation, environmental assessment, consents and land, and physical and price contingency.

7.8.3. Operating Cost

Operating cost is built from the duty of each asset rather than taken as a proportion of its capital cost. For a wastewater system the two largest items are energy and sludge, and neither follows capital value.

Energy — pumping at each lifting station from its duty flow and head, and aeration at the treatment plant from the oxygen demand.

Sludge — thickening, dewatering, transport and disposal, at the quantity the process produces.

Chemicals — coagulant, polymer and disinfectant at the dose rate and the flow treated.

Labour — the establishment required to operate and maintain the plant, the stations and the network.

Maintenance — jetting and closed-circuit television survey of the network, desludging, odour and corrosion control, and planned replacement of mechanical and electrical plant falling within the evaluation period.

Mechanical and electrical plant has a shorter service life than the civil works, so its replacement falls inside the twenty-five year period and is carried as a cost in the year it occurs.¹

¹ PAM-GUD-201 sets a design life of 25 years for assets generally; rotating plant and instrumentation are replaced at least once within that period.

7.8.4. Life-Cycle Cost and Net Present Value

Capital cost, operating cost, replacement cost and revenue arise in different years, so they are not comparable until they are brought to a common date. Each is discounted to present value at the rate the guideline sets, and the options are compared on the resulting totals.¹

$$NPV = \sum_{t=0}^n \frac{C_t}{(1+r)^t} \quad (17)$$

Symbol	Meaning	Unit
NPV	net present value of the option	OMR
C t	net cash flow in year t: revenue and avoided cost less capital, replacement and operating cost	OMR
r	discount rate, 0.05	—
t	year, counted from the base date	—
n	evaluation period, 25 years	—

Total life cycle cost is the present value of capital, replacement and operating cost over the same period, and is the quantity against which the ten per cent band in Section 6.1.4 is applied. Payback is reported alongside it as an indicator, but it does not determine the recommendation: for a wastewater scheme a large part of the benefit is cost avoided rather than revenue earned, and a payback period computed on revenue alone would misstate the case.²

7.9. Risk Management

A risk register is established at this stage and maintained through the project. Each risk carries a description, a likelihood, an impact expressed in cost or time, an owner and the mitigation adopted. Risks that fall on one option and not on another are identified as such, as they affect the comparison rather than only the total.

7.10. Value Engineering

A formal value engineering study is required at the concept and preliminary stages for a treatment plant or pumping station above the stated threshold, carried out by an independent certified consultant. The study is arranged within the concept programme and its outcome reported.³

7.11. Options Appraisal and Recommendation

The options are compared by the method described in Section 6.1.3. The comparison presents, for each option, the capital cost by phase, the operating cost by year, the net present value over twenty-five years, the carbon footprint, and the assessment against each of the remaining criteria, together with the results of the sensitivity tests.

The net present value of each option is computed by the equation of Section 7.8.4, at five per cent over twenty-five years.

Sewer Network Options

The seven sewer network options of Section 6.2 have been compared on their technical results: the plants and the flow each receives, the trunk sewers, the pumping stations with their rising mains, and the energy they use with the lift at the

¹ PAM-GUD-201, page 57, states that the twenty-five year planning life is also the period over which net present value is calculated for the comparison of schemes; the discount rate of five per cent is given at pages 95 to 96.

² Net present value and life cycle cost are not defined by equation in the Nama Water Services guidelines. The formulation above follows ISO 15686-5, with the period and discount rate taken from PAM-GUD-201.

³ PAM-GUD-201, Table 27, page 93, and its accompanying note.

plant inlets (Section 6.2.7). The depth of the network does not separate them. Three are recommended for the cost estimate and the multi-criteria appraisal, each to be costed under the three characters of the guidelines (Section 6.1.1).

Table 52. The three recommended options

	S1	S4	S6
Priority	First	Second	Third
Plants	O1	O1, O4, O9	O1, O3, O4, O9, O16, O22
Largest plant, average 2070, m ³ /d	60,456	36,881	22,867
Sewer at the plants, m deep	10.3	10.0 / 6.9 / 8.8	9.8 / 5.8 / 6.9 / 8.8 / 4.9 / 1.5
Pumping stations	23	21	18
Rising mains, km	43.9	43.0	28.7
Sewers of 1,000 mm and over, km	41	24	18
Largest sewer, mm	2,200	1,800	1,600
Pumping energy with the plant inlets, 2030, MWh a year	1,027	867	705
Pumping energy with the plant inlets, 2070, MWh a year	2,738	2,248	1,788
Rising mains holding the 2030 flow longer than 30 minutes	14	12	10

- S1.** One plant at O1, receiving 24,756 cubic metres a day on average in 2030, 42,635 in 2055 and 60,456 in 2070, so that it is built in phases on one site, with one plant to staff, one sludge line and one source of treated effluent. Its costs are the largest trunk sewers, 41 km of sewer of 1,000 mm and over, up to 2,200 mm, the deepest arrival at a plant, 10.3 m, and the most pumping of the three, 2,738 MWh a year in 2070 with the lift at the plant.
- S4.** Three plants: O1 with 36,881 cubic metres a day in 2070, O4 with 7,935 and O9 with 15,640. The north-eastern subnetworks O4 to O8 and the eastern subnetworks O9 to O14 are treated at their own outfalls instead of being pumped on towards O1. Against S1 it needs 18 per cent less pumping energy, 2,248 MWh a year in 2070, two fewer pumping stations and 17 km less sewer of 1,000 mm and over; the largest sewer is 1,800 mm. Its cost is two further plant sites, each with its buffer, access, power supply and outlet for the treated effluent.
- S6.** Six plants, at O1, O3, O4, O9, O16 and O22. The least pumping of the seven options, 1,788 MWh a year in 2070, 35 per cent below S1, with the fewest pumping stations, 18, and the shortest rising mains, 28.7 km; the largest sewer is 1,600 mm. The plant at O22, 446 cubic metres a day in 2070, is small enough for the nature-based treatment the guideline allows up to about 500 cubic metres a day. Its cost is six sites and six plants to staff, two of them very small in the early years: O22 receives 77 and O16 receives 657 cubic metres a day in 2030.

The three are selected because together they span the choice the options present: all the flow treated at one plant in S1, at three in S4 and at six in S6, so that the appraisal prices the whole range from central to local treatment. S2 is not selected: it treats all the flow at O16 and has to pump the whole of O1's flow there, 4,303 MWh a year in 2070, 57 per cent more than S1. S3, S5 and S7 lie between the recommended options and offer nothing the three do not: S3 and S5 are steps between S1, S4 and S6, and S7 keeps the trunk sewers of S1 while adding two small plants.

Each of the three is designed and costed under each of the three characters of the guidelines, on the same network model: the characters change the treatment process, the energy supply, solar generation included, the reuse and the materials (Section 6.1.2), and the appraisal compares nine cases. The priority rests on the technical results; the cost estimate and the multi-criteria appraisal confirm it or change it. Under the rule of Section 6.1.4, of two cases whose

whole-life costs fall within ten per cent of one another, the more sustainable is preferred. The treated effluent network is designed on the three options, with one source of treated effluent, three or six.¹

¹ PAM-GUD-201, Section 12.1, page 95 (the three characters), and Sections 12.6 to 12.9, pages 104 to 106 (the appraisal).

8. Project Delivery, Conclusions and Recommendations

This chapter covers the delivery of the scheme: the implementation roadmap, the contracting strategy, project integration, and the conclusions and recommendations of this report.

8.1. Implementation Roadmap

An implementation roadmap is prepared for the recommended option, defining the scope of the subsequent design stages, the procurement route for each element, the phasing of construction, and the framework by which performance is monitored once the works are in service.

8.2. Contracting Strategy

The contracting strategy establishes how the works are packaged and procured. It is developed with Nama Water Services in a dedicated workshop, and considers the division of the works into packages, the procurement route for each, and the interfaces between them.

8.3. Project Integration

A project integration plan sets out how the water and wastewater components are coordinated through design, construction and commissioning. It also addresses the rehabilitation or replacement of the existing potable water network where its performance is found to be unsatisfactory.¹

8.4. Conclusions

The design basis for the wastewater and treated effluent systems is established and is set out in Chapter 3. The data supplied has been assessed, and the datasets that are usable, those that require correction and those that relate to areas outside the project have been identified in Chapter 2.

The existing wastewater assets within the study area comprise 111.6 kilometres of constructed gravity sewer and 10.0 kilometres of constructed force main, together with a further 268.2 kilometres of gravity sewer, pumping main and treated effluent main recorded as proposed. The potable water network within the area comprises 647.8 kilometres of mains. The number of properties on each plot and the use of each plot have been established from 33,971 electricity

¹ PAM-GUD-201, Section 13, page 107.

meters and a satellite image, the occupancy rate has been calculated for each settlement, and every plot carries a population and an average sewage flow for 2024 and for every year to the saturation of its settlement. The study area is saturated in 2070 at 349,029 people and 60,099 cubic metres of sewage a day.

The sewer network has been modelled over the whole study area: twenty-four subnetworks, 1,498 km of sewer and 19,083 manholes, designed for the flow of 2070 and within the guideline's limits on velocity and depth of flow in every option and every year. Whichever option is chosen, the plants receive 24,756 cubic metres a day on average in 2030 and 60,456 in 2070, infiltration included, and are designed for 66,502 with the ten per cent margin. The seven options for where the flow is treated, from one plant to six, differ in the trunk sewers, the pumping and the plants, not in the depth of the network: the pumping energy in 2070, with the lift at the plant inlets, runs from 1,788 MWh a year in S6 to 4,303 in S2.

Three are recommended for the appraisal, in order of priority: S1, one plant at O1; S4, three plants at O1, O4 and O9; and S6, six plants. Each is costed under the three characters of the guidelines, nine cases in all (Section 7.11). In every option some long rising mains hold the sewage for hours and need hydrogen sulphide control, the station at O23 pumps a few litres a second against a very high head, and the head pipes do not reach the self-cleansing velocity in any year.

A topographic and utility survey covering the whole study area is in progress. It will establish the levels, diameters and condition that the supplied datasets do not carry, and the assessment of the existing networks follows it.

8.4.1. Recommendations

It is recommended that:

1. options S1, S4 and S6 are taken forward to the cost estimate and the multi-criteria appraisal, in that order of priority, each designed and costed under the three characters of the guidelines, nine cases in all, and the treated effluent network is designed on them (Section 7.11);
2. Nama Water Services takes the six decisions of Section 1.5.2, the design horizon and the overflow first among them, and the decision on the self-cleansing of the head pipes of Section 6.2.8, which extends Decision 6;
3. the connection of O23 and the long rising mains that hold the sewage for more than half an hour are reconsidered at the preliminary design: a shorter route, a nearer receiving subnetwork, local treatment or dosing (Section 6.2.7);
4. Nama Water Services confirms the other matters of Section 1.5.4 and supplies the data requested in Section 1.5.5, so that the options, their cost and the appraisal are completed on an agreed basis.

8.5. Appendices

Appendix	Content
A	Population, land use and flow: the working behind Chapter 4
B	Network options: quantities, pumping stations and depth
C	Design criteria, with references
D	Drawings and figures

The decisions requested, the values adopted and the data requests are set out in Section 1.5.

Appendix A. Population, Land Use and Flow Calculations

A.1. Electricity Tariffs and Consumer Categories

Each tariff in the electricity dataset is assigned to one of the guideline's categories. The large-consumer tariff is resolved account by account in A.2.

Table 53. Tariffs and the category adopted

Tariff	Meters	Category	People
Primary Account Tariff	10,973	Domestic	one property
Primary Account Tariff (with National Subsidy)	5,272	Domestic	one property
Additional Account Tariff	6,344	Domestic	one property
Commercial	9,385	Non-domestic	none
Fisheries	6	Non-domestic	none
Tourism	1	Non-domestic	none
Government	966	Governmental	none
MOD	1	Governmental	none
Agricultural	523	Agricultural	none; no sewage
Industrial	1	Special consumption	none
CRT Seasonal	298	Large consumer, see A.2	none
CRT Time of Use	192	Large consumer, see A.2	none
CRT Fixed Rate	9	Large consumer, see A.2	none

Of the 33,971 meters, 30,931 fall inside a plot, 2,914 were assigned to the nearest plot within fifteen metres, and 126 lie further from any plot; they carry no load (Section 4.1.3).

A.2. Large-Consumer Accounts

The 499 accounts on the Cost Reflective Tariff were placed by clustering them at 120 metres, matching each cluster and each single account against the named features of OpenStreetMap within eighty metres, reverse geocoding the rest, and inspecting the largest clusters on the aerial imagery. Each account carries the use found, the category adopted, the evidence and a confidence.

Table 54. Large-consumer accounts by the use found

Use found	Accounts	Category adopted	Typical sites
Commercial	205	Non-domestic	Bawadi shopping centre, the Al Murtafa souq, the Araqi strip, hypermarkets, banks, fuel stations
Government	67	Governmental	Police headquarters, post office, directorates
Education	37	Governmental	The university campus, the college, schools
Health	15	Governmental	Ibri Hospital, clinics
Religious	16	Governmental	Mosques
Industrial	15	Special consumption	The Tanam estate
Agricultural	28	Agricultural, no load	Farm pumps at Al Aynayn
Telecommunications, utility	6	Non-domestic	Exchanges, water facilities, masts
Unresolved	110	Non-domestic	No recorded feature within eighty metres

Confidence: 119 accounts certain, 194 likely, 186 placed on weaker evidence. The categories adopted affect only where the non-domestic and governmental water is placed, not its total.¹

A.3. Settlement Parameters: Occupancy, Properties per Plot, Home Share and Capacity

Table 55. The settlement rates used in Chapter 4: calculated from the data, and adopted

Settlement	Domestic properties, 2024	Pop. 2024	Empty plots	of which counted as future homes	Occupancy rate calculated	adopted	Properties per home plot calculated	adopted	Home share calculated	adopted	Capacity, people
Ibri	11,052	67,106	14,203	10,469	6.07	6.07	1.33	1.33	0.87	0.87	73,299
Ad Dariz	2,439	11,850	4,888	3,219	4.86	4.86	1.20	1.20	0.90	0.90	16,854
Al Araqi	2,466	10,696	3,360	2,463	4.34	4.34	1.32	1.32	0.84	0.84	11,832
Al Aynayn	889	4,554	1,780	1,362	5.12	5.12	1.14	1.14	0.94	0.94	7,459
Al Wahrah	716	3,351	2,000	1,373	4.68	4.68	1.22	1.22	0.91	0.91	7,109
At Tayyib	833	3,348	10,012	8,640	4.02	4.02	1.09	1.09	0.92	0.92	34,675
Al Jibayyah	524	2,686	2,784	1,966	5.13	5.13	1.26	1.26	0.89	0.89	11,306
Bat	418	2,557	879	355	6.12	6.12	1.17	1.17	0.88	0.88	2,226
Ad Dibayshi	681	2,411	2,430	1,887	3.54	4.00	1.16	1.16	0.90	0.90	7,920
Tanam	403	2,116	2,045	1,556	5.25	5.25	1.24	1.24	0.67	0.67	6,779
Suwayda al Ma	301	1,559	1,593	937	5.18	5.18	1.60	1.60	0.82	0.82	6,369
Hijar	352	1,033	375	211	2.93	4.00	1.69	1.69	0.90	0.90	1,288
Al Ghubayrah	161	631	1,448	1,094	3.92	4.00	1.12	1.00	0.84	0.90	3,940
Al Qurayn	147	549	2,051	1,830	3.73	4.00	1.14	1.00	0.93	0.90	6,588
Sayh al Masarrat	457	466	1,745	1,384	1.02	4.00	2.52	1.00	0.92	0.90	4,984
Al Jahli	344	415	1,235	1,023	1.21	4.00	2.16	1.00	0.78	0.90	3,684
Satwah	27	263	181	79	9.73	4.00	1.23	1.00	0.87	0.90	284
Al Akheedar	120	198	120	45	1.65	4.00	2.12	1.00	0.53	0.90	160
Al Qali	56	191	220	157	3.41	4.00	2.20	1.00	0.92	0.90	564
Shalashil	38	130	3,686	3,250	3.42	4.00	1.11	1.00	0.91	0.90	11,700
Al Makhtibiyah	91	122	1,523	1,192	1.34	4.00	1.15	1.00	0.90	0.90	4,292
Usaybuq	21	91	60	39	4.33	4.00	1.25	1.00	0.87	0.90	140
Wadi al Mankas	8	52	1,580	1,394	6.49	4.00	1.30	1.00	0.87	0.90	5,020

¹ Certain: a named feature within twenty metres, or inspected. Likely: a named feature within eighty metres, or a nearest feature by reverse geocoding. Weaker: the surroundings only.

Settlement	Domestic properties, 2024	Pop. 2024	Empty plots	of which counted as future homes	Occupancy rate calculated	adopted	Properties per home plot calculated	adopted	Home share calculated	adopted	Capacity, people
Ash Shiab	11	47	65	13	4.27	4.00	1.10	1.00	0.87	0.90	48
Miayrid	4	34	246	171	8.49	4.00	1.30	1.00	0.87	0.90	616
Total	22,559	116,452	60,509	46,109							229,136

The occupancy rate adopted is the rate calculated, held to a floor of 4.0 and a cap equal to the highest rate among the settlements of two thousand or more people. A settlement of fewer than a thousand people in 2024 takes an occupancy rate of 4.0, one property per plot and a home share of 0.9, whatever its meters return: it has no sample to measure on, and it does not attract a second property on a plot.¹

A.4. Population Overflow Routes

Every route carrying fifty people or more at saturation. The donor's year is the year its own empty plots are full; the receiver starts in the first year it takes in overflow from any settlement, which can be before this donor is full, and is itself full in the last column.

Table 56. Overflow routes

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Ibri	At Tayyib	22,078	2056	2060	2068
Ibri	Shalashil	11,471	2056	2056	2061
Ibri	Al Qurayn	5,856	2056	2056	2060
Ad Dariz	Wadi al Mankas	4,965	2062	2062	2068
Al Araqi	At Tayyib	4,873	2057	2060	2068
Ibri	Ad Dibayshi	4,078	2056	2062	2063
Ibri	Al Jibayyah	3,927	2056	2068	2070
Bat	Al Ghubayrah	2,493	2052	2052	2070
Ibri	Al Makhtibyah	2,441	2056	2063	2069
Ibri	Tanam	2,010	2056	2063	2069
Al Araqi	Al Aynayn	1,790	2057	2057	2060
Al Aynayn	At Tayyib	1,639	2060	2060	2068
Al Araqi	Al Jibayyah	1,410	2057	2068	2070
Ibri	Sayh al Masarrat	1,225	2056	2063	2069
Ad Dibayshi	Tanam	1,004	2063	2063	2069
Ibri	Al Jahli	915	2056	2054	2063
Ad Dariz	Al Wahrah	863	2062	2062	2070
Al Araqi	Suwayda al Ma	730	2057	2070	2070
Al Aynayn	Al Jibayyah	600	2060	2068	2070
Ad Dariz	Suwayda al Ma	551	2062	2070	2070
Hijar	Al Makhtibyah	519	2053	2063	2069
Hijar	Al Jahli	518	2053	2054	2063
Al Jahli	Al Makhtibyah	507	2063	2063	2069
Ibri	Suwayda al Ma	490	2056	2070	2070
At Tayyib	Miayrid	441	2068	2068	2070
Ibri	Ad Dariz	440	2056	2057	2062
Al Aynayn	Ad Dariz	347	2060	2057	2062

¹ Properties per home plot is measured over built plots whose meters are more than two thirds domestic and fewer than fifteen. Home share is the proportion of built, metered plots of home shape that are residential. Empty plots counted as future homes are those of 200 to 1,000 square metres, compact, not a strip, and not planted, industrial or heritage.

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Al Aynayn	Suwayda al Ma	311	2060	2070	2070
Al Akheedar	Al Qali	297	2036	2036	2054
Ad Dariz	Al Ghubayrah	258	2062	2052	2070
Al Qurayn	At Tayyib	244	2060	2060	2068
At Tayyib	Suwayda al Ma	229	2068	2070	2070
Al Wahrah	Suwayda al Ma	229	2070	2070	2070
Ad Dibayshi	Suwayda al Ma	186	2063	2070	2070
Al Jibayyah	Suwayda al Ma	183	2070	2070	2070
Ad Dibayshi	Sayh al Masarrat	182	2063	2063	2069
Al Akheedar	Al Makhtibyah	177	2036	2063	2069
Al Akheedar	Al Jahli	176	2036	2054	2063
Bat	Suwayda al Ma	175	2052	2070	2070
Ibri	Al Araqi	163	2056	2056	2057
Al Aynayn	Satwah	150	2060	2060	2060
Tanam	Suwayda al Ma	145	2069	2070	2070
Ibri	Miayrid	141	2056	2068	2070
Tanam	Sayh al Masarrat	141	2069	2063	2069
Sayh al Masarrat	Suwayda al Ma	125	2069	2070	2070
Al Qali	Al Jahli	103	2054	2054	2063
Hijar	Al Jibayyah	94	2053	2068	2070
Al Jahli	Al Jibayyah	92	2063	2068	2070
Al Qali	Sayh al Masarrat	83	2054	2063	2069
Al Qurayn	Al Jibayyah	78	2060	2068	2070
Hijar	Ibri	66	2053	2054	2056
Shalashil	At Tayyib	63	2061	2060	2068
Hijar	Usaybuq	60	2053	2053	2054
Hijar	Al Qali	54	2053	2036	2054
Sayh al Masarrat	Al Jibayyah	52	2069	2068	2070

Appendix B. Network Options: Quantities, Pumping Stations and Depth

This appendix gives the quantities of pipe and manholes the options are compared on, the pumping stations of each option, and the depth of each option's 2070 design on a map.

Table 57. Length of sewer by outside diameter, metres

Outside diameter, mm	S1	S2	S3	S4	S5	S6	S7
200	1,280,968	1,279,740	1,283,883	1,287,417	1,289,629	1,292,294	1,283,633
250	56,194	55,284	56,194	56,194	56,194	54,609	54,609
280	25,069	23,523	26,225	26,225	26,573	25,894	24,389
315	9,068	9,068	9,068	9,068	9,845	9,845	9,068
355	15,420	15,420	19,694	19,910	19,910	19,510	15,020
450	10,362	8,801	10,362	10,362	10,362	10,362	10,362
500	14,345	14,345	14,345	14,345	14,345	14,345	14,345
560	13,194	13,108	13,194	13,194	13,194	13,194	13,194
630	7,615	7,615	7,615	7,615	7,615	7,615	7,615
710	12,987	12,987	12,987	14,911	14,911	17,561	15,637
800	6,965	6,965	6,965	9,893	10,919	8,269	4,315
900	4,916	4,916	4,916	4,916	6,849	6,939	5,006
1000	12,540	12,540	9,626	9,626	13,455	13,365	12,450
1100	7,424	7,424	5,332	5,332	1,994	1,994	7,424
1200	500	500	1,154	1,154	2,342	2,342	500
1400	12,053	12,053	14,850	6,249	113	113	12,053
1600	6,789	6,789		1,842	43	43	6,789
1800			1,842	43			
2000	1,842	1,842	43				1,842
2200	43	5,376					43
Total	1,498,295	1,498,295	1,498,295	1,498,295	1,498,295	1,498,295	1,498,295

Lengths are rounded to the metre, so a total can differ by a metre or two from the sum of its rows.

Table 58. Manholes by depth to invert

Depth	S1	S2	S3	S4	S5	S6	S7
0 to 1.5 m	3,315	3,314	3,315	3,316	3,317	3,318	3,316
1.5 to 3 m	8,538	8,526	8,532	8,522	8,514	8,503	8,527
3 to 4.5 m	3,065	3,082	3,046	3,038	3,037	3,038	3,066
4.5 to 6 m	1,698	1,706	1,714	1,704	1,712	1,721	1,707
6 to 9 m	1,886	1,900	1,896	1,923	1,929	1,929	1,886
9 to 12 m	541	515	540	540	534	534	541
over 12 m	40	40	40	40	40	40	40
Total	19,083	19,083	19,083	19,083	19,083	19,083	19,083

Table 59. Length of sewer by depth to invert, metres

Depth	S1	S2	S3	S4	S5	S6	S7
0 to 1.5 m	41,778	41,778	41,778	41,778	41,778	41,778	41,778
1.5 to 3 m	867,034	865,952	866,573	865,476	864,265	863,149	865,919
3 to 4.5 m	251,501	253,109	250,019	249,651	249,926	250,149	251,724
4.5 to 6 m	139,247	139,811	140,151	139,189	140,211	141,103	140,139
6 to 9 m	153,247	154,563	154,349	156,776	157,012	157,012	153,247
9 to 12 m	42,330	39,923	42,267	42,267	41,945	41,945	42,330
over 12 m	3,158	3,158	3,158	3,158	3,158	3,158	3,158

The pumping stations of each option follow, largest duty first. The duty is the 2070 peak flow of everything that reaches the station; the head is the static lift plus the friction in the rising main; the retention is the time the 2030 average flow takes to pass through the main. The figures rest on the concept values of Section 6.2.7.

Table 60. Pumping stations of option S1

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O2	O1-M1519	636.2	900	43	5.1	5.1	49.2	106.0	196.4	0.0
O3	O2-M696	485.6	700	17	5.1	5.2	38.0	68.1	149.9	0.0
O9	O1-M2472	384.8	700	149	7.1	7.3	42.4	27.8	174.6	0.6
O4	O3-M3205	222.5	500	666	6.2	8.2	27.5	44.7	99.4	0.9
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	125.7	400	2,032	8.8	15.0	28.5	50.7	103.9	2.8
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.5	2.7	2.9	4.7	8.2	0.8
O16	STP at O1	70.7	300	4,756	20.2	40.7	43.4	34.0	112.6	14.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O22	O21-M559	17.7	150	2,407	16.7	40.0	10.7	3.3	19.2	18.8
O20	O16-M44	17.7	150	601	8.5	14.3	3.8	4.0	6.5	1.4
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O1	17.7	150	8,785	24.3	109.2	29.1	11.1	77.0	56.3
O18	STP at O1	17.7	150	7,911	19.5	96.0	25.6	6.7	49.1	73.1

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O23	O21-M559	5.0	75	9,455	44.6	300.7	22.6	10.7	60.7	43.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Table 61. Pumping stations of option S2

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O1	O16-M152	1,450.5	1200	2,965	19.9	23.9	523.2	881.3	2,115.8	3.3
O2	O1-M1519	636.2	900	43	5.1	5.1	49.2	106.0	196.4	0.0
O3	O2-M696	485.6	700	17	5.1	5.2	38.0	68.1	149.9	0.0
O9	O1-M2472	384.8	700	149	7.1	7.3	42.4	27.8	174.6	0.6
O4	O3-M3205	222.5	500	666	6.2	8.2	27.5	44.7	99.4	0.9
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	125.7	400	2,032	8.8	15.0	28.5	50.7	103.9	2.8
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.5	2.7	2.9	4.7	8.2	0.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O22	O21-M559	17.7	150	2,407	16.7	40.0	10.7	3.3	19.2	18.8
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O16	17.7	150	7,541	13.4	86.3	23.0	8.7	60.9	48.4
O18	STP at O16	17.7	150	4,418	8.6	51.3	13.7	3.6	26.2	40.8
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O20	O16-M44	10.6	100	601	11.3	27.4	4.4	7.7	12.5	0.6
O23	O21-M559	5.0	75	9,455	44.6	300.7	22.6	10.7	60.7	43.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Table 62. Pumping stations of option S3

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O2	O1-M1519	413.6	700	43	5.1	5.2	32.2	78.4	134.9	0.0
O9	O1-M2472	384.8	700	149	7.1	7.3	42.4	27.8	174.6	0.6
O3	O2-M696	263.1	500	17	5.4	5.5	21.7	42.0	91.9	0.0
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	125.7	400	2,032	8.8	15.0	28.5	50.7	103.9	2.8
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.5	2.7	2.9	4.7	8.2	0.8
O16	STP at O1	70.7	300	4,756	20.2	40.7	43.4	34.0	112.6	14.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O22	O21-M559	17.7	150	2,407	16.7	40.0	10.7	3.3	19.2	18.8
O20	O16-M44	17.7	150	601	8.5	14.3	3.8	4.0	6.5	1.4
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O1	17.7	150	8,785	24.3	109.2	29.1	11.1	77.0	56.3
O18	STP at O1	17.7	150	7,911	19.5	96.0	25.6	6.7	49.1	73.1
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O23	O21-M559	5.0	75	9,455	44.6	300.7	22.6	10.7	60.7	43.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Table 63. Pumping stations of option S4

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O2	O1-M1519	413.6	700	43	5.1	5.2	32.2	78.4	134.9	0.0
O3	O2-M696	263.1	500	17	5.4	5.5	21.7	42.0	91.9	0.0
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	125.7	400	2,032	8.8	15.0	28.5	50.7	103.9	2.8
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.5	2.7	2.9	4.7	8.2	0.8
O16	STP at O1	70.7	300	4,756	20.2	40.7	43.4	34.0	112.6	14.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O22	O21-M559	17.7	150	2,407	16.7	40.0	10.7	3.3	19.2	18.8
O20	O16-M44	17.7	150	601	8.5	14.3	3.8	4.0	6.5	1.4
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O1	17.7	150	8,785	24.3	109.2	29.1	11.1	77.0	56.3
O18	STP at O1	17.7	150	7,911	19.5	96.0	25.6	6.7	49.1	73.1
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O23	O21-M559	5.0	75	9,455	44.6	300.7	22.6	10.7	60.7	43.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Table 64. Pumping stations of option S5

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O2	O1-M1519	150.5	400	43	5.2	5.4	12.2	40.5	50.4	0.0
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	125.7	400	2,032	8.8	15.0	28.5	50.7	103.9	2.8
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.5	2.7	2.9	4.7	8.2	0.8
O16	STP at O1	70.7	300	4,756	20.2	40.7	43.4	34.0	112.6	14.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O22	O21-M559	17.7	150	2,407	16.7	40.0	10.7	3.3	19.2	18.8
O20	O16-M44	17.7	150	601	8.5	14.3	3.8	4.0	6.5	1.4
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O1	17.7	150	8,785	24.3	109.2	29.1	11.1	77.0	56.3
O18	STP at O1	17.7	150	7,911	19.5	96.0	25.6	6.7	49.1	73.1
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O23	O21-M559	5.0	75	9,455	44.6	300.7	22.6	10.7	60.7	43.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Table 65. Pumping stations of option S6

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O2	O1-M1519	150.5	400	43	5.2	5.4	12.2	40.5	50.4	0.0
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	104.8	350	2,032	8.8	17.4	27.5	56.7	108.4	2.2
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.6	2.8	3.0	4.8	8.5	0.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O20	O16-M44	17.7	150	601	8.5	14.3	3.8	4.0	6.5	1.4
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O16	17.7	150	7,541	13.4	86.3	23.0	8.7	60.9	48.4
O18	STP at O16	17.7	150	4,418	8.6	51.3	13.7	3.6	26.2	40.8
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O23	STP at O22	5.0	75	7,048	33.9	224.7	16.9	8.0	45.4	32.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Table 66. Pumping stations of option S7

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O2	O1-M1519	636.2	900	43	5.1	5.1	49.2	106.0	196.4	0.0
O3	O2-M696	485.6	700	17	5.1	5.2	38.0	68.1	149.9	0.0
O9	O1-M2472	384.8	700	149	7.1	7.3	42.4	27.8	174.6	0.6
O4	O3-M3205	222.5	500	666	6.2	8.2	27.5	44.7	99.4	0.9
O5	O4-M1421	125.7	400	103	6.6	7.0	13.2	16.1	43.3	0.2
O21	O1-M445	104.8	350	2,032	8.8	17.4	27.5	56.7	108.4	2.2
O10	O9-M1184	100.1	350	44	11.9	12.0	18.2	11.1	61.4	0.2
O6	O5-M191	71.5	300	64	6.5	6.8	7.4	11.8	25.3	0.1
O24	O1-M445	70.7	300	510	0.6	2.8	3.0	4.8	8.5	0.8
O11	O10-M148	64.4	250	28	8.9	9.1	8.9	7.1	29.9	0.1

Station	Discharges to	Duty, l/s	Main, mm	Main, m	Static lift, m	Head, m	kW	MWh a year 2030	MWh a year 2070	Retention 2030, h
O7	O6-M93	49.1	250	62	10.2	10.5	7.8	10.7	26.2	0.1
O15	O1-M2227	49.1	250	1,218	3.2	9.7	7.2	2.6	17.3	8.3
O12	O11-M96	42.4	200	66	9.8	10.6	6.8	6.3	23.5	0.1
O8	O5-M130	20.4	150	3,375	20.1	62.6	19.2	17.2	61.1	8.0
O20	O16-M44	17.7	150	601	8.5	14.3	3.8	4.0	6.5	1.4
O19	O16-M73	17.7	150	1,101	12.3	22.9	6.1	7.0	12.5	2.4
O17	STP at O16	17.7	150	7,541	13.4	86.3	23.0	8.7	60.9	48.4
O18	STP at O16	17.7	150	4,418	8.6	51.3	13.7	3.6	26.2	40.8
O14	O9-M425	11.8	100	99	12.4	15.7	2.8	3.7	8.2	0.1
O23	STP at O22	5.0	75	7,048	33.9	224.7	16.9	8.0	45.4	32.1
O13	O12-M440	4.4	75	363	3.6	11.5	0.8	0.1	1.2	6.2

Option S1: depth of the sewers to invert

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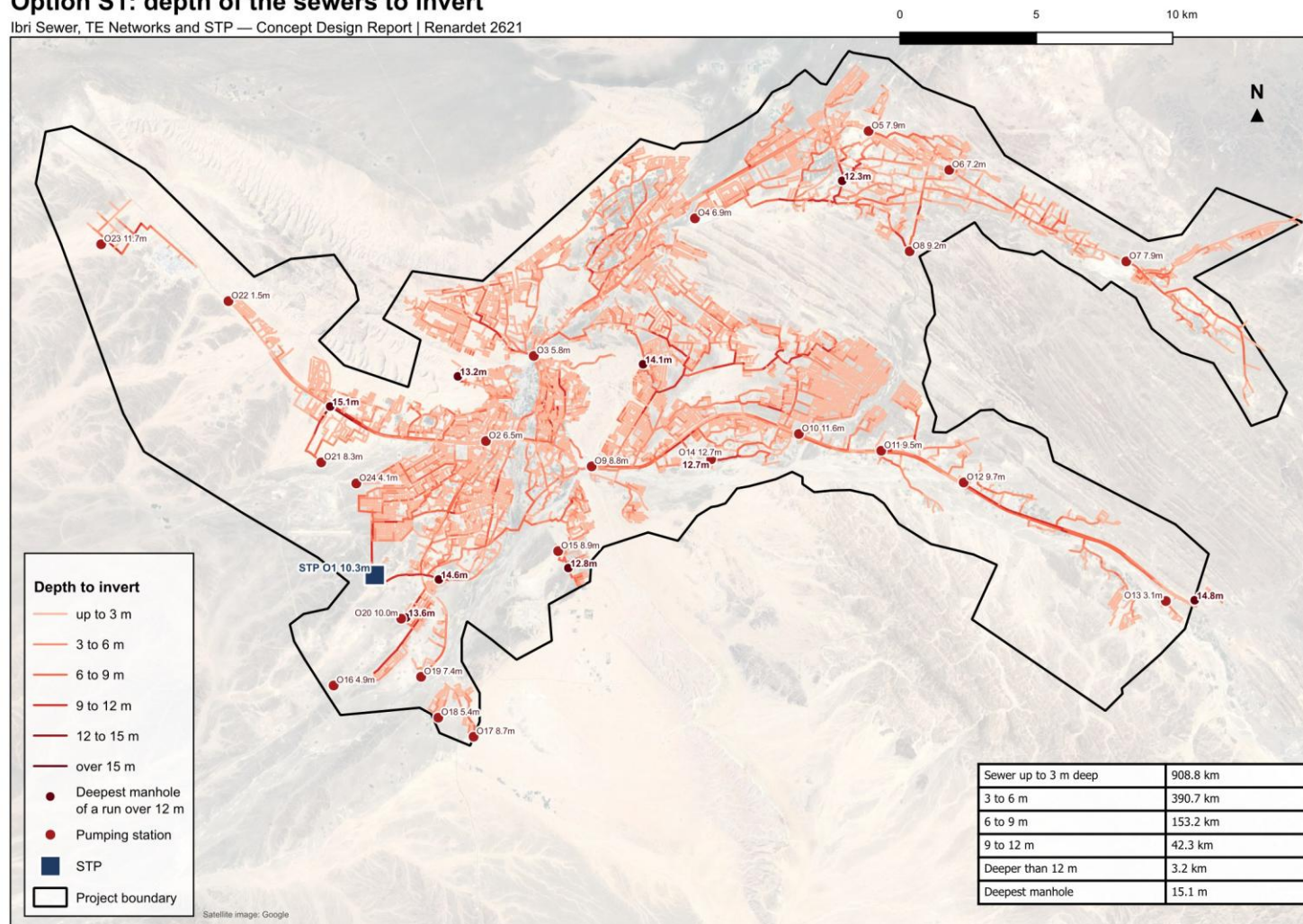


Figure 82. Option S1: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

Option S2: depth of the sewers to invert

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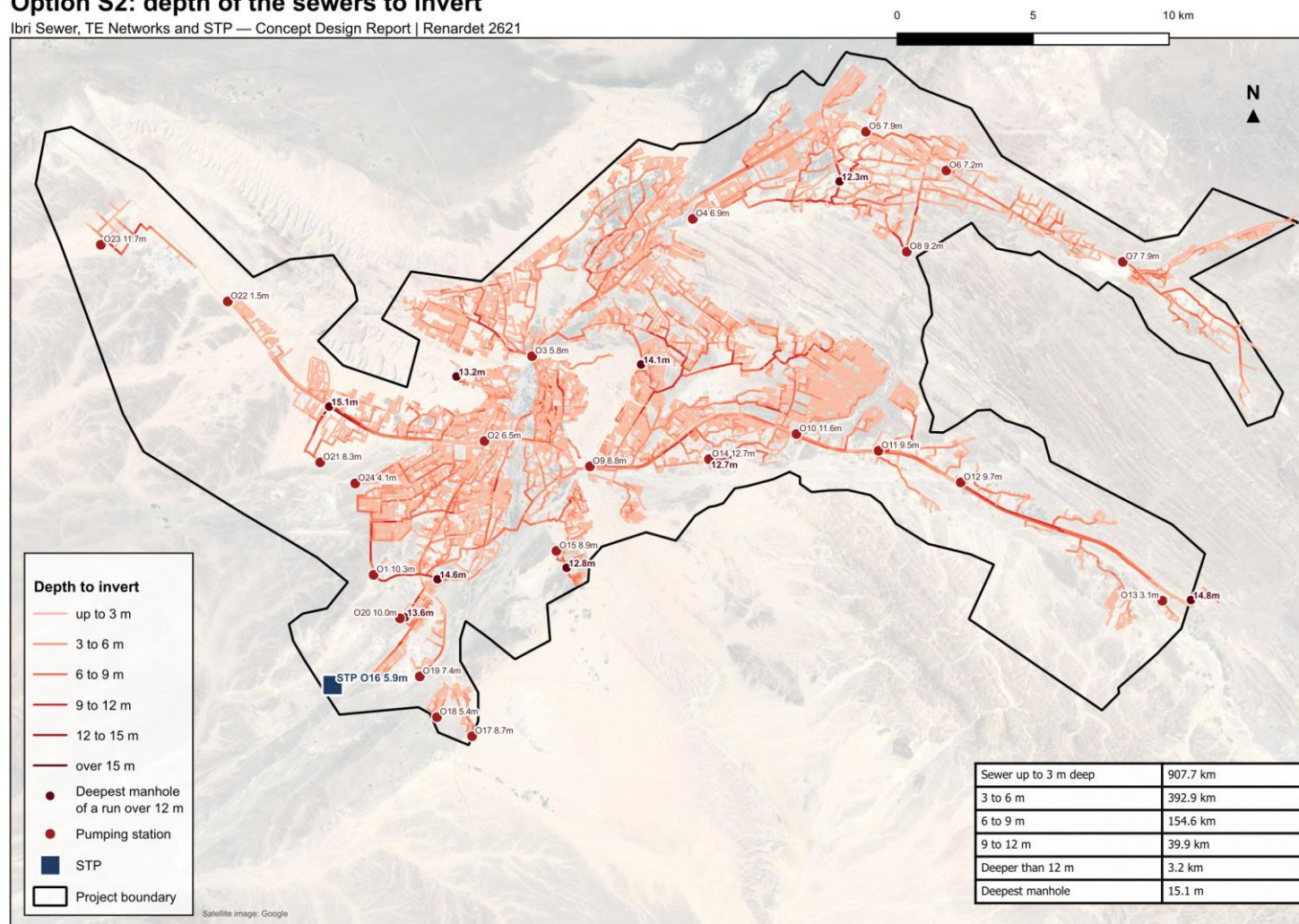


Figure 83. Option S2: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

Option S3: depth of the sewers to invert

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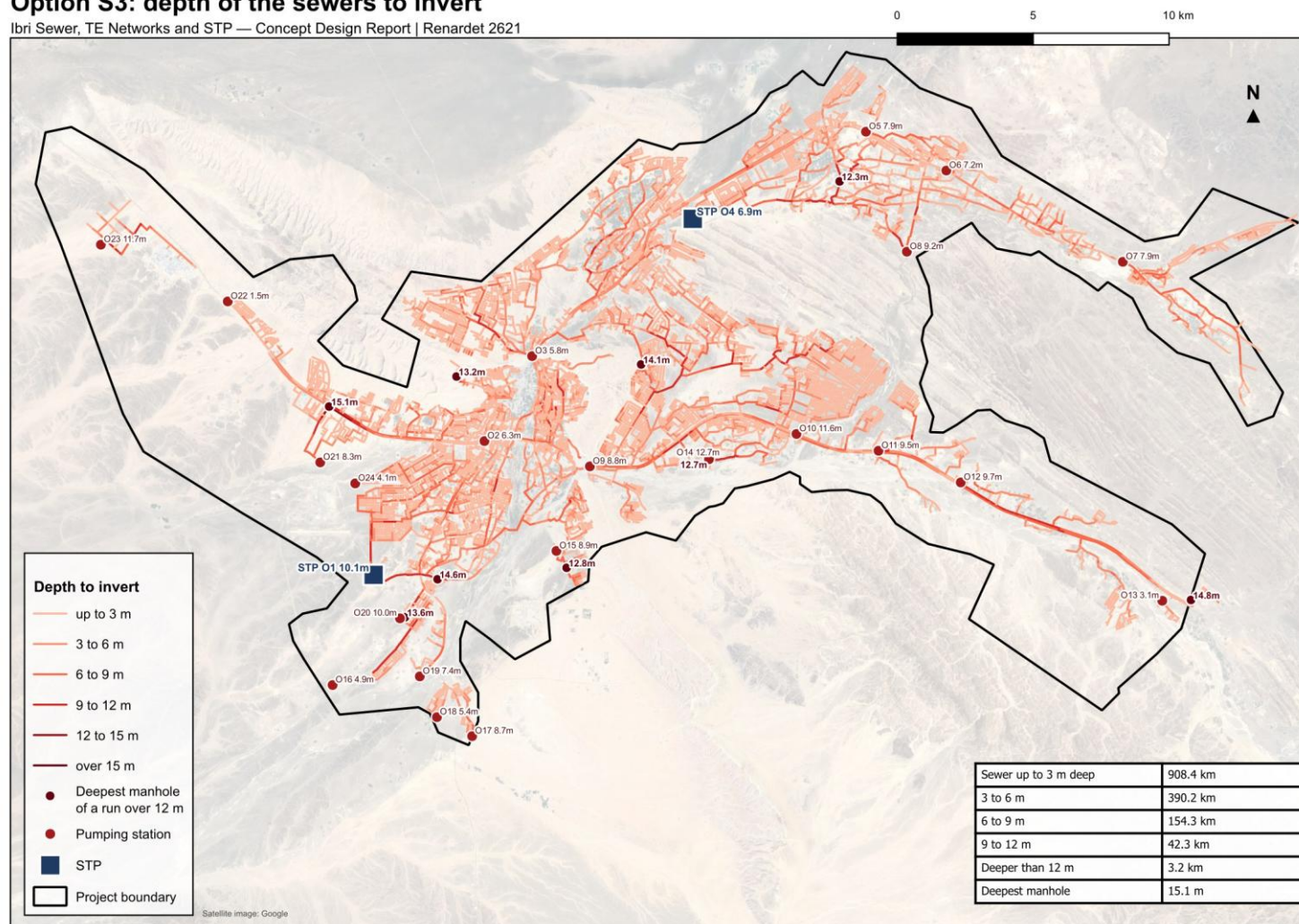


Figure 84. Option S3: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

Option S4: depth of the sewers to invert

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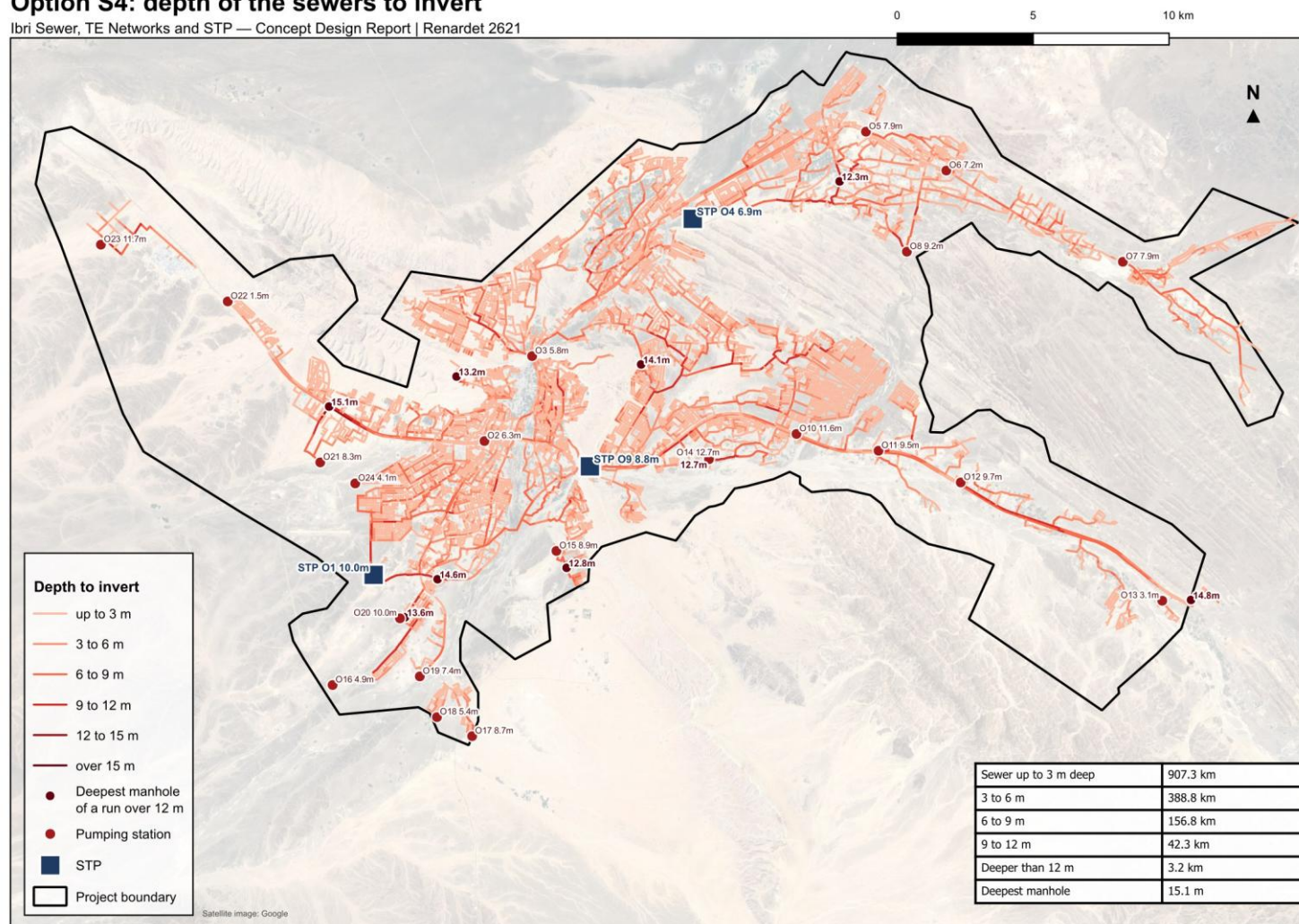


Figure 85. Option S4: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

Option S5: depth of the sewers to invert

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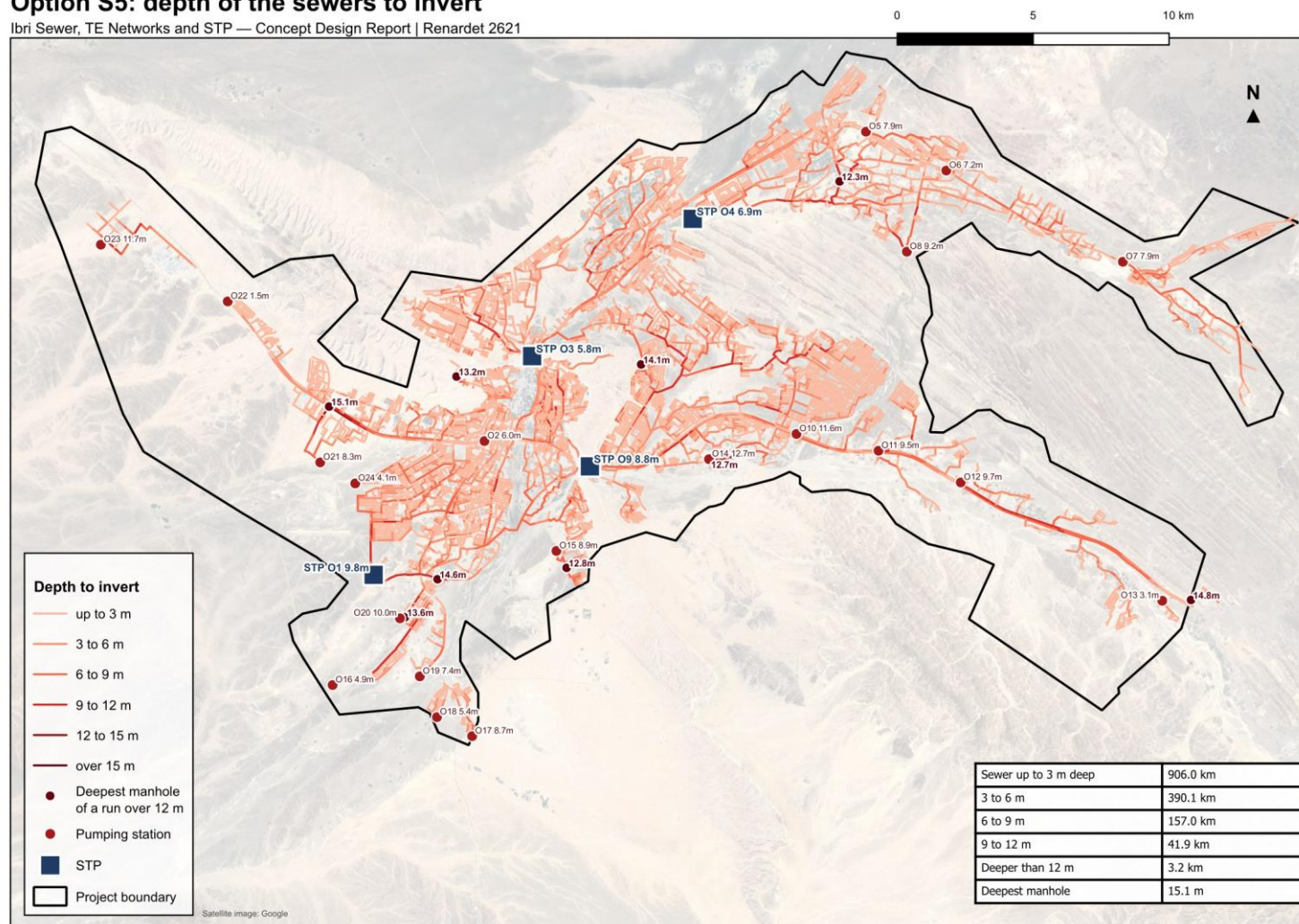


Figure 86. Option S5: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

Option S6: depth of the sewers to invert

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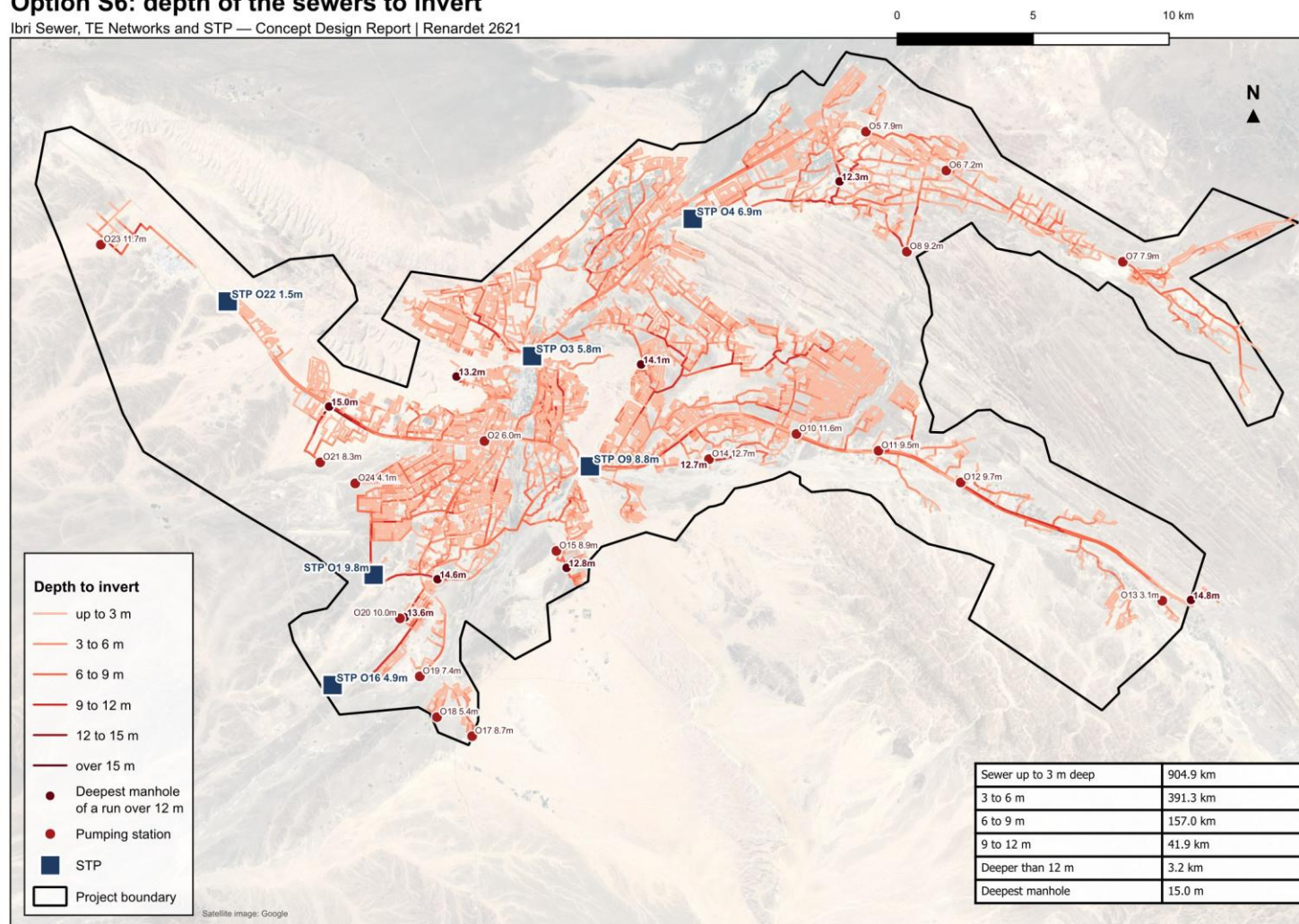


Figure 87. Option S6: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

Option S7: depth of the sewers to invert

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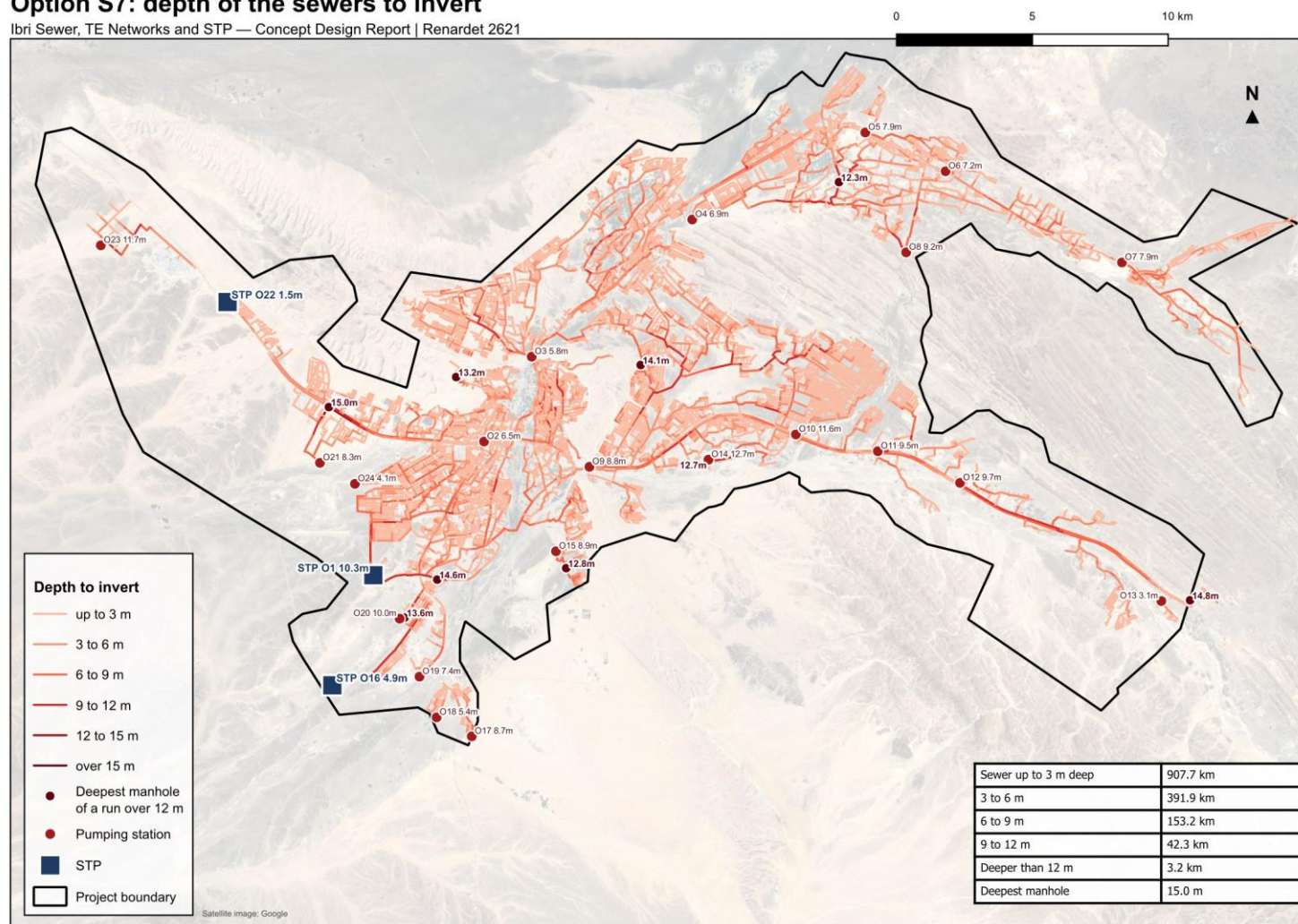


Figure 88. Option S7: depth of the sewers to invert in the 2070 design, the depth of each pumping station's outfall, and the deepest manhole of every run deeper than 12 m.

